BB512 - Population Biology and Evolution

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Welcome to BB512

Welcome to the Population and Evolution course. The course is divided into five teaching parts:

1. Evolution by Natural Selection
2. Population Growth Models
3. Population Genetics and Evolution
4. Interactions Between Species and Community Structure
5. Animal Behaviour, Altruism and Sexual Selection

The recommended textbook is Neal (2019). Other useful textbooks, that are NOT mandatory, but could be useful are Gotelli (2008) and Stearns & Hoekstra (2005).

- Neal, D. (2019). *Introduction to Population Biology* (2nd edition). Cambridge: Cambridge University Press. https://doi.org/10.1017/9781139107976
- Gotelli, NJ (2008) *A Primer of Ecology.* Fourth Edition, Sinauer Associates. ISBN: 978-0878933181
- Stearns, SC & Hoekstra, RF (2005) *Evolution: An Introduction.* 2nd Edition, Oxford University Press. ISBN: 978-0199255634

In addition to this, I will provide a reading list of journal articles that are relevant to the material. These are optional, but highly recommended because they give extra context and background to the course and show you how the theory is used in recent research.

How to use this book

Most chapters contain the same building blocks, so you can move quickly between concepts and tasks:

- Background: the core concepts and equations.
- Learning outcomes: what you should be able to do by the end.
- Your task: the practical exercise or analysis.
- Takeaways: a short summary of the main ideas.

Some chapters also have a worked example that walks through the method before you try it yourself. Answers to the questions are in the Solutions part, so try the questions first and then check.

If you are short on time, read the Learning outcomes and Takeaways first, then do the Your task section.

Parts overview

The table below shows how each part of this website maps onto the major skills you will build through the course. Parts 6 and 7 hold the solutions and some optional extras.

Part	Focus	Skills you build
1. Evolution by Natural Selection	Selection, adaptation, variation	Explain selection in natural contexts; identify selective pressures and outcomes
2. Population Growth Models	Exponential and logistic growth, stochasticity	Build and interpret population growth models; reason about parameter effects
3. Population Genetics and Evolution	Allele frequencies, drift, heritability	Analyse genetic change over time; estimate heritability from data
4. Interactions Between Species and Community Structure	Competition and predator-prey models	Evaluate stability and dynamics of interacting species
5. Animal Behaviour, Altruism and Sexual Selection	Game theory and behaviour	Interpret behaviour through evolutionary payoffs and strategies
6. Solutions/answers	Answer keys and model results	Check work and compare interpretations
7. Appendix - extras	Extensions and advanced examples	Apply methods to additional contexts

This website and other course materials

This website holds most of the materials you need for the practical exercises you will do during the course. You will also find the Schedule here (on the web-version only).

We may not do ALL of these exercises, but you are welcome to do ones we miss in your own time.

You will find other materials via itsLearning.

For most classes you will need your personal laptops: Please bring them to class (and remember a power supply!).

Expectations

There are lectures and practical exercise sessions on the course. The practical sessions are designed to help you understand the subject better and I expect students to attend and actively participate in both. There will also be some multiple-choice quizzes (e-tests) during the semester. These are intended to help you figure out whether you know the material, and whether there are areas you need to revisit. They do not contribute to your final grade, but I hope you will attempt them. They will definitely increase your understanding of the material. Note that the final assessment will be a similar format!

I expect students to make every effort to keep up with the core reading (mainly the textbook chapters), and to ask questions where they don't understand.

Assessment

The assessment for the course will be an electronic exam held in January with multiple-choice and short answer questions. It is worth noting that exam format will be similar to the quizzes mentioned above. The date for the exam will be finalised in December.

Instructors

The instructors of the course are:

- Owen Jones, Associate Professor, jones@biology.sdu.dk
- Thomas Bjørneboe Berg, Associate Prof./Senior Scientist at Naturama, thomas@naturama.dk NULL

If you have any problems accessing materials, or have any questions regarding the course, feel free to send me (Owen) an email. You can also make an appointment to see me via Zoom or in my office if necessary. PLEASE REMEMBER to tell me which course you are referring to if you send an email!

Owen Jones, course coordinator - jones@biology.sdu.dk

Office location: V12-410b-2

Software

We will use Excel and R/RStudio during this course.

Excel

I expect you will already have Excel installed, so there is not much to say here.

Be aware that Excel differs depending on the language it is set up in (for example, Danish vs English). The practicals give formulas in English Excel. In Danish Excel:

- Function names are translated. The ones used most in this course are:

English	Danish
`EXP`	`EKSP`
`SLOPE`	`HÆLDNING`
`RAND`	`SLUMP`
`SQRT`	`KVROD`
`IF`	`HVIS`

- The decimal separator is a comma (`0,05` rather than `0.05`).
- Arguments are separated by semicolons rather than commas, e.g. `=HÆLDNING(C10:C30;A10:A30)` rather than `=SLOPE(C10:C30,A10:A30)`.

See here for more translations.

R and RStudio

R and RStudio are two separate pieces of software. RStudio is a user-friendly interface to talk to R, it cannot work if you have not got R installed. So, when you install these two programs, install R first, then RStudio.

Already have them installed? I strongly recommend updating to the latest versions of R and RStudio, which you can download at the following sites:

- R: https://cran.r-project.org
- RStudio: https://www.rstudio.com/products/rstudio/.

To check which version of R you have, type this in the RStudio console:

```
R.version.string
```

Compare the version number with the latest release listed at https://cran.r-project.org. If yours is more than one minor version behind (for example 4.3 when the latest is 4.5), please update before the course starts. Some practicals install add-on packages, and these often need a recent version of R.

This book includes equations.

Equations sometimes do not render correctly on some web browsers.

From experience, Chrome tends to be the most reliable, Safari (Mac only) also works well.

Can you see the following equation?

$N_{t+1} = N_t + r_m N_t \left(1 - \frac{N_t}{K}\right)$

It should look like this:

$$N_{t+1} = N_t + r_m N_t \left(1 - \frac{N_t}{K}\right)$$

If it doesn't look right, then you should try another browser (maybe even your phone browser) or investigate a solution.

Schedule

The formatted schedule table appears only in the online version of this book.

- View the live schedule page: https://jonesor.github.io/BB512_Book/schedule.html
- Official SDU timetable (always up to date): https://mitsdu.sdu.dk/skema/activity/N100007101/e26
- For dates, times and rooms, also check itsLearning.

Part 1: Evolution by Natural Selection

This part introduces the core logic of natural selection using simple, concrete activities. You will move from a conceptual model of selection to an interactive simulation of adaptation.

By the end of Part 1, you should be able to:

- Explain how selection can generate order from random variation.
- Identify the four conditions for evolution by natural selection (individuals vary in a trait, they vary in reproductive success, the trait affects reproductive success, and the trait is inherited), and the role of the environment in adaptation.
- Describe how selection pressures shape trait distributions over time.

Reading: Neal, chapters *Darwin Questions the Fixity of Species*, *Darwin's Evolutionary Theories* and *Understanding Natural Selection*.

The Blind Watchmaker

Background

If a pocket watch is found on a heath, it is most reasonable to assume that someone dropped it and that it was made by at least one watchmaker, not by natural forces.

This is the "watchmaker analogy", made famous by the theologian William Paley in his book *Natural Theology* (1802), more than 50 years before Darwin published *On the Origin of Species*. Paley argued that, just as finding a pocket watch would naturally lead one to infer a skilled watchmaker, the intricate structures and adaptations of living organisms (such as the eye) must also imply the existence of an intelligent designer. Darwin read Paley as a student at Cambridge and admired the argument; his theory of natural selection was, in part, an answer to it.

More than a century later, the evolutionary biologist Richard Dawkins took up the same image in his 1986 book *The Blind Watchmaker*. Dawkins argued that the complexity and diversity of life can arise through a "blind" process, natural selection, which is unguided and has no foresight. Evolution, like a blind watchmaker, can achieve impressive complexity through small, incremental changes selected over long periods of time.

Randomness and Selection

In this exercise, we will explore how random variation, when paired with selection, can lead to outcomes that would otherwise seem impossible. We will use an R program to simulate the typing of a target sentence through random letter generation. This is a simplified version of a famous computer program described by Dawkins, who used the target phrase "METHINKS IT IS LIKE A WEASEL" (a line from Shakespeare's *Hamlet*).

The key idea is the difference between two kinds of selection that Dawkins described:

- **Single-step selection:** the whole phrase is produced at random in one go, and kept only if it is completely right. Every attempt starts again from scratch.
- **Cumulative selection:** each generation starts from the previous one, and improvements are kept and built on.

By simulating this process, you will discover how random changes alone produce nonsense, but random changes combined with cumulative selection quickly reach the target.

Learning outcomes

Learning outcomes:

- Gain a deeper understanding of adaptive evolution via natural selection.
- Explain the difference between single-step and cumulative selection.
- Recognise how random change can lead to order if paired with selection.
- Run and modify a simple R simulation to explore a biological idea.

Worked example

Inputs

Imagine you are typing random letters on a typewriter. How long would it take to type a sensible sentence by pure chance? Take Dawkins' phrase "`METHINKS IT IS LIKE A WEASEL`" (28 characters, including spaces).

Steps

The R code below picks each character at random from the 26 letters of the alphabet plus a space, giving a pool of 27 possible characters at each position. The chance of randomly typing the first letter correctly is 1 in 27. The probability of getting the first two letters right is $(1/27)^2$, and the chance of typing the entire 28-character phrase correctly in one attempt is $(1/27)^{28} \approx 8 \times 10^{-41}$. On average you would need about $27^{28} \approx 1.2 \times 10^{40}$ attempts. This is single-step selection, and it would never work in practice.

Output and interpretation

But what if every correct letter was kept, and only the wrong ones were typed again in the next generation? That is cumulative selection. Let's find out how much difference it makes.

Your task

0. Make sure you have both R and RStudio installed.
1. Open RStudio.
2. Open a new script (File > New File > New Script)
3. Copy and paste the R code (below) into your script.
4. Save your script with a memorable file name.

Note: only edit the top part of the code (above the line of `#` symbols). The phrase can only contain letters and spaces.

Predict first

Before you run the code, write down your predictions:

- With selection turned on, roughly how many generations will it take to reach the target phrase? 10? 100? 10,000?
- With selection turned off, will it ever get there?

The code

```
phrase <- "Methinks it is like a weasel"
nGenerations <- 200
selection <- TRUE

##################################################
# Do not edit below this line
pool <- c(LETTERS, " ")
phraseSplit <- unlist(strsplit(toupper(phrase), ""))
if (!all(phraseSplit %in% pool)) stop("Use only letters and spaces in the phrase")
output <- data.frame(generation = 1:nGenerations,
```

```
                    phrase = rep(NA, nGenerations))
fixed <- NULL
for (i in 1:nGenerations){
  randomPhrase <- sample(pool, length(phraseSplit), replace = TRUE)
  if(selection == TRUE) {
    randomPhrase[fixed] <- phraseSplit[fixed]
    fixed <- append(fixed, which(randomPhrase == phraseSplit))
    fixed <- sort(unique(fixed))
  }
  output$phrase[i] <- paste(randomPhrase, collapse = "")
}
##################################################

output

# First generation that matches the target (NA if it was never reached)
which(output$phrase == paste(phraseSplit, collapse = ""))[1]
```

How the simulation works

An instructor will explain the code and demonstrate its use.

The program generates a random sequence of characters of the same length as the target phrase (stored in `phrase`). If selection is turned on (`selection <- TRUE`), correct letters are "fixed" in place and kept in every following generation, while incorrect letters continue to be replaced at random. This is cumulative selection: each generation builds on the one before.

Over the generations, the number of correct letters builds up until the sequence matches the target phrase. The last line of the code tells you the generation at which this happened. With selection turned off, each generation is a fresh random guess (single-step selection), and no meaningful sentence appears.

Note that this program is a simplified version of Dawkins' original. Here, each correct letter is locked in place and can never change again. Dawkins' program instead made a population of copies of the current phrase, each with a few random copying errors (mutations), and kept the copy closest to the target. In that version a correct letter can, in principle, be lost again, as in real populations. Both versions show the same thing: cumulative selection is enormously faster than chance alone.

Try it yourself

1) **Change the phrase**: make it shorter or longer (letters and spaces only). How does this change the number of generations needed?
2) **Change the number of generations**: experiment with different values for `nGenerations`. With 100 generations, is the target always reached?
3) **Repeat the simulation**: run the code again with the same settings. Does it always take the same number of generations?
4) **Turn selection off**: set `selection <- FALSE`. What happens?

Questions

- What happens if selection is turned OFF?
- Does the number of generations affect whether the target phrase is reached?
- Does the speed of reaching the target phrase depend on the length of the phrase?
- How does this simulation differ from real natural selection?

Takeaways

- Cumulative selection, which keeps and builds on small random improvements, is enormously faster than single-step selection (pure chance).
- Random variation alone is extremely unlikely to generate complex outcomes, but selection changes the odds.
- Real natural selection has no target; it favours traits that improve fitness in the current environment.

Bug hunt camouflage (NetLogo)

Background

Adaptive evolution via natural selection is a fundamental concept in evolutionary biology and is one of the central mechanisms driving the diversity of life on Earth. This process is shaped by the interplay between variation in heritable traits within a population, differential survival and reproduction rates, and the environmental challenges faced by organisms. One illustrative example of natural selection is the phenomenon of camouflage in predator-prey systems.

In an environment where predators rely on visual cues to locate their prey, individuals with better camouflage that matches their surroundings are more likely to avoid detection and capture, leading to higher survival rates. Over generations, the frequency of camouflage traits increases in the prey population due to higher survival and increased reproduction of well-camouflaged individuals.

On the other hand, predators with better detection skills, like keen eyesight, are more successful at capturing prey, leading to increased survival and reproduction of these individuals. Consequently, traits associated with effective predation also become more prevalent in the predator population. This dynamic interplay between camouflage in prey and predator detection abilities exemplifies the power of natural selection in driving the adaptation and evolution of species to their ecological niches.

Learning outcomes

Learning outcomes:

- Understand adaptive evolution via natural selection.
- Identify the four conditions for evolution by natural selection in a model.
- Explain how fitness depends on the environment.
- Use the terms adaptation, selection, selection pressure, and heritability correctly.

Your task

In this exercise, you will use an individual-based simulation to examine adaptive evolution and natural selection, exploring how living organisms can undergo transformations to better survive and thrive in their environments. You step into the role of a predatory bird and hunt bugs living in a virtual landscape. You will see how the colours of the bugs ("traits") influence their survival. The bugs that blend into their surroundings have a higher chance of avoiding detection, leading to greater survival and more opportunities to pass on their advantageous traits to future generations.

The exercise is done using one of the built-in models in a computer program called NetLogo (https://ccl.northwestern.edu/netlogo/).

There are two ways to use NetLogo: (1) You can log in to our SDU UCloud computing system at https://cloud.sdu.dk/, and use it there; (2) You can download and install NetLogo from the site above.

I recommend the UCloud option: Please follow these steps: (i) search for NetLogo in the search bar at the top of the screen; (ii) click on Applications and then on the NetLogo icon; (iii) click on "Run Application";

(iv) Give your job a name (e.g., "BB512") and under "Hours" enter 1 hour; (v) under Machine Type, select "u1-standard-1"; (vi) click on Submit. After a few moments while UCloud is preparing your job, a new screen appears. Click on Open Interface to open a virtual desktop. There, you can open NetLogo and use it as if it was on your own computer by clicking on its icon.

Open NetLogo and click File > Models Library. You can then use the search box at the bottom of the screen to find the model called "Bug Hunt Camouflage".

This is a model of natural/artificial selection that shows how a population hunted by a predator can develop camouflaging. For example, in a forest with green leaves, green bugs may emerge as the predominant bug colour.

In the simulation you assume the role of a predatory bird that feeds on insects, which can be different colours. You will see what effect your hunting has on the colour traits/genetics of these simulated insects, and also how this evolution affects your hunting efficiency.

How the simulation works

Read this before you start, so that you know what the model does (and does not) include.

The simulation starts with bugs of random colours. Each bug has three pigment "genes" (red, green and blue) that together set its colour (see "Details about colours" at the end).

The model makes these assumptions:

- **Constant population size.** Each time you catch a bug, one of the surviving bugs is chosen at random to produce one offspring, so the number of bugs stays at the carrying capacity.
- **Asexual reproduction.** An offspring has a single parent and inherits its colour genes.
- **Small mutations.** Each of the offspring's three colour genes can differ slightly from the parent's. The "max-mutation-step" slider sets the largest possible difference: each gene can move up or down by any amount up to that value, with small changes more likely than large ones.
- **Offspring appear near their parent**, within the distance set by the "offspring-distance" slider.
- **Newborn bugs start very small** and grow to full size over time.
- **Bugs do not move**, and the predator (you) is the only cause of death.

So if you consistently catch (e.g.) non-reddish bugs, the reddish bugs that remain will have offspring that are also reddish, and the average colour of the population will shift towards red.

In the lecture, evolution was defined as: "*change in the heritable characteristics [colour traits] of biological populations [bugs] over successive generations [time].*"

Predict first

Before you run the model, write down your predictions:

1. What will happen to the average colour of the bugs after you have hunted for two minutes on one background?
2. What will happen to the time it takes you to catch 10 bugs?
3. What will happen when you then switch to a different background?

You will check these predictions against what you observe.

Getting started

1. When you open the model, you will see a screen with various purple and green "sliders", buttons and menus that control parts of the model.
2. Start with 30 bugs, by moving the "carrying-capacity" slider until it reads 30.

3. Now click "setup". You should see the environment appear on the right hand side, and you should see a few "bugs" on the environment. Start the simulation by clicking "go".
4. After you click go, you need to click on bugs as fast as you can using your mouse/track pad. You can also keep the mouse button depressed, and move the cursor around the world to catch the bugs. Try to consume bugs as fast as possible to remove any "deliberation" on your part as a predator. The camouflaging effect will emerge more clearly if you aren't taking your time trying to find bugs that typically would be more difficult to find.

You can track your progress by looking at the graphs on the left. The most useful are (1) the number of bugs caught through time, (2) the time taken to catch each batch of 10 bugs, and (3) the average colour values of the bugs.

In the "Bugs Caught vs. Time" plot the slope of the curve gives a good idea of your hunting efficiency: the faster you can catch bugs, the steeper the slope. If you didn't catch any bugs for a while, the slope would be horizontal.

You should notice that bugs with contrasting colours (e.g. black on white) are easier to catch. Keep hunting for 2 minutes, then pause the simulation by clicking "go" again. Now take a look at the graphs and see what effect your hunting has had on the colours in the population.

Measure your hunting efficiency

Use the "Time to catch 10 bugs" plot to put numbers on your hunting efficiency. Record:

	Time to catch 10 bugs
First 10 bugs	
Last 10 bugs before you switch background	
First 10 bugs after you switch background	
Last 10 bugs on the new background	

Compare your numbers with your neighbour's. They will differ (you are different predators!), but is the pattern the same?

Your catch time does not depend only on the bugs. Apart from bug evolution, what else could change how long it takes you to catch 10 bugs? Which of the plots gives more direct evidence that the bugs themselves have evolved?

Questions

Answer the following questions, then make sure you confirm your understanding and have the right answers with an instructor.

- What happens to the average colour of the bug population with time as you hunt? Was your prediction correct?
- What happens to your hunting efficiency? Use your measurements.
- Would you say that the bug population becomes worse or better adapted to their environment?
- Can you explain how this happens?
- After simulating in one environment (e.g. poppy field) for a few minutes, pause then switch to another environment. Are the bugs now well- or poorly-adapted to their environment?
- Do the genotypes of individuals change (e.g. with individual age)? Switch on "show-genotype?" and watch a few individual bugs for 30 seconds to find out.
- Increasing the "max-mutation-step" makes bug offspring less like their parents. How do you think this will influence the speed of adaptation of the bugs?
- In the lecture we met four conditions for evolution by natural selection: (1) individuals vary in a trait, (2) individuals vary in reproductive success, (3) the trait affects reproductive success, and (4) the trait is inherited. Where do you see each of these in the model? Where does new variation come from?

- Newborn bugs start very small. Could this let some bugs escape for reasons that have nothing to do with their colour? How might that affect the results?
- How does this model differ from a real predator–prey system? Give at least two differences.

Some useful keywords:

- Selection
- Adaptation
- Selection pressure
- Heritability (how much of the variation in a trait among individuals is passed on from parents to offspring; you will meet a more precise definition in Part 3)

Optional extension: gene flow and local adaptation

In the lecture we saw that gene flow can swamp local adaptation. You can explore this with the "offspring-distance" slider, which sets how far from its parent an offspring can appear.

1. Choose the background with the most distinct patches of different colour.
2. Set "offspring-distance" to a small value and hunt for a few minutes. Use "flash" or "clear-background" to see where the bugs are. Do bugs in different patches come to match their own patch?
3. Repeat with a large "offspring-distance". What happens now, and why?

Details about colours (optional)

Each bug has three pigment genes: red, green and blue. Each gene has a value from 0 to 100, which sets how strongly that pigment is expressed. Switch on "show-genotype?" to see each bug's values; for example, 40-100-24 means red at 40%, green at 100% and blue at 24%. The three pigments are mixed, as in the pixels of a TV or computer screen, to give the bug's colour. A bug with a lot of red but little green or blue looks red.

The plots summarise colour in a different way, as **hue**, **saturation** and **brightness** (Fig @ref(fig:colours)):

- **Hue** is the position around the colour wheel: red, through orange, yellow, green and blue, and back to red.
- **Saturation** is how vivid the colour is. Low saturation gives washed-out colours (greys, and white at full brightness).
- **Brightness** is how light or dark the colour is. A brightness of 0 is black, whatever the hue.

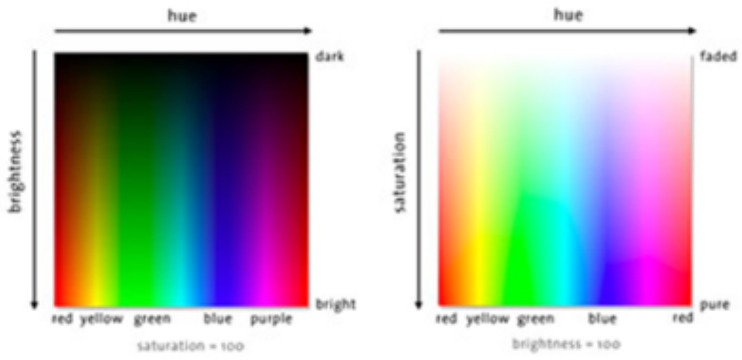


Figure 1: Hue, brightness and saturation

If this figure does not display in your browser, open it directly here: Hue, brightness and saturation diagram.

Read more about the model here: http://ccl.northwestern.edu/netlogo/models/BugHuntCamouflage

Takeaways

- Selection changes trait distributions when survival depends on the environment.
- All four conditions for evolution by natural selection are present in the model; mutation supplies the new variation.
- Adaptation is context-dependent: a well-adapted population can become poorly adapted when conditions change.
- Heritability and mutation govern how quickly populations can respond to selection pressure.

Part 2: Population Growth Models

This part builds the core toolkit for describing and predicting population change. You will move from basic growth models to stochasticity, life tables, and structured population models.

By the end of Part 2, you should be able to:

- Distinguish exponential, geometric, and logistic growth models.
- Interpret how parameters like growth rate and carrying capacity shape dynamics.
- Estimate λ from time-series data.
- Explain how environmental stochasticity and small population size affect extinction risk.
- Use life tables and matrix models to analyse structured populations.

Notation used in Part 2 (following Neal)

Different books use different symbols for growth rates. In this course we follow Neal (chapters *Exponential Growth*, *Logistic Growth* and *Life Tables*):

- λ: the *multiplication rate* (also called the finite rate of increase). Over one time step, $N_{t+1} = \lambda N_t$, so $N_t = N_0 \lambda^t$.
- $R_m = \lambda - 1$: the per capita rate of increase per time step, used in discrete-time (*geometric*) growth: $\frac{\Delta N}{\Delta t} = R_m N_t$, i.e. $N_{t+1} = N_t + R_m N_t$. It equals births minus deaths over the time step, per individual: $R_m = (B - D)/N$.
- $r_m = \ln \lambda = b - d$: the *intrinsic rate of increase*, the per capita rate in continuous time (*exponential* growth): $\frac{dN}{dt} = r_m N$, so $N_t = N_0 e^{r_m t}$ and $\lambda = e^{r_m}$. The doubling time is $t_d = \ln 2 / r_m$.
- $r = r_m \left(1 - \frac{N}{K}\right)$: the *realised* per capita growth rate in the logistic model. It falls from r_m (when N is small) to 0 (when $N = K$).
- $r_c = \ln(R_0)/T$: an approximation of r_m calculated from a life table (you will meet R_0 and T in the life tables chapter).

Note that $dN/dt = r_m N$ is *exponential* (continuous) growth, while $N_{t+1} = \lambda N_t$ is *geometric* (discrete) growth. R_m and r_m are both per capita births minus deaths, but R_m is measured over a whole time step and r_m over an infinitesimally small one. Some other books use R for λ, so check the definitions when you compare equations from different sources. The discrete logistic model used in the Excel practicals applies r_m once per time step, so a small population is multiplied by about $1 + r_m$ per step rather than by e^{r_m}.

Geometric growth

Background

The geometric growth model is a simple model of unconstrained population growth. In other words, the population can grow to infinite size and is not affected by competition (population density). In the model, the rate at which a population grows is described by the population growth rate, often represented by the symbol lambda (λ). Higher values of λ indicate rapid population growth, while smaller values suggest slow growth. If $\lambda < 1$, the population shrinks in size.

Geometric growth is often used interchangeably with exponential growth, but there is an important distinction. Exponential growth pertains to "continuous time" scenarios, whereas geometric growth involves models where the population changes in discrete time steps, such as yearly intervals.

The key quantities involved in this model are as follows, although it's important to note that different books or sources may use varying symbols:

- N_t = population size at time t; and N_0, N_1 is population size at time 0, time 1 etc.
- λ = population multiplication rate over a single time step, Δt. This is sometimes called the *finite rate of increase*.
- R_m = per capita rate of increase per time step, Δt (the change in N per individual over one time step).
- r_m = per capita rate of increase over an infinitesimally small time step. This is called the *intrinsic rate of increase* (or *instantaneous rate of increase*), and it equals the per capita birth rate minus the per capita death rate, $r_m = b - d$.

R_m is related to λ such that $R_m = \lambda - 1$ (and $\lambda = R_m + 1$).

r_m is related to λ such that $\lambda = e^{r_m}$, so $r_m = \ln \lambda$. This is why the slope of $\ln(N_t)$ against t (see below) is $r_m = \ln \lambda$. It also gives the *doubling time*, the time a growing population takes to double in size: $t_d = \ln 2 / r_m = \ln 2 / \ln \lambda$.

These symbols are summarised in the notation box in the Part 2 introduction, and we use them throughout Part 2.

The relevant equations are in the Neal textbook chapter *Exponential Growth*, but here we will use the following:

$N_{t+1} = N_t + (R_m N_t)$,

which leads to,

$N_{t+1} = N_t(1 + R_m)$ and, equivalently, $N_{t+1} = N_t \lambda$.

Learning outcomes:

- Competence in using Excel formulae for mathematical modelling.
- Understanding the parameters of exponential/geometric growth.
- Competence in using mathematical models in Excel to strengthen own understanding of biological processes.
- Awareness of rearranging of mathematical formulae to produce different forms of models.
- Knowing that the slope of the $ln(N_t)$ vs. t relationship can tell you about population growth rate (it is $\ln(\lambda)$).

Worked example

Inputs

- Initial population: $N_0 = 10$
- Population multiplication rate: $\lambda = 1.1$
- Time steps: $t = 5$

Steps

1. Use the geometric growth equation: $N_t = N_0\lambda^t$.
2. Substitute values: $N_5 = 10 \times 1.1^5$.

Output and interpretation

$N_5 = 16.1051$

Plotting N_t against t gives a curve, while plotting $\ln(N_t)$ against t gives a straight line with slope $\ln(\lambda) = r_m$.

Your task

Download and open the Excel file `GeometricGrowth.xlsx`.

Danish Excel: `LN` is the same, `EXP` is `EKSP` and `SLOPE` is `HÆLDNING`; decimals use a comma (`0,0953`) and arguments are separated by `;` rather than `,`.

1. Starting with an initial population size (`N`) of 10 [at time (`t`) =0], and with a λ of `1.1`, use Excel's equation functions to work out the population size from `t=1` through to `t=20`. In the template, λ is in cell B8 and N_0 in B10, so the formula for $t = 1$ (cell B11) is `=B10*$B$8`, filled down.

2. Use charts to plot the results. On the horizontal axis (x-axis) you should have time, and on the vertical axis (y-axis) you should have population size (`N`). A scatterplot would be best for this.

3. Fill in the `logN` column with the natural logarithm of population size. The Excel formula for natural log is `LN()` (the same in Danish Excel), e.g. `=LN(B10)` in cell C10, filled down the column. Then make another scatterplot with time on the x-axis and `logN` on the y-axis.

4. What do you notice about the curves in these different versions of plotting the same data?

5. Try altering the population growth rate. Try values of 0.8, 1 and 1.2 for example. What happens to the curves? What is special about the value of 1?

6. Add a linear trendline with its equation to the `logN` plot from step 3. Identify the slope. The slope is $\ln(\lambda) = r_m$, so you can work out λ by "reversing" the ln by taking its exponential ($\lambda = e^{r_m}$), e.g. `=EXP(0.0953)` in Excel (`=EKSP(0,0953)` in Danish Excel).

   Note: do not use Excel's "logarithmic scale" axis option for this step. That option only changes how the axis is drawn (and uses base-10 logs); a trendline added to that chart is still fitted to the untransformed N, so its slope is not $\ln(\lambda)$.

7. With $\lambda = 1.1$, read off your chart roughly how many time steps the population takes to double (from 10 to 20, and from 20 to 40). Check your answer with the doubling time, $t_d = \ln 2 / \ln \lambda$. Does it depend on the starting population size?

Now reset the population growth rate to 1.1. Now let's see if the mathematical rules of so-called "geometric series" work…

1. If the population starts at t=0, $N_t = N_0\lambda^t$. In words, the population at time t is the initial population size multiplied by the multiplication rate (λ) raised to the power of t.
2. This works because each time step multiplies by λ again: $N_1 = \lambda N_0$, $N_2 = \lambda N_1 = \lambda^2 N_0$, $N_3 = \lambda^3 N_0$, and so on.
3. Following these rules, the population size at time $t = 5$ (N_5), is $10 * 1.1^5$, or $10 * 1.61051$, which is 16.1051. Check that this matches what you got earlier.
4. This approach can be really useful because it can save lots of time. For example, what is the population after 900 generations? It would be tedious to work this out using the first, "simple" approach, but very easy if you use the geometric series rules and have access to a calculator/computer.

Questions

- In geometric growth, what does a constant growth rate imply for the population's future growth? Explain with an example.
- What makes the geometric growth model "unrealistic" for real-world scenarios? Explain your reasoning.
- Give an example of real-world populations where geometric growth might be a suitable model. Explain your reasoning.
- Suppose a population's size decreases by 25% every year due to environmental factors. If there are initially 800 individuals, determine the population size after 4 years.
- A population has $\lambda = 1.1$ per year. What is λ per half-year? (Hint: convert to r_m, which scales with time, then convert back.)

Takeaways

- Geometric growth assumes discrete time steps and a constant growth multiplier λ.
- Log-transforming N_t linearises the growth pattern and reveals $\ln(\lambda) = r_m$ as the slope.
- The model is useful for short-term or early-phase growth but ignores density dependence.

Estimating Population Growth Rate

Background

In population biology, one important measure of population dynamics is the population growth rate (λ), called the multiplication rate in Neal and also known as the finite rate of increase.

The finite rate of increase (λ) is defined as the ratio of population size at time t + 1 to that at time t (N_{t+1}/N_t), and it is related to the intrinsic rate of increase r_m by the expression $r_m = \ln \lambda$ (so $\lambda = e^{r_m}$; see the notation box in the Part 2 introduction).

This rate tells us how fast a population is growing or shrinking over time. When $\lambda > 1$, the population grows exponentially; when $\lambda = 1$, it stays constant, and when $\lambda < 1$, the population declines.

In this exercise, you will estimate the population growth rate (λ) from real population data. The method involves plotting the population size over time, applying a log transformation to linearise the data, and fitting a linear regression model to estimate λ. This method is widely used in ecology to analyse growth trends and make predictions.

Learning outcomes:

- Competence in using Excel formulae for data transformation and regression analysis.
- Understanding the role of λ in population growth and its estimation through log-transformed data.
- Competence in applying mathematical models in Excel to analyse real biological data.
- Awareness of how log transformations can linearise exponential growth data for easier interpretation.
- Knowing that the slope of the $\ln(N)$ vs. time relationship represents $\ln(\lambda) = r_m$ and can be used to estimate population growth rate.

Worked example

Inputs

- Regression slope from $\ln(N_t)$ vs time: 0.08

Steps

1. The slope is $\ln \lambda = r_m$, so use $\lambda = e^{\text{slope}}$.
2. Substitute the slope: $\lambda = e^{0.08}$.

Output and interpretation

$\lambda = 1.083$

This indicates a population growing by about 8.3% per time step.

Your task

Step 1: Download and Open the Data

- Download the provided Excel file: `EstimatingGrowth.xlsx`.
- Open the file in Excel to view the population data for a species recorded annually over a 25-year period.
- Danish Excel: `LN` is the same, `EXP` is `EKSP` and `SLOPE` is `HÆLDNING`; decimals use a comma and arguments are separated by `;`. The trendline menu items are "Tilføj tendenslinje" and "Vis ligning i diagram".

Step 2: Plot the Population Size Over Time

- In Excel, create an x-y scatter plot of the **population size** (N_t) on the y-axis and **time** (Year) on the x-axis.

Step 3: Log-Transform the Population Size

- Add a new column in Excel for the natural logarithm of the population size using the formula `=LN(cell)`. This transformation helps linearise the exponential growth data.
- Create a new scatter plot with **log-transformed population size** ($\ln(N_t)$) on the y-axis and **time** (Year) on the x-axis.

Step 4: Fit a Linear Regression Model

- In the log-transformed scatter plot, add a trendline by right-clicking on the data points and selecting "Add Trendline."
- Choose "Linear" and ensure you check the box to "Display Equation on Chart."
- The slope of the trendline represents $\ln(\lambda) = r_m$. The trendline label rounds the slope; for the exact value use `=SLOPE(ln-column, year-column)` (Danish: `=HÆLDNING(ln-kolonne; år-kolonne)`). Strictly, the slope is the *realised* average growth rate over the census years; it equals the intrinsic rate r_m only if growth was not limited by density during that time.

Step 5: Calculate λ

- Use the slope from the regression equation to calculate λ with the formula: $\lambda = e^{\text{slope}}$. In Excel, you can do this with the equation `=EXP(cell)`

Questions

- What does the log-transformed plot of population size over time tell you about the population's growth trend? Does the population appear to grow exponentially?
- What is the estimated population growth rate (λ) based on your linear regression analysis?
- What assumptions does this model make about population growth? Discuss any potential real-world factors that might affect the accuracy of your estimate for λ.

Takeaways

- Estimating λ from data relies on a log-transform and a linear trend over time.
- The slope of $\ln(N_t)$ vs. time is $\ln(\lambda) = r_m$, so λ is the exponential of the slope.
- Measurement error and changing environments can bias simple growth estimates.

Stochastic population growth

Background

In contrast to the deterministic nature of the geometric growth model, a stochastic version of the model introduces an element of randomness, making it more reflective of real-world scenarios. Instead of assuming fixed and predictable growth rates, the stochastic model incorporates uncertainties and fluctuations that occur due to various factors, such as environmental variability, chance events, and individual-level variations in reproductive success. By embracing this stochastic approach, the population dynamics become more realistic, allowing us to better understand and predict the inherent variability and uncertainties associated with natural populations. This stochastic framework provides a valuable tool for ecologists and researchers to explore the dynamics of real populations under more complex and dynamic conditions.

Remember: this model still allows for unbounded population growth (the populations are not influenced by population density).

We'll focus on the discrete (Geometric) growth model: $N_{t+1} = \lambda N_t$.

Because $\lambda = \mathrm{e}^{r_m}$ (and $r_m = \ln(\lambda)$; see the notation box in the Part 2 introduction), we can rearrange the equation to express it as $N_{t+1} = \mathrm{e}^{r_m} N_t$.

Therefore, we can predict next year's population size from this year's population if we know either r_m or λ.

We will create some models in Excel, and then in R, to explore the effect of stochastic variation in the population growth rate.

We can simulate variation in r_m by drawing a random number from a normal distribution with a particular mean ($\bar{r_m}$) and variance ($\sigma^2_{r_m}$) (Fig. @ref(fig:stochGrowthRate)). The $\bar{r_m}$ value determines the long-term average while the $\sigma^2_{r_m}$ estimates how much "spread" there is in the data from year to year.

The figure shows 1,000 values drawn from a normal distribution with mean 0.05 and variance 0.1 (the settings you will use in this practical). The peak of the r_m distribution is just above 0 (at 0.05), so on average the population will tend to grow. However, in poor years r_m is < 0, so the population will decline in those years. With this much variation, declines are almost as common as increases: about 44% of years have $r_m < 0$. Both the mean value and the spread of the distribution (i.e. variance) determine the fate of the population.

Remember that we can convert r_m to λ by taking the exponential, because $r_m = \ln(\lambda)$.

When using the equation above to calculate population at time $t+1$ (N_{t+1}) from the population at time t (N_t), one would draw a random r_m value from this distribution. Sometimes r_m will be high, other times it will be low, most of the time it will be from around the middle of the distribution.

Average growth: why the mean λ is misleading

In this model $r_m = \ln \lambda$ is drawn afresh each year. After t years, $\ln N_t$ is $\ln N_0$ plus the sum of t random r_m values, so the *median* (typical) trajectory grows at exactly the mean, $\bar{r}_m$, per year: $\lambda = e^{0.05} = 1.051$ with our settings. The *arithmetic mean* of λ is larger, $e^{\bar{r}_m + \sigma^2/2} = e^{0.05+0.05} = 1.105$, because a few very good years pull the average up. A population projected with the average λ would therefore grow about twice as fast (on the log scale) as a typical stochastic population. Long-run growth follows the *geometric*

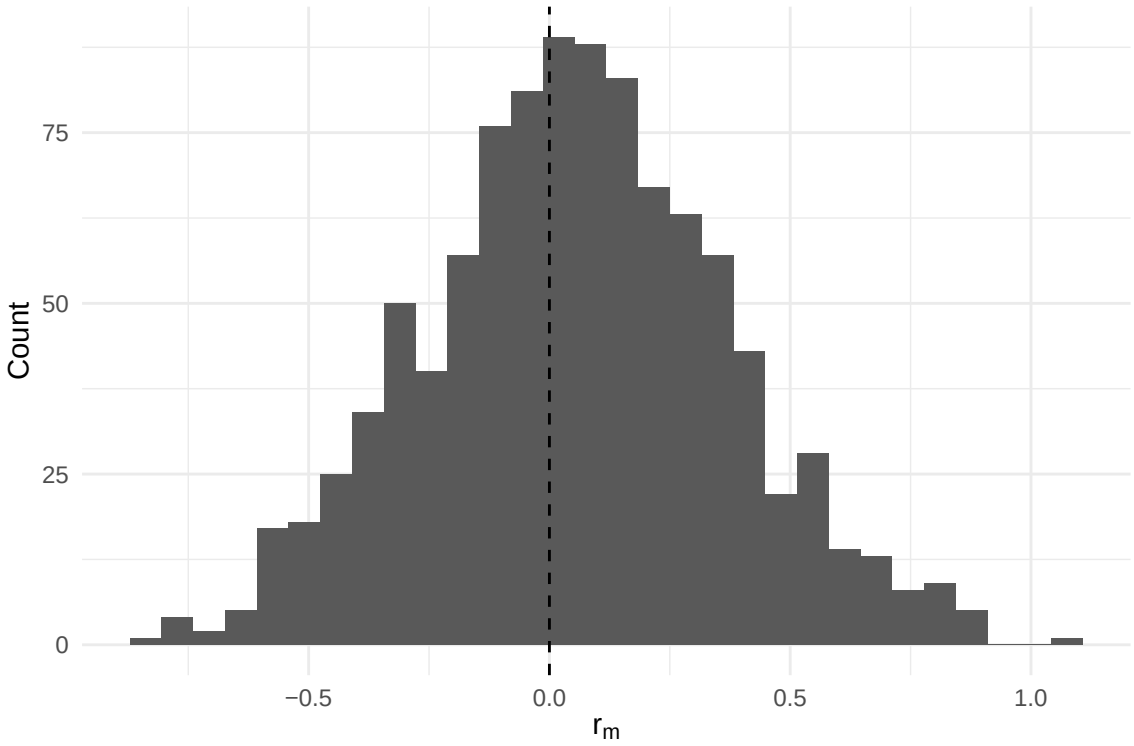


Figure 2: 1,000 values of r_m drawn from a normal distribution with mean 0.05 and variance 0.1. The dashed line marks $r_m = 0$.

mean of λ ($e^{\bar{r}_m}$), not the arithmetic mean. Put the other way round: if the arithmetic mean λ stays the same, more year-to-year variation lowers long-run growth.

You may meet the rule that extinction is (eventually) almost certain when $\sigma^2 > 2\bar{r}$. That rule takes $\bar{r}$ as the average growth rate on the λ scale, so its long-run growth is $\bar{r} - \sigma^2/2$. In this practical r_m is drawn on the log scale, so long-run growth is $\bar{r}_m$ itself, and the population is doomed in the long run only if $\bar{r}_m \leq 0$. Both say the same thing: a population is doomed if its long-run (geometric-mean) growth rate is zero or below. The rule is also about the very long run; over 50 generations, extinction risk can be well below 1.

Learning outcomes:

- Increased competence in using Excel formulae for mathematical modelling.
- Understanding the concept of stochasticity in simple population models.
- Competence in using mathematical models in Excel to strengthen own understanding of biological processes.
- Understanding how stochasticity relates to (i) uncertainty in the prediction of population trajectories and (ii) probability of (local) extinction.

Worked example

Inputs

- Population this year: $N_t = 100$
- r_m drawn at random for this year: -0.2 (a poor year)

Steps

1. Convert r_m to λ: $\lambda = e^{-0.2} = 0.819$.

2. Project one year: $N_{t+1} = 100 \times e^{-0.2} = 81.9$.

Output and interpretation

$N_{t+1} = 81.9$: in this poor year the population shrinks by about 18%. Next year a new r_m is drawn (for example +0.3, giving $81.9 \times e^{0.3} = 110.5$), and so on. Repeating this step with a fresh random r_m each year produces one stochastic trajectory.

Your task

Use the Excel worksheet, StochasticPopulationGrowth.xlsx, to study how stochastic population growth works with this simple model.

In this practical we use the same settings throughout: r_m is drawn from a normal distribution with **mean 0.05 and variance 0.1**. Years of growth then only slightly outnumber years of decline (about 56:44). This means a population can grow on average yet still crash by chance, especially when starting size is small.

In the Excel template, first set the mean of r_m to **0.05** and the variance to **0.1** (the template may contain other values), and use a starting population of **100**.

1) Use Excel formulae to calculate deterministic population size through time (25 generations, t = 0 to 25, with starting population of 100), using the multiplication rate for the mean r_m, $\lambda = e^{\bar{r}_m}$.
2) Use charts to plot the results. (you already did this last time!)
3) Use a formula to generate a column of stochastic r_m values, based on a chosen mean and variance. The formula refers to the cells that hold the mean and the variance of r_m; check where these are in your sheet (in the template they are G10 and G11). In English versions of Excel the function may be written either as `=NORM.INV(RAND(), $G$10, SQRT($G$11))` (newer versions) or `=NORMINV(RAND(), $G$10, SQRT($G$11))` (older versions), while in Danish Excel it appears as `=NORMINV(SLUMP(); $G$10; KVROD($G$11))`. If you encounter errors, check whether your copy of Excel expects commas or semi-colons as argument separators and adjust the formula accordingly.
4) Use the same procedure as before, to create a stochastic population size vector (stochastic N). Remember to convert r_m to λ by taking the exponential (`=EXP(...)`; Danish Excel: `=EKSP(...)`).
5) Compare the two trajectories using a chart.
6) Try altering initial population size, mean population growth, and the amount of stochasticity (Variance).
7) In this model N is multiplied by e^{r_m}, which is always positive, so N can never reach exactly 0. Instead, we treat the population as (quasi-)extinct if N falls below 1 at any point, i.e. fewer than one individual is left (the R code below uses the same rule). What happens to extinction risk as stochasticity (variance) increases? What happens when initial N is small? Does the population ALWAYS survive if the mean population growth rate, $\bar{r_m}$, is > 0?

Note: Excel re-randomises the random numbers every time you change any cell in the sheet. This is OK, and allows you to explore a stochastic simulation many times.

Simulations in R

Excel is of limited use to really get a feel for this. For the next part we'll use *R*.

The simulation below repeats the stochastic projection many times, so you can estimate extinction risk and see how it is related to starting population size, mean growth rate, and the amount of stochasticity.

You can copy/paste the code below into *R*.

The output of the modelling is shown in Fig. @ref(fig:stochProjection)

Copy-and-paste the code below into a text file (or directly into *R*).

The final line of the code (`nExtinct/nTrials`) gives you an estimate of extinction probability - the proportion of trials in which the population falls to 1 or fewer individuals at some point, the same rule as in the Excel exercise. The R settings use the same mean (0.05) and variance (0.1) of r_m as the Excel exercise, but a smaller starting population (10), so that extinctions are common enough to see.

Modify the simulation settings to explore what happens to (i) the plot of population growth and (ii) extinction risk, when you vary `mean.r` ($\bar{r_m}$), the amount of stochasticity ($\sigma^2_{r_m}$) (`var.r`), and the number of generations (`nGen`).

```
#Simulating stochastic geometric population growth rate

#Simulation settings (try changing these)
mean.r = 0.05 # the mean value of r_m
var.r = 0.1 # the variance in r_m (stochasticity)
startPop = 10 # pop size at start
nGen = 50 # number of generations
nTrials = 1000 # number of repeated simulations

############################################################
#If you are unfamiliar with R, don't edit below this line!
############################################################

pseudoExtinction = 1

# First randomly generate some r values
rValues <- matrix(rnorm(nTrials * nGen,
                        mean = mean.r,
                        sd = sqrt(var.r)),
                  ncol = nTrials,
                  nrow = nGen)

# Use a histogram to see what they look like
# (uncomment the line below)
# hist(rValues,col="grey",main="")

# Now run the simulations to see what the resulting
#population growth looks like
trials = matrix(data = NA, nrow = nGen, ncol = nTrials)
for (j in 1:nTrials) {
  popSize = startPop
  for (i in 2:nGen) {
    stoch.r = rValues[i, j]
    newN = popSize[i - 1] * exp(stoch.r)
    # once the population falls below the threshold it is extinct,
    # so set it to 0 (it cannot recover)
    if (newN < pseudoExtinction) newN = 0
    popSize = append(popSize, newN)
  }
  trials[, j] = popSize
  rm(popSize)
}

#Calculate probability of (pseudo)extinction
minvals <- apply(trials, 2, min)
nExtinct <- length(minvals[minvals <= pseudoExtinction])
probExtinct <- nExtinct / nTrials
cat("Pseudo-extinction probability:", round(probExtinct, 3), "\n")
```

```
## Pseudo-extinction probability: 0.058
```

Example stochastic trajectories

```
#Make a plot of the population trajectories
yMax <- max(100000, max(trials)) # upper limit of the y-axis
plot(
  1:nGen,
  log(seq(0.1, max(trials), length.out = nGen)),
  type = "n",
  axes = F,
  xlab = "Time",
  ylab = "log(N)",
  ylim = log(c(0.1, yMax))
)
matlines(log(trials[, 1:min(100, nTrials)]), # plot up to 100 trajectories
         col = "#FF234520",
         lty = 1,
         lwd = 3)
axis(1)
axis(2,
     at = log(10^(-1:ceiling(log10(yMax)))),
     labels = 10^(-1:ceiling(log10(yMax))))
abline(h = log(pseudoExtinction), lty = 2)
```

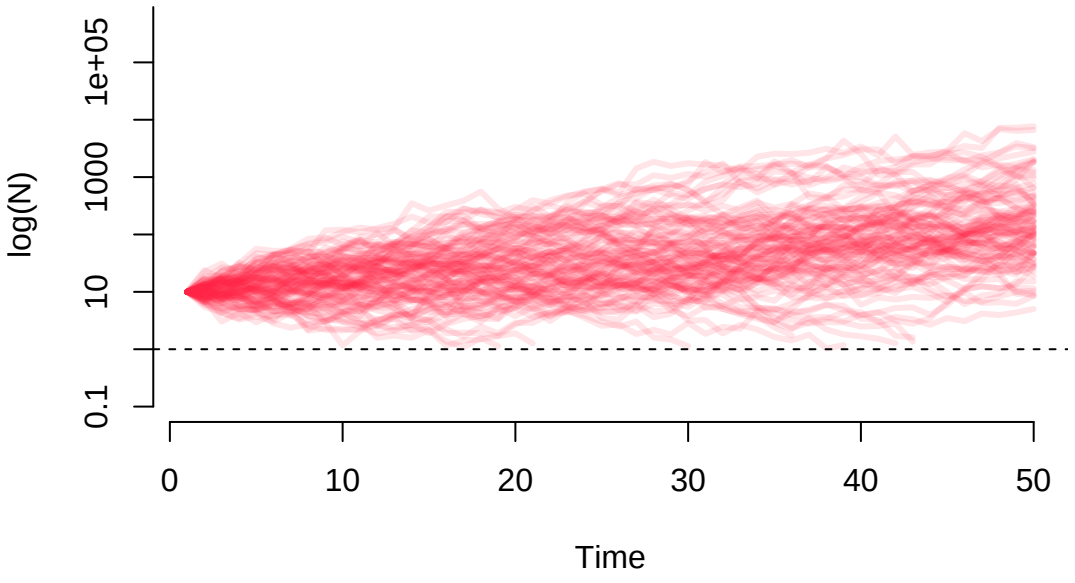


Figure 3: An example of stochastic population projection (100 of the 1,000 simulations, 50 generations)

Things to try

Work through these experiments and note what happens to (a) the plot and (b) the extinction probability each time.

1. Run the code as-is. Note the extinction probability and the spread of trajectories.
2. Keep `mean.r = 0.05` but increase `var.r` to **0.5**. How does extinction risk change?
3. Reset `var.r = 0.1`, then reduce `startPop` from 10 to **3**. Does starting population size matter?
4. Set `mean.r = 0.0` (population neither growing nor shrinking on average). Is extinction certain, unlikely, or somewhere in between?
5. **Challenge:** find a combination of `mean.r` and `var.r` where `mean.r > 0` but extinction probability exceeds 0.5. What does this tell you?

Questions

- What is the main difference between deterministic and stochastic population growth models?
- Describe how incorporating randomness into the stochastic model makes it more realistic for understanding real-world populations.
- Simulate a scenario where two populations with identical growth rates experience different outcomes due to stochastic factors. Explain the implications of these findings.
- What can this stochastic model tell us about extinction risk and population size?
- What can this stochastic model tell us about extinction risk and environmental variation?

Takeaways

- Stochasticity introduces variability in trajectories.
- Long-run growth follows the mean of r_m (the geometric mean of λ), not the arithmetic mean of λ, so projecting with the average λ overestimates typical growth.
- Even when the average growth rate is positive, stochastic variation can create a real risk of extinction.
- Small populations are more vulnerable to random declines and extinction.
- The variance of r_m can be as important as its mean for long-term population persistence. This is why conservationists worry about increasing environmental variability, not just declining average conditions.
- Running a model many times (as in the R simulation) reveals the *distribution* of possible outcomes, a much richer picture than any single trajectory provides.

Basic logistic population growth

Background

In nature, populations can't grow indefinitely. They eventually reach an upper limit, known as the *carrying capacity* (K), which represents the maximum number of individuals an environment can support. Factors like available resources, competition, predation, and disease set this limit.

The **logistic growth model** accounts for this limitation and provides a more realistic picture of population growth. Unlike simpler models, such as exponential growth, that assume unlimited resources, the logistic model reflects how populations slow their growth as they approach the carrying capacity. When the population is small, it can grow quickly, but as it gets closer to the limit, the growth rate slows down. When the population is at carrying capacity the population will not grow (birth rates = death rates).

The model is described by the following equation:

$N_{t+1} = N_t + r_m N_t \left(1 - \frac{N_t}{K}\right)$

Where:

- N_{t+1} is the population size at the next time step.
- N_t is the current population size.
- r_m is the intrinsic growth rate, or the maximum possible per capita population growth rate.
- K is the carrying capacity.

In this equation, the term $\left(1 - \frac{N_t}{K}\right)$ slows population growth as the population size N_t approaches K. If $N_t = K$, the population stops growing.

The quantity $r = r_m \left(1 - \frac{N_t}{K}\right)$ is the *realised* per capita growth rate (Neal, *Logistic Growth*): the growth rate per individual actually achieved at population size N_t. It equals r_m when the population is very small and falls to 0 at K. In the spreadsheet it is the column "Per capita growth rate (r)", which you will plot in Graph 2. (The symbols are summarised in the notation box in the Part 2 introduction.)

This is the discrete-time form of the logistic model used by Neal. Here r_m is applied once per time step, so when the population is very small (and $\left(1 - \frac{N_t}{K}\right) \approx 1$) it grows by a factor of about $1 + r_m$ each time step. If you remember from earlier exercises that $\lambda = e^{r_m}$, note that this model treats λ as approximately $1 + r_m$ instead. For example, with $r_m = 0.8$, the model gives 1.8 where $e^{0.8} \approx 2.23$. The approximation is good when r_m is small and poor when it is large (the exponential growth lecture showed the same thing: 1.20 vs. $e^{0.2} = 1.22$ for $r_m = 0.2$). Strictly, a rate applied once per time step is what Neal calls R_m in the geometric model ($\lambda = 1 + R_m$). Neal (and we) keep the symbol r_m in the discrete logistic, but remember that here it is applied once per step. Stepping forward a whole time unit at a time is also what allows the overshoot, oscillations and chaos you will explore below. The continuous-time logistic model always approaches K smoothly.

As growth rates increase beyond a certain point, the population can exhibit more complex behaviours. At higher growth rates, populations may overshoot the carrying capacity, leading to oscillations as they fluctuate around it. In some cases, the population may show what's called deterministic chaos. This means that while the population fluctuates unpredictably, the system itself is not random: small changes in initial conditions or growth rates can lead to vastly different outcomes.

This Excel-based exercise will help you explore how changes in the growth rate and carrying capacity affect population dynamics. Though simple, the logistic model provides valuable insights into how populations stabilise, fluctuate, and respond to environmental limits.

Learning outcomes:

- Increased competence in using Excel formulae for mathematical modelling.
- Understanding the parameters of the logistic population growth model.
- Understanding how strikingly different types of population dynamics can result from the same (logistic) model simply by varying the population growth rate parameter.
- Understanding the concept of *deterministic chaos* and how it is different from randomness.
- Competence in using mathematical models in Excel to strengthen own understanding of biological processes.

Worked example

Inputs

- `Initial N = 10`
- `r_m = 0.8`
- `K = 200`

Steps

1. Enter these values in the parameter cells.
2. Recalculate the model through time.
3. Inspect both graphs: population size through time and per-capita growth vs population size.

Output and interpretation

The population grows quickly at first, then slows as it approaches `K`. This produces a sigmoidal curve and a per-capita growth rate that declines linearly with `N`.

Your Task

In this exercise, you will explore how changes in key parameters affect population dynamics using the logistic growth model. Follow the steps below to guide your analysis.

1. **Download the Excel File**
   Download and open the Excel file: `Basic Logistic Growth.xlsx`.
2. **Familiarise Yourself with the Spreadsheet**
   In the file, you'll see three sections:
   - **Pink Block**: This is where you will input and adjust the model parameters.
   - **Data Table**: This calculates the population size (N) and per capita growth rate at each time step. You do not need to edit it.
   - **Graphs**: These display how the population behaves over time and in relation to the parameters you adjust.
3. **Understand the Parameters**
   In the *pink block*, you will find the key parameters of the logistic growth model:
   - `Initial N`: The starting population size at time 0 (default = 10).

- `r_m`: The maximum per capita population growth rate (default = 0.8).
  - If $r_m > 0$, the population grows.
  - If $r_m < 0$, the population shrinks.
  - Biologically, population size cannot fall below zero (but see the footnote in Step 5: the spreadsheet does not enforce this).
- `K`: The carrying capacity of the population (default = 200).

4. **Explore the Model**
   Use the following steps to guide your exploration of population dynamics:

Graph 1: Population Size Through Time

- **Step 1**: Examine the default settings in Graph 1, where `Initial N` = 10, `r_m` = 0.8, and `K` = 200.
  - What is the maximum population size in relation to the carrying capacity?
  - At what time does the population reach its maximum size?

- **Step 2**: Increase the carrying capacity (`K`) to 300.
  Prediction: What do you expect to happen to the population size over time?

  Test: How does increasing `K` affect the maximum population size and the time taken to reach it?

- **Step 3**: Halve the growth rate (`r_m`) to 0.4.
  Prediction: How will this change affect the population's growth and the time it takes to reach the maximum population size?

  Test: What do you observe in the graph?

- **Step 4**: Increase `r_m` to 1.8.
  - How does the population size change over time?
  - How does the maximum population size compare to the carrying capacity?
  - How would you describe the dynamics of this population (e.g., overshoot, oscillations)?

- **Step 5**: Experiment with higher values for `r_m` (e.g., 2.0, 2.4, 2.5, 2.8, 2.9, 3.0, 3.1).
  - How do the population dynamics change as `r_m` increases?
  - Is there a limit to how high `r_m` can be?
  - What happens when `r_m` is 2.8, 3.0 or 3.1[1]? Are the dynamics predictable or chaotic?

- **Step 6**: Compare the population trajectory for populations with `r_m` = 2.8 and `r_m` = 2.81, and then fix `r_m` at 2.8 but vary the initial population size by a small amount.
  - How sensitive are these populations to small changes in `r_m` or the initial population size?
  - Would these populations be easy or hard to predict?
  - What kinds of species have high population growth rates like this? How might this relate to managing pest species or diseases?

[1]The behaviour where `r_m` values above 3 cause the population to become very negative, is a mathematical artefact caused by numerical instability in the discrete-time logistic growth model. When the growth rate is too high, the population can overshoot its carrying capacity, and the formula's negative feedback term can drive the population to unrealistic, negative values. This issue arises because the model doesn't impose biological constraints, such as preventing population sizes from falling below zero. In real biological systems, such negative values would not occur.

Graph 2: Per Capita Growth Rate vs. Population Size

- **Step 1**: Examine Graph 2 with the default settings (`Initial N` = 10, `r_m` = 0.8, `K` = 200).
  - How does the per capita growth rate change as the population size increases?
  - Where does the per capita growth rate cross the x-axis (population size)?
- **Step 2**: Modify the carrying capacity (`K`) to 300 and observe the changes in Graph 2.
  - How does increasing the carrying capacity affect where the per capita growth rate crosses the x-axis?
  - Does the slope of the per capita growth rate change when you alter `K`?
- **Step 3**: Change `r_m` to 1.8 and observe how the graph is affected.
  - How does increasing `r_m` affect the shape of the curve?
  - Does the intercept on the x-axis change when you increase `r_m`?
- **Step 4**: Experiment with different values for `K` and `r_m`.
  - How do changes in `K` and `r_m` together affect the graph?

Reflection on Graph 2

- **Step 5**: Compare the per capita growth rate in **Graph 2** to the population size over time in **Graph 1**. **Questions**:
  - How are the two graphs connected?
  - What would the per capita growth rate look like for a population experiencing exponential growth instead of logistic growth?

These steps help you visualise how population size and per capita growth rate are interrelated, reinforcing the concepts behind logistic population growth.

These two graphs are different ways to visualise the same model.

It is important that you can make the connections between these graphs.

How would the same plots look for regular exponential growth?

Some useful keywords:

- Oscillation
- Damped oscillation
- Cycle/cyclic dynamics
- Stable-limit cycle (2-point, 4-point, 8-point … limit cycles)
- Chaos/Chaotic dynamics
- Unpredictable/predictable

Takeaways

- Logistic growth captures density dependence through the carrying capacity K.
- High growth rates can cause overshoot, oscillations, or chaos in discrete-time models.
- Per-capita growth declines linearly with population size in the logistic model.

Questions

- In the logistic growth model, what happens to the population growth rate as the population size approaches the carrying capacity (K)? How does this lead to population stability?
- In the logistic growth model, what factors can cause a population to overshoot its carrying capacity temporarily? How does the population respond to such overshooting?
- How can a ballpark knowledge of maximum per capita population growth rate (r_m) be useful to a population manager?
- What are the key assumptions of the logistic growth model, and how might these assumptions limit its real-world application?

Optional: Do this in R.

You can implement this model in R as follows. In this code, I have added a line which ensures that population size cannot be negative.

```
# Parameters
r_m <- 2.9  # intrinsic growth rate
K <- 1000   # carrying capacity
N0 <- 10    # initial population size
timesteps <- 50  # number of time steps

# Initialise population vector
population <- numeric(timesteps)
population[1] <- N0

# Initialise per capita growth rate vector
# Length of one less than population, as it's based on changes
per_capita_growth <- numeric(timesteps - 1)

# Loop to calculate population at each time step
for (t in 2:timesteps) {
  # Logistic growth model equation
  population[t] <- population[t-1] +
    (r_m * population[t-1]) * (1 - (population[t-1] / K))

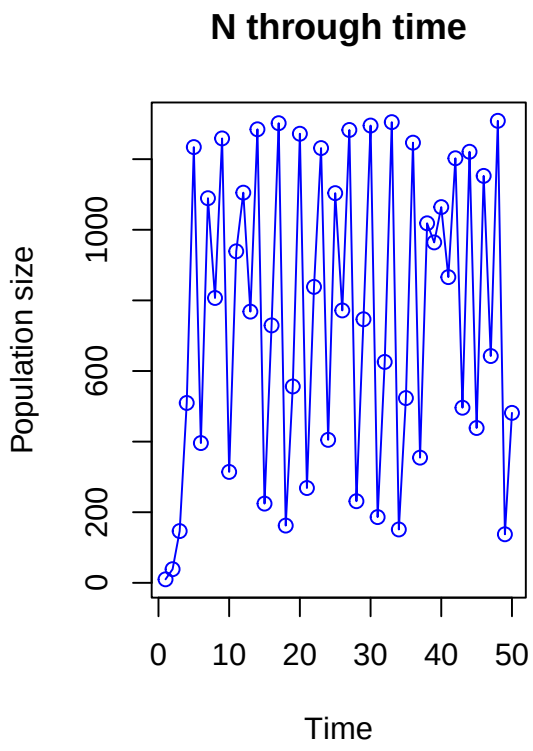
  # Ensure population doesn't go negative
  if (population[t] < 0) {
    population[t] <- 0
  }

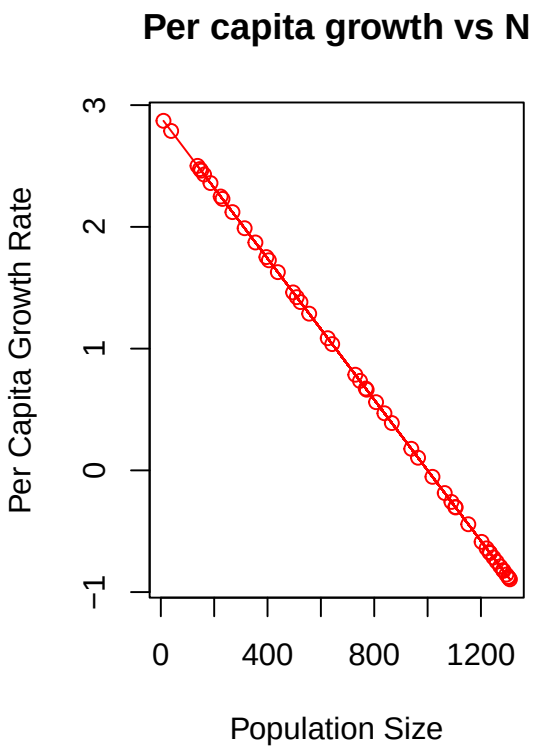
  # Calculate per capita growth rate for the previous time step
  # (undefined if the previous population was zero)
  if (population[t-1] > 0) {
    per_capita_growth[t-1] <- (population[t] -
                                 population[t-1]) / population[t-1]
  } else {
    per_capita_growth[t-1] <- NA
  }
}

par(mfrow=c(1,2))
# Plot 1: Population size through time
plot(1:timesteps, population, type = "o", col = "blue",
```

```
     xlab = "Time", ylab = "Population size",
     main = "N through time")

# Plot 2: Per capita growth rate vs population size
plot(population[1:(timesteps-1)], per_capita_growth,
     type = "o", col = "red",
     xlab = "Population Size",
     ylab = "Per Capita Growth Rate",
     main = "Per capita growth vs N")
```





Deeper into Logistic Growth

Background

The logistic and exponential growth models are closely related and serve as fundamental tools in understanding population dynamics.

- The **exponential** growth model (continuous time; its discrete-time counterpart is *geometric* growth) assumes that populations grow without any environmental constraints, leading to unlimited growth.
- The **logistic** growth model, on the other hand, incorporates environmental limits, specifically the **carrying capacity** (K), which represents the maximum number of individuals the environment can support.

Linking Logistic and Exponential Growth Models

The logistic growth model is represented by the following equation (see the Neal textbook chapter *Logistic Growth*):

$\frac{dN}{dt} = r_m N \left(1 - \frac{N}{K}\right)$

In this model, as the population size N approaches the carrying capacity K, the term $\left(1 - \frac{N}{K}\right)$ decreases, slowing the population growth rate until it reaches zero when $N = K$.

However, if we remove the constraint of a carrying capacity (i.e., set $K = \infty$), the model simplifies to the exponential growth equation:

$\frac{dN}{dt} = r_m N \left(1 - \frac{N}{\infty}\right)$

which simplifies to

$\frac{dN}{dt} = r_m N \left(1 - 0\right)$

which simplifies to

$\frac{dN}{dt} = r_m N$

This is the familiar **exponential growth equation** (Neal chapter *Exponential Growth*), which assumes unlimited resources and continuous population growth. (Its discrete-time counterpart, $N_{t+1} = \lambda N_t$, is called *geometric* growth.)

The Excel workbook uses Neal's *discrete-time* versions of these models, in which the population is updated once per time step: $N_{t+1} = N_t + r_m N_t \left(1 - \frac{N_t}{K}\right)$ for logistic growth, and $N_{t+1} = N_t + r_m N_t$ for unlimited growth. The second is geometric growth with $\lambda = 1 + r_m$: here r_m is playing the role of Neal's per-step rate R_m (see the notation box in the Part 2 introduction). These use the same r_m as the continuous equations, but apply it once per whole time step, so dN/dt is approximated by the change in population size over one time step, $N_{t+1} - N_t$. The same simplification applies: setting $K = \infty$ turns the discrete logistic model into the discrete (geometric) model.

The discrete models are approximations of the continuous ones. For a small population, the discrete model multiplies the population by $1 + r_m$ each time step, whereas the continuous model gives $\lambda = e^{r_m}$ (e.g. 1.8 vs. 2.23 for $r_m = 0.8$, or 1.20 vs. 1.22 for $r_m = 0.2$, as in the exponential growth lecture). The approximation is good when r_m is small and poor when it is large. Stepping forward a whole time unit

at a time is what makes the overshoot, oscillations, cycles and chaos you will see in Task 1 possible. Because the population cannot "see" that it is approaching K until the next time step, it can overshoot. The continuous logistic model never does this: it always approaches K smoothly.

Take-home Message: The logistic and exponential growth models are closely related. By setting $K = \infty$, we transition from the logistic model to the exponential model.

Learning outcomes:

- Increase competence in using Excel for mathematical modelling.
- Understand the relationship between exponential and logistic growth models.
- Learn how models can be adjusted to explore different biological phenomena.
- Develop skills in visualising and interpreting model outputs from different perspectives.
- Strengthen understanding of biological processes by applying mathematical models.

Worked example

Inputs

- `r_m = 1.2`
- `K = 200`
- Worksheet: `BasicLogistic`

Steps

1. Set `r_m` and `K` in the parameter block.
2. Observe Figure 1 (population through time).
3. Observe Figure 2 (per-capita growth vs `N`).
4. Observe Figure 3 (total growth vs `N`).

Output and interpretation

In Figure 1 the population rises quickly and settles at K, overshooting it only very slightly (to about 201) before levelling off. Figures 2 and 3 show the same model from two other perspectives. You will explore what their shapes and axis intercepts tell you in Tasks 2 and 3.

Your Task

Download the Excel file `Deeper Into Logistic Growth.xlsx`.

Look at the `BasicLogistic` worksheet and work through the following tasks:

Task 1: Population Dynamics (Figure 1)

1. Experiment with different values of r_m (e.g., 0.8, 1.2, 1.8, 2.4, 2.7) and observe how the population dynamics change over time in **Figure 1**.
2. Use the following terms to describe the dynamics you see:
    - **Oscillation**, **damped oscillation**, **stable cycle**, **2-point cycle**, **4-point cycle**, **chaotic**, **unpredictable**, **predictable**.

Task 2: Per Capita Growth Rate vs. Population Size (Figure 2)

1. Examine **Figure 2**, which shows the per capita growth rate as a function of population size at time t.

2. Notice where the line intercepts the x- and y-axes.

    - What are these intercepts?
    - How do these intercepts relate to the values of r_m and K that you have set?

3. Try varying the values for r_m and K, and note how the graph changes.

4. On paper, sketch a graph of the per capita growth rate vs. population size for a logistic model with $r_m = 1.5$ and $K = 250$. Then, verify your sketch by entering these values into the Excel model.

Task 3: Population Growth Rate vs. Population Size (Figure 3)

1. **Figure 3** shows the overall population growth rate (dN/dt), the change in population size per unit time. Adjust the values for r_m and K and observe how **Figure 3** changes.

2. Answer the following questions:

    - At what population sizes is the population growth rate 0 ($dN/dt = 0$)?
    - At what population size is the growth rate maximised?

Task 4: Comparison with Exponential Growth

1. Now, consider the differences between **logistic growth** and **exponential growth**. How would Figures 1, 2, and 3 change when considering exponential growth?

Sketch equivalent graphs for the exponential growth model (unlimited growth, with no K), including:

- **Fig 1**: Population size (N) over time (t).
- **Fig 2**: Per capita growth rate ($\frac{1}{N}\frac{dN}{dt}$) vs. population size (N).
- **Fig 3**: Population growth rate ($\frac{dN}{dt}$) vs. population size (N)

2. Look at the **`Exponential` worksheet** to see how close you were. In this sheet the parameter cell is labelled "r_m intrinsic growth rate" (default 0.1). Because the sheet applies it once per time step, $N_{t+1} = N_t(1 + 0.1)$, it is really acting as the per-step rate R_m, giving geometric growth with $\lambda = 1.1$ (the continuous model would give $e^{0.1} = 1.105$). The note on the sheet that dN/dt "increases linearly" means linearly with N (Figure 3); through time, dN/dt increases exponentially.

Task 5: Adding a Time Lag

1. In the **`TimeLag` worksheet**, explore how adding a time lag to the logistic model affects population dynamics. This is the equation we are using: $\frac{dN}{dt} = r_m N_t \left(1 - \frac{N_{t-\tau}}{K}\right)$, where τ is the length of the time lag. With $\tau = 1$, density dependence acts on the population size one time step ago rather than the current population size.

2. Adjust the formula in the Excel sheet to incorporate a time lag in the population size ($N_{t-\tau}$). The `dN/dt` column (column C) currently uses the current population size. For a 1-step lag, change the formula in cell C11 to `=$B$5*B11*(1-(B10/$B$6))`, so that the density-dependent term uses the population size from the previous time step. Then fill this formula down to the bottom of the column. Before you do, check that B5 holds r_m, B6 holds K and row 11 is the second row of data in your copy of the sheet (note that this sheet starts at Time = 1). The formula works in Danish Excel as written.

3. Start with the default $r_m = 0.6$, which gives smooth convergence to K in the ordinary logistic model. With the 1-step lag, you should see the population overshoot K and then show damped oscillations before settling. Now increase r_m to 1.2. What happens to the oscillations?

4. Compare this with the ordinary logistic model (no lag) at the same r_m values (e.g. using the `BasicLogistic` worksheet). This exercise demonstrates how a simple life history trait (such as a time lag) can make a population overshoot and oscillate at growth rates that would otherwise give smooth convergence to K. With a large enough lag or growth rate, the oscillations become sustained cycles.

5. Note that the lecture's thresholds for time-lag models ($r_m\tau$ below $e^{-1} = 0.37$ gives a smooth approach, up to $\pi/2 = 1.57$ damped oscillations, above that sustained cycles with a period of about 4τ) are for the *continuous-time* lag model. In this *discrete* version with a one-step lag, the approach to K is smooth only for r_m below about 0.25, there are damped oscillations from about 0.25 to 1, and sustained cycles above $r_m = 1$, with a period of about 6 to 7 time steps. So do not expect the Excel results to match the lecture's numbers exactly; the qualitative message (lags cause overshoot and cycles) is the same.

Questions

- How does increasing or decreasing r_m affect the shape and behaviour of the population time series in Figure 1? How does it change the per capita growth rate curve in Figure 2 and the population growth rate in Figure 3?

- The peak of the parabola in Figure 3 gives the *maximum sustainable yield* (MSY): the largest number of individuals that could be harvested each time step without the population declining. What is the MSY for a population with $r_m = 0.8$ and $K = 200$, and at what population size should the population be kept to achieve it? Why might harvesting at exactly the MSY be risky in practice?

- In the discrete model, $N_{t+1} = N_t + dN/dt$. Using what you know about Figure 3, what shape would a plot of N_{t+1} against N_t have? Where would it cross the line $N_{t+1} = N_t$, and why?

- What happens to the population dynamics when a time lag is introduced? A time lag could be caused by a long gestation period, or by the need to take a year out before breeding again.

Takeaways

- Logistic and exponential growth are linked by the carrying capacity term.
- Figures of N vs. time, per-capita growth rate vs. N and dN/dt vs. N are three views of the same model; you should be able to move between them.
- In discrete time, high r_m causes overshoot, oscillations, cycles and chaos.
- Time lags can destabilise otherwise stable dynamics.

Life tables and survivorship types

Background

Life tables are powerful analytical tools used in various fields to understand and quantify mortality patterns and life expectancy within populations. Originating in population biology and demography, life tables have found extensive applications in epidemiology, human mortality studies, and even product life-cycle management. By systematically tracking and analysing the survival and mortality rates of individuals within specific cohorts, life tables reveal essential information about population dynamics, age-specific risks, and the factors that influence longevity. When they also record reproduction, they tell us how fast a population grows.

The basic algebra used in life tables is explained in Neal, in the chapters *Life Tables* and *Growth of Age-Structured and Stage-Structured Populations*, and in the lecture "Life tables and age-structured population growth". This practical uses the same notation as Neal and the lecture (see the table below). Gotelli's *A Primer of Ecology* (optional) covers the same material with different symbols: g_x for p_x, b_x for m_x and G for T.

In addition to providing valuable insights into mortality patterns and life expectancy, life tables also allow us to consider mortality patterns in the context of distinct survivorship types that result from the way mortality risk changes over the life course. Three primary survivorship types are commonly identified, based on survivorship curves (l_x) plotted on a log scale:

- Type I (convex curve): mortality is low early in life and **increases** with age. Most individuals survive to old age, and most deaths occur in older age groups. Type I survivorship is seen in humans and other long-lived species with strong parental care and low early-life mortality.
- Type II (straight line): mortality is **constant** with age, so individuals have about the same chance of dying at any age. Some small mammals and many birds come close to this pattern.
- Type III (concave curve): mortality is high early in life and **decreases** with age. Survival is low in the early stages of life, but those that survive the vulnerable period are likely to live a long time. Type III survivorship is typical of species that produce many offspring with little parental care, such as many fish and trees.

Real species are often mixtures: for example, Dall sheep and many birds have high juvenile mortality (Type III-like), followed by mortality that rises with age in adults (Type I-like).

By identifying and understanding these survivorship types, researchers gain critical insights into the demographic structure of populations and the selective pressures that influence survival and longevity.

Learning outcomes:

- Increased competence in using Excel formulae for mathematical modelling.
- Competence in using mathematical models in Excel to strengthen your own understanding of biological processes.
- Building a cohort life table, and interpreting survivorship (l_x), survival probability (p_x), probability of death (q_x) and life expectancy (e_x).
- Calculating the net reproductive rate (R_0), generation time (T) and the capacity for increase ($r_c = \ln(R_0)/T$), an approximation to the intrinsic rate of increase r_m.

- Understanding the three types of survivorship curve and how they relate to the way mortality changes with age.
- Understanding why changes in reproduction early in life matter more than changes late in life (the declining "force of selection").

Notation

Symbol	Meaning	Calculation
x	Age at the start of the interval ($x = 0$ means age 0 to 1)	
S_x	Number alive at the start of age x	counted
l_x	Survivorship: probability of surviving from birth to age x	S_x/S_0
p_x	Survival probability: probability of surviving from age x to $x+1$	S_{x+1}/S_x, or l_{x+1}/l_x
q_x	Probability of dying between age x and $x+1$	$1-p_x$
e_x	Life expectancy: further years expected at age x	$\left(\sum_{y\geq x} l_y\right)/l_x - 0.5$
μ_x	Force of mortality: the instantaneous death rate of individuals still alive at age x (a rate, not a probability)	if μ is constant over the year, $q_x = 1 - e^{-\mu_x}$
m_x	Female offspring per female aged x	
R_0	Net reproductive rate: female offspring per female per lifetime (per generation)	$\sum l_x m_x$
T	Generation time: mean age of mothers when their daughters are born	$\sum x l_x m_x / R_0$
r_c	Capacity for increase: approximate population growth rate per year (an approximation to r_m)	$\ln(R_0)/T$
r_m	Intrinsic rate of increase per year (exact, from the Euler-Lotka equation); $\lambda = e^{r_m}$	solve $\sum e^{-r_m x} l_x m_x = 1$

Worked example

The *Worked example* sheet in the spreadsheet is the life table from the lecture: a cohort of 500 newborn females, with S_x = 500, 400, 200, 50 and 0 at ages 0 to 4, and m_x = 0, 2, 3, 1 and 0. Click on the cells to see the formulas. For example:

- Survivorship: $l_1 = 400/500 = 0.8$
- Survival probability from age 1 to 2: $p_1 = 200/400 = 0.5$, so $q_1 = 1 - 0.5 = 0.5$
- Life expectancy at birth: $e_0 = (1 + 0.8 + 0.4 + 0.1)/1 - 0.5 = 1.8$ years
- Net reproductive rate: $R_0 = 0 + 1.6 + 1.2 + 0.1 = 2.9$
- Generation time: $T = (1 \times 1.6 + 2 \times 1.2 + 3 \times 0.1)/2.9 = 4.3/2.9 = 1.48$ years

- Capacity for increase: $r_c = \ln(2.9)/1.48 = 0.72$ per year, so $\lambda \approx e^{r_c} = 2.05$

Note that $r_c = \ln(R_0)/T$ is an approximation to the intrinsic rate of increase, r_m. The exact value for this life table ($r_m = 0.776$, $\lambda = 2.17$) comes from the Euler-Lotka equation or from a matrix model (next practical). The spreadsheet labels this quantity "r (approx.)".

Your task

Download and open the Excel file `Life_Tables_Exercises.xlsx`.

The file has three worksheets: "*Life table*", "*Survivorship Curves*" and "*Worked example*".

Life table

Let's start with "**Life table**".

The aim now is to use Excel as a modelling tool to produce a life table.

I have provided some initial data collected from a cohort of animals. I know how many individuals start each year (i.e. how many "**enter** an interval") and therefore I can see how many survive each year (i.e. how many "enter the **next** interval"). I also know how many female offspring (on average) are produced by each female of each age (m_x).

Start by calculating survivorship (l_x). Survivorship is the **probability of surviving from birth to a particular age**. Therefore, at age 0, $l_0 = 1$, since everyone is alive at this point. The next value (l_1) must be calculated based on the number alive at that point. In this case it is $352/500 = 0.704$. You must generalise this calculation into a formula that can be dragged to fill column `D` in the worksheet. In algebraic form, the equation is $l_x = S_x/S_0$.

Tip You need to understand the use of the `$` symbol in Excel, and how to drag the selected area to place the formula in the column.

Next, you can calculate the age-specific **survival probability**, p_x. Note that this is different from l_x. Survival probability is simply the probability that an individual will survive its current age class, i.e. the probability that an individual currently aged 2 will survive to become age 3. In this case, it is $254/298 = 0.852$. The calculation is $p_x = l_{x+1}/l_x$, or S_{x+1}/S_x.

From there you can calculate the age-specific **probability of death**, q_x. This is simply $1 - p_x$: the probabilities of survival and death **must** sum to 1.

Then calculate **life expectancy**, e_x, in column `I`: add up the l_x values from age x to the end of the table, divide by l_x, and subtract 0.5 (because, on average, individuals that die during a year die halfway through it). For example, in the worked example $e_1 = (0.8 + 0.4 + 0.1)/0.8 - 0.5 = 1.125$ years. **Tip** `=SUM(D3:$D$13)/D3-0.5` in cell `I3` can be dragged down the column.

Now complete the remaining columns, $l_x m_x$ and $l_x m_x x$, and use them to calculate (a) the net reproductive rate, $R_0 = \sum l_x m_x$; (b) the **generation time**, $T = \sum x l_x m_x / R_0$; and (c) the capacity for increase, $r_c = \ln(R_0)/T$, which approximates the population growth rate per year. Is this population growing, stationary or shrinking?

Refer to the sheet "*Worked example*" if you get stuck (click on the cells to see the formulae used there).

Declining force of selection (optional, if time)

This experiment is optional here: you will do a fuller version in the practical *Trade-offs and the declining force of selection*. Do it if you have finished the rest of the life table and the survivorship curves.

The force of selection describes the relative importance of events that happen during the life course for fitness (measured, for example, by R_0 or r_m).

To investigate this you can do a computer "experiment" by altering reproduction at different ages. You can think of these as "what if?" experiments, e.g. "What if there was a mutation that added 1 to reproduction at age 1, or at age 2, etc.?"

Before you start, **predict**: will adding one daughter at age 2 or at age 8 have the bigger effect on R_0? Why?

Add 1 to m_x at one age at a time, from age 1 to age 10 (put it back before moving to the next age), and record the new value of R_0. Do not include age 0: newborns cannot breed at birth. Graph your results as age vs. change in R_0.

What you should see is that changes early in life matter more than changes later in life. (Adding one daughter at age x raises R_0 by exactly l_x: a female must survive to age x to have that extra daughter.)

Use R_0, not r_c, for this experiment. The approximation $r_c = \ln(R_0)/T$ can even fall when reproduction is added late in life, because T increases; the exact r_m always increases, but by less and less with age.

Survivorship curves

In the second part of this class we focus on the **Survivorship Curves** worksheet.

The aim here is to explore how different types of organisms with different ways of life ("**life history strategies**") can have qualitatively different kinds of life tables. The most important thing to observe is the difference in **survivorship curves** (l_x). These differences become very obvious when you plot the log-transformed survivorship against age.

In the Excel worksheet, I have placed tables showing the fate of cohorts of three populations of different species. Your job now is to calculate the survivorship curve (l_x) for these species, and take the natural log (using the formula `=LN(C3)` for the first population, `=LN(J3)` for the second population, etc.).

You should see that the graphs automatically fill up with lines. These show Type I, II and III survivorship.

Next, calculate the age-specific probability of death (q_x) and of survival (p_x). These are the proportions of individuals that **enter** the age interval that **do not** and **do** survive to the end of the interval, respectively. You can calculate them as `=1-(B4/B3)` and `=B4/B3` for the first row of the first population.

Plot graphs of these.

Make sure that you are (1) able to "diagnose" survivorship type from looking at a graph of $\ln(l_x)$ vs x; and (2) able to sketch a cartoon of the mortality trajectory if shown one of these survivorship curves.

Questions

1. What are the three main types of survivorship curves, and which organisms typically exhibit each type?

2. What does a net reproductive rate (R_0) of less than 1 signify for a population?

3. How does the shape of a survivorship curve reflect trade-offs in life history strategies, typically?

4. How is survivorship (l_x) different from survival probability (p_x)?

5. How does altering reproduction or survival/mortality at different ages affect R_0 and population growth rate?

6. Two populations have the same R_0, but one has a shorter generation time. Which grows faster per year, and why?

7. The probability of dying in a year, q_x, can never be greater than 1, but the force of mortality, μ_x (see the Notation table and the lecture), can. Why?

Takeaways

- Life tables summarise age-specific survival and reproduction in a compact, comparable way.
- Survivorship type reflects how mortality changes with age, and so different life history strategies.
- R_0 is growth per generation; dividing $\ln(R_0)$ by the generation time T gives the capacity for increase, r_c, an approximation to the growth rate per year, r_m.
- Early-life changes usually have the greatest impact on R_0 and population growth.

Matrix population modelling

Background

Matrix population models are powerful and widely used tools in population ecology for studying the dynamics of structured populations. Unlike simple growth models that assume every individual is the same, they take into account variation in vital rates (survival, growth and reproduction) across different life stages or age classes. They are particularly well suited to species with distinct life stages, such as plants with seedlings, juveniles and adults, or animals with different age classes.

A matrix population model uses a transition matrix to describe how individuals move between stages, and how many new individuals each stage contributes, over a fixed time step. By incorporating demographic data and life-history traits, these models give a more realistic picture of population dynamics. They are widely used to *project* the future trajectory of a population under stated assumptions, and to assess how vulnerable it is to environmental change and management actions. In this practical, we will assume that one matrix projection corresponds to one year. Each matrix element therefore represents survival, growth, or reproduction over a one-year interval.

This practical aims to give you a good understanding of the basics of their construction and use. The background reading is Neal's chapter *Growth of Age-Structured and Stage-Structured Populations.*

Learning outcomes:

- Competence in constructing life cycle diagrams to represent the life history of real (or theoretical) organisms.
- Understanding how to parameterise life-cycle diagrams and use them to produce a matrix population model.
- Competence in using R to calculate a population growth rate and project a population.
- Understanding how to connect these results to a management question.
- Understanding the logic of "*in silico* experiments" to investigate a biological question (mathematical modelling).

Your task

1) First, think of an organism you would like to model the dynamics of. It could be a mammal, a bird, a fish, insect or tree … real or fantasy.

2) Think about their life cycle, and draw it as a life cycle diagram with circles indicating the stages and arrows representing transitions between stages (e.g. growth) and reproduction. Keep it simple: aim for **2-3 stages** and no more than about **7 arrows**. Your diagram must include at least one reproduction arrow and at least one survival arrow. If your model is stage-based, consider whether individuals can stay in the same stage from one year to the next (an arrow that loops back to the same circle).

Things to think about:

- Is it age based or stage based?

- How many stages are there? Can you simplify (e.g. instead of age in years, you could use life stage)
- If using stages, how are stages defined? E.g. by size, by development, etc.
- Are the survival and fecundity higher in earlier or later life?
- Does the organism ever skip stages?
- Can the organism move "backwards" through the life cycle?

3) Next to the arrows, write values for survival probability and fecundity using your biological knowledge. It is fine to use "ballpark" estimates. Read the box below before you choose your fecundity values: a reproduction arrow is not simply the number of babies.

What goes on a reproduction arrow?

A matrix entry tells you how many individuals will be in a stage at the *next* census (next year's count) for each individual in a stage *now*. So a reproduction arrow is the number of newborns that will be counted next year per individual this year. That includes whatever survival is needed for those newborns to be counted:

- If you count the population just **after** breeding (a post-breeding census), the parent must survive the year to breed again before the next count. The entry is the parent's survival × the number of offspring it then produces. For example, an individual that survives the year with probability 0.8 and then has 2 offspring contributes $0.8 \times 2 = 1.6$ newborns to next year's count.
- If you count just **before** breeding (a pre-breeding census), the newborns must survive from birth until the next count. The entry is the number of offspring × their first-year survival.

Putting the raw number of babies straight into the matrix is the most common mistake in building matrix models (see the lecture, and the chapter "Pre- and Post-reproduction census" for a worked comparison). Demographers call this matrix entry *fertility* (Neal: the fertility coefficient); here we use the everyday word fecundity for the same thing.

The model also follows females only (counting daughters), and assumes there is no shortage of mates.

Sanity check your numbers

- Survival and stage-transition probabilities must be between 0 and 1.
- Fecundity (new individuals counted at the next census, per individual per year) can be greater than 1.
- For individuals starting in a given stage, the survival and transition probabilities leaving that stage cannot add up to more than 1. For example, 0.8 staying in the same stage plus 0.7 moving to the next stage is impossible, because it adds up to 1.5. (Fecundity does not count towards this total, because newborns are *added* to the population rather than being a fate of the parent.)

Here's an example for a fictional organism.

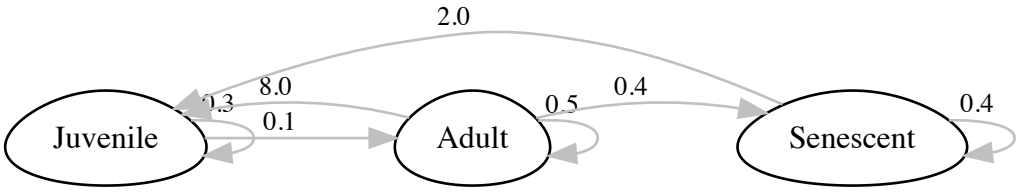


Figure 4: Life cycle diagram of the example organism.

4) Now you can turn this diagram into a matrix population model by filling in a square of survival/fecundity values.

The life cycle shown above looks like this (below). The matrix describes transitions from time t to time $t+1$, where one time step corresponds to one year.

Interpretation note: **columns are stages at time t (from)** and **rows are stages at time $t+1$ (to)**.

To place each arrow in the matrix, ask two questions:

- **FROM which stage does the arrow start?** This gives the **column**.
- **TO which stage does the arrow point?** This gives the **row**.

Before you touch R, fill in the blank table below on paper (or in your notes) for your own organism. Name your stages along the top and down the side, then write each arrow's value in the correct cell. Leave a cell as 0 if there is no arrow.

TO ↓ / FROM →	Stage 1	Stage 2	Stage 3
Stage 1			
Stage 2			
Stage 3			

Check your work: every arrow is one non-zero matrix element.

Count the arrows in your life cycle diagram, then count the non-zero cells in your table. The two numbers must match. If they don't, you have either missed an arrow or put a value in the wrong cell. Also check the column totals against the sanity check above.

For the example organism, the diagram has 7 arrows (including the loops), and the matrix below has 7 non-zero cells.

$$A = \begin{bmatrix} 0.3 & 8 & 2 \\ 0.1 & 0.5 & 0 \\ 0 & 0.4 & 0.4 \end{bmatrix}$$

Using R for matrix modelling

Working with matrices is very tedious in Excel. However, in R you can use this information to predict the future dynamics of the population, and estimate the population growth rate.

Open up **RStudio**, and let's see if we can predict future dynamics. First you will need to install a package called `popdemo`.

Package setup

If you are running this chapter in a fresh R session, you might need to install and/or load the `popdemo` package.

```
install.packages("popdemo")
```

You only need to install packages once. After that you can load the package for use by using the `library` function.

```
library(popdemo)
```

You can put your matrix into R like in the example below (change the numbers to match YOUR model). If your model has fewer, or more, stages then you will need to modify the code a bit. Ask for help if you get stuck.

```r
A <- matrix(c(
  0.3, 8.00, 2.00,
  0.1, 0.50, 0.00,
  0.0, 0.40, 0.40
), ncol = 3, byrow = TRUE)

stage_names <- c("Juvenile", "Adult", "Senescent")
dimnames(A) <- list(stage_names, stage_names)
A
```

```
##           Juvenile Adult Senescent
## Juvenile       0.3   8.0       2.0
## Adult          0.1   0.5       0.0
## Senescent      0.0   0.4       0.4
```

Important: interpret matrix elements over the chosen time step

In matrix population models, all transitions occur over a fixed time interval. In this practical, we use a 1-year time step, so:

- survival probabilities are annual
- fecundity is offspring per individual per year
- λ is growth per year

Projecting the population

And now you can use the project function to project what happens to the population, then plot it. Look at what happens if you log or don't log the y-axis. First you need to define an initial starting population structure.

In my example, I have 3 stages, so I have 3 values for the initial population sizes.

What the projection does

Let $\mathbf{n}_t$ be the vector of stage abundances at time t. The population at the next time step is found by multiplying the matrix by that vector:

$$\mathbf{n}_{t+1} = \mathbf{A}\,\mathbf{n}_t$$

Each row of $\mathbf{A}$ answers the question "where do the individuals in this stage next year come from?": multiply each entry in the row by the number of individuals in the matching *from* stage (column), then add them up. Using the example matrix and starting population of 10, 5 and 3 individuals in the juvenile, adult and senescent stages:

$$\begin{bmatrix} n_J \\ n_A \\ n_S \end{bmatrix}_{t=1} = \begin{bmatrix} 0.3 & 8.0 & 2.0 \\ 0.1 & 0.5 & 0.0 \\ 0.0 & 0.4 & 0.4 \end{bmatrix} \begin{bmatrix} 10 \\ 5 \\ 3 \end{bmatrix} = \begin{bmatrix} (0.3 \times 10) + (8.0 \times 5) + (2.0 \times 3) \\ (0.1 \times 10) + (0.5 \times 5) + (0.0 \times 3) \\ (0.0 \times 10) + (0.4 \times 5) + (0.4 \times 3) \end{bmatrix} = \begin{bmatrix} 49 \\ 3.5 \\ 3.2 \end{bmatrix}$$

So after one year there are 49 juveniles (mostly newborns produced by adults), 3.5 adults and 3.2 senescent individuals. To project two years ahead, we simply repeat the multiplication using the new vector. R does this for us with the `%*%` operator for matrix multiplication:

```
initial <- c(10, 5, 3)
A %*% initial
```

```
##           [,1]
## Juvenile  49.0
## Adult      3.5
## Senescent  3.2
```

The `popdemo` function `project` repeats this step as many times as we ask. Here I use it to do a population projection for 10 years (10 time steps). Each step represents one annual cycle of survival, growth, and reproduction.

Predict first. Before you run the projection for your own organism, write down your answers to these questions:

- Do you expect your population to increase, decrease or stay roughly stable? Why?
- Which stage do you expect to become the most abundant?

Then run the projection and compare the result with your prediction. If it surprises you, work out why: look at which matrix elements are large and which are small.

```
pr <- popdemo::project(A, vector = initial, time = 10)
```

Take a look at `pr`, the projected population. This gives you the total population size, and below that the population sizes in each stage.

```
pr
```

```
## 1 deterministic population projection over 10 time intervals.
##
##  [1]  18.0000  55.7000  58.4300  85.2570 111.5643 151.4946 203.2634 273.8278
##  [9] 368.4012 495.8572 667.3121
```

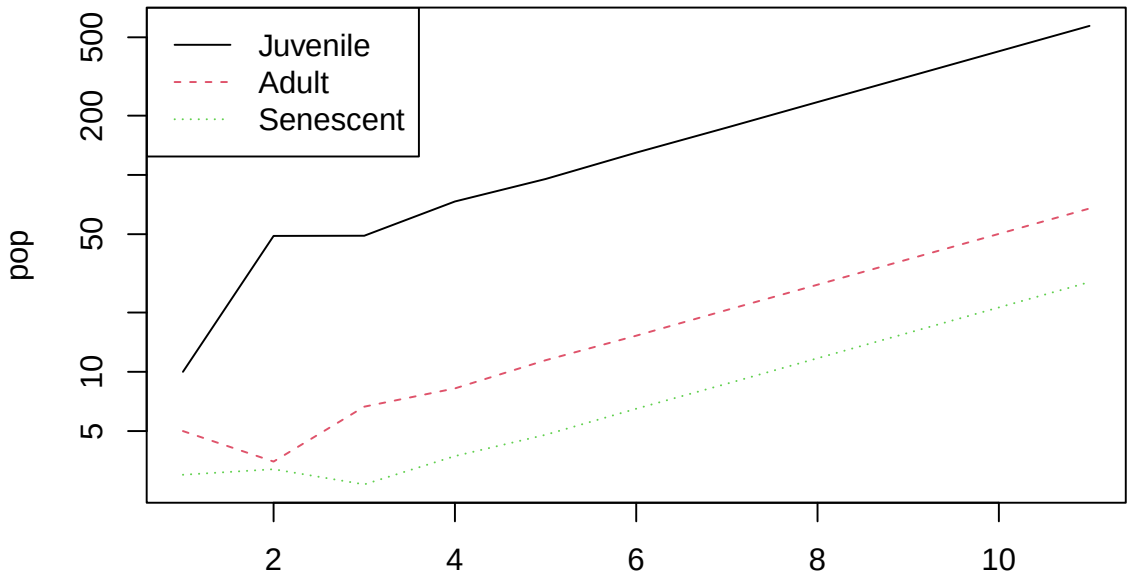
You can access the population sizes of the different stages using `vec(pr)`.

```
vec(pr)
```

```
##       Juvenile    Adult Senescent
##  [1,]  10.0000  5.00000  3.000000
##  [2,]  49.0000  3.50000  3.200000
##  [3,]  49.1000  6.65000  2.680000
##  [4,]  73.2900  8.23500  3.732000
##  [5,]  95.3310 11.44650  4.786800
##  [6,] 129.7449 15.25635  6.493320
##  [7,] 173.9609 20.60267  8.699868
##  [8,] 234.4093 27.69742 11.721013
##  [9,] 315.3442 37.28964 15.767375
## [10,] 424.4552 50.17924 21.222808
## [11,] 571.2161 67.53514 28.560821
```

Let's plot this…

```
pop <- vec(pr)
matplot(pop, type = "l", log = "y")
legend("topleft", legend = colnames(pop),
       col = 1:ncol(pop), lty = 1:ncol(pop))
```



For the example matrix, the population increases exponentially. Does your own population behave as you predicted? The population growth rate is the so-called "*dominant eigenvalue*" of the matrix **A**.

Does the starting population matter?

Now try the projection again, but with a very different starting population, for example nearly all individuals in the youngest stage, or nearly all in the oldest stage. Keep the matrix the same and change only `initial`:

```
initial_2 <- c(100, 0, 0) # change to suit your own life cycle
pr_2 <- popdemo::project(A, vector = initial_2, time = 10)
pop_2 <- vec(pr_2)
matplot(pop_2, type = "l", log = "y")
```

Predict, then compare.

- Before you run it: will the *early* part of the trajectory look the same as before? Will the *late* part?
- After you run it: what changes? What eventually becomes the same?

Hint: look at the slopes of the lines on the log scale, and at the vertical gaps between them, once the first few years have passed.

Growth rate and stable stage distribution

We can ask R for the *eigenvalues* and *eigenvectors* of the matrix. The dominant eigenvalue is the population growth rate (λ). The corresponding eigenvector gives the stable stage distribution (*SSD*): the expected *proportion* of individuals in each stage class once the population has settled into its long-term pattern. This is exactly what your projections with different starting populations were converging towards.

Predict first. Look at your projection plot. Is λ above 1, below 1, or about 1? Roughly how steep is the line on the log scale? Which stage do you think will have the highest *SSD*?

Because our projection interval is one year, λ represents the annual population growth rate. You can see that in this case, using my example values the population is growing 34.6% per year.

```
eigs(A, what = c("lambda", "ss"))
```

```
## $lambda
## [1] 1.345824
##
## $ss
## [1] 0.85599732 0.10120281 0.04279987
```

Elasticity

Elasticity analysis asks how responsive the population growth rate (λ) is to a small *proportional* change in each transition of the matrix. A transition with a high elasticity is one where a given percentage change produces a relatively large percentage change in λ. The elasticities of all the matrix elements sum to one, so each can be read as that transition's share of the total influence on λ. This is useful in management and conservation when we ask questions like, "which parts of the lifecycle should we focus on to preserve the population?". The mathematics are beyond this course, and R can calculate the elasticities for us.

Predict first. Looking at your own matrix, which single element do you think has the highest elasticity, i.e. which one would change λ most, in proportional terms, if you changed it by the same percentage? Write it down before running the code.

```
popdemo::elas(A)
```

```
##              Juvenile      Adult  Senescent
## Juvenile  0.09517264 0.30005511 0.03172421
## Adult     0.33177932 0.19612796 0.00000000
## Senescent 0.00000000 0.03172421 0.01341654
```

Which transition has the highest elasticity, and what does that tell you about your organism's life cycle? Was your prediction correct? If not, why do you think the biology of your organism led you to the wrong guess? Note that a high elasticity means λ responds strongly to a *proportional* change in that transition, which is not the same as saying it is easy or practical to change in the real world.

Elasticity is not the same as *sensitivity*. Sensitivity measures the change in λ for a small *absolute* change in a matrix element (e.g. adding 0.01), whereas elasticity measures the proportional change in λ for a proportional change in the element (e.g. a 1% increase). Because elasticity rescales by the size of each element, small probabilities tend to have a high sensitivity but a lower elasticity, and large fecundities the reverse, so the two rankings can differ. You can compare them:

```
popdemo::sens(A)
```

```
##           [,1]       [,2]       [,3]
## [1,] 0.426952 0.05047766 0.02134760
## [2,] 4.465165 0.52790728 0.00000000
## [3,] 0.000000 0.10673799 0.04514076
```

For the example matrix, adult fecundity has a low sensitivity (0.05) but one of the highest elasticities (0.30), because it is a large number (8), so a 1% change in it is a large absolute change. Juvenile to adult (0.1) has the highest sensitivity (4.47) and also the highest elasticity (0.33).

In silico experiments: changing the life cycle

Now that we have a model, we can use it as an experimental system. The idea is very simple: you run "experiments" on your matrix model (*in silico*, i.e. on the computer) by asking "what if" questions. For example, what would happen if we could increase survival of the juveniles by 20%? what would happen if adults are hunted more, and thus have a decreased survival by 60%? what would happen if we provided supplemental food to reproducing females, and increase fecundity by 50%? etc. Use your imagination!

For each "what if" question, follow the same pattern: **predict** what will happen to λ, **change** the matrix, **observe** the new λ, and **explain** the difference in biological terms.

In practice, we do that simply by modifying the matrix model. In the following, I am looking at the effect of increasing fecundity in adult and senescent individuals by 50%:

```
A_mod <- matrix(c(
  0.3, 8.00*1.5, 2.00*1.5,
  0.1, 0.50, 0.00,
  0.0, 0.40, 0.40
),
byrow = TRUE, nrow = 3,
dimnames = list(stage_names, stage_names)
)

popdemo::eigs(A, what = "lambda")     # original
```

```
## [1] 1.345824
```

```
popdemo::eigs(A_mod, what = "lambda") # modified
```

```
## [1] 1.546586
```

```
popdemo::elas(A_mod)
```

```
##              Juvenile      Adult   Senescent
## Juvenile  0.08514005 0.32540118 0.028380016
## Adult     0.35378119 0.16901684 0.000000000
## Senescent 0.00000000 0.02838002 0.009900705
```

Your investigation

Now it is time to work with your own organism. Rather than just running every function once, you will carry out a small investigation of a biological question.

1. **Project and describe.** Do a projection for your organism, and calculate and interpret λ, the stable stage distribution and the elasticities (remember to write down your predictions first).
2. **Choose a question.** Pick a "what if" question about your organism. For example:
   - What happens if juvenile survival improves?
   - What happens if adults reproduce more?
   - What happens in a bad year (e.g. lower survival in all stages)?
   - What if reproduction increases at the cost of adult survival?
3. **Predict.** Before changing the model, write down your prediction for what will happen to λ, and why.
4. **Change the model.** Modify the appropriate matrix element(s) in a copy of your matrix (e.g. `A_mod`), so that you keep the original for comparison.
5. **Compare.** Calculate λ for both matrices. Was your prediction right?
6. **Explain.** Explain the result in biological terms. Use your elasticity results to help: does the change you made involve a transition with high or low elasticity?

An evolutionary experiment (homework, optional)

If you run out of time in class, do this section at home. A model answer is in the Solutions.

You can think of lambda (population growth rate) as being a measure of fitness. Imagine that some of your population had a mutation that caused them to have, say, 1 extra baby, but at the expense of reduced survival in one of the younger stages. Would this mutation persist in the population?

First, predict: increasing fecundity is beneficial and decreasing survival is costly. Which effect do you think will win? Then modify the matrix, calculate λ for the mutant life history, and compare it with λ for the original. Does the mutant have a higher or lower λ? What does that suggest about selection on this trade-off? (Comparing λ like this treats it as fitness, which works when growth is density-independent, as it is in this model. With density dependence, as in the logistic growth chapters, things are more complicated.)

Questions

1. In the graph showing log-transformed population size through time, what is the significance of the lines being straight after the transient phase?
2. Explain how an elasticity analysis of a matrix model can be used to inform the management of a threatened species.
3. What are some of the assumptions of a matrix population model? (Hint: some are similar to the assumptions of exponential/geometric growth models. Think about the environment, vital rates, migration and population density.)

Takeaways

- Matrix models link life history to population growth through structured vital rates.
- The dominant eigenvalue λ summarises long-term growth, while its eigenvector gives the stable stage distribution.
- Elasticity analysis and *in silico* experiments turn models into management and evolutionary experiments.

Pre- and Post-reproduction census

Background

Population growth estimates depend on when the census is taken relative to reproduction. A pre-breeding census counts individuals just before reproduction; a post-breeding census counts them just after reproduction. This changes how survival and fertility are represented in the projection matrix, even if the underlying biology is the same.

Provided the conversion between the two census types is done correctly, the population growth rate (λ) they produce is identical: census timing is just a bookkeeping convention, not a biological difference. However, it is easy to get the conversion wrong, and doing so *does* change your estimate of λ. This chapter shows both: a correct conversion (same λ), and a common mistake (different λ).

The background reading is the chapter *Growth of Age-Structured and Stage-Structured Populations* in Neal, which builds a Leslie matrix for a post-breeding census. This chapter follows on from the fecundity box in "Matrix population modelling".

Learning outcomes

Learning outcomes:

- Explain the difference between pre-breeding and post-breeding census models.
- Convert a post-breeding matrix to a pre-breeding matrix.
- Recognise that a correct conversion leaves λ unchanged, and explain why.
- Identify how a mis-specified survival or fertility term changes the resulting λ estimate.

Worked example

The life history

We use a simple bird-like life history, with the same notation as in the life tables practical. Individuals breed for the first time at age 2, breed again at age 3, and none survive beyond age 3. We follow females only, so m_x counts daughters.

Parameter	Value	Meaning
p_0	0.2	Survival from birth (age 0) to age 1 (first-year survival)
p_1	0.9	Survival from age 1 to age 2
p_2	0.6	Survival from age 2 to age 3
m_1	0	Daughters per female aged 1 (age-1 birds do not breed)
m_2	3.0	Daughters per female aged 2
m_3	6.0	Daughters per female aged 3

Breeding happens once a year, in a short breeding season (a "birth pulse"). The only question is whether we count the population just after or just before that breeding season. Each choice gives a different set of age classes:

	Post-breeding census (`A1`)	Pre-breeding census (`A2`)
When we count	Just **after** breeding: this year's newborns are included	Just **before** breeding: this year's newborns are not born yet
Age classes in the matrix	Ages 0, 1, 2	Ages 1, 2, 3
Survival arrows (sub-diagonal)	Age 0 $\to$ 1 (p_0), age 1 $\to$ 2 (p_1)	Age 1 $\to$ 2 (p_1), age 2 $\to$ 3 (p_2)
Fertility arrows (top row)	From age 1: $p_1 m_2$; from age 2: $p_2 m_3$	From age 2: $m_2 p_0$; from age 3: $m_3 p_0$
Whose survival is in the fertility	The **parent's**: it must survive the year and then breed at the next breeding season	The **newborns'**: they must survive from birth to the next census (p_0)
Life cycle	0 $\to$ 1 $\to$ 2, with fertility arrows from 1 and 2 back to 0	1 $\to$ 2 $\to$ 3, with fertility arrows from 2 and 3 back to 1

Notice that in the post-breeding census the oldest class is age 2: those individuals survive (with p_2), breed at age 3 just before the next census, and then die, so they appear only through their fertility. In the pre-breeding census the youngest class is age 1: newborns are never counted at age 0.

Post-breeding census

We start with the **post-breeding census** matrix. Individuals are counted right after breeding, so the fertility of an individual counted at age x is $p_x m_{x+1}$: it must survive the year (p_x), and then breed at age $x+1$, just before the next census, where its newborns are counted at age 0. For example, a bird counted at age 1 contributes $p_1 m_2 = 0.9 \times 3 = 2.7$ newborns to next year's count.

```
p0 <- 0.2
p1 <- 0.9
p2 <- 0.6
m2 <- 3.0
m3 <- 6.0

A1 <- matrix(c(0.0, m2 * p1, m3 * p2,
               p0, 0.0, 0.0,
               0.0,p1,0), byrow = TRUE, nrow = 3)
A1
```

```
##      [,1] [,2] [,3]
## [1,]  0.0  2.7  3.6
## [2,]  0.2  0.0  0.0
## [3,]  0.0  0.9  0.0
```

Pre-breeding census

Now let's correctly convert this to a **pre-breeding census** matrix. Individuals are counted just *before* breeding, so two things change:

- The age classes are now ages 1, 2 and 3, so the survival arrows are the survival from age 1 to 2 (p_1) and from age 2 to 3 (p_2). First-year survival, p_0, is no longer on the sub-diagonal.
- The fertility term is no longer scaled by the *parent's* survival. A parent counted at age x is about to breed, so it is alive at breeding by construction (its survival up to this point is already in the sub-diagonal). What must be added instead is the survival of its *newborns* from birth to the next census, p_0. So the fertility of an individual counted at age x is $m_x p_0$.

```
A2 <- matrix(c(0.0, m2 * p0, m3 * p0,
               p1, 0.0, 0.0,
               0.0,p2,0), byrow = TRUE, nrow = 3)
A2
```

```
##      [,1] [,2] [,3]
## [1,]  0.0  0.6  1.2
## [2,]  0.9  0.0  0.0
## [3,]  0.0  0.6  0.0
```

Either way, first-year survival p_0 appears once in every route from birth to reproduction. In `A1` it is on the sub-diagonal; in `A2` it is inside the fertilities.

Comparing the two

Compare population growth rates. This uses the `eigs` function from the `popdemo` R package (install it with `install.packages("popdemo")` if you have not already done so in "Matrix population modelling").

```
(popdemo::eigs(A1, what = "lambda"))
```

```
## [1] 1.070261
```

```
(popdemo::eigs(A2, what = "lambda"))
```

```
## [1] 1.070261
```

Notice that λ is exactly identical for `A1` and `A2`. This is not a coincidence or a rounding effect. Both matrices have the same characteristic equation,

$$\lambda^3 = p_0 p_1 m_2 \lambda + p_0 p_1 p_2 m_3,$$

so they must have the same λ. Each term is one route through the life cycle: survive to age 2 and breed ($p_0 p_1 m_2$), or survive to age 3 and breed ($p_0 p_1 p_2 m_3$). The conversion only changes *how* the same survival and reproduction are arranged in the matrix; it does not change the biology, so it cannot change the population's growth rate.

The net reproductive rate, $R_0 = \sum l_x m_x$, is also identical, because it is calculated from the same life table ($l_1 = p_0$, $l_2 = p_0 p_1$, $l_3 = p_0 p_1 p_2$):

```
l2 <- p0 * p1
l3 <- p0 * p1 * p2
(R0 <- l2 * m2 + l3 * m3)
```

```
## [1] 1.188
```

Not everything is invariant, though. Because the two matrices count different age classes, the stable age distribution and the reproductive values are different: for example, in a post-breeding census most of the population is newborns (age 0), whereas a pre-breeding census never counts newborns at all. The sensitivities of λ to individual matrix elements also differ. Only λ (and quantities derived from it) and R_0 are guaranteed to be the same.

```
popdemo::eigs(A1, what = "ss") # stable age distribution, ages 0, 1, 2
```

```
## [1] 0.7440408 0.1390391 0.1169202
```

```
popdemo::eigs(A2, what = "ss") # stable age distribution, ages 1, 2, 3
```

```
## [1] 0.4324619 0.3636641 0.2038740
```

A common mistake

It is easy to get the pre-breeding conversion partly right and partly wrong. A very common error is to correctly shift the *survival* transitions to the new age classes, but forget to also change the fertility term, i.e. to keep scaling fertility by the parent's own survival (p_1, p_2) instead of switching to the newborns' survival (p_0):

```
A3 <- matrix(c(0.0, m2 * p1, m3 * p2,
               p1, 0.0, 0.0,
               0.0,p2,0), byrow = TRUE, nrow = 3)

(popdemo::eigs(A3, what = "lambda"))
```

```
## [1] 1.863632
```

This looks like a small, easy-to-miss slip: only the fertility row is "wrong", and the matrix still looks like a sensible pre-breeding matrix. But it produces a substantially different (and incorrect) estimate of λ, compared to the correctly converted `A2`.

Your task

- Confirm that `A1` and `A2` give exactly the same λ. Explain, in your own words, why a *correct* change of census timing should not change the underlying population growth rate.
- Compare `A2` (correct pre-breeding matrix) and `A3` (the common mistake). Which specific matrix entries differ, and why does that particular mistake push λ so far from the correct value? (Hint: look for p_0 in each matrix.)
- Modify one survival probability (e.g. `p1`) and re-build `A1`, `A2` and `A3`. Does the correct conversion (`A1` vs `A2`) still agree? Does the size of the error from the mistake (`A2` vs `A3`) grow, shrink, or stay about the same? Now try changing `p0` instead. What do you notice?

Takeaways

- Census timing (pre- vs post-breeding) changes which age classes are counted, and how fertility and survival are encoded in the matrix. A *correctly done* conversion between the two gives exactly the same λ and R_0: timing alone is a bookkeeping choice, not a biological difference.
- In a post-breeding census the fertility includes the parent's survival over the year ($p_x m_{x+1}$); in a pre-breeding census it includes the newborns' first-year survival ($m_x p_0$).
- The stable age distribution, reproductive values and sensitivities do depend on census timing, so always say which census a matrix uses.
- Small errors in how survival or fertility terms are calculated for a given census type are easy to make and can substantially bias your λ estimate. The general lesson extends beyond census timing: applying the wrong calculation will bias your estimates, even if the matrix "looks" reasonable.

Life Table Response Experiments

Background

A Life Table Response Experiment (LTRE), in matrix population modelling, is a technique used to analyse how differences in demographic parameters between two or more populations or "treatments" contribute to the observed difference in their population growth rate (λ). It answers the question: *why* do these populations have different growth rates? Which of the differences in survival, growth or fertility account for most of the difference in λ?

The method compares matrix models of the populations in question and decomposes the difference in growth rate between the models into contributions from the individual matrix elements. Each contribution is (roughly) the difference in that element multiplied by how sensitive λ is to that element.

LTRE is a **retrospective** analysis (Caswell's terminology): it looks back at differences that have already happened, and explains an observed difference in λ. Sensitivity and elasticity analysis, and *in silico* experiments like those in "Matrix population modelling", are **prospective**: they ask what *would* happen to λ if we changed a vital rate in the future. The two kinds of question need different tools, and we come back to this distinction at the end of the chapter.

The background reading is the chapter *Growth of Age-Structured and Stage-Structured Populations* in Neal. Sensitivity was introduced in the lecture; we compute it below.

Learning outcomes

Learning outcomes:

- Explain the purpose of an LTRE, and the difference between a retrospective analysis (LTRE) and a prospective one (sensitivity, elasticity, *in silico* experiments).
- Calculate the sensitivity of λ to each matrix element, and the LTRE contribution of each element as (difference in element) $\times$ (sensitivity at the midpoint matrix).
- Check that the contributions add up (approximately) to the observed difference in λ, and explain why the match is not exact.
- Interpret LTRE output biologically, including the sign of each contribution.

Set up

For the core LTRE analysis you need `popbio` (for the `LTRE` function) and `popdemo` (for the `eigs` function used to compare growth rates). If you have not installed them already, do so now.

```
install.packages(c("popbio", "popdemo"))
library(popbio)
library(popdemo)
```

A worked example

In this example, we consider a two-stage matrix model to analyse, say, squirrel populations in two different geographic locations. The life cycle has juveniles (J) and adults (A). Each year, a juvenile can survive and stay a juvenile, or survive and mature into an adult; an adult can survive (and stay adult) and also produces juveniles (the fertility: juveniles counted at the next census per adult).

The reference population, from Location A, lives in an area with favourable environmental conditions. The treatment population, from Location B, experiences worse conditions in spring. This particularly affects juveniles, which are much less likely to survive and mature, and adults produce fewer young (mothers in poor condition produce fewer young). Adult survival, on the other hand, is a little higher at Location B.

Matrix element	Meaning	Reference (Location A)	Treatment (Location B)
a_{11} (top left)	Juvenile survives and stays juvenile	0.1	0.1
a_{21} (bottom left)	Juvenile survives and matures to adult	0.5	0.2
a_{12} (top right)	Fertility: juveniles produced per adult	2.3	1.3
a_{22} (bottom right)	Adult survives	0.6	0.7

The two matrices can be created in R as follows. Compare the values in the matrices with the values in the life cycle diagrams below.

```
A_ref <- matrix(c(0.1,2.3,
                  0.5,0.6), nrow = 2, ncol = 2, byrow = TRUE)

A_treat <- matrix(c(0.1,1.3,
                    0.2,0.7), nrow = 2, ncol = 2, byrow = TRUE)
```

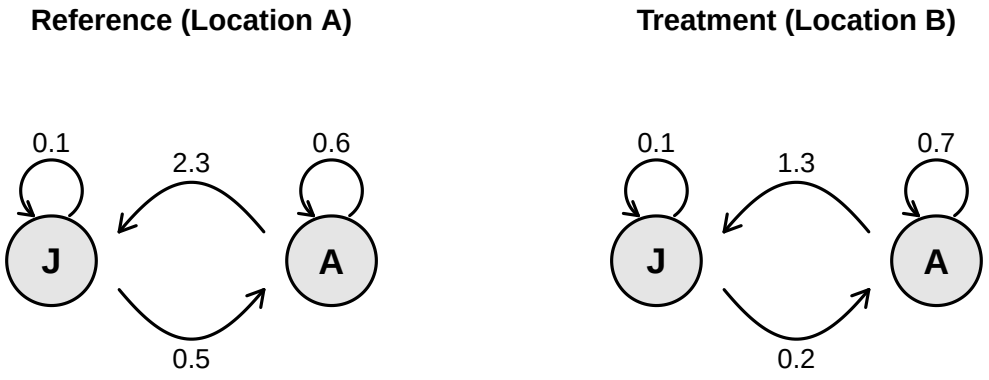


Figure 5: Life cycle diagrams for the reference population (Location A, left) and the treatment population (Location B, right). J = juvenile, A = adult. Arrows show survival and maturation probabilities, and the fertility (arrow from A to J).

Comparing matrices

Before we dig into the formal LTRE, it is useful to compare the two matrices, `A_ref` and `A_treat`, to see how they are different from each other.

We can check what overall difference there is between the matrices in a couple of ways.

Firstly, we can examine the differences in the individual transition rates between the matrices by subtracting one from the other.

```
A_ref - A_treat
```

```
##      [,1] [,2]
## [1,]  0.0  1.0
## [2,]  0.3 -0.1
```

Next, we can ask what impact these differences have on the population growth rate (lambda). Note that putting the command in parentheses tells R to both create the new object (e.g. `lambda_A_ref`) **and** print the result to the screen.

```
(lambda_A_ref <- popdemo::eigs(A_ref, what = "lambda"))
```

```
## [1] 1.451136
```

```
(lambda_A_treat <- popdemo::eigs(A_treat, what = "lambda"))
```

```
## [1] 0.991608
```

The reference population is growing ($\lambda > 1$), whereas λ for the treatment population is just below 1, so the Location B population is slowly declining.

We could express this as a difference, or as a proportional difference.

The difference in growth rates between the matrices is 0.46.

```
(lambdaDiff <- lambda_A_ref - lambda_A_treat)
```

```
## [1] 0.4595278
```

This is a 32% reduction, compared to the reference matrix.

```
(lambdaDiff/lambda_A_ref) * 100
```

```
## [1] 31.66677
```

So far, so good. We can see that λ is lower at Location B, but we don't know exactly why. There are 3 changes in the matrix. Which of these changes is most instrumental in driving the change in the population growth rate?

Sensitivity: how much does λ change when one element changes?

To answer this, we need to know how strongly λ responds to each matrix element. This is the **sensitivity** of λ to element a_{ij}, written s_{ij}: the change in λ for a small *absolute* change in a_{ij} (the slope of λ against a_{ij}). Elasticity, which you met in "Matrix population modelling", is the *proportional* version of the same idea.

Sensitivities can be calculated from two eigenvectors of the matrix: the stable stage distribution $\mathbf{w}$ (the right eigenvector) and the reproductive values $\mathbf{v}$ (the left eigenvector, which is the right eigenvector of the transposed matrix). The formula is

$$s_{ij} = \frac{v_i \, w_j}{\sum_k v_k w_k}.$$

So λ is most sensitive to transitions *into* stages with high reproductive value (v_i) *from* stages that are common in the stable stage distribution (w_j). In base R:

```
sensitivity <- function(M) {
  # Right eigenvector: stable stage distribution w
  eig_right <- eigen(M)
  w <- Re(eig_right$vectors[, which.max(Re(eig_right$values))])
  # Left eigenvector: reproductive values v (eigenvector of the transposed matrix)
  eig_left <- eigen(t(M))
  v <- Re(eig_left$vectors[, which.max(Re(eig_left$values))])
  # s_ij = v_i * w_j / sum(v * w)
  outer(v, w) / sum(v * w)
}

sensitivity(A_ref)
```

```
##           [,1]      [,2]
## [1,] 0.3864808 0.2270383
## [2,] 1.0443762 0.6135192
```

(The `popbio` package has a function `sensitivity()` that does the same thing; here you can see what it does.)

Contributions from individual matrix elements

The LTRE splits the observed difference in λ into one contribution per matrix element. The contribution of element a_{ij} is

$$c_{ij} = \left(a_{ij}^{\text{treat}} - a_{ij}^{\text{ref}}\right) \times s_{ij}\left(\mathbf{A}_{\text{mid}}\right), \qquad \mathbf{A}_{\text{mid}} = \frac{\mathbf{A}_{\text{ref}} + \mathbf{A}_{\text{treat}}}{2},$$

that is, how much the element differs, multiplied by how much λ responds to that element. The sensitivity is calculated from the matrix halfway between the two populations (the midpoint matrix), because the sensitivities of the two populations are not the same, and the midpoint is a fair compromise. A large contribution can come from a large difference in an element, from a high sensitivity, or from both.

We can calculate the contributions by hand:

```
A_mid <- (A_ref + A_treat) / 2
(contributions <- (A_treat - A_ref) * sensitivity(A_mid))
```

```
##            [,1]        [,2]
## [1,]  0.0000000 -0.20832964
## [2,] -0.3214229  0.06636876
```

The `LTRE` function in the `popbio` package does exactly this calculation. Note the order of the arguments: treatment first, then reference.

```
popbio::LTRE(A_treat, A_ref)
```

```
##             [,1]        [,2]
## [1,]  0.0000000 -0.20832964
## [2,] -0.3214229  0.06636876
```

The contributions should add up to the observed difference in λ (treatment minus reference):

```
sum(popbio::LTRE(A_treat, A_ref))
```

```
## [1] -0.4633838
```

```
lambda_A_treat - lambda_A_ref
```

```
## [1] -0.4595278
```

The two numbers are very close, but not identical (-0.463 vs -0.460). The LTRE treats λ as if it changed in a straight line with each element, with a slope equal to the sensitivity at the midpoint matrix. In reality the sensitivities change as the elements change, so this linear approximation is only approximately right. It is very good when the differences between the matrices are small, and gets worse when they are large.

Interpreting the contributions

Contributions are in the same units as λ. A **negative** contribution means that difference in the element *lowers* λ in the treatment population relative to the reference; a **positive** contribution means it raises λ.

Here, the juvenile stasis element contributes nothing, because it is the same (0.1) in both populations. The other three differences all contribute:

- The lower probability of juveniles maturing into adults ($0.5 \rightarrow 0.2$, bottom left) makes the largest contribution (-0.321). The difference is only 0.3, but λ is very sensitive to this element.
- The lower fertility ($2.3 \rightarrow 1.3$, top right) contributes -0.208. The difference in the element is much larger (1.0), but λ is much less sensitive to it.
- The slightly higher adult survival at Location B ($0.6 \rightarrow 0.7$, bottom right) makes a small *positive* contribution (0.066), which partly offsets the other two.

So the main reason the Location B population has a lower growth rate (and is declining) is that fewer juveniles survive and mature into adults; the reduced fertility is the second most important reason.

Retrospective vs prospective analyses

It is tempting to read an LTRE as a management recommendation ("juvenile maturation made the largest contribution, so managers should improve juvenile maturation"). That is not what an LTRE tells you. Following Caswell, it helps to separate two kinds of question:

- **Retrospective** (LTRE): *Why do these observed populations differ in λ?* The LTRE takes differences in vital rates that have already happened and shares out the observed difference in λ among them. A contribution is large if the element *did* differ a lot, if λ is sensitive to it, or both.

- **Prospective** (sensitivity, elasticity, *in silico* experiments): *What would happen to λ if we changed a vital rate?* These analyses do not depend on how much the rates have varied in the past. They are the tools for exploring management options, together with knowledge of which rates can actually be changed, and by how much.

The answers can differ. A vital rate can make a large LTRE contribution simply because it happened to vary a lot between places (for example because of weather), even if managers cannot change it. And a vital rate with a high sensitivity or elasticity can make no LTRE contribution at all, if it did not differ between the populations: in our example λ has a sensitivity of 0.336 to juvenile stasis at the midpoint matrix, but juvenile stasis contributes nothing because it is 0.1 in both populations.

The two approaches complement each other. LTRE is useful for understanding what has caused observed differences between populations, years or experimental treatments (and so for generating hypotheses about the causes), and it can handle several vital rates differing at the same time. To ask what management actions *would* do, use prospective tools.

Your task

- Before running any more code, look at the table of matrix elements and the sensitivities. Predict the sign of each LTRE contribution, and which one will be largest. Then compare your prediction with the output above.
- Check for yourself that the contributions sum approximately to $\lambda_{\text{treat}} - \lambda_{\text{ref}}$.
- Modify one element in `A_treat` (for example, set the treatment fertility to 1.8 instead of 1.3, or the treatment maturation probability to 0.4 instead of 0.2). Predict which LTRE contribution will change most, and in which direction. Re-run the contributions (by hand and with `popbio::LTRE`) and check your prediction. Explain the change using "difference in element $\times$ sensitivity".
- Explain, in two or three sentences, why the LTRE result does not on its own tell a manager which vital rate to target.

Takeaways

- LTRE is retrospective: it decomposes an *observed* difference in λ into contributions from specific matrix elements.
- Each contribution is (difference in the element) $\times$ (sensitivity of λ to that element, at the midpoint matrix), and the contributions sum approximately to the observed difference in λ.
- Sensitivity and elasticity analyses are prospective: they ask what would happen if a vital rate changed, and are the right tools for exploring management options.

How many eggs should a bird lay?

Background

Trade-offs are inherent compromises in biology, where allocating limited resources to one trait may come at the expense of another. Organisms face these trade-offs when making crucial life-history decisions. One classic example is the British ornithologist David Lack's clutch size trade-off. Parents can feed only so many chicks, so the more eggs a bird lays, the less food each chick gets and the lower its chance of surviving. Lack proposed that clutch size evolves to the number from which, on average, the most young survive. In other words, although it may be beneficial to have many offspring, the survival of those offspring will decline if they cannot be cared for.

Understanding such trade-offs sheds light on the fascinating strategies that organisms employ to optimise their fitness and evolutionary success. Further reading: Neal's chapter *Evolution of Life Histories*.

Learning outcomes:

- Calculate the expected number of surviving offspring from clutch size and chick survival.
- Explain how a trade-off between offspring number and offspring survival produces an intermediate optimum.
- Predict how the optimum changes when the environment, or the cost of breeding to the parents, changes.

Before you start: predict

Before you calculate anything, make a prediction and write it down: will a big bird that lays **10 eggs** raise more surviving chicks than one that lays **4 eggs**, or fewer? Why? You will check your prediction below.

Your task

The big bird (*Bigus canarius*) (Fig @ref(fig:bigbird)) can lay up to 10 eggs per breeding season. If there is only 1 egg, the probability that the parents can adequately feed the chick and ensure it survives is very high (0.9). However, as the number of siblings increases, the amount of energy and food that the parents can dedicate to caring for *each* chick decreases, and the probability of survival therefore declines . With a clutch size of 10 eggs, there is so little food *per chick* that the survival rates are close to zero.

A recent study gathered data on chick survival probability as a function of number of eggs in the nest. These are given in the table below.

Eggs	Survival probability
1	0.90
2	0.85
3	0.82
4	0.70

Eggs	Survival probability
5	0.50
6	0.30
7	0.22
8	0.15
9	0.10
10	0.05

Use these data to plot a graph in Excel with number of eggs on the x-axis and survival probability on the y-axis.

Now, in another column in Excel, calculate, given the chick survival probability, what the expected number of surviving chicks will be for a big bird laying between 1 and 10 eggs[2].

Plot your result on another graph with number of eggs on the x-axis and number of surviving offspring on the y-axis.

What do you notice? What is the optimum number of eggs to lay?

Advanced: What happens to the optimum as you change the relationship between clutch size and survival?

Field observations of the big bird show that most females lay **3 eggs**. Keep this in mind for the questions below.

Assumptions

This simple model assumes that every egg hatches, that every chick in a nest has the same chance of surviving, that we count only one breeding season, and that laying and raising more eggs has no cost to the parents' own survival. Question 3 relaxes the last assumption.

Questions

1. What is Lack's hypothesis? Does the big bird's optimum support it, given that most big birds lay 3 eggs?
2. In a good food year, each chick's survival probability in clutches of 5 or more eggs is 0.2 higher (for example, 0.50 becomes 0.70). Recalculate. What is the best clutch size now, and why?
3. Raising chicks is costly for the parents. Suppose the probability that a parent survives to breed again next year is $0.8 - 0.08 \times$ eggs (so 0.72 with 1 egg, and 0.48 with 4 eggs). On average, a parent then breeds $1/(1 -$ parent survival) times in its life. Calculate the number of surviving chicks over a parent's whole life for each clutch size. Does the best clutch size change? What does this mean for Lack's hypothesis?
4. (Discussion) Birds also trade off the number of eggs against the size of each egg. What would you need to add to this model to study that trade-off?

Takeaways

- Trade-offs often produce intermediate optimal strategies.
- Expected reproductive success depends on both offspring number and survival probability.
- If raising young costs the parents' future survival, the best clutch is smaller than the one that produces the most surviving young in a single season.

[2]This may require some thought!



Figure 6: Big bird, *Bigus canarius*

Trade-offs and the declining force of selection

Why does evolution care less and less about you as you age? Because fewer individuals survive to late ages, selection acts more weakly on late life, and trade-offs then let early benefits win over late costs.

Background

An important kind of trade off are those which occur between processes in early and late life. For example, it may be beneficial to increase reproduction at younger ages, but this might lead to increased risk of death at older ages. A mechanism for this could be that limited resources are allocated to producing offspring rather than maintaining/repairing the body.

The balance of the trade off between these early and late life processes is shaped by the **force of selection**: how much a small change in survival or reproduction at a given age changes fitness. Crucially, from the age of first reproduction onwards, the force of selection tends to decline with age, meaning that late-life processes are not as important as early-life processes. Medawar (1952) pointed out that because few individuals survive to old age, harmful mutations whose effects appear only late in life are hardly removed by selection, so they can build up in a population: this is **mutation accumulation**. Hamilton (1966) showed mathematically why the force of selection on survival and reproduction declines with age once reproduction has begun. It also means that a mutation giving a modest reproductive advantage at an early age, alongside a harmful or even deadly effect late in life, can be "seen" by natural selection as advantageous. This idea is called **antagonistic pleiotropy** (Williams 1957): one gene with a good effect early in life and a bad effect later. Further reading: Neal's chapter *Evolution of Life Histories*.

References: Medawar, P. B. (1952) *An Unsolved Problem of Biology*. H. K. Lewis, London. Williams, G. C. (1957) Pleiotropy, natural selection, and the evolution of senescence. *Evolution* 11: 398-411. Hamilton, W. D. (1966) The moulding of senescence by natural selection. *Journal of Theoretical Biology* 12: 12-45.

This highlights the significance of early-late life trade-offs and their role in shaping ageing patterns in various organisms. Understanding this concept is essential for comprehending life history evolution, ageing, and longevity across species.

The aim of this exercise is to gain an understanding of early-life late-life trade offs and to understand why events early in life tend to be much more important to evolution than those that happen later in life.

Learning outcomes:

- Calculate how a change in fertility or survival at a given age changes fitness (R_0).
- Explain why the force of selection declines with age, and why this lets genes with early benefits and late costs spread.
- Explain why survival after reproduction has ended has little or no effect on fitness in this model.

Before you start: predict

Recall that fitness can be measured as $R_0 = \sum l_x m_x$, where l_x is the probability of surviving from birth to age x, and m_x is the number of daughters produced per female at age x (as in *Life tables and survivorship types*).

Before you open the spreadsheet, make a prediction and write it down: will **one extra offspring at age 1** or **one extra offspring at age 20** raise R_0 more? Why? You will check your prediction in the tasks below.

Your task

Open the Excel file TradeOffsAndForceOfSelection.xlsx.

This file shows a simplified life table, following a cohort of 1000 individuals for a fictional creature.

Survival rates from year to year are set to be 0.8 (i.e. 80% make it through to the next year). This fixed, constant survival rate leads to an exponentially declining survival curve, illustrated with a chart in the Excel file. Fertility (m_x, the number of babies produced per year; assume these are daughters per female) is set to be 10 per year.

In the spreadsheet, the **Survival** value in each row is the probability of surviving *into* that age from the age before. So setting Survival to 0 in the age-25 row means that nobody reaches age 25. (The Survival value in the age-1 row is not used: the cohort starts at age 1.)

The product of survivorship (l_x) and fertility (m_x), $l_x m_x$, is the expected number of offspring produced at age x by a newborn. The sum of these values, $\sum l_x m_x$, is the net reproductive rate, R_0: the population's growth per generation (see the practical *Life tables and survivorship types*). R_0 is a good measure of the fitness of a life history strategy when the population is roughly stable in size (stationary, or regulated around an equilibrium).

A caveat. This fictional life table has $R_0 \approx 50$, which would make the population grow about 10.8-fold per year ($\lambda \approx 10.8$, from the Euler-Lotka equation). In a growing population the right fitness measure is r_m (or λ), and it weights early reproduction even more heavily than R_0 does: an offspring produced early is itself breeding while later offspring are still to come. For this life table, matching the effect of one extra offspring at age 1 would need about 2.4 extra offspring at age 5 if fitness is R_0, but about 46,000 if fitness is r_m. So treat the numbers below as an illustration: the qualitative message (early life matters more than late life) holds, and is even stronger, in growing populations.

Note that the initial R_0 is 49.81

Exploring different life history strategies

We will now use this data to explore how alternative life history strategies affect fitness.

Consider a trade-off between early reproduction and late life survival (i.e. via resources allocated to body maintenance). In this scenario the species could increase reproduction early in life by allocating more energy to making babies. However, resources are limited and this will come at the expense of survival at a later date. A mechanism for this could be that the body no longer fixes cancers so effectively.

- Simulate this by adding 1 to fertility (m_x) at age 1 (the benefit), but setting Survival to 0 in the age-25 row, so that all die before age 25 (the cost). *What is the fitness of this strategy?*
- By setting Survival to 0 in earlier rows instead, determine how many years of life could be lost before this cost is no longer worth bearing. Is this surprising?
- Now reset everything (m_x =10; survival = 0.8). Recall what fitness was when you added 1 to m_x at age 1 (50.81).

- If, instead of adding to m_x at age 1 you were to increase m_x at age 25, how much would you need to increase m_x to reach this figure?
- What about at age 20? Age 15? Age 10? Age 5? Plot the increase required vs. age (make a new worksheet/ark in Excel). What do you notice?
- Reset everything again (m_x =10; survival = 0.8). Set m_x from age 15 onwards to be 0. Now alter survival rate after this point (at ages 15-25). What happens to fitness?

Questions

- Explain the concept of the declining force of selection in the context of life history evolution and ageing. How does this phenomenon contribute to the evolution of life history strategies in different organisms?
- Use concepts of life history evolution, including the force of selection, to explain why age-related diseases are common in humans.
- How would predation risk influence the early-late life trade off in resource allocation to reproduction vs. body maintenance?
- In this model, survival after the last reproduction has no effect on fitness. Why might survival after reproduction still matter for killer whales or humans?

Takeaways

- The force of selection declines with age, making early-life traits more influential.
- Small early-life advantages can outweigh large late-life costs.
- Trade-offs between reproduction and survival shape ageing and life history strategies.

Matrix Population Models (MPMs): Projection and Simulation

This short tutorial introduces **matrix population models (MPMs)** and shows two closely related uses:

1. **Projection**: how a population vector is projected forward through time using a matrix.
2. **Simulation**: how the same structure can be used to explore scenarios (e.g. **culling**) and quantify uncertainty.

The examples use a simple 3-stage life cycle. The goal is to be clear about:

- the **projection interval** (time step, Δt),
- how **survival/transition** and **reproduction** appear in the matrix,
- how to project a **population vector** $\mathbf{n}_t$,
- how to convert the deterministic model into a basic **stochastic simulation**.

The background reading is the chapter *Growth of Age-Structured and Stage-Structured Populations* in Neal. The chapter "Matrix population modelling" covers building a matrix, projection and λ in more detail, using the `popdemo` package; here we do everything in base R.

Learning outcomes

Learning outcomes:

- Explain why the vital rates in a matrix must match the projection interval Δt.
- Build a stage-structured projection matrix in base R, project a population forward and calculate λ with `eigen()`.
- Explain the difference between a deterministic projection and a stochastic simulation with demographic stochasticity.
- Write a simulation once as a function, and use it to run replicate simulations and summarise their uncertainty.
- Use simulation to compare management scenarios (proportional harvest, fixed quotas, contraception) and explain the results in terms of the life cycle.

What is a matrix population model?

An MPM describes how a structured population changes over discrete time steps.

The projection interval (Δt)

MPMs are defined for a particular **projection interval** Δt (the time step between updates). Common choices are:

- $\Delta t = 1$ year (very common in ecology)
- $\Delta t = 1$ month (or other shorter steps)

Key rule: the vital rates you put in the matrix must match the chosen Δt.

- If Δt is one year, use **annual survival** and **annual fertility/recruitment**.
- If Δt is one month, use **monthly survival** and **monthly recruitment**.

A common mistake is mixing time scales (e.g. monthly survival with yearly reproduction).

The population vector

Let $\mathbf{n}_t$ be a vector of abundances at time t (measured in units of Δt):

$$\mathbf{n}_t = \begin{bmatrix} n_{1,t} \\ n_{2,t} \\ \vdots \\ n_{k,t} \end{bmatrix}$$

Each element is the number (or density) in a class (age/stage/size/etc.).

The projection matrix

Let $\mathbf{A}$ be a square matrix of **vital rates**. The basic model is:

$$\mathbf{n}_{t+\Delta t} = \mathbf{A}\,\mathbf{n}_t$$

Interpretation of entries:

- **Survival / transition**: individuals moving between classes (including remaining in the same class)
- **Reproduction**: production of new individuals (often into the first class)

A simple 3-stage model

We will use three stages:

1. Juvenile (J)
2. Subadult (S)
3. Adult (A)

The life-cycle assumptions (per $\Delta t = 1$ year) are:

- Adults produce juveniles: F_A juveniles per adult per year (recruits to J). This is a fertility: the number of new juveniles *counted at the next census* per adult, so it already includes the survival needed to reach that count, not just the number born (see the fecundity box in the chapter "Matrix population modelling").

- Juveniles that survive and grow move to Subadult with probability G_J.
- Subadults either:
  - survive and remain subadult with probability P_S, or
  - survive and grow to adult with probability G_S.
- Adults survive and remain adult with probability P_A.

A common stage-structured (Lefkovitch) matrix is:

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & F_A \\ G_J & P_S & 0 \\ 0 & G_S & P_A \end{bmatrix}$$

Projection: projecting $\mathbf{n}_t$ forward

Define vital rates and build A

```
# Vital rates for a 1-year projection interval
F_A <- 1.2   # recruits (juveniles) per adult per year
G_J <- 0.35  # juvenile -> subadult (survive + grow)
P_S <- 0.55  # subadult -> subadult (survive + stay)
G_S <- 0.25  # subadult -> adult (survive + grow)
P_A <- 0.80  # adult -> adult (survive + stay)

# Build the projection matrix (rows = stage at t+1, cols = stage at t)
A <- matrix(
  c(0,   0,   F_A,
    G_J, P_S, 0,
    0,   G_S, P_A),
  nrow = 3, byrow = TRUE
)

dimnames(A) <- list(
  c("Juvenile", "Subadult", "Adult"),
  c("Juvenile", "Subadult", "Adult")
)

# Print matrix so we can check entries
A
```

```
##          Juvenile Subadult Adult
## Juvenile     0.00     0.00   1.2
## Subadult     0.35     0.55   0.0
## Adult        0.00     0.25   0.8
```

Define an initial population vector $\mathbf{n}_0$

```
# Initial population vector (year 0)
n0 <- c(Juvenile = 50, Subadult = 30, Adult = 20)
n0
```

```
## Juvenile Subadult    Adult
##       50       30       20
```

One-step projection

```
# Deterministic one-step update
n1 <- A %*% n0
n1
```

```
##          [,1]
## Juvenile 24.0
## Subadult 34.0
## Adult    23.5
```

Given the stage abundances at time t, multiplying by $\mathbf{A}$ gives the expected abundances at time $t+1$ (because $\Delta t = 1$ year here).

Project for many years (deterministic)

```
# Number of projection steps
years <- 30

# Matrix to store yearly stage abundances
nmat <- matrix(NA_real_, nrow = years + 1, ncol = length(n0))
colnames(nmat) <- names(n0)

# Set initial year
nmat[1, ] <- n0

# Project forward one year at a time
for (t in 1:years) {
  nmat[t + 1, ] <- as.numeric(A %*% nmat[t, ])
}

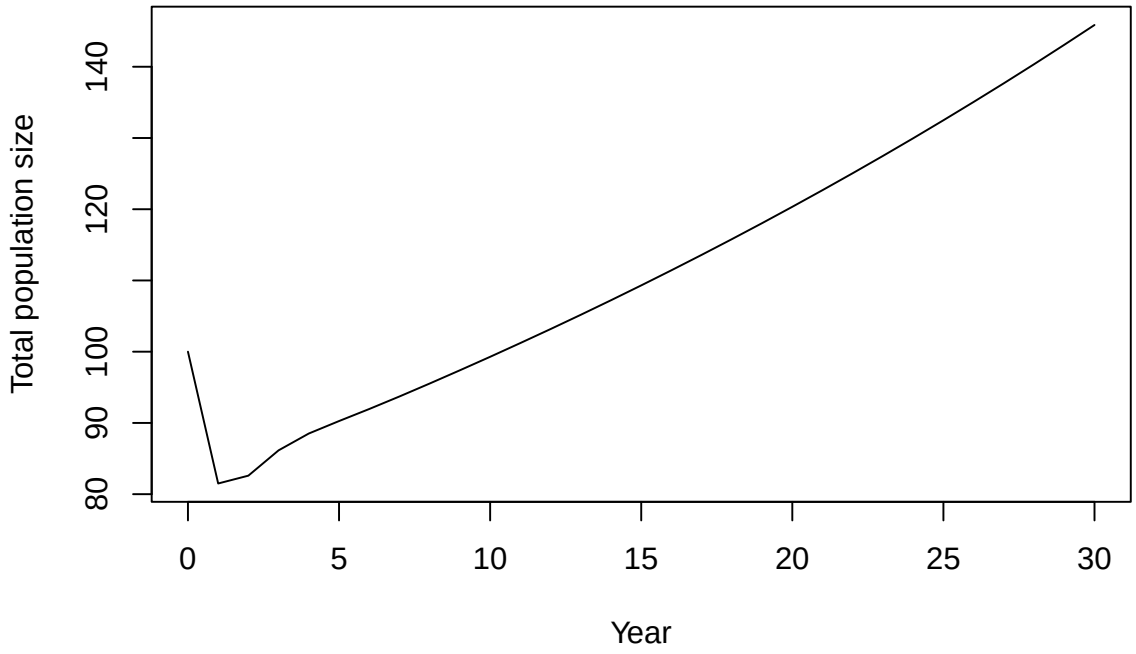
head(nmat, 6)
```

```
##      Juvenile Subadult    Adult
## [1,]  50.0000 30.00000 20.00000
## [2,]  24.0000 34.00000 23.50000
## [3,]  28.2000 27.10000 27.30000
## [4,]  32.7600 24.77500 28.61500
## [5,]  34.3380 25.09225 29.08575
## [6,]  34.9029 25.81904 29.54166
```

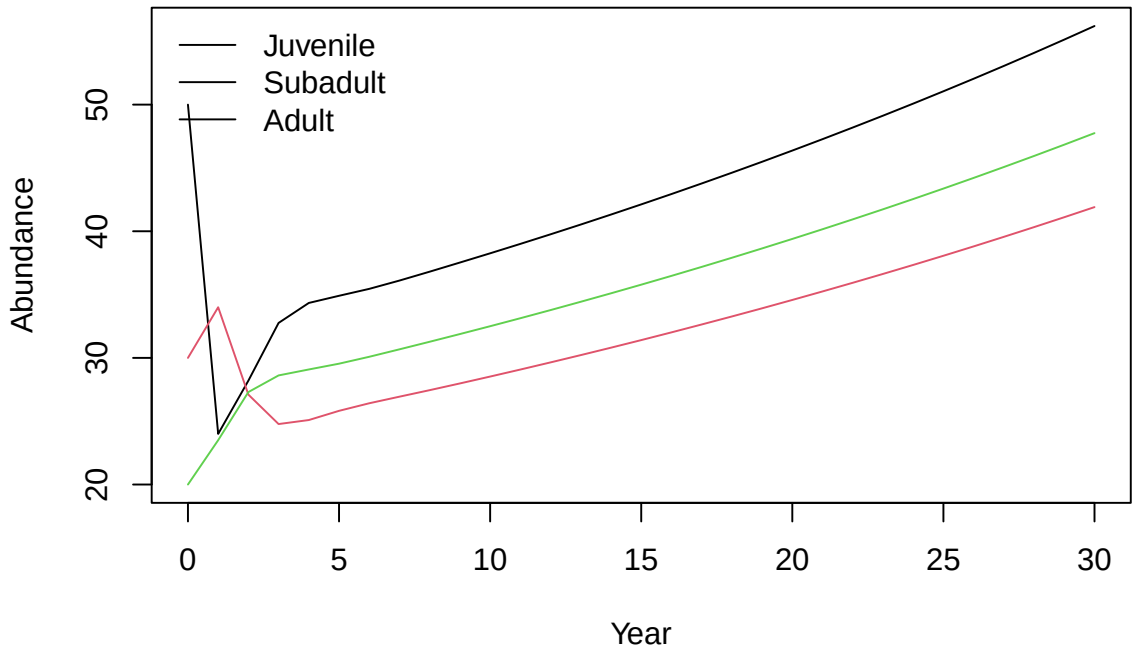
Plot totals and stage trajectories

```r
# Total abundance each year
totalN <- rowSums(nmat)

# Plot total abundance
plot(
  0:years, totalN, type = "l",
  xlab = "Year", ylab = "Total population size"
)
```




```r
# Plot stage-specific abundances
matplot(
  0:years, nmat, type = "l", lty = 1,
  xlab = "Year", ylab = "Abundance"
)
legend("topleft", legend = colnames(nmat), lty = 1, bty = "n")
```



Long-term growth rate (λ)

For a constant matrix **A**, the asymptotic growth rate is the **dominant eigenvalue** λ.

```
# Eigenvalues of the projection matrix
eig <- eigen(A)

# Dominant eigenvalue is asymptotic growth rate lambda
lambda <- Re(eig$values[which.max(Re(eig$values))])
lambda
```

```
## [1] 1.01942
```

- If $\lambda > 1$: growth (on average) per time step
- If $\lambda = 1$: stationary
- If $\lambda < 1$: decline

Here λ is only just above 1 (1.019): the population is expected to grow by about 2% per year. Keep this in mind for the management scenarios below: a population that is only just growing does not need much extra mortality to tip it into decline.

Simulation: adding stochasticity and management actions

Deterministic projection treats the model output as exact expectations. In reality:

- births and deaths vary (demographic stochasticity),

- vital rates vary across years (environmental stochasticity),
- management can intervene (culling, supplementation, etc.).

Below is a transparent stochastic simulation that:

- simulates reproduction with Poisson variation,
- simulates survival/transitions with binomial variation,
- allows a simple **culling** intervention.

This simulation includes **demographic stochasticity only**: the matrix (the vital rates) is the same every year, and the randomness comes from the fact that a whole number of individuals survive, grow or are born, by chance. Environmental stochasticity (good and bad years, with different vital rates) is covered in the chapter "Stochastic population growth", and in an optional exercise at the end of this chapter.

Writing the simulation once, as a function

We are going to run the same simulation many times, with and without culling, so we write it once as a function. The function:

1. Reads the vital rates from the matrix `A` (so if you change the matrix, the simulation changes too).
2. Creates an output matrix with one row per year and one column per stage, and fills row 1 with the initial stage abundances (`n0`).
3. For each year, reads the current stage counts (`nJ`, `nS`, `nA`). We do not call these `J`, `S` and `A`: the name `A` is already taken by the projection matrix, and overwriting it would silently break everything that uses the matrix afterwards.
4. If the year has reached `cull_start`, removes individuals from the target stage: a proportion `cull_prop` (each individual is removed with that probability), plus a fixed number `cull_n`. Both are 0 by default (no culling).
5. Draws the number of births and transitions using random distributions:
   - births from adults: Poisson,
   - survival/transition events: Binomial.
6. Writes the resulting stage counts into the next row, and returns the whole trajectory.

The stored counts are the census counts before any cull. The cull is applied to the working counts only, just before breeding and survival.

```
simulate_once <- function(A, n0, years,
                          cull_stage = "Adult", cull_prop = 0, cull_n = 0,
                          cull_start = 10) {
  # Step 1: read the vital rates from the projection matrix
  F_A <- A["Juvenile", "Adult"]
  G_J <- A["Subadult", "Juvenile"]
  P_S <- A["Subadult", "Subadult"]
  G_S <- A["Adult", "Subadult"]
  P_A <- A["Adult", "Adult"]

  # Step 2: a matrix to store the trajectory, with year 0 in row 1
  out <- matrix(0L, nrow = years + 1, ncol = 3)
  colnames(out) <- c("Juvenile", "Subadult", "Adult")
  out[1, ] <- as.integer(n0[c("Juvenile", "Subadult", "Adult")])

  for (t in 1:years) {
    year <- t - 1 # row t holds year t - 1
```

```
    # Step 3: current stage counts at the census
    n_now <- out[t, ]

    # Step 4: culling, once the start year has been reached
    if (year >= cull_start) {
      removed <- rbinom(1, size = n_now[cull_stage], prob = cull_prop) + cull_n
      removed <- min(removed, n_now[cull_stage]) # cannot remove more than are there
      n_now[cull_stage] <- n_now[cull_stage] - removed
    }
    nJ <- n_now["Juvenile"]
    nS <- n_now["Subadult"]
    nA <- n_now["Adult"]

    # Step 5: births, survival and transitions
    newJ <- rpois(1, lambda = F_A * nA)          # births from adults
    J_to_S <- rbinom(1, size = nJ, prob = G_J)    # juvenile -> subadult
    S_stay <- rbinom(1, size = nS, prob = P_S)    # subadult stays subadult
    S_remaining <- nS - S_stay                    # the rest either grow or die
    p_grow_cond <- if ((1 - P_S) > 0) G_S / (1 - P_S) else 0
    S_to_A <- rbinom(1, size = S_remaining, prob = p_grow_cond)
    A_surv <- rbinom(1, size = nA, prob = P_A)    # adult survival

    # Step 6: write next year's abundances
    out[t + 1, "Juvenile"] <- newJ
    out[t + 1, "Subadult"] <- J_to_S + S_stay
    out[t + 1, "Adult"] <- S_to_A + A_surv
  }
  out
}
```

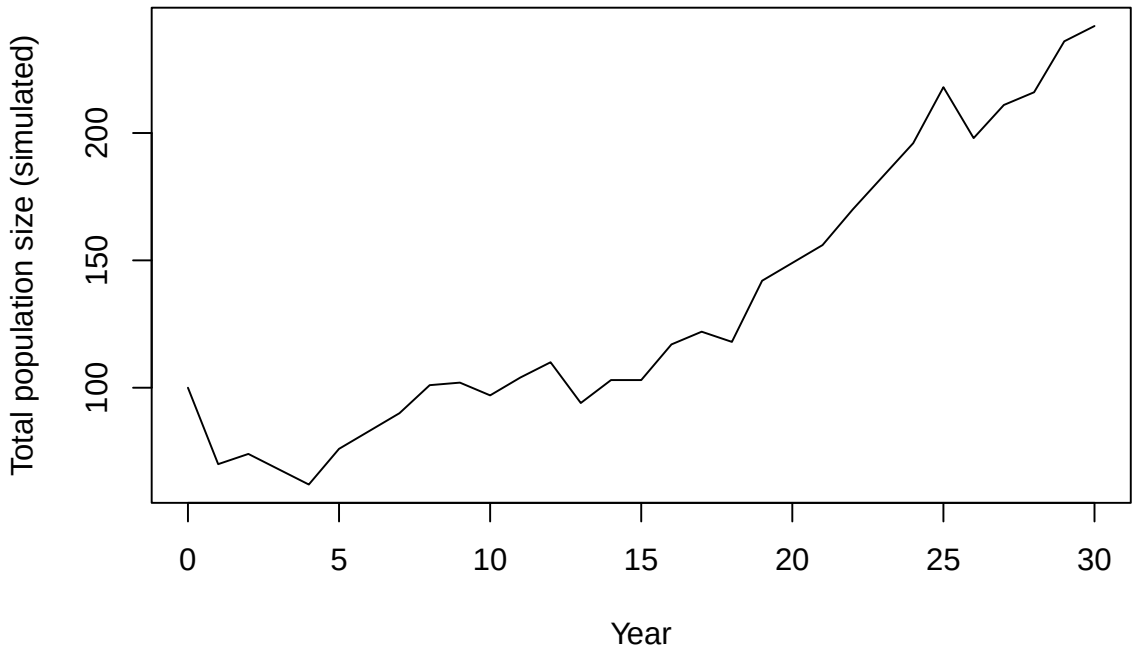
The subadult step needs a little explanation. A subadult has three possible fates: stay subadult (P_S), grow to adult (G_S) or die ($1 - P_S - G_S$). We first draw how many stay. Among the rest, the probability of growing (rather than dying) is $G_S/(1 - P_S)$, so on average a fraction G_S of all subadults grow, as in the matrix.

One simulated trajectory (no culling)

Each run draws new random numbers, so your trajectory will differ from the one shown here. Run the chunk a few times to see how much it varies.

```
sim1 <- simulate_once(A, n0, years)

# Plot the total population trajectory
plot(0:years, rowSums(sim1), type = "l",
     xlab = "Year", ylab = "Total population size (simulated)")
```



We can calculate the realised λ for this trajectory. This is the average yearly multiplication rate over the whole run, $(N_T/N_0)^{1/T}$, where N_0 and N_T are the total population sizes at the start and at the end, after T years.

```
tot_no <- rowSums(sim1)
(realised_lambda_no <- (tot_no[years + 1] / tot_no[1])^(1 / years))
```

```
## [1] 1.029897
```

This is a finite-horizon, trajectory-specific estimate. Different runs give different values, scattered around the deterministic λ (1.019). (In many ecology texts the long-run stochastic growth rate is written λ_s; that is related to, but not the same as, a single-run realised value.) If the population goes extinct, $N_T = 0$ and so the realised λ is 0. It is better to report extinction separately, as we do below with many replicates.

Add a culling intervention

Example: from year 10 onwards, harvest 10% of the adults each year (each adult has a probability of 0.1 of being removed). This is a **proportional harvest**: the number removed goes up and down with the number of adults.

A proportional harvest of adults has the same effect as multiplying the adult column of the matrix by 0.9 (only 90% of adults are left to breed and survive). So we can predict its long-term effect with a deterministic calculation before we simulate it:

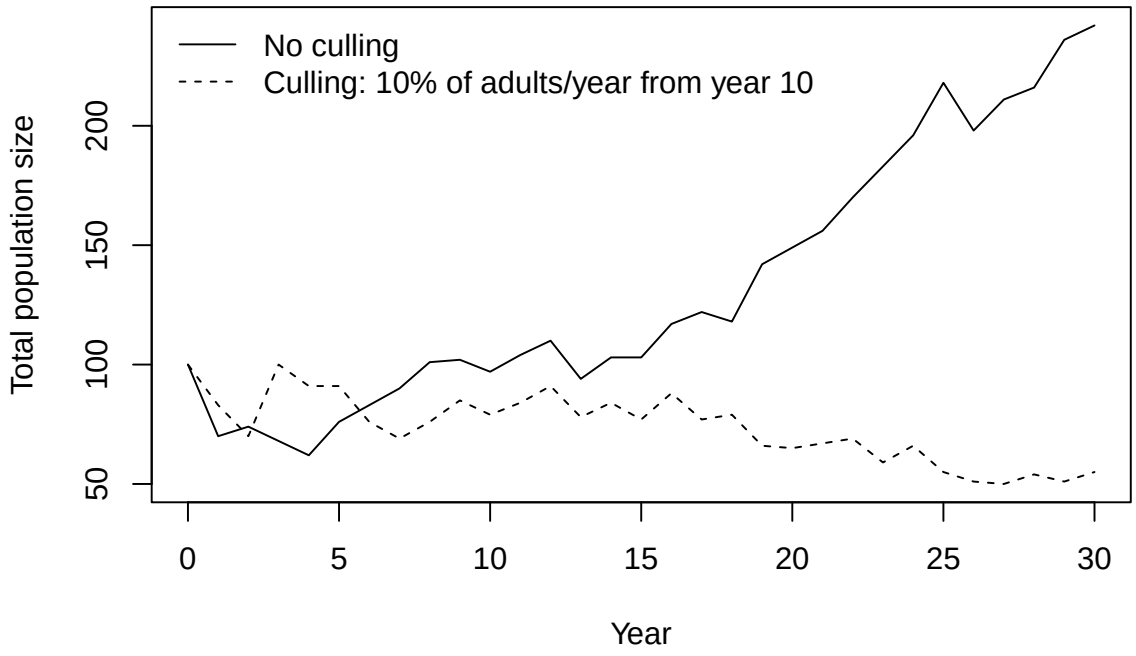
```
A_harvest <- A
A_harvest[, "Adult"] <- A_harvest[, "Adult"] * 0.9
(lambda_harvest <- max(Re(eigen(A_harvest)$values)))
```

```
## [1] 0.9600498
```

With a 10% adult harvest, λ falls from 1.019 to 0.96, so once harvesting starts the population should change from slowly growing to declining by about 4% per year. Now the simulation:

```
sim_cull <- simulate_once(A, n0, years,
                          cull_stage = "Adult", cull_prop = 0.1, cull_start = 10)

# Compare trajectories: solid = no culling, dashed = culling
plot(0:years, rowSums(sim1), type = "l",
     xlab = "Year", ylab = "Total population size",
     ylim = range(c(rowSums(sim1), rowSums(sim_cull))))
lines(0:years, rowSums(sim_cull), lty = 2)
legend("topleft", legend = c("No culling", "Culling: 10% of adults/year from year 10"),
       lty = c(1, 2), bty = "n")
```




```
# Realised lambda from each simulated trajectory
tot_cu <- rowSums(sim_cull)
realised_lambda_cu <- (tot_cu[years + 1] / tot_cu[1])^(1 / years)

data.frame(
  scenario = c("No culling", "Culling"),
  final_N = c(tot_no[years + 1], tot_cu[years + 1]),
  realised_lambda = c(realised_lambda_no, realised_lambda_cu)
)
```

```
##     scenario final_N realised_lambda
## 1 No culling     242       1.0298971
## 2    Culling      55       0.9802693
```

Interpretation:

- realised $\lambda > 1$: the population grew over the simulated period,
- $0 <$ realised $\lambda < 1$: it declined,
- realised $\lambda = 0$: it went extinct at some point in the simulated period.

These are just two runs, and with only about 100 individuals chance plays a large part. The realised λ for the culling run is also averaged over the whole 30 years, including the first 10 years without culling. To compare the scenarios properly, we need many replicates.

Many replicates (uncertainty bands)

A single run is one possible realisation. We typically run many replicates and summarise. Here we run two sets of replicates (no culling vs culling), then compute the median and 10-90% interval for each year. Note: different studies may use different interval widths (for example, 95% intervals) depending on purpose and reporting conventions.

```
# Number of Monte Carlo replicates
nreps <- 200

# Store total abundance per year (rows) for each replicate (columns)
totals_no <- matrix(NA_real_, nrow = years + 1, ncol = nreps)
totals_cull <- matrix(NA_real_, nrow = years + 1, ncol = nreps)

for (rep in 1:nreps) {
  # Scenario A: no culling
  totals_no[, rep] <- rowSums(simulate_once(A, n0, years))
  # Scenario B: 10% of adults harvested each year from year 10
  totals_cull[, rep] <- rowSums(simulate_once(A, n0, years,
                                              cull_stage = "Adult",
                                              cull_prop = 0.1))
}

# Summarise each year across replicates (median and 10-90% interval)
summarise_reps <- function(totals) {
  data.frame(
    year = 0:(nrow(totals) - 1),
    median = apply(totals, 1, median),
    lo = apply(totals, 1, quantile, probs = 0.1),
    hi = apply(totals, 1, quantile, probs = 0.9)
  )
}
sum_no <- summarise_reps(totals_no)
sum_cu <- summarise_reps(totals_cull)

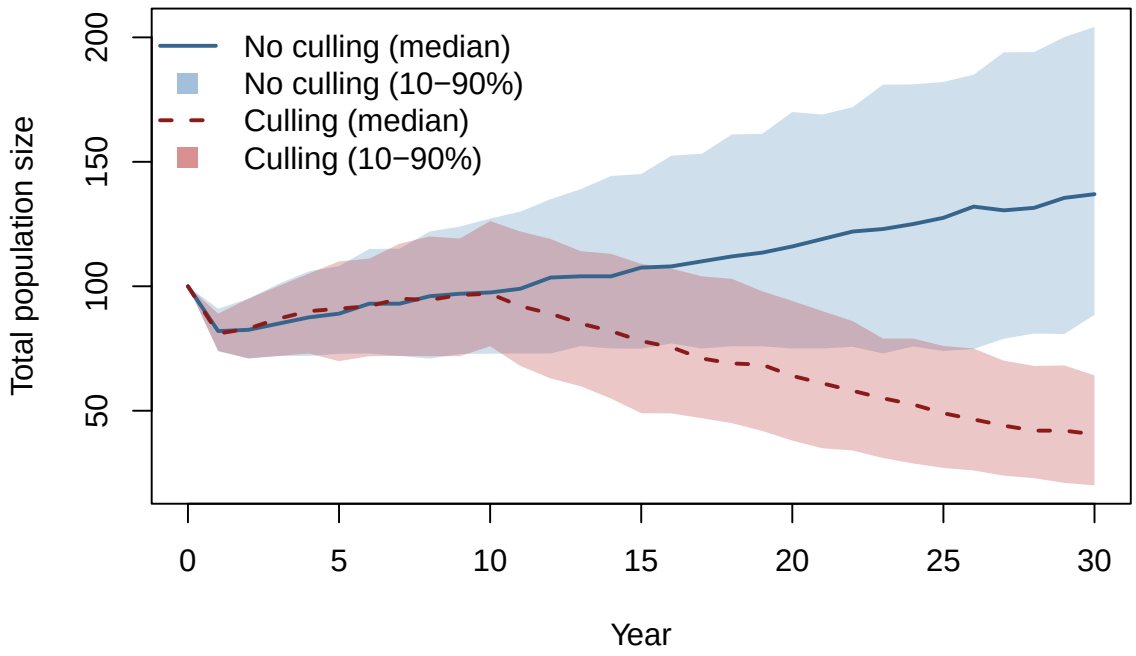
# Scenario A vs Scenario B: compare summaries in one figure
plot(sum_no$year, sum_no$median, type = "n",
     xlab = "Year", ylab = "Total population size",
     ylim = range(c(sum_no$lo, sum_no$hi, sum_cu$lo, sum_cu$hi)))

# Ribbon for no-culling uncertainty (10-90%)
polygon(
  x = c(sum_no$year, rev(sum_no$year)),
  y = c(sum_no$lo, rev(sum_no$hi)),
  col = grDevices::adjustcolor("steelblue", alpha.f = 0.25),
  border = NA
)
```

```
# Ribbon for culling uncertainty (10-90%)
polygon(
  x = c(sum_cu$year, rev(sum_cu$year)),
  y = c(sum_cu$lo, rev(sum_cu$hi)),
  col = grDevices::adjustcolor("firebrick", alpha.f = 0.25),
  border = NA
)

# Median trajectories on top of ribbons
lines(sum_no$year, sum_no$median, lwd = 2, col = "steelblue4")
lines(sum_cu$year, sum_cu$median, lwd = 2, lty = 2, col = "firebrick4")

legend("topleft",
       legend = c("No culling (median)", "No culling (10-90%)",
                  "Culling (median)", "Culling (10-90%)"),
       lty = c(1, NA, 2, NA),
       lwd = c(2, NA, 2, NA),
       pch = c(NA, 15, NA, 15),
       col = c("steelblue4", grDevices::adjustcolor("steelblue", alpha.f = 0.5),
               "firebrick4", grDevices::adjustcolor("firebrick", alpha.f = 0.5)),
       pt.cex = c(1, 1.5, 1, 1.5),
       bty = "n")
```



Two useful summaries of the replicates are the median final population size and the proportion of replicates that went extinct (an estimate of the probability of extinction within 30 years):

```
data.frame(
  scenario = c("No culling", "Culling 10% of adults"),
  median_final_N = c(median(totals_no[years + 1, ]), median(totals_cull[years + 1, ])),
```

```
  prob_extinct = c(mean(totals_no[years + 1, ] == 0), mean(totals_cull[years + 1, ] == 0))
)
```

```
##                 scenario median_final_N prob_extinct
## 1            No culling          137.0            0
## 2 Culling 10% of adults           40.5            0
```

Notice that even without culling a few replicates can go extinct by chance, although $\lambda > 1$: demographic stochasticity matters in small populations.

Your task

Exercise 1: Projection interval sanity check

1. In the tutorial, $\Delta t = 1$ year. Write one sentence explaining what that means.
2. Suppose you wanted $\Delta t = 1$ month. What changes would you need to make to the interpretation of:
   - P_A?
   - F_A?
3. Name one real reason you might prefer monthly time steps over yearly ones.

Exercise 2: Change one vital rate and re-run projections

1. Reduce adult survival from $P_A = 0.80$ to $P_A = 0.70$. Rebuild `A`, then recompute:
   - the deterministic projection plot,
   - λ.
2. Does the population grow or decline now? How can you tell?
3. Run `simulate_once()` with the new matrix. Do you need to change the function? Why not?

Hint: change `P_A` in the chunk that defines the vital rates, then re-run the chunks from there onwards (building `A`, the projection loop and `eigen()`).

Exercise 3: Culling as management

1. Change the harvest from 10% to 20% of adults per year. Before you run it, predict λ under the new harvest using the deterministic calculation (multiply the adult column of `A` by 0.8). Then run the replicates and compare the median trajectory, the uncertainty bands and the probability of extinction.
2. Now use a **fixed quota** instead: remove exactly 5 adults per year from year 10 (`cull_prop = 0`, `cull_n = 5`). How does the probability of extinction compare with the 10% harvest? Why is a fixed quota riskier for a small population than a proportional harvest?
3. To identify which stage is most impactful, run the same culling intensity for each stage separately (juveniles, subadults, adults: change `cull_stage`), then compare results. Do this both for a fixed quota of 5 individuals per year and for a 10% harvest. Which stage has the largest effect, and why?

Hint for question 3: ask how many future juveniles an individual in each stage is "worth". The left eigenvector of `A` (the reproductive values) does this for you: `v <- Re(eigen(t(A))$vectors[, 1]); v / v[1]`.

Exercise 4: Contraception treatment (reduced fertility)

Design a second management treatment where fertility is reduced instead of removing individuals.

1. Keep survival/transition rates unchanged, but reduce fertility from year 10 with a 90% effective contraceptive treatment (multiply `F_A` by 0.1 from year 10 onwards).
2. Implement this by adding an argument `fert_mult` to `simulate_once()` (with default `fert_mult = 1`), and using `F_A * fert_mult` instead of `F_A` in the births step from year `cull_start` onwards.
3. Run replicate simulations for:
   - baseline (no treatment),
   - culling treatment (10% of adults per year),
   - contraception treatment.
4. Compare all three scenarios in one figure and discuss which treatment gives the largest reduction in population growth. Also predict the long-term λ for the contraception treatment with a deterministic calculation.

Exercise 5 (optional): Environmental stochasticity

So far the vital rates are the same every year. Make adult survival vary between years: in each year, with probability 0.5 the year is "good" ($P_A = 0.9$) and otherwise it is "bad" ($P_A = 0.7$). The average is still 0.8. Modify a copy of `simulate_once()` to draw the type of year at the start of each year, and compare the spread of the replicate trajectories with the original simulation.

Takeaways

- Every rate in a matrix model refers to the projection interval Δt: survival, growth and fertility must all be per Δt.
- Deterministic projection gives the expected trajectory and, for a constant matrix, the long-term growth rate λ (the dominant eigenvalue).
- A stochastic simulation with the same vital rates shows the range of outcomes that chance alone produces. In small populations this includes extinction, even when $\lambda > 1$.
- Writing the simulation once as a function makes it easy to run many replicates and to compare scenarios fairly.
- A proportional harvest is equivalent to changing the matrix, so its long-term effect on λ can be predicted before simulating. A fixed quota is much riskier for a small population, because the proportion removed rises as the population shrinks.
- Removing individuals from different stages has very different effects: individuals in stages with high reproductive value (here adults) matter most.

Notes and extensions

- If you include juvenile stasis (juveniles surviving but remaining juveniles), add a P_J term and include it in the matrix and simulation.
- The deterministic matrix model is most useful for *structure* and *long-term behaviour* (λ, stable stage structure, elasticities).
- Stochastic simulations are most useful for *short-term variability* and *scenario exploration* (management, uncertainty, risk).

Part 3: Population Genetics and Evolution

This part introduces the genetic mechanisms that shape evolutionary change. You will connect allele frequencies to evolutionary forces such as mutation, drift, migration and selection, and to the heritability of traits.

By the end of Part 3, you should be able to:

- Use Hardy-Weinberg logic to interpret allele and genotype frequencies.
- Simulate allele frequency change under genetic drift, including bottlenecks and founder effects.
- Distinguish traits likely to evolve adaptively from those likely to evolve neutrally.
- Estimate heritability and interpret its evolutionary implications.

Reading: Neal, chapters *The Hardy-Weinberg Principle*, *Genetic Drift and Effective Population Size* and *Polygenic Inheritance and Quantitative Genetics*.

Hardy-Weinberg equilibrium

Learning outcomes. By the end of this practical you should be able to:

- Explain the difference between alleles and genotypes.
- Calculate expected genotype frequencies from allele frequencies, and allele frequencies from the frequency of a recessive genotype.
- Estimate allele frequencies from observed genotype counts and compare the observed counts with those expected under Hardy-Weinberg equilibrium (HWE).
- List the assumptions of HWE and explain what a departure from HWE might, and might not, tell you.

Background

Hardy-Weinberg equilibrium is a fundamental concept in population genetics for understanding the genetic dynamics of populations. Named after British mathematician G.H. Hardy and German physician Wilhelm Weinberg, this principle describes the stable genetic proportions in a population under specific conditions of no evolution.

In an idealised population, where mating is random, genetic mutations are absent, natural selection is not acting, and there is no gene flow or genetic drift, the frequencies of alleles and genotypes remain constant across generations.

The Hardy-Weinberg equilibrium provides a baseline against which real populations can be compared, allowing researchers to detect factors that may cause deviations from this equilibrium and, in turn, understand the forces driving evolution and genetic changes within populations.

In this practical we will learn more about HWE and practise solving some HWE-related problems.

Reading: Neal, chapter *The Hardy-Weinberg Principle*. Neal writes the two alleles as A_1 and A_2; here we use A and a.

The key formulae of the Hardy-Weinberg Equilibrium are:

Equation 1: $p + q = 1$

and

Equation 2: $p^2 + 2pq + q^2 = 1$.

Let's first consider where these equations come from. First, Equation 1.

In a diploid organism, each individual carries two alleles for a particular gene. Let's consider a gene with two alleles, A and a. In a given population:

Symbol	Meaning
p	Frequency of allele A
q	Frequency of allele a

The sum of the frequencies of these two alleles in the population must equal 1, as they are the only two possibilities for this gene, so $p + q = 1$.

Note that dominance affects the *phenotype*, not the allele frequency. A dominant allele can be rare and a recessive allele can be common.

When genotype counts are available, we can calculate allele frequencies directly by counting allele copies:

$$p = \frac{2n_{AA} + n_{Aa}}{2N}, \qquad q = \frac{2n_{aa} + n_{Aa}}{2N}.$$

Here, n_{AA}, n_{Aa} and n_{aa} are the numbers of individuals with each genotype, and $N = n_{AA} + n_{Aa} + n_{aa}$. Each homozygote contributes two copies of the same allele, while each heterozygote contributes one copy of each allele. These calculations do not require HWE. Once we know p and q, we can calculate the expected genotype counts under HWE as Np^2, $N(2pq)$ and Nq^2, and compare them with the observed counts.

Equation 1 serves as the foundational stepping stone for Equation 2. In Equation 1, we establish the sum of allele frequencies, p and q, in the gene pool. This equation sets the stage for the more complex Equation 2, where we translate these allele frequencies into genotype frequencies for the population.

We can derive Equation 2 from the gene pool and random mating. Consider a zygote formed by two gametes, one from the father and one from the mother, each drawn at random from the gene pool (the *random union of gametes*). A gamete carries A with probability p and a with probability q, so the possible combinations (father's allele first) are:

- $p \times p = p^2$ (AA)
- $q \times q = q^2$ (aa)
- $p \times q = pq$ (Aa)
- $q \times p = pq$ (aA)

Notice that we have two pq terms (Aa and aA). The order of alleles doesn't matter, so they are summed together: $2pq$.

So, the total probability for all possible combinations is:

$p^2 + 2pq + q^2$.

This must equal 1 (100% probability), which gives us Equation 2:

$p^2 + 2pq + q^2 = 1$.

Note that this identity is true for *any* value of p: it is just $(p+q)^2 = 1$. What HWE claims is something stronger, namely that the genotype frequencies in the population actually *equal* p^2, $2pq$ and q^2.

In these formulae, p^2 is the frequency of the homozygous AA genotype, q^2 is the frequency of the homozygous aa genotype and $2pq$ is the frequency of the heterozygous Aa genotype. From these equations we can produce the following plot, which shows the expected genotype frequencies at HWE for any allele frequency. HWE can occur at any allele frequency, but once the allele frequency is known, the expected genotype frequencies are fixed by Equation 2. (It is a common misconception that the three genotypes should occur in a 1:2:1 ratio, or that allele frequencies should tend towards 50% p and 50% q.)

Given a small amount of information it is possible to figure out what the allele and genotype frequencies should be *if the population is at HWE.*

If a population is NOT at HWE, that is a signal worth investigating. It may mean that one of the assumptions below is violated, but apparent departures can also arise from genotyping errors or simple chance in a small sample.

Assumptions of Hardy-Weinberg Equilibrium

For a population to be in Hardy-Weinberg Equilibrium, certain assumptions must be met:

- Random mating: There is no preference in mate selection based on genotype.
- No mutation: Alleles do not change into other alleles, so mutation does not alter allele frequencies.

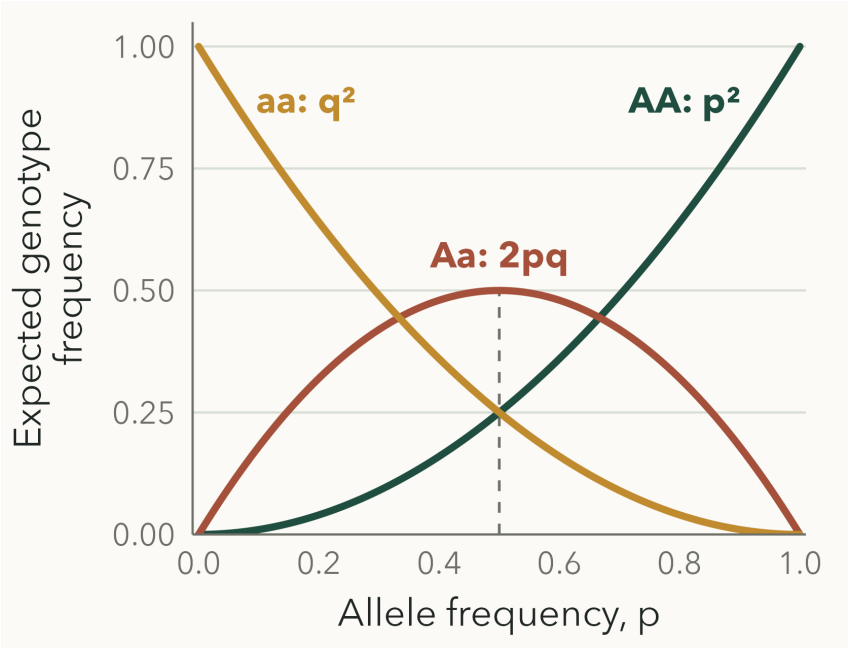


Figure 7: The Hardy-Weinberg Equilibrium. The lines show the expected genotype frequencies at HWE for each allele frequency, p.

- Very large population size: Random sampling has a negligible effect on allele frequencies (genetic drift).
- No gene flow: Migration does not transfer alleles into or out of the population's breeding gene pool.
- No natural selection: All genotypes in the population have an equal chance of reproductive success.

These assumptions do two different jobs. Random mating is what produces the Hardy-Weinberg genotype proportions ($p^2 : 2pq : q^2$) in each generation. The other four (no mutation, no drift, no gene flow and no selection) are what keep the allele frequencies themselves constant from one generation to the next.

Any deviation from these assumptions could result in a population that is not in Hardy-Weinberg Equilibrium.

Worked example

Unless a problem states otherwise, assume that the population is at HWE. This matters: working back from the frequency of a recessive genotype to an allele frequency (as below) is only valid if the population is at HWE.

Suppose 16% of individuals have the recessive genotype, aa, so $q^2 = 0.16$. What proportion of the population would we expect to be heterozygous?

1. Calculate $q = \sqrt{0.16} \approx 0.4$.
2. Calculate $p = 1 - q \approx 0.6$.
3. Calculate the heterozygote frequency: $2pq \approx 2(0.6)(0.4) \approx 0.48$.

We therefore expect about 48% of individuals to be heterozygous (Aa), assuming the population is at HWE.

Your task

Tackle the following problems, using the HWE theory outlined above. Worked solutions are given in Section @ref(solutions-hardy-weinberg-equilibrium).

Problem #1.

A population has two alleles, A and a, at a gene of interest. The frequency of allele A is 0.8. Assuming the population is at HWE, calculate the following:

- The frequency of allele a.
- The expected frequencies of the genotypes AA, Aa and aa.
- The frequencies of the two possible phenotypes if A is completely dominant over a.

PROBLEM #2.

Sickle-cell disease is caused by the HbS allele of the haemoglobin gene; the normal allele is HbA. People with genotype AA have normal red blood cells but are vulnerable to severe malaria. People with genotype SS have sickle-cell disease, which is often fatal without treatment. Carriers (AS) are usually healthy and are protected against severe malaria, so in malarial regions they survive better than either homozygote.

If 9% of babies in a population are born with sickle-cell disease (SS), what percentage are carriers (AS)?

Problem #3.

You genotype a sample of 200 individuals from a population at a gene with two alleles, A and a. You find 90 AA, 70 Aa and 40 aa individuals. This time, do not assume the population is at HWE: assess whether the observed counts match HWE expectations. A descriptive comparison is sufficient for this exercise; a formal statistical test is an optional extension (see the box below).

- Calculate the frequencies of alleles A and a from the genotype counts. (Hint: 200 diploid individuals carry 400 allele copies, and each heterozygote carries one copy of each allele.)
- Calculate the expected number of each genotype if the population were at HWE.
- Compare the observed and expected counts. Is there an excess or a deficit of heterozygotes? Suggest some reasons why a population might show this pattern.

Optional extension: a formal test. Neal tests for departures from HWE with a χ^2 (chi-square) test. For each genotype, calculate $(O-E)^2/E$, where O is the observed and E the expected count, and add the three values to get χ^2. Compare your value with the critical value of the χ^2 distribution for the correct degrees of freedom (3.84 for $P = 0.05$ and 10.83 for $P = 0.001$ with 1 degree of freedom; in R, `pchisq(your_value, df = 1, lower.tail = FALSE)` gives the P value).

Think carefully about the degrees of freedom. There are three genotype classes, but how many independent pieces of information did you take from the data to calculate the expected counts? Why is the answer 1 degree of freedom, not 2?

Problem #4.

Within a population of butterflies, brown (B) is completely dominant over white (b), and 40% of all butterflies are white. Assuming the population is at HWE, calculate the following:

- The percentage of butterflies in the population that are heterozygous.
- The frequency of homozygous dominant individuals.
- *Extension:* Of the brown butterflies, what proportion would you expect to be heterozygous? (Think carefully about what the denominator should be.)

Takeaways

- Hardy-Weinberg equilibrium provides a baseline for detecting evolutionary forces.
- Allele frequencies (p, q) determine genotype frequencies (p^2, $2pq$, q^2).
- Comparing observed genotype counts with HWE expectations can reveal a departure from HWE. A departure identifies a pattern worth investigating (for example, a violated assumption, genotyping error or chance in a small sample), not its cause.

The Gene Pool Model

Learning outcomes. By the end of this practical you should be able to:

- Explain why genetic drift is stronger in small populations than in large ones.
- Estimate the probability that an allele is fixed from replicate simulations, and show that for a neutral allele it equals the allele's starting frequency.
- Predict the fate of a new neutral mutation (starting frequency $1/(2N)$).
- Describe how bottlenecks and founder events change allele frequencies and genetic diversity.
- Use a simple simulation in R to explore a biological process.

This practical comes *before* the lecture on genetic drift, on purpose: you will discover the main results of drift by simulation, and the lecture will then explain and quantify them. Sections marked "optional" are best revisited after the drift lecture.

Reading: Neal, chapter *Genetic Drift and Effective Population Size*.

Background

A central goal in evolutionary biology is to understand variation (including genetic variation) and how it changes through time. One important idea is the "gene pool," which contains all the different versions of genes (alleles) within a group of organisms. These alleles influence phenotypic traits and other characteristics. To grasp how populations evolve, we need to understand how allele frequencies change in a population. Over time, genetic diversity changes as the relative numbers of alleles change. Some alleles may be lost from the population while others may become "fixed": the allele is the only one left, so every individual in the population carries it.

When no other process is acting, allele frequencies still change by chance. This is **genetic drift**. The main factor that determines the strength of drift is population size. In small populations, "sampling error" means that, just by chance, large changes often occur in allele frequencies. In large populations, dramatic changes are much less likely.

Two related concepts are critical in understanding genetic drift: bottlenecks and founder effects. A **genetic bottleneck** occurs when a population is dramatically reduced in size, for example by an environmental catastrophe. During a bottleneck, allele frequencies (and genetic diversity) can change dramatically because of sampling effects. A **founder effect** is a similar scenario, where a small group (a sample from the population) establishes a new population. Because the new population starts from a small sample of the ancestral population, it can have a rather different genetic composition from the ancestral population.

Your task

In this practical, you will use R to simulate how allele frequencies change in a population over generations. By changing population size, starting frequency and the history of the population, you will see how drift behaves in different situations.

The practical has four parts. The **core path** (about 45 minutes) is Parts 1 to 3 plus the bottleneck simulation in Part 4; the rest is optional, for the end of the session or at home.

1. **The gene pool** (5 minutes). Build a gene pool and see that the gene pool in the next generation is a *sample* of the current one.
2. **Drift through time** (10 minutes). Project allele frequency through time and investigate the importance of population size.
3. **Many populations** (15 minutes). Repeat the simulation many times to estimate the probability that an allele is fixed or lost, including for a rare (new) allele.
4. **Bottlenecks** (15 minutes). Modify the code to simulate a population that passes through a period of small size.

Optional: tracking genetic diversity (Part 3), effective population size and founder effects (Part 4), and the Challenge at the end.

Before you start, make a prediction. Imagine two populations, each starting with allele A at a frequency of 0.5. One has 20 individuals and the other has 1000. Sketch what you expect the frequency of A to look like over 100 generations in each population. Which population do you expect to lose one of its alleles first? Keep your sketch, and compare it with your simulations in Part 2.

The model and its assumptions

Our model is a simple version of the **Wright-Fisher model** of population genetics. It assumes:

- a single locus with two alleles, A and a;
- a constant number of diploid individuals, N, so the gene pool contains 2N alleles;
- non-overlapping generations: all adults are replaced by their offspring each generation;
- random mating, so each new allele is a random draw from the parents' gene pool;
- no selection, no mutation and no migration.

So the *only* thing that changes allele frequencies is chance. Keep these assumptions in mind: we come back to them at the end.

Part 1: A simple model

We will first aim to get an understanding of what a gene pool is, using a simple model. We will use R to make a gene pool, then look at how allele frequency in the gene pool changes over a single generation.

The gene pool

To establish the gene pool we first set the population size. I set this to be reasonably large, at 500 individuals.

```
pop_size <- 500
```

There are two alleles (gene variants) per individual: one allele came from the father, and one from the mother.

Therefore, the gene pool contains 2 x N alleles (in this case, 1000 alleles).

```
pool_size <- pop_size * 2
```

I now set the initial allele frequency to be 0.5. You can use other values, but 0.5 is best to start with.

```
initial_frequency_A <- 0.5
initial_frequency_a <- 1 - initial_frequency_A
```

We can now calculate the **number** of A and a alleles, by multiplying the frequency by the pool size and rounding the result to the closest whole number.

```
number_A <- round(initial_frequency_A * pool_size)
number_a <- round(initial_frequency_a * pool_size)
```

Following this, the gene pool can be filled like this. What we are saying here is "repeat `A`, `number_A` times, then repeat `a` `number_a` times".

```
gene_pool <- c(rep("A", number_A), rep("a", number_a))
```

We can count the alleles in the initial gene pool with `table`:

```
table(gene_pool)
```

```
## gene_pool
##   a   A
## 500 500
```

We can express these numbers as allele frequencies (i.e. as proportions of the total) by wrapping the `table` command with `prop.table` like this:

```
prop.table(table(gene_pool))
```

```
## gene_pool
##   a   A
## 0.5 0.5
```

Projecting allele frequency over one time step

Let's assume the population size remains constant at N = 500.

During the next time step (i.e. 1 generation), the individuals in the population mate with each other randomly. We can use the `sample` function to simulate this.

This code is saying "**sample, from the gene pool, 1000 alleles, with replacement**". We need to sample 2× the population size because each individual contains two alleles. We sample *with replacement* because any allele in the parents' gene pool can be passed on to more than one offspring (or to none).

```
new_gene_pool <- sample(gene_pool, 2 * pop_size, replace = TRUE)
```

Let's look at the allele frequency in the new gene pool. It should be similar, but probably not exactly the same as the initial gene pool.

```
prop.table(table(new_gene_pool))
```

```
## new_gene_pool
##     a     A
## 0.491 0.509
```

There are still 500 individuals, but the allele frequencies are (probably) not exactly the same as in the initial gene pool. The reason is that the new gene pool is a random sample, not simply a carbon copy.

Try to vary the population size and do this several times at small and large population sizes. You should notice that the frequencies are more similar to the starting frequency, and to each other, when you repeat this for large population sizes than for small population sizes.

Part 2: Simulation of allele frequency through time

To simulate the passage of time, we need to do this sampling procedure many times.

We can do this using a loop in R. In our loop, we repeat the gene pool sampling process once per generation. Each time we go through the loop, we sample alleles from the *old* gene pool at time t to create a *new* gene pool for time $t + 1$. Each time through the loop we also calculate the allele frequency of A, so we can track how it changes through time.

Let's set up a simulation that runs for 100 generations, starting with a small population of 20 individuals. You can copy and paste this code, which is a complete gene pool model.

```
# Set the population size and number of generations
pop_size <- 20
n_gen <- 100

# Set up an empty vector of length n_gen to contain results
allele_freq_A <- rep(NA, n_gen)

# Put in an initial value for frequency of A
allele_freq_A[1] <- 0.5

# Establish the initial gene pool and name it gene_pool_time_n
# This part looks complicated, but it just condenses the code
# from Part 1 into one line.
gene_pool_time_n <- c(
  rep("A", round(2 * allele_freq_A[1] * pop_size)),
  rep("a", round(2 * (1 - allele_freq_A[1]) * pop_size))
)

for (i in 2:n_gen) {
  # Sample 2N alleles, with replacement, to make the next generation
  gene_pool_time_n_plus_1 <- sample(gene_pool_time_n,
    pop_size * 2,
    replace = TRUE
  )
  # Calculate the frequency of A in the new gene pool
  gene_pool_table <- table(factor(gene_pool_time_n_plus_1,
                                  levels = c("A", "a")))
  allele_freq_A[i] <- gene_pool_table["A"] / sum(gene_pool_table)

  # Replace the old gene pool with the new one.
  gene_pool_time_n <- gene_pool_time_n_plus_1
}
```

Now we can plot this result (Figure @ref(fig:simulationplot)).

```r
plot(1:n_gen, allele_freq_A, type = "l",
     xlab = "Generation",
     ylab = "Frequency of allele A",
     ylim = c(0, 1))
```

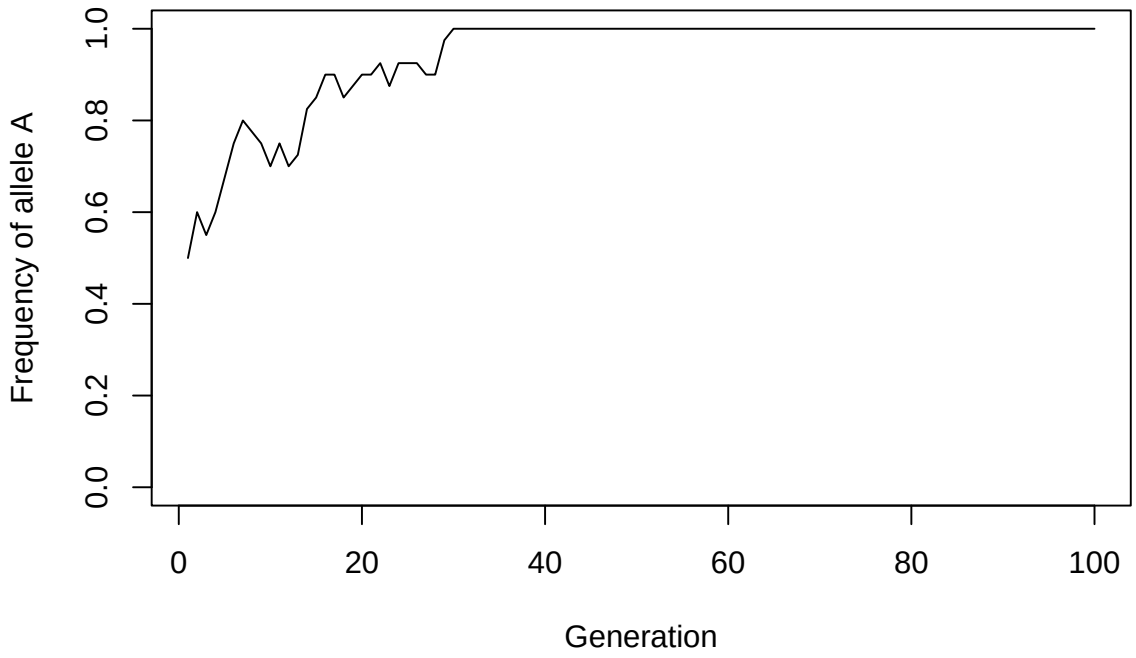


Figure 8: Simulation of allele frequency through time in a population of 20 individuals

Tip: make a new script and paste the code loop along with the plot command so that you can run both together easily.

Run the code several times with `pop_size <- 20`, then several times with `pop_size <- 1000`. Compare the results with the prediction you made at the start.

- How does population size affect the variation through time in allele frequencies? Why do you see these patterns?
- With 20 individuals, does A always end up fixed (frequency 1), or is it sometimes lost (frequency 0)?
- When the population size is large (say 1000), can you see an allele become fixed within 100 generations? Do you think it could happen if you ran the simulation for much longer? (Try `n_gen <- 3000`. It may take a little while to run.)

Part 3: Many populations

A single run of the simulation is like watching a single population. To answer questions about *probability* (for example, "how likely is it that A is fixed?") we need to watch many populations: we run the same simulation many times and count the outcomes.

To make this easy, we first wrap the loop from Part 2 into a **function**. A function is a reusable piece of code: you give it some inputs (here, population size, number of generations and the starting frequency of A), and it returns the frequency of A in every generation.

```
simulate_gene_pool <- function(pop_size, n_gen, initial_freq_A) {
  allele_freq_A <- rep(NA, n_gen)
  allele_freq_A[1] <- initial_freq_A

  gene_pool_time_n <- c(
    rep("A", round(2 * initial_freq_A * pop_size)),
    rep("a", round(2 * (1 - initial_freq_A) * pop_size))
  )

  for (i in 2:n_gen) {
    gene_pool_time_n_plus_1 <- sample(gene_pool_time_n,
      pop_size * 2,
      replace = TRUE
    )
    gene_pool_table <- table(factor(gene_pool_time_n_plus_1,
                                    levels = c("A", "a")))
    allele_freq_A[i] <- gene_pool_table["A"] / sum(gene_pool_table)
    gene_pool_time_n <- gene_pool_time_n_plus_1
  }
  return(allele_freq_A)
}
```

Now we can use `replicate` to run the function many times. Here we simulate 20 populations, each of 20 individuals, for 100 generations. The result is a matrix with one row per generation and one column per population. The `matplot` function plots each column as a separate line (Figure @ref(fig:replicateplot)).

```
n_gen <- 100
results <- replicate(20, simulate_gene_pool(pop_size = 20, n_gen = n_gen,
                                            initial_freq_A = 0.5))

matplot(1:n_gen, results, type = "l", lty = 1,
        xlab = "Generation",
        ylab = "Frequency of allele A",
        ylim = c(0, 1))
```

We can count the outcomes by looking at the last row of the matrix (the final generation):

```
final_freq <- results[n_gen, ]
sum(final_freq == 1)   # number of populations where A is fixed
```

```
## [1] 11
```

```
sum(final_freq == 0)   # number of populations where A is lost
```

```
## [1] 7
```

Use this code to investigate the following questions:

- What proportion of the populations have lost one of their alleles (i.e. A is fixed or lost) by generation 100 when N = 20? When N = 100? When N = 1000?

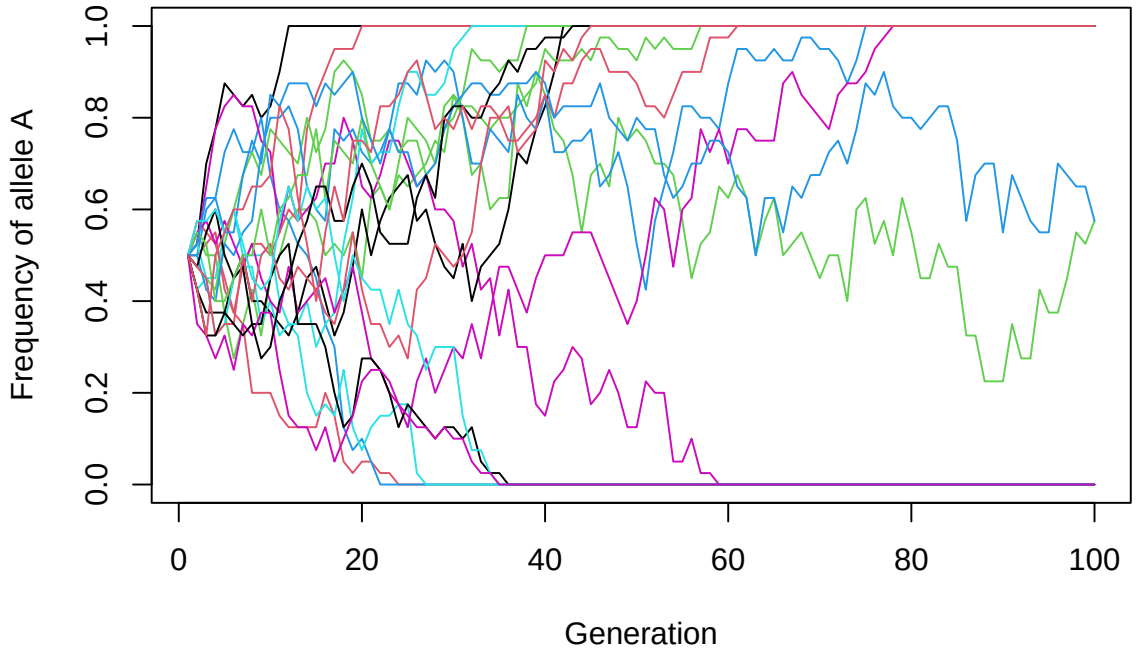


Figure 9: Twenty replicate populations of 20 individuals, each starting with allele A at a frequency of 0.5

- Of the populations where one allele was fixed, in what proportion was it A? Does this proportion depend on population size? (You may need more than 20 replicates to see this clearly.)
- So what does population size change: *which* allele is fixed, or *how quickly* variation is lost?

Rare alleles and new mutations

A brand-new mutation starts as a single copy in the gene pool, so its starting frequency is $1/(2N)$. Let's see what happens to it. Here we run 200 populations of 20 individuals, each starting with one copy of A (frequency 1/40), for 300 generations (long enough for nearly all of them to be fixed or lost).

```
pop_size <- 20
results_rare <- replicate(200, simulate_gene_pool(pop_size = pop_size, n_gen = 300,
                                                   initial_freq_A = 1 / (2 * pop_size)))
final_freq_rare <- results_rare[300, ]
mean(final_freq_rare == 1)   # proportion of populations where A was fixed
```

```
## [1] 0.015
```

```
mean(final_freq_rare == 0)   # proportion of populations where A was lost
```

```
## [1] 0.985
```

- Are rare alleles more or less likely to be lost than common ones?
- Compare the proportion of populations where the new allele was fixed with $1/(2N)$. What does this tell you about the fate of most new mutations?

- Try a few different starting frequencies (e.g. 0.1, 0.25, 0.75). Can you find a simple rule linking the starting frequency of a neutral allele to its probability of fixation? You can use this template: change `p0` and run it again. Record your results in a small table (starting frequency, proportion of populations where A was fixed).

```
p0 <- 0.25
results_p0 <- replicate(200, simulate_gene_pool(pop_size = 20, n_gen = 300,
                                                initial_freq_A = p0))
mean(results_p0[300, ] == 1)   # proportion of populations where A was fixed
```

With 200 replicates, expect your proportions to differ from the "true" probability by a few percentage points just by chance.

Going further (optional): tracking genetic diversity. In the mutation lecture we measured genetic diversity with expected heterozygosity, $H_e = 2pq$. Theory predicts that drift reduces H_e by a factor of $1 - 1/(2N)$ every generation, so that $H_t = H_0(1 - 1/(2N))^t$. The code below calculates H_e for each generation of each population, averages it across 500 populations of 20 individuals (an average over only 20 populations is too noisy), and adds the theoretical curve in red. Note that the first row of `results` is generation $t = 0$, so the theoretical curve uses `t <- 0:(n_gen - 1)`. Does your simulation agree with the theory?

```
n_gen <- 100
results <- replicate(500, simulate_gene_pool(pop_size = 20, n_gen = n_gen,
                                             initial_freq_A = 0.5))
He <- rowMeans(2 * results * (1 - results))   # mean He in each generation

t <- 0:(n_gen - 1)
plot(t, He, type = "l", ylim = c(0, 0.5),
     xlab = "Generation", ylab = "Mean expected heterozygosity")
lines(t, 0.5 * (1 - 1 / (2 * 20))^t, col = "red")
```

Part 4: Bottlenecks and founder effects

Next, modify the code to simulate a **genetic bottleneck**, when a population goes through a period of small population size. Start from the code in Part 2, set `pop_size <- 1000` at the top *before* the initial gene pool is built (otherwise the gene pool starts with only 40 alleles, which is itself a hidden founder event), and add this code at the *start* of the loop (inside the curly brackets), so that the population size is reset each generation:

```
if (i %in% 30:50) {
  pop_size <- 10
} else {
  pop_size <- 1000
}
```

This code means, "*...if the generation is between 30 and 50, make the population size 10, otherwise make the population size 1000*". In other words, the population goes through a bottleneck of 10 individuals for 21 generations (30 to 50 inclusive). You can modify this code to change the length and severity of the bottleneck.

Figure @ref(fig:bottlenecksimulationplot) shows an example of what your plot of a bottleneck simulation might look like. I have added some lines to show where the bottleneck starts and ends. Here is the complete code, which you can also simply copy:

```
# Set the starting population size and number of generations
pop_size <- 1000
n_gen <- 100

# Set up an empty vector of length n_gen to contain results
allele_freq_A <- rep(NA, n_gen)

# Put in an initial value for frequency of A
allele_freq_A[1] <- 0.5

# Establish the initial gene pool and name it gene_pool_time_n
gene_pool_time_n <- c(
  rep("A", round(2 * allele_freq_A[1] * pop_size)),
  rep("a", round(2 * (1 - allele_freq_A[1]) * pop_size))
)

for (i in 2:n_gen) {
  # Set the population size for this generation
  if (i %in% 30:50) {
    pop_size <- 10
  } else {
    pop_size <- 1000
  }
  gene_pool_time_n_plus_1 <- sample(gene_pool_time_n,
    pop_size * 2,
    replace = TRUE
  )
  gene_pool_table <- table(factor(gene_pool_time_n_plus_1,
                                  levels = c("A", "a")))
  allele_freq_A[i] <- gene_pool_table["A"] / sum(gene_pool_table)

  # Replace the old gene pool with the new one.
  gene_pool_time_n <- gene_pool_time_n_plus_1
}
plot(1:n_gen, allele_freq_A, type = "l",
     xlab = "Generation",
     ylab = "Frequency of allele A",
     ylim = c(0, 1))
abline(v = 30, col = "red", lty = 3)
abline(v = 50, col = "red", lty = 3)
text(40, 0.95, "bottleneck", col = "red")
```

Run the bottleneck simulation several times, and answer the following questions:

- How does a genetic bottleneck influence the genetic composition of the population?
- How might a genetic bottleneck affect the probability that one allele is fixed and the other lost?
- What happens to allele frequency *after* the bottleneck, when the population is large again? Does the population "recover" its original allele frequency?
- Does the severity of the bottleneck (i.e. its length and the population size during it) matter?

Optional (revisit after the drift lecture): effective population size. Over the 100 generations of the bottleneck run, the census population size averages about 790 (79 generations at 1000 and 21 at 10). Yet the population loses diversity like a much smaller one. For drift, what matters is the *effective population size*, N_e, the size of an idealised population that would lose diversity at the same rate. When N changes over time, N_e is approximately the *harmonic mean* of N, which is dominated by the small values:

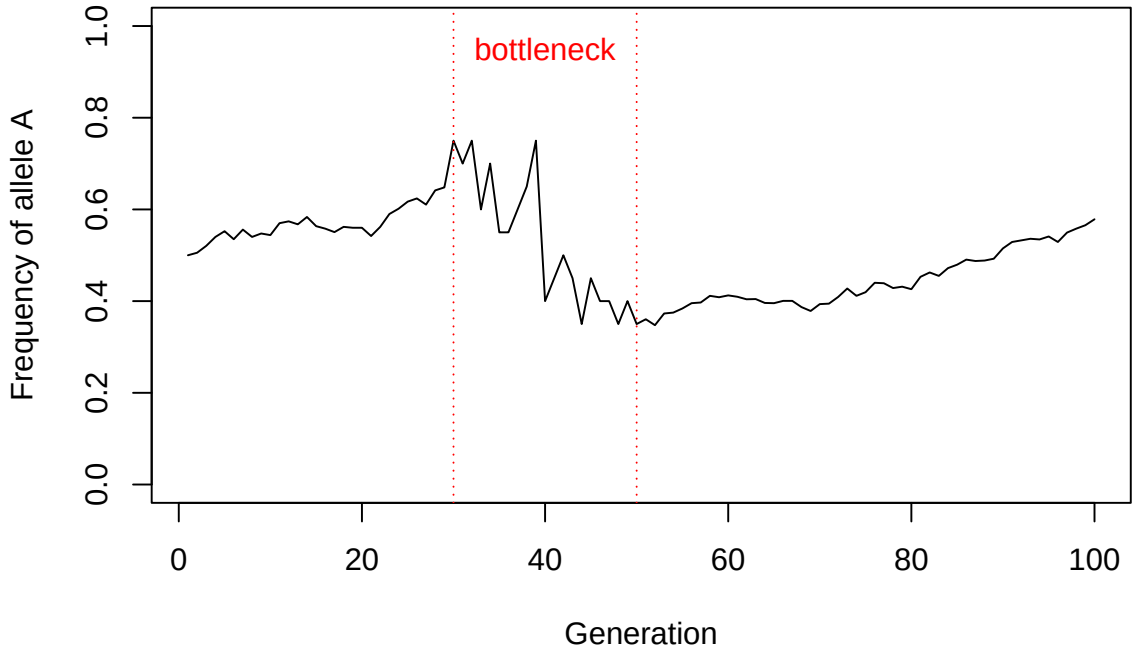


Figure 10: Simulation of a genetic bottleneck: 1000 individuals, falling to 10 individuals in generations 30 to 50

$$N_e \approx \frac{t}{\sum_{i=1}^{t} 1/N_i}$$

Calculate the arithmetic mean and the harmonic mean of N for your bottleneck run. Here is one way to do it in R:

```
N <- rep(1000, 100)     # population size in each of the 100 generations
N[30:50] <- 10          # the bottleneck
mean(N)                 # arithmetic mean
1 / mean(1 / N)         # harmonic mean, approximately Ne
```

What does the difference between the two tell you about the importance of short periods of small population size?

Founder effects (optional)

A founder effect is like a bottleneck at the very start of a new population. To simulate it, take a large source population (1000 individuals, frequency of $A = 0.5$) and found a new population from just 10 individuals (20 alleles) drawn at random from it. Repeat this 50 times, as if 50 islands were each colonised by a different group of 10 founders:

```
source_pool <- c(rep("A", 1000), rep("a", 1000))

founder_freq_A <- replicate(50, {
```

```
  founders <- sample(source_pool, 2 * 10, replace = FALSE)
  mean(founders == "A")
})

hist(founder_freq_A, breaks = seq(0, 1, by = 0.05),
     xlab = "Frequency of allele A in the founded population",
     main = "")
abline(v = 0.5, col = "red", lty = 3)
```

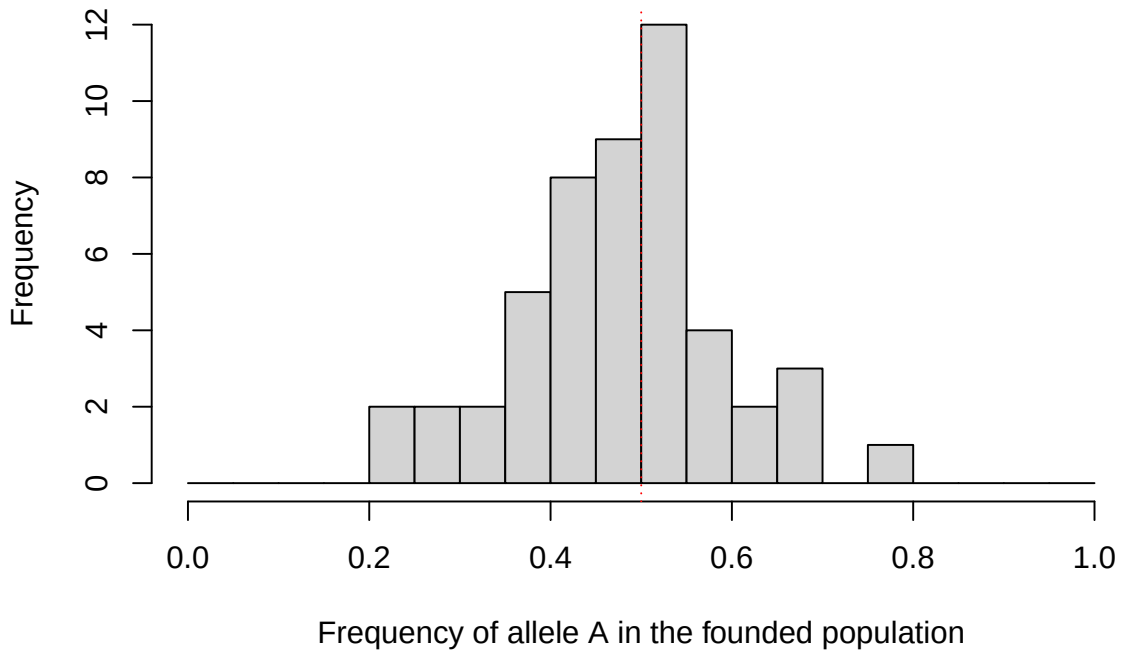


Figure 11: Frequency of allele A in 50 new populations, each founded by 10 individuals from a source population where the frequency of A is 0.5

- How different are the new populations from the source population, and from each other?
- How would this change with 50 founders instead of 10?
- Optional: use `simulate_gene_pool()` with `initial_freq_A` set to one of these founder frequencies to follow a founded population forward in time.

Conclusions

You should now have a good idea of what a gene pool model is, the importance of population size, the concept of sampling from a population, and how bottlenecks and founder effects change genetic composition.

Let's finish with some more general questions:

- What are the limitations of this simulation? Look back at the list of assumptions in "The model and its assumptions". Which real-world processes and complexities are missing?

- Why is genetic diversity important to a population, for example in a conservation context?
- Can you think of real-life scenarios where understanding gene pool dynamics would be valuable, such as in conservation biology or medical genetics?

Takeaways

- Genetic drift is stronger in small populations because sampling error is larger.
- Drift does not favour either allele: a neutral allele's probability of fixation equals its starting frequency. Population size changes how *quickly* variation is lost.
- Most new mutations (starting at $1/(2N)$) are lost by chance.
- Bottlenecks and founder effects can rapidly reshape allele frequencies, and the change persists after the population becomes large again.

Challenge (optional): Can you think of a way to simulate the genetics of a population that is steadily shrinking through time? Try it.

Neutral or Adaptive Evolution in Humans: What Drives Evolution of Our Traits?

Background

Understanding the role of natural selection and genetic drift in human traits requires a look at several key aspects:

- Heritable variation: a trait can only evolve (by drift or by selection) if individuals differ in it *and* some of those differences are inherited.
- Selection vs drift: selection acts when variation in a trait is correlated with variation in survival or reproduction. Drift is chance change in allele frequencies, and it acts in every finite population.
- Role of environment: the environment can change the strength, and even the direction, of selection on a trait.

First ask: is the variation heritable? Some traits vary a lot but are hardly inherited at all. For example, the fine texture of the iris and the detailed ridge patterns (minutiae) of fingerprints differ even between identical twins: they are mostly the result of chance events during development. Variation like this cannot evolve by drift or by selection, because it is not passed on.

Neutral and adaptive evolution are two ends of a continuum. In the lecture on genetic drift we defined neutral evolution. Suppose a trait varies, individuals vary in reproductive success, and the trait is inherited, *but* the trait is not (or only very weakly) correlated with reproductive success. The trait still evolves, but by chance: this is neutral evolution, caused by genetic drift. When the correlation between the trait and reproductive success is strong, we get adaptive evolution, caused by natural selection. Most traits lie somewhere in between.

Population size also decides where a trait sits on this continuum. Drift acts on the alleles that affect a trait, and its strength per generation is about $1/(2N_e)$, where N_e is the effective population size. Selection on those alleles is effective only when the selective advantage, s, is clearly larger than roughly $1/(2N_e)$. So weak selection that is effective in a large population can be swamped by drift in a small one. Whether a trait evolves "neutrally" therefore depends on population size as well as on the environment.

When selection is moderate or strong, a trait tends to be held close to some optimal value (stabilising selection) or to change in a consistent direction (directional selection): adaptive evolution. When selection is weak or absent, the alleles that affect the trait change in frequency at random, and the trait drifts: neutral evolution. Both require heritable variation to begin with: if there is little heritable variation, the trait will not respond much even if selection is strong.

Finally, the strength of selection depends on the environment. A trait may be important in some environments but not others, and may be selected for in one environment but against in another. For example, the skin pigment melanin protects against intense ultraviolet (UV) radiation, including UV damage to the vitamin folate, so darker skin is favoured where UV is intense. Where UV is weak, lighter skin allows more vitamin D to be made in the skin (Jablonski & Chaplin 2010, *PNAS*). Ancient DNA shows that some of the main light-skin alleles became common in Europe only within roughly the last 8,000 years

(Mathieson et al. 2015, *Nature*). Exactly which selective pressures mattered most, and when, is still debated.

The objective of this exercise is to develop an intuitive understanding of the effect of selection and genetic drift on traits.



Figure 12: Two present-day human environments: Hadza men returning from a hunt in the dry savanna woodland of northern Tanzania (left) and Nyhavn in Copenhagen (right). Both photographs show people living today: all living human populations are equally evolved, and all are still subject to natural selection. How would selection on each trait differ between these environments? Photos: Andreas Lederer (left) and Jorge Láscar (right), both CC BY 2.0, via Wikimedia Commons.

Learning outcomes:

- Decide whether a trait can evolve at all, by asking whether its variation is heritable.
- Place traits on the continuum from neutral evolution (genetic drift) to adaptive evolution (natural selection), and justify each placement with evidence or reasoning.
- Explain how the environment (foraging societies vs modern industrialised countries such as Denmark) and population size change the strength of selection on a trait.
- Predict which differences between human populations gene flow would erase, and which local selection could maintain.
- Use correct scientific terminology when arguing for your categorisation in a group.

Key idea

Traits like resistance to infectious disease are likely to face strong selection in environments where the disease is common. Heritable variation with little or no effect on survival or reproduction (for example, many DNA differences at synonymous sites, which do not change the protein) is more likely to change by drift.

Your task (30 minutes)

You are given a short list of eight human traits (below). They were chosen to cover the whole range, from traits that are probably drifting, through traits with little heritable variation, to traits under strong selection in some environments but not others. In small groups:

1. Write the traits on PostIt notes.
2. Discuss and categorise the traits based on the strength of natural selection in foraging (hunter-gatherer) societies, like those in which most of human evolution took place (10 minutes).
3. Repeat the exercise for modern industrialised societies like Denmark (10 minutes). In Denmark today, most people survive to adulthood, so ask how each trait might affect the *number of children* people have: selection still acts, but now mainly through differences in reproduction rather than survival.

4. **Gene flow between populations** (10 minutes). Human populations exchange migrants, and gene flow pulls their allele frequencies together. In the lecture on migration and gene flow we saw that very few migrants (about one per generation) are enough to stop drift making populations very different, but that local selection can keep populations different if it is strong compared with the migration rate (s larger than m). For each trait, ask:

    - Does the trait differ between human populations living in different environments?
    - Would gene flow erase those differences (as expected for neutral traits), or could local selection hold them apart? Good places to start: skin pigmentation and UV; lactase persistence (digesting milk sugar as an adult), which became common in populations that keep milk-producing animals; and the sickle-cell allele (HbS), which protects carriers against malaria.
    - Remember that most human genetic variation is found *within* populations rather than between them.

For steps 2 and 3, order the traits from those likely influenced by **GENETIC DRIFT** to those evolving through **ADAPTIVE EVOLUTION** via **NATURAL SELECTION**. For each trait, start with the question "is the variation heritable?".

The traits

Trait	Description
Metabolic efficiency	How effectively your body converts food into energy.
Eyesight acuity	Sharpness of vision.
Height	Standing height.
Birth canal (pelvic) shape	The shape of the bony opening through which a baby is born.
Resistance to bubonic plague	Immunity or resilience against bubonic plague.
Eye colour	The colour of the iris (brown, green, blue and so on).
Earwax type	Wet or dry earwax, set by a single DNA change; dry earwax is common in East Asia and rare in Europe and Africa.
Face symmetry	The balance and proportionality of facial features.

Discussion (10 minutes)

After the exercise, discuss as a class:

- Were there any traits that were categorised differently between the two types of societies? Why?
- Which traits did you decide had little or no heritable variation, and so could not evolve by drift or by selection?
- Which differences between human populations would gene flow erase, and which are maintained by local selection?
- Did the exercise challenge any preconceived notions you had about trait evolution?
- Can you think of any other traits that would be interesting to add to this exercise?

Takeaways

- A trait can only evolve if its variation is heritable.
- Neutral and adaptive evolution are two ends of a continuum: where a trait sits depends on how strongly it affects survival or reproduction, and on population size (s compared with $1/(2N_e)$).
- Selection strength depends on environment and context, not just on the trait itself, and different environments can reverse which traits are favoured.

- Humans are still evolving: in industrialised societies, selection acts mainly through differences in the number of children.
- Gene flow erases neutral differences between populations, but local selection can maintain differences in traits such as skin pigmentation.

Writing Short Answers

The final exam has two parts: multiple-choice questions and short-answer questions. A short-answer question needs a paragraph or two of written text. It tests whether you understand the key terms and concepts, and whether you can communicate that understanding clearly. Knowing the material is not enough: you need to show the examiner that you know it.

In this session you will look at what makes a good short answer, mark three example answers yourself, and then write and mark an answer of your own.

Learning outcomes:

- Break a question into its parts, and use its verbs (define, describe, explain, compare) to decide what each part needs.
- Plan an answer before writing it.
- Write paragraphs with a topic sentence, supporting sentences and a concluding sentence.
- Use precise terminology and specific examples.
- Mark an answer against a rubric, and use the rubric to improve your own writing.

What examiners look for

- **Read the question, then read it again.** Most questions have several parts. Underline each one and answer every part: a missing part costs marks however good the rest is.
- **Look at the verbs.** *Define* asks for a precise meaning. *Describe* asks what happens. *Explain* asks why or how it happens. *Compare* asks for similarities and differences, side by side. *Contrast* asks for differences only.
- **Plan first.** Spend a couple of minutes listing the points you will make for each part. A plan stops you writing everything you know about the topic and missing what was asked.
- **Answer first.** Your first sentence should answer the question directly. Background can come afterwards, if it is needed at all. An examiner should not have to hunt for your point.
- **Use the right terms, and use them correctly.** Precise terms are clearer than everyday descriptions, and they show the examiner you understand the concept.
- **Give specific examples.** "A bird" or "a plant" is not a specific example. "Copper-tolerant common bent grass on a small copper mine in Wales" is. The examples from the lectures and the textbook are ideal, because you know what was found and why it matters.
- **Keep to time.** A long answer to one question can cost you a whole question elsewhere.

Compare these two definitions:

> Gene flow is the movement of alleles from one population into the gene pool of another, through immigrants that breed there or gametes, such as pollen, that fertilise there.

> Gene flow is when animals move to a new place and take their genes with them.

The first is precise and complete. The second describes migration, misses the point that immigrants must breed, and leaves out plants altogether.

Paragraph structure

Paragraphs in scientific writing follow a simple pattern:

1. A **topic sentence** that states the main idea of the paragraph, and tells the reader what to expect. An examiner who reads only your topic sentences should be able to see your whole answer.
2. One or more **supporting sentences** that explain, give evidence or give examples. Each one should back up the topic sentence: if it does not, it belongs in another paragraph.
3. A **concluding sentence** that says what follows from the point, qualifies it, or links to the next paragraph. Do not just repeat the topic sentence: that costs time and earns nothing. In a short paragraph you can leave the concluding sentence out.

One paragraph should make one main point. Make it long enough to make and support that point, and no longer.

A worked example

> **Question.** Compare how haploid and diploid genetic systems affect the response to selection. Give an example of a trait under selection in each, and explain how the genetic system affects the speed of evolutionary change.

> (1) Haploid systems respond to selection more quickly than diploid systems, because in a diploid an allele can be hidden from selection in heterozygotes. (2) In a haploid organism every allele is expressed, so selection acts on a new advantageous mutation as soon as it appears. (3) For example, drug resistance in *Mycobacterium tuberculosis*, a haploid bacterium, can evolve within a single patient, helped by short generation times and very large population sizes. (4) In a haploid, then, the speed of change is set by the strength of selection and the supply of new mutations, not by dominance.
>
> (5) In a diploid, the speed of change depends on dominance, and a favoured dominant allele spreads fastest while it is rare. (6) This is because it is expressed in heterozygotes, which carry almost all the copies of a rare allele: the dominant *carbonaria* allele of the peppered moth rose from about 1% to 99% of moths near Manchester in about 50 generations, which needs selection of about $s = 0.3$ against the pale form. (7) However, a dominant allele slows down near fixation, because the last copies of the recessive allele hide in heterozygotes.
>
> (8) A favoured recessive allele shows the opposite pattern. (9) While it is rare it occurs almost only in heterozygotes, where it is not expressed, so it can take thousands of generations to start spreading, but once it is common it goes to fixation quickly. (10) Dominance therefore decides whether change in a diploid is slow at the start or at the end, a delay that haploids never have.

How the answer is built:

- **The first sentence answers the question:** haploids respond faster, and it says why.
- **One point per paragraph:** haploids; dominant alleles in diploids; recessive alleles in diploids.
- **Topic sentences (1, 5 and 8):** read on their own, they outline the whole answer.
- **Supporting sentences** back up their topic sentence, and each claim has a specific example with numbers (tuberculosis; the peppered moth).
- **Concluding sentences (4, 7 and 10)** each add something: what follows (4), a qualification (7), and a link back to the comparison (10). None of them just repeats its topic sentence.

Together, the three paragraphs answer every part of the question: they compare the two systems, give an example of each, and explain how each system affects the speed of change.

Mark three answers yourself

Here is a short-answer question with its marking rubric, followed by three student answers. **Mark each answer using the rubric on your own, then agree marks with a partner.** The examiner's marks are shown in the practical session, and a full explanation is posted on itslearning (Resources › Practice Exams) after the session. Where your marks differ from the examiner's, work out why.

> **Question.** Define gene flow and explain how it differs from migration. Explain how gene flow can both help and hinder a population's adaptation to local conditions, giving a specific example. Explain why very few migrants per generation are enough to stop populations diverging by genetic drift, but many more are needed to swamp local adaptation.

Rubric (10 points)

Criterion	2 points	1 point	0 points
1. Definition	Gene flow is the movement of alleles into another population's gene pool, by immigrants that breed there or by gametes (such as pollen) that fertilise there	Partly right: talks about genes moving, but vague about how they enter the gene pool	Incorrect, or describes only the movement of individuals
2. Differs from migration	Explains that migration is the movement of individuals, and that it causes gene flow only if the migrants or their gametes take part in breeding	Says the two differ, but not why	Treats them as the same thing, or missing
3. Local adaptation	Explains both how gene flow can help (bringing in variation, genetic rescue) and how it can hinder (bringing in alleles suited to other environments), with a specific example	Only one side, or no specific example	Incorrect or missing
4. Drift versus selection	Explains that a *number* of migrants (about one per generation) is enough to stop divergence by drift whatever the population size, but that gene flow swamps selection only when the migration *rate* m is similar to or larger than s	States the result without explaining it, or explains only one half	Incorrect or missing
5. Structure	The first sentence of each part answers it; one point per paragraph, each with a topic sentence; correct terminology	Mostly clear, but loosely structured, repetitive or imprecise	Disorganised

A factual error in a criterion caps that criterion at 1 point. Material the question does not ask for earns nothing, however accurate it is.

Answer A

Gene flow is the process where animals from one area move to another area and bring their genes with them.

One of the main ways gene flow happens is through movement, for example when birds fly to a different place and then mate with the local birds.

Gene flow can make a population less unique, because when animals from different areas mix, the local traits get diluted. For example, if birds with colourful feathers mate with birds with dull feathers, you might end up with fewer colourful birds.

Drift is random, so it can be stopped by a few migrants, but selection is stronger, so more migrants are needed to stop it.

Answer B

Gene flow is the movement of alleles from one population into the gene pool of another. It happens only when the incoming individuals or gametes take part in reproduction: a migrant that arrives but does not breed adds nothing to the gene pool. Migration, the movement of individuals, is therefore not the same as gene flow. Sockeye salmon migrate thousands of kilometres at sea, yet because they return to breed in the stream where they hatched, there is little gene flow between river populations.

Gene flow happens by two main routes. Animals that disperse from where they were born and breed in another population carry their alleles with them. Plants do not move as adults, so their gene flow happens mainly through pollen, carried by wind or insects, and through seeds.

Gene flow can both help and hinder local adaptation. It helps by bringing back genetic variation that a small, isolated population has lost: when eight female pumas from Texas were released into the inbred Florida panther population in 1995, heterozygosity increased, signs of inbreeding depression became rarer and the population grew (genetic rescue). It hinders by bringing in alleles that suit other environments. Common bent grass on a small copper mine in Wales is copper tolerant, but non-tolerant pollen blows onto the mine, so seed produced there is less tolerant than the adults, and copper then kills the non-tolerant seedlings. Downwind, tolerant pollen blowing off the mine produces tolerant seedlings on normal soil, where tolerance is costly.

Drift and selection respond to gene flow differently. The strength of drift is about $1/(2N)$ per generation, so drift is weaker in large populations, while gene flow pulls a population back towards the others by the proportion m each generation. The two balance when Nm is about 1, so about one migrant per generation is enough to stop drift making populations very different, whatever their size. Selection, in contrast, changes allele frequencies by a proportion set by the selection coefficient s, so gene flow swamps local adaptation only when m is similar to or larger than s. Populations can therefore exchange enough migrants to stop divergence by drift and still stay locally adapted, as the grass on the copper mine shows.

Answer C

Gene flow is about how animals move to new places, like birds flying south for winter.

So, the mechanisms are migration and movement, because that's how animals and people move around.

Gene flow doesn't let animals adapt, because they are always moving, so they can't get used to one place. For example, if a penguin moves to a desert, it can't adapt because it's too hot and there's no water.

Drift happens in small populations and selection happens in big ones, so how many migrants you need depends on the size of the population.

Now write your own (15 minutes)

Answer the question below without notes, in about 15 minutes, as you would in the exam. Aim for about 250 words: three short paragraphs is enough for full marks. Plan for a couple of minutes first. Write in a document on your laptop. There is nothing to hand in.

> **Question.** Adaptive and neutral evolution are two fundamental modes of evolutionary change. (i) Explain the difference between them, giving two ways in which they differ. (ii) Give one example of each, and explain why it fits.

Rubric (10 points)

Criterion	2 points	1 point	0 points
1. Adaptive evolution	A change in allele frequencies caused by natural selection, consistent in direction, that produces adaptations	Vague (for example, "organisms adapt to their environment"), or the cause is missing	Incorrect or missing
2. Neutral evolution	A change in allele frequencies caused by genetic drift, in variants with little or no effect on fitness, random in direction	Partly right, or the cause is missing	Incorrect (for example, "neutral means no change") or missing
3. Where they differ	States at least two differences: in direction, in outcome, or in what controls them (the selection coefficient s versus drift, about $1/(2N_e)$)	One difference, or differences implied but never stated	None
4. Examples, explained	A specific example of each, in a heritable trait, each with an explanation of why it fits	Both examples named but not explained, or only one explained	No suitable examples
5. Structure	The first sentence answers the question; one point per paragraph, each with a topic sentence; correct terminology	Mostly clear, but loosely structured, repetitive or imprecise	Disorganised

A factual error in a criterion caps that criterion at 1 point.

When the time is up, swap laptops with a neighbour (or email them your answer) and mark each other's answer with the rubric. Give a mark for each criterion, and add one sentence on the change that would gain the most marks. If you would rather not share your answer, mark your own instead. After the session, compare both answers with the model answer posted on itslearning (Resources › Practice Exams). This practice does not count towards your final grade.

Takeaways

- Answer every part of the question, and let the verbs tell you what each part needs.
- Plan before you write.
- Answer the question in your first sentence.
- One main point per paragraph: a topic sentence, support, and an ending that adds something rather than repeating the topic sentence.
- Use precise terms, and back every general claim with a specific example.
- Marking other people's answers against a rubric is one of the quickest ways to improve your own.

Heritability from a linear regression

Background

Heritability measures how much of the variation in a trait **within a population** is due to additive genetic differences between individuals. It is a proportion, between 0 and 1, and it tells us how closely offspring resemble their parents. Heritability matters in plant and animal breeding, in medicine and in conservation, because it determines how fast a trait can respond to selection.

Narrow-sense heritability (Neal writes it h^2_N) is a ratio of two variances: the additive genetic variance divided by the total phenotypic variance.

$h^2 = \frac{V_A}{V_P}$

Only the additive part of the genetic variance counts, because parents pass on alleles, not whole genotypes. Effects that depend on particular combinations of alleles (dominance and interactions between genes) are broken up every generation, so they do not make offspring resemble their parents in a predictable way.

The rest of the phenotypic variance is mainly environmental. If we ignore non-additive genetic effects, as we do in this practical, $V_P = V_A + V_E$, where V_E is the environmental variance.

A heritability close to 1 means that most of the variation among individuals is additive genetic, so offspring closely resemble their parents. A heritability close to 0 means that parents' trait values tell us little about their offspring's: the variation is mostly due to the environment, to chance, or to non-additive genetic effects.

Heritability is not a fixed property of a trait. It belongs to a particular population in a particular environment, because it changes whenever either variance changes:

- **More environmental variation** (a patchier habitat, more variable weather or food) increases V_P and so lowers h^2, even if V_A is unchanged.
- **Selection** removes unfavourable alleles and so reduces V_A. A trait under strong, persistent selection can end up with low heritability. (Traits closely related to fitness also tend to have low heritability because their environmental variance is large; see the lecture.)
- **Genetic architecture** (how many genes affect the trait, and how large their effects are) differs between traits and species.

We can estimate heritability from the resemblance between parents and offspring. If we plot the offspring value in each family (one offspring, or the mean of several) against the **mid-parent value** (the mean of the two parents), the slope of the regression line estimates h^2. If you only know one parent, the slope estimates $h^2/2$, because each parent passes on only half of its alleles. Using the mean of several offspring per family gives a more precise estimate, but one offspring per family (as in this practical) also works. This is the method used in the lecture for East's tobacco plants and for Darwin's finches.

Reading: Neal, chapter *Polygenic Inheritance and Quantitative Genetics*.

Reference

Wray, N. & Visscher, P. (2008) Estimating trait heritability. Nature Education 1(1):29

Learning Outcomes:

- Estimate narrow-sense heritability, with its uncertainty, from a mid-parent–offspring regression in R.
- Use heritability and phenotypic variance to estimate additive genetic variance, and explain why heritability can differ between populations.
- Evaluate the assumptions behind regression estimates of heritability.

Before you start: predict

Imagine plotting offspring wing length against mid-parent wing length.

- What slope would you expect if offspring were, on average, exactly like their parents?
- What slope would you expect if offspring were no more like their own parents than like any other bird in the population?
- What does a slope between these two tell you?

Write down your answers, then check them against your results at the end.

Your Task

Estimating heritability (15 minutes)

You are provided with two datasets for wing length (mm) in a small songbird. Each row is one family: `Midparent_value` is the mean wing length of the two parents, and `Offspring_value` is the wing length of one of their offspring, measured as an adult. Each dataset contains 100 families.

Data set 1

Data set 2

1) Download the first data set and import it into R (RStudio). The starter code assumes you have saved it in a folder called `data` inside your project; change the path if you saved it somewhere else.
2) Plot the data, with mid-parent value on the x-axis and offspring value on the y-axis.
3) Fit a linear regression model of the form `y ~ x`.
4) Look at the slope of the relationship: this is your estimate of the heritability. Use `confint()` to find its 95% confidence interval.

Starter code in R:

```
pop_1 <- read.csv("data/heritability_pop1.csv")
plot(pop_1$Midparent_value, pop_1$Offspring_value,
     xlab = "Mid-parent wing length (mm)", ylab = "Offspring wing length (mm)")
mod1 <- lm(Offspring_value ~ Midparent_value, data = pop_1)
abline(mod1, col = "red")
summary(mod1)
coef(mod1)[2]      # slope = heritability estimate
confint(mod1)[2, ] # 95% confidence interval for the slope
```

Reading a confidence interval: the 95% confidence interval is the range of values for h^2 that are compatible with your data. The narrower it is, the more precise your estimate. If the interval includes zero, the data cannot rule out $h^2 = 0$, but that does not show that h^2 is zero: the upper limit tells you how large it could plausibly be.

Questions

- What is your estimate of h^2, and how precise is it?
- What does this heritability tell us about the variation in wing length in this population? What does it *not* tell us about any one bird?
- What else might explain the remaining variation?
- Calculate V_P as the variance of the offspring values (`var(pop_1$Offspring_value)`). Now calculate `var(pop_1$Midparent_value)` and compare the two. Why is the variance of the mid-parent values only about half of V_P, and so why should we not use it as V_P? (Hint: each mid-parent value is the mean of two birds.)
- Calculate the additive genetic variance, $V_A = h^2 \times V_P$.
- Imagine a breeder selects parents whose mean wing length is 2 mm above the population mean. Use the breeder's equation, $R = h^2 S$, to predict the change in mean wing length in the next generation.

Comparative Analysis (10 minutes)

Now analyse the second data set. It is for the same trait in the same species, but from a different population living in a different environment. Repeat steps 1 to 4, and calculate V_P and V_A for this population too. Copy this table into your notes and fill it in for both populations.

	Population 1	Population 2
h^2		
95% confidence interval		
V_P (mm^2)		
V_A (mm^2)		
$V_P - V_A$ (mm^2), an estimate of V_E		

Questions

- How does the heritability compare with population 1? Do the two confidence intervals overlap?
- Use your table to compare V_A and $V_P - V_A$ between the two populations. Which differs more, and what does that suggest about the cause of the difference in heritability?
- Can you suggest environmental differences between the two populations that could explain this?
- For the same selection differential ($S = 2$ mm), which population would respond faster to selection?
- Can you think of any real-world applications where understanding heritability would be important?

Assumptions (10 minutes)

Estimating heritability from a mid-parent–offspring regression assumes the following.

- **Additive genetic effects:** the slope estimates the additive genetic variance only. Dominance and interactions between genes do not make offspring resemble parents, and the method does not account for genotype-by-environment interactions.
- **No shared environment:** parents and offspring do not resemble each other for non-genetic reasons. If they share a territory, food supply or parental care, the heritability will be overestimated. Cross-fostering experiments (swapping eggs or chicks between nests) break this link.
- **Correct parentage:** the measured parents are the genetic parents. In many birds, some chicks are sired by a male other than the social father (extra-pair paternity), which makes the heritability look lower than it is.
- **Linearity:** the relationship between parent and offspring trait values is a straight line.
- **Measurement accuracy:** measurement error in the parents' values makes the slope shallower, so heritability is underestimated; error in the offspring values makes the estimate less precise.
- **Random mating:** mates do not choose each other on the basis of the trait (no assortative mating).
- **Representative sample:** the families measured are a random sample of the population. If the offspring measured are only those that survived a harsh winter, for example, selection on wing length can bias the estimate.

- **Independence:** each family contributes one data point. Using the same parents in several families, or several offspring from one family as separate data points, violates this.

Questions

- For **shared environment**, **correct parentage** and **measurement accuracy**: give a specific example of how each assumption could be violated in a wild bird population, and explain why that would make the heritability estimate too high or too low.
- **Optional extension:** do the same for the other five assumptions. Some have no clear direction, or cause a different problem (for example, a confidence interval that is too narrow), and one may surprise you.

Takeaways

- The slope of offspring on mid-parent estimates narrow-sense heritability, $h^2 = V_A/V_P$; report it with its uncertainty.
- Heritability belongs to a population in an environment: two populations with the same additive genetic variance can have very different heritabilities if their environments differ.
- High heritability does not mean a trait is fixed or unaffected by the environment.

Part 4: Interactions Between Species and Community Structure

This part explores how species interactions shape population and community dynamics, using competition and predator-prey models.

By the end of Part 4, you should be able to:

- Predict the outcome of competition from the isocline intercepts (K_1, K_1/α, K_2, K_2/β).
- Interpret competition and predator-prey dynamics from model outputs.
- Explain why Lotka-Volterra predator-prey cycles are neutrally stable in continuous time, and why the discrete-time version spirals outwards.
- Reason about stability and equilibrium in multi-species systems.

Reading: Neal, chapters *Interspecific Competition*, *Predator-Prey Interactions* and *Species Interactions and Community Structure*.

Notation used in Part 4 (following Neal)

- Competition: $\frac{dN_1}{dt} = r_1 N_1 \frac{K_1 - N_1 - \alpha N_2}{K_1}$ and $\frac{dN_2}{dt} = r_2 N_2 \frac{K_2 - N_2 - \beta N_1}{K_2}$. α is the per capita effect of species 2 on species 1, and β the effect of species 1 on species 2. The Excel worksheet calls these α_{12} and α_{21}.
- Predator-prey: $\frac{dH}{dt} = rH - aHP$ and $\frac{dP}{dt} = cHP - dP$, with isoclines $P = r/a$ (prey) and $H = d/c$ (predator). The Excel worksheet and the R chapters use V (prey), C (predator), R, a, f and q, where $V = H$, $C = P$, $R = r$, $af = c$ and $q = d$.

Lotka-Volterra competition

The Lotka-Volterra competition model is a foundational tool in ecology for understanding interspecific competition, or competition between different species for shared resources such as food, space, and nesting sites. It examines how population sizes change in a shared environment by considering factors like interaction strength and environmental carrying capacities.

The model is closely linked to the idea of "competitive exclusion": two species with identical ecological roles cannot coexist indefinitely, and eventually one will exclude the other. This principle comes mainly from experiments, especially Gause's work with yeasts and *Paramecium* in the 1930s, rather than from the model itself. In the basic model, two truly identical species do not exclude each other at all (see the fifth case below).

The model also explores the role of both inter- and intra-specific competition. It identifies conditions for species coexistence, influenced by parameters like competition coefficients and carrying capacities. These insights are vital for understanding community structure, species interactions, and ecological stability.

In a two-species system there are four general outcomes:

1. *Predictable Exclusion of Species 1 by Species 2:* Species 2 outcompetes and excludes species 1 from the environment.
2. *Predictable Exclusion of Species 2 by Species 1:* Species 1 outcompetes and excludes species 2 from the environment.
3. *Exclusion Dependent on Initial Densities:* Depending on the starting numbers of each species, either one could exclude the other.
4. *Stable Coexistence:* Both species may reach a stable equilibrium where they coexist without one excluding the other.

Neal also describes a fifth, knife-edge case, in which the two isoclines lie exactly on top of each other (see below).

The model suggests that stable coexistence is possible if each species limits its own population growth more than it affects the other species. Overlapping niches could lead to competitive exclusion, influenced by their competition coefficients and carrying capacities.

Learning outcomes:

- Competence in constructing an Excel-based population model.
- Understanding how the Lotka-Volterra competition model works.
- Understanding the competition coefficients α and β, and using the isocline intercepts (K_1, K_1/α, K_2, K_2/β) to predict the outcome of competition.

Background: the model and its isoclines

We use the notation of Neal (*Interspecific Competition*). Each species grows logistically, but its growth is also slowed by the other species:

$$\frac{dN_1}{dt} = r_1 N_1 \frac{K_1 - N_1 - \alpha N_2}{K_1}$$

$$\frac{dN_2}{dt} = r_2 N_2 \frac{K_2 - N_2 - \beta N_1}{K_2}$$

- α is the per capita effect of species 2 on species 1, measured relative to the effect of species 1 on itself. If $\alpha = 0.5$, two individuals of species 2 have the same effect on species 1 as one individual of species 1.
- β is the per capita effect of species 1 on species 2, measured in the same way.

Species 1 stops growing when $N_1 = K_1 - \alpha N_2$. This straight line is the *zero isocline* (zero net growth isocline) of species 1. On a graph with N_1 on the x-axis and N_2 on the y-axis it crosses the x-axis at K_1 (no species 2 present) and the y-axis at K_1/α (so many individuals of species 2 that species 1 cannot grow at all). In the same way, the isocline of species 2, $N_2 = K_2 - \beta N_1$, crosses the y-axis at K_2 and the x-axis at K_2/β. Each species increases when the populations are below (to the left of) its own isocline and decreases above it.

The outcome depends on how the two isoclines are arranged (Fig. @ref(fig:lv-isocline-cases)):

Arrangement of the intercepts	Outcome
$K_1 > K_2/\beta$ and $K_1/\alpha > K_2$ (species 1's isocline lies outside species 2's)	Species 1 excludes species 2
$K_2 > K_1/\alpha$ and $K_2/\beta > K_1$ (species 2's isocline lies outside species 1's)	Species 2 excludes species 1
$K_1 < K_2/\beta$ and $K_2 < K_1/\alpha$ (the isoclines cross)	Stable coexistence at the crossing point
$K_1 > K_2/\beta$ and $K_2 > K_1/\alpha$ (the isoclines cross the other way)	Unstable equilibrium: either species can win, depending on the starting densities

In the fifth case (Neal's "coexistence at a range of equilibrium densities"), $K_1 = K_2/\beta$ and $K_2 = K_1/\alpha$, so the two isoclines lie exactly on top of each other. Every point on the shared line is an equilibrium, and the species coexist at whatever ratio they happen to reach. Two identical species ($\alpha = \beta = 1$, $K_1 = K_2$) are an example. This case needs exactly matching values, so it is unlikely in nature. It also shows that the basic model does not, on its own, predict the exclusion of identical species: Gause observed exclusion in his *Paramecium* experiments, and Neal shows that an extra source of mortality (a "removal factor") has to be added to the model to explain it.

Key idea

Whether a coefficient is bigger or smaller than 1 does not decide the outcome on its own. What matters is how each competition coefficient compares with the ratio of the carrying capacities. The easiest way to see this is to ask whether each species can invade when it is rare and the other species is sitting at its carrying capacity.

- Species 1 can invade a population of species 2 at K_2 if $K_1 > \alpha K_2$, i.e. if $\alpha < K_1/K_2$ (equivalently $K_2 < K_1/\alpha$).
- Species 2 can invade a population of species 1 at K_1 if $K_2 > \beta K_1$, i.e. if $\beta < K_2/K_1$ (equivalently $K_1 < K_2/\beta$).

Species 1 can invade?	Species 2 can invade?	Outcome
Yes	No	Species 1 excludes species 2
No	Yes	Species 2 excludes species 1
Yes	Yes	Stable coexistence

Species 1 can invade?	Species 2 can invade?	Outcome
No	No	Exclusion, but which species wins depends on the starting densities

These are the same conditions as in the isocline table above, written in a different way.

For example, with $K_1 = 3000$, $\alpha = 1.25$, $K_2 = 1000$ and $\beta = 0.5$, species 2 has the stronger per capita effect ($\alpha > 1$), yet species 1 wins: species 1 can invade ($1.25 < 3000/1000 = 3$) but species 2 cannot ($0.5 > 1000/3000 = 0.33$).

Your task

In this class (over TWO sessions) you will be working from a PDF available here to make a model exploring this idea.

This PDF walks you through the creation of an Excel spreadsheet exploring interspecific competition and competitive exclusion using the Lotka-Volterra model.

Notation and time steps in the PDF

- The PDF writes the competition coefficients as α_{12} and α_{21}. In Neal's notation (used in this book), $\alpha_{12} = \alpha$ (effect of species 2 on species 1) and $\alpha_{21} = \beta$ (effect of species 1 on species 2).
- The spreadsheet is a **discrete-time** version of the model: $N_{1,t+1} = N_{1,t} + R_1 N_{1,t}(K_1 - N_{1,t} - \alpha N_{2,t})/K_1$, and the same for species 2. Its R is a rate per time step, not Neal's continuous r. Keep R_1 and R_2 at 1 or below, so that you see the Lotka-Volterra behaviour rather than the oscillations and chaos of the discrete logistic model (compare *Deeper into logistic growth*).

- Create the Excel spreadsheet by following the instructions in the PDF.

Note on the PDF: the labels added to Figure 5 have the subscripts the wrong way round. The correct isocline intercepts are K_1 and K_1/α_{12} for species 1, and K_2 and K_2/α_{21} for species 2 (in Neal's notation, K_1, K_1/α, K_2 and K_2/β; see the Background above).

- In the first session, build the sheet and check it: with the PDF's default values ($R_1 = R_2 = 1$, $K_1 = 1200$, $\alpha = 0.75$, $K_2 = 1000$, $\beta = 0.5$, starting at $N_1 = 100$, $N_2 = 50$) the two species should settle at $N_1 = 720$, $N_2 = 640$. Label the four isocline intercepts on your graph.
- In the second session, use the model to answer the questions below. For each parameter set you try, **sketch the isoclines and predict the outcome first**, then run the model to check.

The questions

1. What parameter values will cause species 1 to exclude species 2 from the habitat? What do these values mean in ecological terms?
2. What parameter values will reverse this outcome? What do these values mean in ecological terms?
3. What parameter values will allow the two species to coexist indefinitely and stably? What do these values mean in ecological terms?
4. Are there parameter values under which the outcome depends on initial population sizes or rates of population growth? What do these values mean in ecological terms?
5. (Optional) Neal describes two pairs of flour beetles. Predict the outcome for each from the isoclines, then check with your sheet: (a) *Rhizopertha* (species 1, $K_1 = 330$) and *Oryzaephilus* (species 2, $K_2 = 440$), with $\alpha = 0.235$ and $\beta = 0.12$; (b) *Tribolium confusum* (species 1, $K_1 = 220$) and *T. castaneum* (species 2, $K_2 = 340$), with $\alpha = 1$ and $\beta = 2.2$. For (b), try several different starting densities.

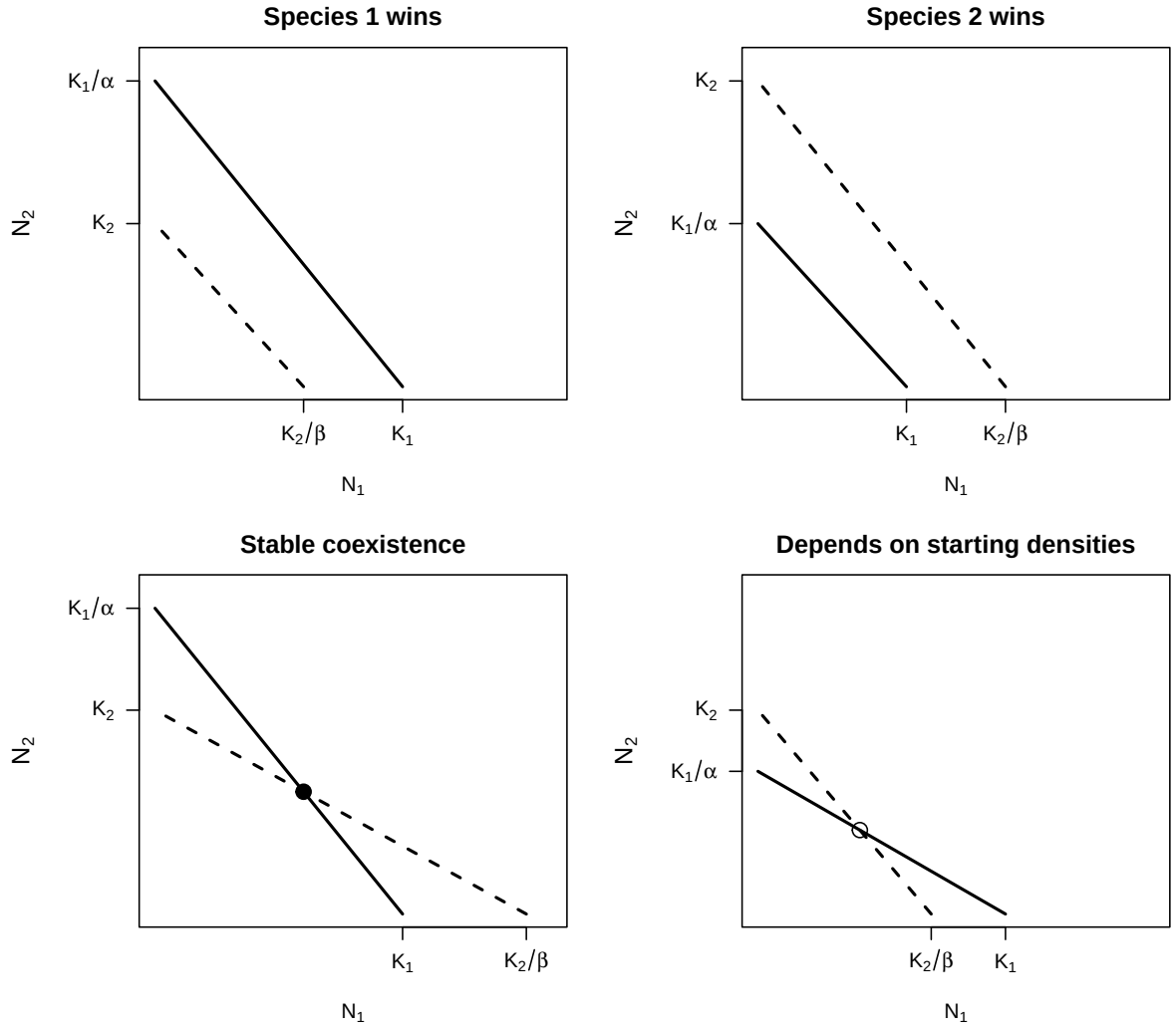


Figure 13: The four general arrangements of the Lotka-Volterra competition isoclines. Solid line: species 1 (intercepts K_1 and K_1/α). Dashed line: species 2 (intercepts K_2 and K_2/β). Filled circle: stable equilibrium; open circle: unstable equilibrium.

Takeaways

- Competition outcomes depend on relative self-limitation vs. interspecific effects.
- The competition coefficients α and β encode how much one species suppresses the other.
- Coexistence occurs when each species limits itself more than it limits the other, in the scaled sense that $\alpha < K_1/K_2$ and $\beta < K_2/K_1$ (each species can invade when rare). Comparing a coefficient with 1 is not enough when the carrying capacities differ.
- The four isocline intercepts (K_1, K_1/α, K_2, K_2/β) are all you need to predict the outcome.

Lotka-Volterra predator-prey dynamics

Background

The Lotka-Volterra predator-prey model (Lotka 1925; Volterra 1926) is a fundamental concept in ecological dynamics, used to study the dynamics between predator and prey populations. Rosenzweig and MacArthur (1963) later extended this basic model to incorporate prey carrying capacity and predator functional response. The model assumes that the predator's population growth is directly influenced by the availability of its prey, while the prey's population growth is affected by predation pressure. As predator numbers increase, the prey population declines, which, in turn, leads to a decrease in predator numbers due to reduced food availability. This cyclical pattern continues as predator and prey populations oscillate over time.

The Lotka-Volterra predator-prey model gives us a framework for understanding the dynamics of species interactions.

In this class you will build and explore a Lotka-Volterra predator-prey model in Excel to gain insight into the ecology of interacting predators and prey.

Learning outcomes:

- Competence in constructing an Excel-based population model.
- Understanding how the Lotka-Volterra predator-prey model works, and how it is visualised.

Key idea

The two isoclines (the lines where prey or predator numbers stop changing) are set by different parameters: the prey isocline by the prey's growth rate and the predator's attack rate, the predator isocline by the predator's death rate and how efficiently it turns prey into offspring. Changing a parameter moves an isocline, and so moves the equilibrium point where they cross. Whether the populations then cycle more or less widely depends on how far the starting numbers are from that new equilibrium.

Important: the spreadsheet is a discrete-time model

In the lecture, the Lotka-Volterra model is written in continuous time (dH/dt, dP/dt). In that model the populations cycle for ever: the cycles neither grow nor shrink (they are *neutrally stable*), and their size depends only on the starting numbers.

The spreadsheet instead updates the populations once per time step (N(t+1) = N(t) + change over one step), like the discrete logistic model in *Deeper into logistic growth*. Because each step uses the population sizes at the start of the step, the model always reacts a little too late, and the oscillations **grow** until one or both populations crash. This is a property of the discrete-time version (a time lag), not of predator-prey interactions in general.

Keep this in mind when you answer the questions below, especially question 6. You can compare the two versions in the R chapters Continuous-time Lotka-Volterra predator-prey model in R and Discrete-time Lotka-Volterra predator-prey model in R.

Your task

Follow the PDF worksheet (here), which guides you to build and explore a Lotka-Volterra predator-prey model in Excel. The model has parameters for the prey and for the predator, and you will explore how these parameters influence the dynamics of the populations.

The worksheet is divided into four parts. We will mainly **focus on Part 1** - the basic model - in the class.

Symbols: worksheet vs Neal

The worksheet uses different symbols from Neal (*Predator-Prey Interactions*), whose continuous model is $dH/dt = rH - aHP$ and $dP/dt = cHP - dP$. The worksheet's equations are $V_{t+1} = V_t + RV_t - aC_tV_t$ and $C_{t+1} = C_t + afV_tC_t - qC_t$.

Worksheet	Neal	Meaning
V ("victim")	H	Prey population
C ("consumer")	P	Predator population
R	r	Prey growth rate (in the worksheet, per time step)
a	a	Attack rate
af (f = conversion efficiency)	c	Rate at which prey eaten become new predators
q	d	Predator death (starvation) rate

So the prey isocline $C = R/a$ is Neal's $P = r/a$, and the predator isocline $V = q/(af)$ is Neal's $H = d/c$.

Correct formula for step 7 of the PDF

The predator formula printed in the PDF is missing a minus sign. In cell C15 use:

- English Excel: `=IF(C14+$H$8*$H$9*B14*C14-$H$7*C14>0,C14+$H$8*$H$9*B14*C14-$H$7*C14,0)`
- Danish Excel: `=HVIS(C14+$H$8*$H$9*B14*C14-$H$7*C14>0;C14+$H$8*$H$9*B14*C14-$H$7*C14;0)`

The prey formula in cell B15 (step 6) in Danish Excel is `=HVIS(B14+$F$7*B14-$H$9*C14*B14>0;B14+$F$7*B14-$H$9*C14`
and `MAX` is `MAKS`. Your first rows should read 1050.000, 19.600 (time 1) and 1106.700, 19.286 (time 2).

The sheet runs for 100 time steps. With the default values neither species goes extinct within that time, although the cycles clearly grow. To see the end of the story, copy the formulas further down (to row 314 gives 300 time steps): the prey hit zero at time step 220.

After completing the Excel sheet, try to answer the following questions:

1. In the spreadsheet model, does a larger prey population growth rate (R) increase or decrease the stability of the predator-prey interaction?
2. What happens if the predators starve more quickly? Less quickly?
3. What happens if the predator is more efficient at converting prey into offspring? Less efficient?
4. What happens if the predator is better at finding prey? Worse?
5. Is the behaviour of the model sensitive to starting populations? Begin with populations near the point where the isoclines cross, and move slowly farther out.
6. What is the ultimate outcome of the predator-prey interaction in the spreadsheet, regardless of parameter values? Is this outcome caused by the biology or by the model being discrete-time (see the box above)? How does this compare to real predator and prey populations? What factors not included in the model may explain the differences between model predictions and reality?

Optional extras

Part 2 modifies the basic model to include a refuge for the prey, part 3 adds a carrying capacity for the prey, and part 4 adds carrying capacities for both prey and predator. Feel free to continue to work on

these if there is time, and if you are interested. Part 3 is particularly worth trying: prey self-limitation turns the expanding oscillations into damped ones that settle at an equilibrium.

R-version

Two R versions are available: a continuous-time version (the model in the lecture) and a discrete-time version (the same model as this spreadsheet).

References

Lotka, A. J. 1925. *Elements of Physical Biology.* Williams & Wilkins.

Volterra, V. 1926. Variazioni e fluttuazioni del numero d'individui in specie animali conviventi. *Memorie della Reale Accademia Nazionale dei Lincei* 2: 31–113.

Rosenzweig, M. L. and R. H. MacArthur. 1963. Graphical representation and stability conditions of predator-prey interactions. *American Naturalist* 97: 209–223.

Takeaways

- Predator-prey feedbacks can generate oscillations without external forcing.
- Parameter changes move the isoclines and the equilibrium; in the continuous model the size of the cycles is set by the starting numbers.
- The spreadsheet's growing oscillations come from its discrete time steps; the continuous model gives neutrally stable cycles.
- Simple models highlight mechanisms but omit real-world constraints.

Continuous-time Lotka-Volterra Predator-Prey Model in R

Background

The Lotka-Volterra model describes the interaction between predators and prey. This document demonstrates how to program the continuous-time version of the model in R. Then, by varying the parameters you can explore its features, and visualise the dynamics of predator and prey populations over time.

Learning outcomes:

- Use of R to investigate an ecological model.
- Understanding how the Lotka-Volterra predator-prey model works, and how it is visualised.

What you should see

With the default parameters, you should see:

- repeating oscillations in predator and prey abundances through time, with the predator peaks following the prey peaks
- a closed orbit in the phase plane around the coexistence equilibrium (the same loop is traced again and again)
- straight-line isoclines (ZNGIs) whose intersection is the coexistence equilibrium

This is the same model as the Excel spreadsheet and the discrete-time R chapter, with the same parameter values, but solved in continuous time. Comparing the three shows which behaviour belongs to the biology and which to the time steps.

Your task

In R/RStudio, you will create an R script that programs the continuous time version of the Lotka-Volterra model. You will then use this model to investigate how the model works and how varying the parameters affects the system.

Step 0: Create a new script

- Open RStudio and create a new script (File > New File > R Script).
- Save it with a useful name (File > Save).
- Type or copy/paste the following code, step-by-step, into the Script.
- Run the code, or parts of it, selecting the parts you want to run and clicking “Run”.

Step 1: Load Necessary Libraries

We will use the `deSolve` package to solve the system of differential equations numerically. Note that you may need to install the package first using the command `install.packages("deSolve")`.

```
library(deSolve)
```

Step 2: Define the Model

The system of equations is defined as follows:

- Victim (prey) population growth: $\frac{dV}{dt} = RV - aVC$
- Consumer (predator) population growth: $\frac{dC}{dt} = afVC - qC$

The function below implements these equations in R.

```
predator_prey_model <- function(t, state, parameters) {
  with(as.list(c(state, parameters)), {
    dV <- R * V - a * V * C  # Victim (prey) growth equation
    dC <- a * f * V * C - q * C  # Consumer (predator) growth equation

    list(c(dV, dC))
  })
}
```

Step 3: Set Parameters and Initial Conditions

We set the parameters and initial conditions for the model. I use the same nomenclature as the PDF from the Excel exercise, and the same values. The parameters are as follows: R is the victim population growth rate, a is the attack rate of the predator, f is the conversion efficiency of the predator and q is the predator starvation (death) rate. Here R is an instantaneous rate (Neal's r), because the model is in continuous time.

The Neal textbook (chapter *Predator-Prey Interactions*) writes the same model as $dH/dt = rH - aHP$ and $dP/dt = cHP - dP$. The translation is: $V = H$ (prey), $C = P$ (predators), $R = r$, $a = a$, $a \cdot f = c$ and $q = d$.

```
parameters <- c(R = 0.25, # Victim (prey) growth rate
                a = 0.01, # Attack rate
                f = 0.008, # Conversion efficiency
                q = 0.1)  # Predator death rate

initial_state <- c(V = 1000,  # Initial victim population
                   C = 20)   # Initial consumer population
```

Step 4: Simulate the Model

We simulate the model over a specified time period and store the results.

```
time <- seq(0, 200, by = 0.1)  # Time sequence
output <- ode(y = initial_state, times = time, func = predator_prey_model, parms = parameters)
output_df <- as.data.frame(output)
```

Step 5: Visualise the Dynamics

How does the size of the two populations change through time? You should see that the **amplitude** and **period** change as the parameters change.

- **Amplitude**: Half the difference between a population's peak and trough. It shows how much the population fluctuates.
- **Period**: The time it takes for a population to complete one full cycle of oscillation. i.e. the peak-to-peak distance. It indicates the speed of the population cycles.

Population Dynamics Over Time

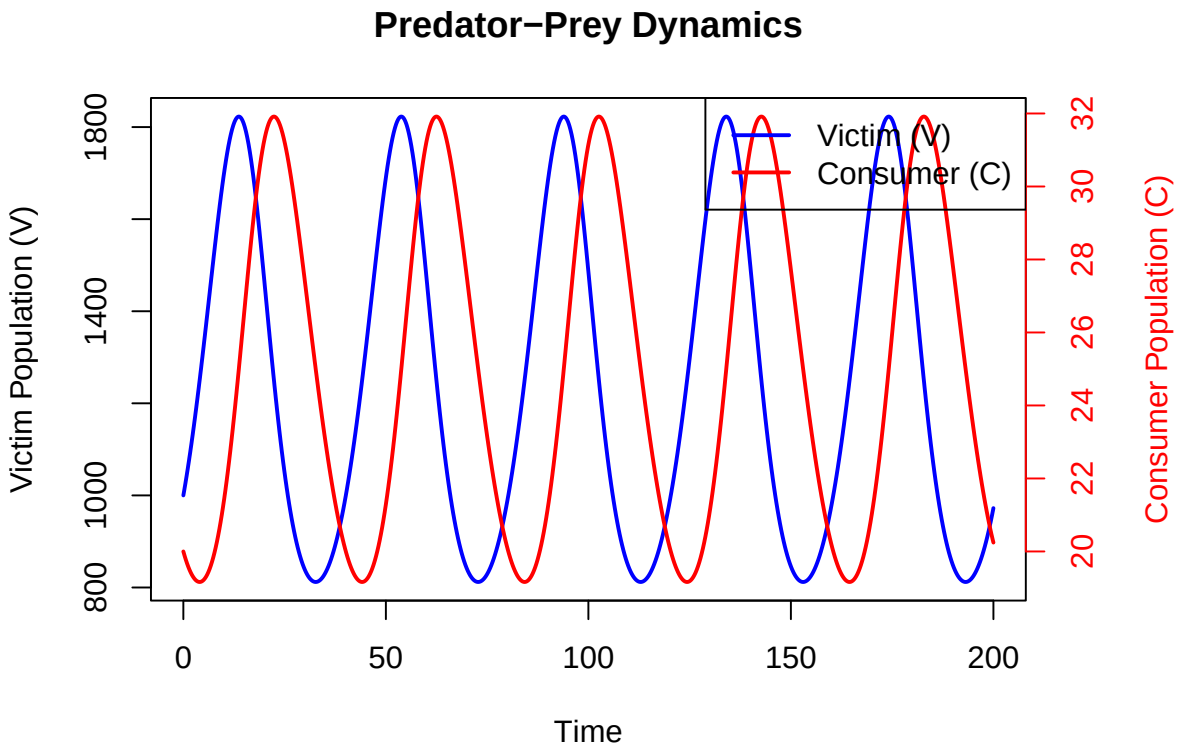
The plot below shows the changes in consumer and victim populations over time.

```
# Make room in the right-hand margin for the second axis label
par(mar = c(5, 4, 4, 5))

# Plot victim population on the primary y-axis
plot(output_df$time, output_df$V, type = "l", col = "blue", lwd = 2,
     ylab = "Victim Population (V)", xlab = "Time", main = "Predator-Prey Dynamics")

# Add consumer population on a secondary y-axis
par(new = TRUE)
plot(output_df$time, output_df$C, type = "l", col = "red", lwd = 2,
     axes = FALSE, xlab = "", ylab = "")
axis(side = 4, col = "red", col.axis = "red")
mtext("Consumer Population (C)", side = 4, line = 3, col = "red")

# Add legend
legend("topright", legend = c("Victim (V)", "Consumer (C)"),
       col = c("blue", "red"), lwd = 2)
```



Phase Plot with ZNGIs

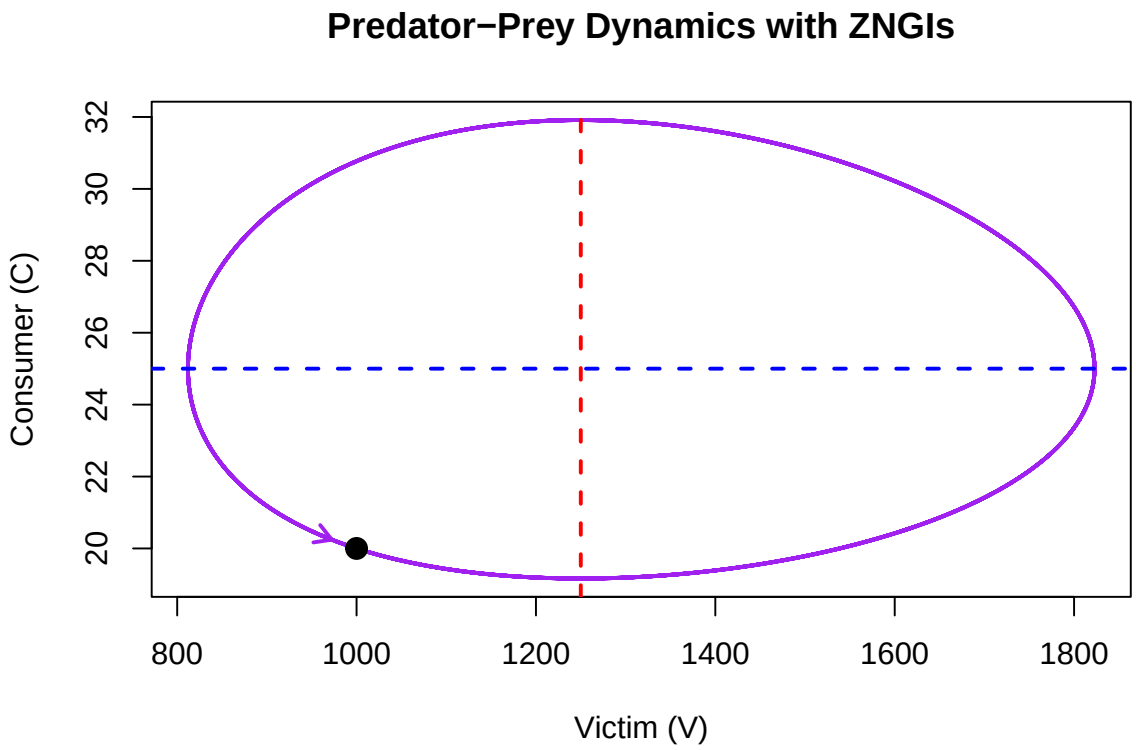
The phase plot shows the relationship between victim and consumer populations. Zero Net Growth Isoclines (ZNGIs) indicate where the growth rate of each population is zero.

```
plot(output_df$V, output_df$C, type = "l", col = "purple", lwd = 2,
     xlab = "Victim (V)", ylab = "Consumer (C)",
     main = "Predator-Prey Dynamics with ZNGIs")

# Consumer ZNGI: dC/dt = 0 when V = q / (a * f) (a vertical line)
abline(v = parameters["q"] / (parameters["a"] * parameters["f"]), col = "red",
       lwd = 2, lty = 2)
# Victim ZNGI: dV/dt = 0 when C = R / a (a horizontal line)
abline(h = parameters["R"] / parameters["a"], col = "blue", lwd = 2, lty = 2)

# Add point for initial population sizes
points(output_df$V[1], output_df$C[1], pch = 19, col = "black", cex = 1.5)

# Add a direction arrow only if the final step has visible movement.
n <- nrow(output_df)
step_inches <- c(
  diff(grconvertX(output_df$V[c(n - 1, n)], from = "user", to = "inches")),
  diff(grconvertY(output_df$C[c(n - 1, n)], from = "user", to = "inches"))
)
if (all(is.finite(step_inches)) && sqrt(sum(step_inches^2)) > 1e-3) {
  arrows(x0 = output_df$V[n - 1], y0 = output_df$C[n - 1],
         x1 = output_df$V[n], y1 = output_df$C[n],
         col = "purple", length = 0.1, lwd = 2)
}
```



Questions

Before you change anything, write down a prediction for each question (a sentence or a quick sketch), then run the code to check it.

1. Oscillations

- Run the simulation with the default parameters. Describe how the two populations change through time. Which peaks first, prey or predators, and why?
- In the phase plot, which way does the orbit go round (clockwise or anticlockwise)? Explain the direction using the four regions marked out by the isoclines.

2. Neutral stability and starting numbers

- *Predict:* what will happen if you start exactly at the equilibrium, $V = 1250$, $C = 25$? And just next to it, at $V = 1300$, $C = 25$?
- Now try starting values further away, e.g. $V = 2000, C = 20$ and $V = 1000, C = 40$. What happens to the *amplitude* and the *period*? Do the oscillations grow, shrink or stay the same over time?

3. Equilibrium and time-averages

- Calculate the equilibrium for the default parameters, using

$$V^* = \frac{q}{a \cdot f}, \quad C^* = \frac{R}{a}.$$

- The populations never settle at these values. Instead, compare them with the *average* population sizes over the simulation: `mean(output_df$V)` and `mean(output_df$C)`. What do you find?

4. Moving the isoclines

- *Predict* which isocline (prey or predator) moves, and in which direction, if you double each of R, a, f and q in turn. Check your predictions on the phase plot.
- A pesticide kills both prey and predators: it lowers R and raises q. Try $R = 0.2$ and $q = 0.12$. What happens to the *average* numbers of prey and predators? Is this what you would have guessed? (This is Volterra's principle; see Neal's *Predator-Prey Interactions*.)

5. Extinction and reality

- Start far from the equilibrium, at $V = 1000$, $C = 100$. What are the smallest prey and predator numbers reached? Can either population reach exactly zero in this model? What would happen to a real population at these numbers?
- Which assumptions of the model are unrealistic? Which real-world factors might stop the cycles continuing for ever?

Conclusion

This document demonstrates how to implement and explore the Lotka-Volterra predator-prey model in continuous time using R. By varying parameters and initial conditions, you can investigate how predator-prey dynamics are influenced by ecological factors. Notice that, unlike the discrete-time model (which has an implicit time lag of one time step and gives oscillations of increasing amplitude that end in extinction), the continuous-time model gives *neutrally stable* cycles: they neither grow nor shrink, and their size depends on the starting populations. To see the discrete-time version, take a look at the Excel exercise (here) or the R exercise here.

The main findings of the continuous time model are as follows:

1. **Oscillatory Dynamics**: Predator and prey populations exhibit cyclical oscillations. The prey population increases when predator numbers are low, while the predator population grows when prey is abundant, so predator peaks follow prey peaks.
2. **Neutral Stability**: The oscillations are neutrally stable. They persist indefinitely without damping or amplification, and each pair of starting numbers gives its own orbit: the further from equilibrium you start, the larger the cycles. In the model neither population can ever reach exactly zero, but large orbits can take a population down to a fraction of one individual (for example, starting at $V = 1000$, $C = 100$ the predators fall to about 2 and the prey to about 8; with $q = 1$ the predators fall to about 0.01). In a real population that would mean extinction.
3. **Equilibrium Points**: There are two equilibria:
    - The trivial equilibrium, where both predator and prey populations are zero.
    - The coexistence equilibrium, $V^* = q/(af)$ and $C^* = R/a$. The populations cycle around it rather than settling at it, but the average population sizes over a cycle equal the equilibrium values.
4. **Parameter Dependence**: The parameters set the position of the isoclines, and so the equilibrium: R and a move the prey isocline, q, a and f move the predator isocline. The *amplitude* of the cycles is set by how far the starting numbers are from the equilibrium. For small cycles the *period* is approximately $2\pi/\sqrt{Rq}$ (about 40 time units with the default parameters); large cycles take longer.
5. **Phase Plane Behaviour**: Predator and prey populations follow closed orbits in the phase plane around the coexistence equilibrium, anticlockwise when prey are on the x-axis, under the assumption that the system remains undisturbed.

Takeaways

- Continuous-time predator-prey models generate sustained, neutrally stable cycles: their size is set by the starting numbers, not by the parameters alone.

- Equilibria and isoclines help explain why cycles form and how parameters shift them; the average population sizes over a cycle equal the equilibrium.
- Real ecosystems add damping, delays, and noise that change these dynamics.

Discrete-time Lotka-Volterra Predator-Prey Model in R

Learning outcomes:

- Implement and explore a discrete-time predator-prey model in R.
- Understand how time steps introduce delays and affect stability.
- Compare discrete-time dynamics to continuous-time dynamics.

Key idea

With $\Delta t = 1$, the phase plane typically shows expanding spirals, reflecting instability introduced by discrete time steps.

Introduction

The Lotka-Volterra model describes the interaction between predators and prey. This document demonstrates how to program the discrete-time version of the model in R, explore its features, and visualise the dynamics of predator and prey populations over time. The time step (Δt) can be adjusted to explore the impact of a time delay in the system. In ecological systems, time delays can represent biological processes such as the time it takes for predators to respond to changes in prey density, for prey populations to recover after predation. If Δt is set to 1, this model is equivalent to a year-by-year model (as we have fitted in Excel in class), where the impact of predators on prey has a 1 year lag. i.e. the growth of predators, depends on the availability of prey 1 year ago and the growth of prey depends on the amount of predators 1 year ago.

There are two ways to think about Δt. With $\Delta t = 1$ this is a model of populations that really do change in steps (e.g. one breeding season per year). With Δt very small, nothing biological has changed: we are simply calculating the continuous-time model more accurately, in many small steps. The parameters and the starting numbers are the same as in the Excel spreadsheet and the continuous-time R chapter.

Step 1: Define the Model

The discrete-time equations are as follows:

- Victim (prey) population growth:
  $V_{t+1} = V_t + \Delta t \cdot (R \cdot V_t - a \cdot C_t \cdot V_t)$
- Consumer (predator) population growth:
  $C_{t+1} = C_t + \Delta t \cdot (a \cdot f \cdot V_t \cdot C_t - q \cdot C_t)$

Note that, when $\Delta t = 1$, the model is as follows:

- Victim (prey) population growth:
  $V_{t+1} = V_t + R \cdot V_t - a \cdot C_t \cdot V_t$
- Consumer (predator) population growth:
  $C_{t+1} = C_t + a \cdot f \cdot V_t \cdot C_t - q \cdot C_t$

Compare these to Equations 1 and 2 in the Excel exercise PDF.

The function below implements these equations in R.

```
discrete_predator_prey_model <- function(time_steps, dt, initial_state, parameters) {
  # Initialise vectors to store populations
  time_steps <- round(time_steps)
  V <- numeric(time_steps + 1)
  C <- numeric(time_steps + 1)
  time <- seq(0, time_steps * dt, by = dt)

  # Set initial populations
  V[1] <- initial_state["V"]
  C[1] <- initial_state["C"]

  # Extract parameters
  R <- parameters["R"]
  a <- parameters["a"]
  f <- parameters["f"]
  q <- parameters["q"]

  # Loop through time steps to calculate populations
  for (t in 1:time_steps) {
    # max(0, ...) stops populations going negative (like IF() in Excel)
    V[t + 1] <- max(0, V[t] + dt * (R * V[t] - a * V[t] * C[t]))
    C[t + 1] <- max(0, C[t] + dt * (a * f * V[t] * C[t] - q * C[t]))
  }

  # Combine results into a data frame
  data.frame(time = time, V = V, C = C)
}
```

Step 2: Set Parameters and Initial Conditions

We set the parameters, initial conditions, the total time to simulate (`total_time`) and the time step (Δt). The number of steps is calculated from these, so that changing Δt always covers the same total time and the runs can be compared fairly.

```
parameters <- c(R = 0.25,  # Victim (prey) growth rate
                a = 0.01,  # Attack rate
                f = 0.008, # Conversion efficiency
                q = 0.1)   # Consumer (predator) death rate

initial_state <- c(V = 1000,  # Initial victim population
                   C = 20)    # Initial consumer population

total_time <- 300             # Total time to simulate
dt <- 1                       # Time step (dt), 1 = a standard discrete model
time_steps <- total_time / dt # Number of time steps needed to cover total_time
```

Step 3: Simulate the Model

We simulate the model for the specified parameters and time steps.

```
output_df <- discrete_predator_prey_model(time_steps = time_steps, dt = dt,
                                          initial_state = initial_state,
                                          parameters = parameters)
```

Step 4: Visualise the Dynamics

Population Dynamics Over Time

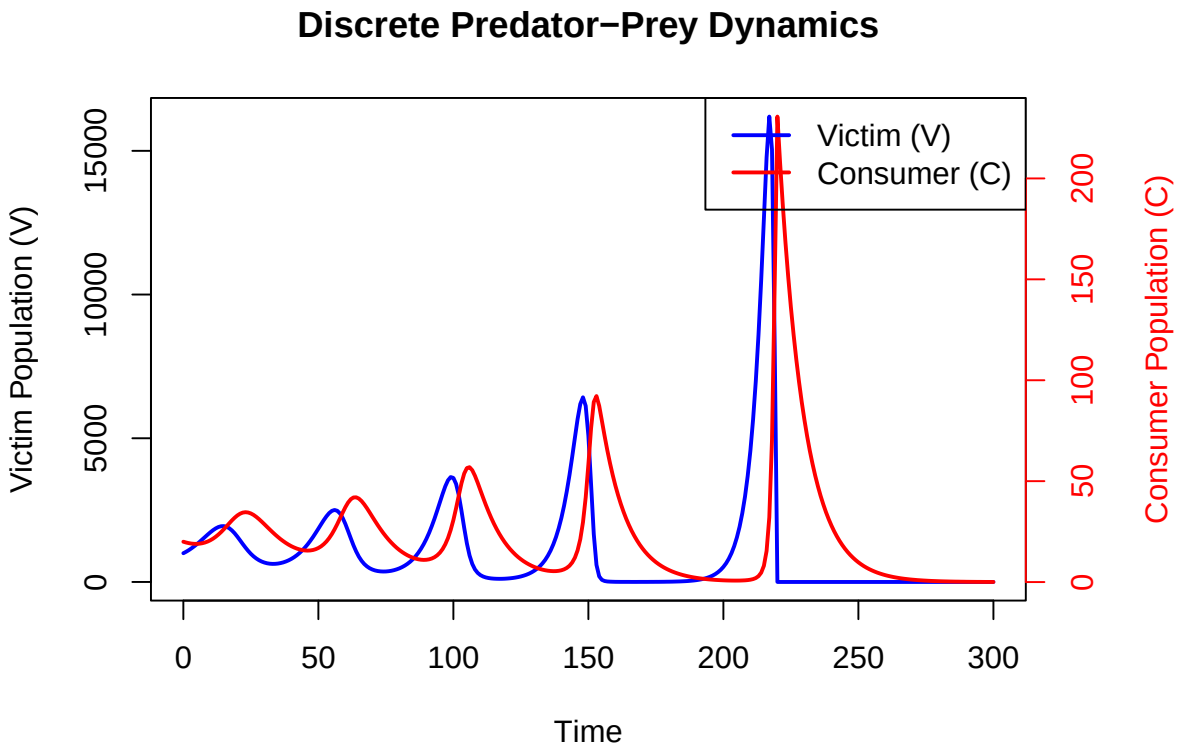
The plot below shows the changes in consumer and victim populations over time.

```
# Make room in the right-hand margin for the second axis label
par(mar = c(5, 4, 4, 5))

# Plot victim population on the primary y-axis
plot(output_df$time, output_df$V, type = "l", col = "blue", lwd = 2,
     ylab = "Victim Population (V)", xlab = "Time", main = "Discrete Predator-Prey Dynamics")

# Add consumer population on a secondary y-axis
par(new = TRUE)
plot(output_df$time, output_df$C, type = "l", col = "red", lwd = 2,
     axes = FALSE, xlab = "", ylab = "")
axis(side = 4, col = "red", col.axis = "red")
mtext("Consumer Population (C)", side = 4, line = 3, col = "red")

# Add legend
legend("topright", legend = c("Victim (V)", "Consumer (C)"),
       col = c("blue", "red"), lwd = 2)
```



Phase Plot with ZNGIs

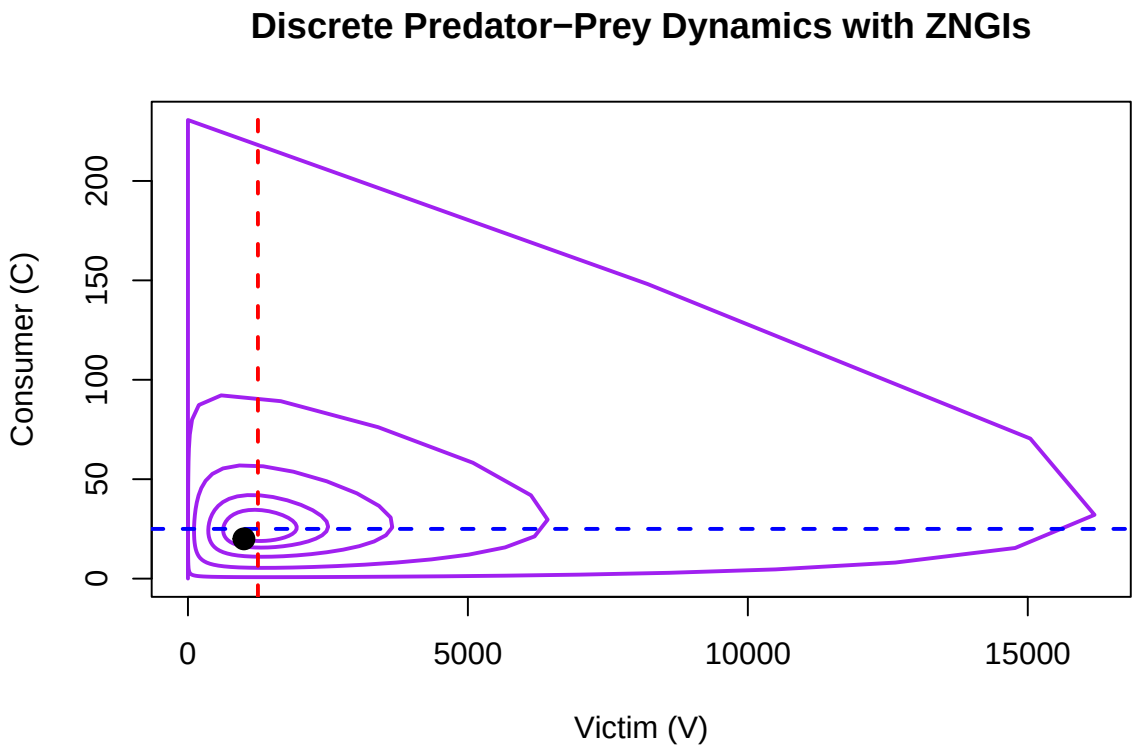
The phase plot shows the relationship between victim and consumer populations.

```
### Phase Plot with ZNGIs, Initial Point, and Arrow
plot(output_df$V,
     output_df$C,
     type = "l",
     col = "purple",
     lwd = 2,
     xlab = "Victim (V)", ylab = "Consumer (C)",
     main = "Discrete Predator-Prey Dynamics with ZNGIs")

# Add Consumer ZNGI: no change in C when V = q / (a * f) (a vertical line)
abline(v = parameters["q"] / (parameters["a"] * parameters["f"]),
       col = "red", lwd = 2, lty = 2)

# Add Victim ZNGI: no change in V when C = R / a (a horizontal line)
abline(h = parameters["R"] / parameters["a"], col = "blue", lwd = 2, lty = 2)

# Add point for initial population sizes
points(output_df$V[1], output_df$C[1], pch = 19, col = "black", cex = 1.5)
```




```r
# Add a direction arrow only if the final step has visible movement.
n <- nrow(output_df)
step_inches <- c(
  diff(grconvertX(output_df$V[c(n - 1, n)], from = "user", to = "inches")),
  diff(grconvertY(output_df$C[c(n - 1, n)], from = "user", to = "inches"))
)
if (all(is.finite(step_inches)) && sqrt(sum(step_inches^2)) > 1e-3) {
  arrows(x0 = output_df$V[n - 1], y0 = output_df$C[n - 1],
         x1 = output_df$V[n], y1 = output_df$C[n],
         col = "purple", length = 0.1, lwd = 2)
}
```

Your task

Before each run, predict what you will see (sketch the phase plot), then check.

- Run the model with the default settings ($\Delta t = 1$, `total_time = 300`). How does the phase plot differ from the closed orbit in the continuous-time chapter? When does the first population hit zero? Find it with `output_df$time[which(output_df$C == 0 | output_df$V == 0)[1]]`.
- Decrease Δt to 0.1 (keep `total_time = 300`; the number of steps is recalculated for you, so re-run the parameter chunk as well as the simulation) and compare the phase plane behaviour to $\Delta t = 1$. Then try $\Delta t = 0.01$.
- Increase Δt to 2 and observe whether extinction happens faster.
- Change R or q (e.g. double one of them) with $\Delta t = 1$. Does the spiral grow faster or slower? Now double a or f instead. Which parameters control how quickly the spiral grows? (Careful: changing a or f also moves the equilibrium, so the default start is then nearer to or further from it. Compare how fast the spiral widens, not only when it ends.)

Conclusion

This document demonstrates how to implement and explore the discrete-time Lotka-Volterra predator-prey model. By adjusting the time step (Δt) and other parameters, you can investigate how predator-prey dynamics are influenced by ecological factors. With the default settings ($\Delta t = 1$), the time lag of one step makes the phase plot spiral outwards: the cycles grow until the prey are wiped out (they hit zero at $t = 220$), after which the predators starve and dwindle towards zero. Smaller time steps make the spiral grow more slowly, and with $\Delta t = 0.01$ the orbit is almost closed, like the continuous-time model. Larger time steps make it grow faster: with $\Delta t = 2$ the prey hit zero at $t = 142$. Contrast this with the perpetual (neutrally stable) oscillations you get with the continuous-time model.

Takeaways

- Discrete time steps destabilise the predator-prey cycles of the continuous-time model: the populations react one step too late, so each cycle overshoots a little more than the last.
- Each time step, the distance from the equilibrium grows by a factor of about $\sqrt{1 + \Delta t^2 Rq}$. This depends on the prey growth rate, the predator death rate and the step length, but not on the attack rate or conversion efficiency. With the default values it is about 1.012 per step (roughly 1 % per step).
- Smaller Δt values bring the discrete model closer to the continuous-time behaviour.

Part 5: Animal behaviour, altruism and sexual selection

This part uses game theory to explore how selection shapes behavioural strategies such as aggression and restraint. Sexual selection and mating systems are covered in the lectures and the reading; there is no practical for them.

By the end of Part 5, you should be able to:

- Explain how payoffs influence behavioural strategies.
- Use game-theoretic reasoning to interpret conflicts and cooperation.
- Connect behavioural models to evolutionary outcomes.

Reading: Neal, chapters *Animal Behaviour, Altruism and Limiting Aggression* and *Sexual Selection and Mating Systems*.

Game theory: Hawks and doves

Background

Game theory analyses interactions in which the payoff of a choice depends on what others choose. Maynard Smith and Price (1973) brought it into biology as *evolutionary* game theory, which does not assume that animals are rational. Instead, each individual follows a strategy, strategies are heritable, and payoffs are measured in fitness, so strategies that earn more spread through the population. One particularly fascinating example is the Hawk and Dove game. Representing two distinct strategies, the Hawk embodies aggression and assertiveness, always seeking to escalate conflicts and dominate opponents. On the other hand, the Dove adopts a more peaceful and conciliatory approach, aiming to avoid confrontations and achieve cooperative outcomes.

By studying the Hawk and Dove strategies we can gain insights into understanding the delicate balance between competition and cooperation, shedding light on the dynamics of conflicts and the evolution of cooperative behaviour in both the natural world and human societies.

The simple model unravels the intricacies of strategic decision-making, with profound implications across various fields, from biology and social sciences to economics and politics.

In this activity, you play the role of the strategies: you will play the hawk and dove game with a single partner and then with multiple partners to observe the hypothetical costs and benefits of antagonistic or cooperative behaviour, respectively. The whole class enters results into one shared, live spreadsheet as you play, so we can watch the class-wide patterns emerge on screen in real time, round by round, instead of waiting until after class to see what happened.

Learning outcomes:

- Understanding of the fundamental concepts of game theory and its application in analysing strategic interactions in competitive environments.
- Ability to apply game theory concepts to real-world contexts in biology, or politics.
- Appreciation of how simple models can be used to address complex biological or political decision making.
- Practice generating a testable prediction from a simple model *before* collecting data, then checking it against real, live class results.

Worked example

Inputs

- Benefit: $B = 4$
- Cost: $C = 3$

Steps

1. Compute Hawk-Hawk payoff: $(B - C)/2$.
2. Compute Hawk-Dove payoff to Hawk: B (and 0 to Dove).
3. Compute Dove-Dove payoff: $B/2$ to each player.

Output and interpretation

Hawk-Hawk gives 0.5 each, Hawk-Dove gives `4` to Hawk and `0` to Dove, and Dove-Dove gives `2` each.

Link to the lecture: when does a mix of Hawks and Doves evolve?

The class game uses the lecture's payoff matrix with the display cost set to zero ($D = 0$). This is also the matrix in Neal's chapter "Animal Behaviour, Altruism and Limiting Aggression", where Dove-Dove pays $B/2 - D$.

- If $B > C$ (as today, $B = 4$, $C = 3$), Hawk earns more than Dove whatever the opponent plays, so in a one-off encounter the evolutionarily stable strategy (ESS) is pure Hawk.
- If $C > B$, neither pure strategy is stable. The ESS is a mix, with Hawks at frequency $p^* = B/C$, where Hawks and Doves earn the same on average. For example, with $B = 4$ and $C = 8$, $p^* = 0.5$. (With a display cost, as in the lecture's worked example, $p^* = (B/2 + D)/(C/2 + D)$, which gives 7/12.)

Your task

Before the activity starts, review the cost-benefit payoff table below to understand how to score the game.

The payoff table

The general payoff table looks like this (payoff to **Player 1**):

Player 1	Player 2	Hawk	Dove
Hawk		$(B - C)/2$	B
Dove		0	$B/2$

You can determine the benefit by looking at the left column and choosing the row corresponding to the card you played. The box that corresponds to your partner's choice is in the top row. For this game, B equals four (4), and C equals three (3). So depending on the cards played, player one (i.e., you) would gain a score of two (2), four (4), zero (0), or one-half (0.5). Like this:

Player 1	Player 2	Hawk	Dove
Hawk		0.5	4
Dove		0	2

What does theory predict?

Before you play, work through this with your partner.

Because B (4) is bigger than C (3), Hawk beats Dove *no matter what your opponent plays*: against a Dove you get 4 (vs. 2 for playing Dove yourself), and against a Hawk you still get 0.5 (vs. 0 for playing Dove). So in a single, anonymous encounter, Hawk is the dominant strategy: theory predicts everyone should end up playing Hawk.

But notice that mutual Hawk-Hawk (0.5 each) is *worse for both of you* than mutual Dove-Dove (2 each). That means this game has the same structure as the famous Prisoner's Dilemma, with Hawk playing the role of "Defect". One twist: if you and your partner take turns (Hawk/Dove one round, Dove/Hawk the next), you each average $(4 + 0)/2 = 2$ per round, exactly as much as mutual Dove-Dove. So with these payoffs, taking turns is just as good a way to "cooperate".

- **Prediction for Game Two** (a new partner every round, i.e. effectively a series of one-off encounters): theory says Hawk should dominate and pay off best.
- **Prediction for Game One** (the *same* partner, 15 times in a row): think about what happens if you play Hawk against your partner. Might they retaliate next round? If you get stuck trading Hawk-Hawk for the rest of the game, is that better or worse than settling into mutual Dove-Dove? Could you and your partner do as well as mutual Dove in another way? Write down what you think will happen to the *frequency of Hawk play* and to *mean payoff per round* as the 15 rounds go on, before you start.
- **Prediction for round 15:** you both know the game ends after round 15, so there is no next round in which to be punished. What do you expect people to play in the last round?

Keep your predictions: we'll come back to them once the live results are in.

Hypotheses

Building on the predictions above, the experimental hypothesis for this exercise is that, compared with changing partners (Game Two), playing a repeated partner (Game One) leads to:

1. a *lower frequency of Hawk play* (overall, and declining over the 15 rounds), and
2. a *higher mean payoff per player per round*, because pairs avoid costly Hawk-Hawk encounters.

The null hypothesis is that neither the frequency of Hawk play nor the mean payoff per player differs between the repeated-partner and changing-partner games.

Note what we are *not* testing: whether the more hawkish player in a pair earns more. Within any single round Hawk always earns more than Dove ($0.5 > 0$ against a Hawk, $4 > 2$ against a Dove), so the more hawkish player in a pair always ends up with at least as high a total. What repeated play can change is how your partner responds, and so how much *both* of you earn.

Instructor setup (before class)

(Read this section if you are running the session; students can skip ahead to "How today's data collection works".)

To let the whole class see results build up live, set up one shared spreadsheet before class instead of the old per-round Google Form:

1. Duplicate the "Hawk-Dove Live Results" spreadsheet template (keep one copy per year; reuse and re-share it each time you teach the course). It needs two data tabs and one dashboard tab:
   - **Game1_Data**: one row per pair, pre-filled with `Pair ID` = 1, 2, 3, … (enough rows for `number of students / 2`). For each of the 15 rounds, two entry columns (`R1_P1`, `R1_P2`, `R2_P1`, `R2_P2`, … using `H`/`D`), plus a formula-computed benefit column per round and a `Total_P1_benefit` / `Total_P2_benefit` column, so students never have to do the arithmetic themselves. Keep `B` and `C` in two named cells that the benefit formulas refer to, so you can change the payoffs in future years without editing every formula.
   - **Game2_Data**: one row per student, pre-filled with `Player ID` = 1, 2, 3, … up to the class size. For each of the 15 rounds, two entry columns (`R1_self`, `R1_opponent`, …), the same formula-computed per-round and total benefit columns.
   - **Live_Dashboard**: charts wired to the data tabs: a line chart of "% Hawk played, by round" for each game, and a scatter chart of "mean hawkishness vs. total benefit" per pair/player for each game (these are the same two charts reproduced from archived data in the Results of the hawk-dove games chapter after class).
2. Set sharing so students can edit without individually logging in (e.g. "anyone with the link can edit"), and share the link at the start of class (projected, or via itsLearning).

3. Assign Pair IDs (Game 1) and Player IDs (Game 2) before class (e.g. by seat number or an attendance list) rather than letting students choose their own names or team labels. This is what lets everyone's data line up automatically and avoids the messy name-matching problems free-text team names caused in past years.
4. Project the **Live_Dashboard** tab throughout the activity.
5. After class, keep the spreadsheet (or an exported copy) as this year's archived dataset for the Results chapter.

How today's data collection works

Instead of filling in a new form after every round, the whole class shares **one live spreadsheet**, projected on the screen as you play:

- You'll be given a **Pair ID** (Game 1) or a **Player ID** (Game 2): use it to find your row. Don't invent your own label; the ID is what lets the spreadsheet combine everyone's results automatically.
- You only ever type into your own row.
- Enter cards as single letters: `H` for Hawk, `D` for Dove. The spreadsheet calculates the benefit for you from the payoff table, so there's no arithmetic to do mid-game.
- Watch the projected screen: two live charts update automatically as the class enters data, so you can see the round-by-round pattern emerge as you play (see "Watching the results live" below).

GAME ONE: repeated partner

1. Find your assigned partner and note your shared **Pair ID**.
2. Collect a set of two game cards (Hawk and Dove), one each.
3. Agree who is recorded as "Player 1" and who is "Player 2" for your pair (it doesn't affect scoring).
4. Conceal your cards. When the instructor gives the signal ("reveal!"), simultaneously show either card to your partner.
5. Agree on which card each of you showed, then enter both (`H/D`) into your pair's row for that round (one phone or laptop per pair is enough).
6. Repeat for 15 rounds, waiting for the instructor's signal each time. Try to strategise how to get the best benefit from every round, but don't communicate your strategy with your partner beyond the cards themselves.
7. At the end, check your row: the spreadsheet shows each player's total benefit across all 15 rounds. Divide by 15 to get your mean payoff per round, and compare it with the class average and, later, with Game Two. (Do not just ask who "won" within your pair: the more hawkish player always does, for the reason given under Hypotheses.)

No wifi? Jot your results on paper first (round, your card, partner's card) and enter them all at once at the end; the live chart will simply catch up when you do.

GAME TWO: changing partners every round

To swap partners quickly with a large class, use a **rotating circle**: half the class forms an inner circle facing outward, the other half forms an outer circle facing inward, so everyone starts facing one partner.

1. Find your assigned **Player ID** and starting position in the circle.
2. Play a round exactly as in Game One: conceal your card, reveal on the instructor's signal, agree on both cards played.
3. Enter your own card and your opponent's card (`H/D`) into your row for that round.
4. When the instructor calls "rotate", everyone in the **outer circle** shifts one position to their left. You now face a new partner for the next round.
5. Repeat for 15 rounds. Because you rotate through the circle, you meet a new person every round without any time lost searching for a free partner.

A double circle gives each person (class size / 2) different partners, so 15 distinct partners needs **at least 30 students**. With a smaller class, either play only (class size / 2) rounds, or carry on and note that some partners will be met twice (and discuss whether that matters).

Watching the results live

As you play, two charts update automatically on the projected screen:

- **% playing Hawk, by round**: lets you see whether hawkish play increases, decreases, or stays flat as the rounds go by.
- **Total benefit vs. mean hawkishness**, one point per pair/player: lets you see, once the game finishes, whether being more hawkish or more dovish paid off better.

We'll pause after each game to look at these together before moving on to the next part.

Summarise and discuss results

Before looking at the final numbers, revisit the predictions you wrote down earlier.

- Did the % Hawk trend in Game One and Game Two match what you predicted?
- Did the total-benefit-vs-hawkishness pattern support the one-shot theory prediction (Hawk pays off best), or the repeated-game reciprocity argument (Dove pays off best when you keep meeting the same partner)?
- Which game, if either, better matches what you'd expect from a species whose individuals encounter each other only once versus a species that lives in small, stable social groups?
- Did you or your partner take turns (Hawk/Dove, then Dove/Hawk)? What happened in round 15?
- Discuss as a class what would likely happen if you increased or decreased the relative value of the benefit (B) compared with the cost (C): would that change which strategy dominates in a one-off encounter? (Hint: see the box above on $p^* = B/C$.)

The full analysis of the actual class data, including how the live charts you just watched connect to the underlying theory, is worked through in detail in the Results of the hawk-dove games chapter.

Acknowledgement

This exercise is taken from https://www.jove.com/science-education/10611/group-behaviour

Takeaways

- With B > C, a single anonymous encounter is won by playing Hawk, but repeated encounters with the same partner change the incentives, because mutual Hawk-Hawk is worse for both players than mutual Dove-Dove.
- Collecting and displaying data live, as a class, makes it possible to test a theoretical prediction in real time rather than waiting until after the session.
- Game outcomes depend on the structure of the interaction (one-off vs. repeated), not just on the payoffs themselves.
- When the cost of fighting exceeds the value of the resource (C > B), the ESS is a mix of Hawks and Doves, with Hawk frequency p* = B/C.

Optional: running a live dashboard in R

If you're comfortable with R, you can draw the "% playing Hawk, by round" chart yourself from the shared spreadsheet using the `googlesheets4` package, instead of (or alongside) the spreadsheet's own charts. This is optional: the built-in spreadsheet charts are enough to run the session. The code assumes the column names in the instructor setup above (`R1_P1`, `R1_P2`, … for Game 1).

```
library(googlesheets4)
library(dplyr)
library(tidyr)
library(ggplot2)

# A publicly-editable sheet can be read without signing in:
gs4_deauth()
sheet_url <- "PASTE_THIS_YEAR'S_LIVE_SHEET_URL_HERE"

# Re-run from here every 10-15 seconds during class to refresh the chart.
game1 <- read_sheet(sheet_url, sheet = "Game1_Data")

# One row per pair, round and player, with the card played (H or D)
game1_long <- game1 %>%
  select(matches("^R[0-9]+_P[12]$")) %>%
  pivot_longer(everything(),
               names_to = c("round", "player"),
               names_pattern = "R([0-9]+)_(P[12])",
               values_to = "card") %>%
  filter(card %in% c("H", "D")) %>%
  mutate(round = as.numeric(round))

hawk_by_round <- game1_long %>%
  group_by(round) %>%
  summarise(percent_hawk = 100 * mean(card == "H"))

ggplot(hawk_by_round, aes(x = round, y = percent_hawk)) +
  geom_point() +
  geom_line() +
  ylim(0, 100) +
  xlab("Round") +
  ylab("% playing Hawk")
```

Part 6: Solutions/answers

This part provides model answers and worked solutions. Use it to check your reasoning after attempting the tasks.

By the end of Part 6, you should be able to:

- Compare your work to model solutions.
- Diagnose mistakes in reasoning or calculation.
- Reflect on alternative interpretations when results differ.

Solutions and “take-home” messages

In this section you will find the solutions and/or main take home messages of the practical exercises used in this course.

How to use this section

Use these solutions to check your reasoning after you attempt the tasks. If your answer differs, focus on the logic rather than the final number.

Solutions: The blind watchmaker

- *Predictions and the default run*

With selection on, one example run using Dawkins' 28-character phrase and 27 possible characters reached the target at generation 66. The number varies between runs: in 5,000 simulated runs the median was 98 generations, and 90% of runs finished between about 60 and 166 generations. Compare this with single-step selection, where you would expect to need about $27^{28} \approx 10^{40}$ attempts.

- *What happens if selection is turned OFF?*

Each generation is then a completely new random guess (single-step selection), and nothing is kept from one generation to the next. The output stays random nonsense. The chance of hitting the whole phrase in any one generation is $(1/27)^{28} \approx 8 \times 10^{-41}$, so in practice it never happens.

- *Does the number of generations affect whether the target phrase is reached?*

Yes. With 100 generations, only about half of runs (about 52%) reach this phrase; with 200 generations almost all do (about 99%). The last letters to be fixed are the slow part: each wrong letter has only a 1 in 27 chance of being right in any one generation.

- *Does the speed of reaching the target phrase depend on the length of the phrase?*

Yes, but much less than you might expect. Longer phrases take more generations, because there are more letters that all have to be fixed, but the increase is small: “I LOVE SCIENCE” (14 characters) takes a median of about 81 generations, the 28-character weasel phrase about 98, and the 53-character “WITH YOUR FEET ON THE AIR AND YOUR HEAD ON THE GROUND” about 117. With single-step selection, every extra character makes success 27 times less likely.

- *How does this simulation differ from real natural selection?*

1. Natural selection has no long-term goal or target. Here, every phrase is judged by how close it is to a fixed target that is decided in advance. In nature, individuals are “judged” only by how well they survive and reproduce in their current environment.

2. The intermediate phrases have no use of their own: "METHINKS IT IS LIKE A WXASEL" is only better because it is closer to the target. In evolution, every intermediate form (for example, each stage in the evolution of an eye) must itself work well enough to survive and reproduce.
3. The simulation judges each letter separately, with perfect knowledge of the right answer, and a correct letter can never be lost again. There is no population, no reproduction and no inheritance. Real selection acts on whole organisms in a population, and favourable variants can still be lost by chance.

The important lesson still holds: random variation combined with cumulative selection can produce outcomes that would be impossible by chance alone.

Solutions: Bug hunt camouflage

- *Predictions*

Typical good predictions: the average colour will move towards the background colour; it will take longer to catch 10 bugs as time goes on; after switching background, catching will suddenly become easy again, and the bugs will then start to evolve towards the new background. Students who predicted that individual bugs would change colour have met a key misconception (see the genotype question below).

- *What happens to the average colour of the bug population with time as you hunt? Was your prediction correct?*

The colour evolves to become more similar to the background colour because you (the hunter) find it harder to find these better-camouflaged individuals.

- *What happens to your hunting efficiency? Use your measurements.*

Your hunting efficiency tends to decrease with time because the bugs are evolving to be harder to see. They are evolving camouflaged colours. In your measurements, the time to catch 10 bugs should increase over the first two minutes, drop sharply when you switch background (the bugs are suddenly conspicuous), and then increase again as the bugs adapt to the new background. The exact times differ between students, because each of you is a different predator, but the pattern should usually be the same.

- *Apart from bug evolution, what else could change how long it takes you to catch 10 bugs? Which of the plots gives more direct evidence that the bugs themselves have evolved?*

You, the predator, also change: you learn where to look and get faster (catch times fall), you get tired or bored (catch times rise), and a new background can be distracting at first, or make you concentrate harder. So catch time mixes changes in the bugs with changes in you, and a student whose times did not rise has not shown that evolution failed. The direct evidence that the bugs evolved is in the colour plots (the histograms of the colour values and the change in average genotype): the distribution of colours in the population shifts towards the background.

- *Would you say that the bug population becomes worse or better adapted to their environment?*

The population becomes better-adapted to their environment with time.

- *Can you explain how this happens?*

You (the hunter) kill the most obvious bugs first (i.e. those with contrasting colours to the background). These are the individuals that are not well-adapted to the environment. The survivors have offspring that are similar to them, while the ones you kill leave no offspring. So as time goes on, the population becomes dominated by individuals that are well-camouflaged.

- *After simulating in one environment (e.g. poppy field) for a few minutes, pause then switch to another environment. Are the bugs now well- or poorly-adapted to their environment?*

Changing the environment means that the individuals are now poorly-adapted to their environment. This is because the individuals now find themselves in an environment that they did not evolve in. Their evolved camouflage does not work so well in a new environment.

- *Do the genotypes of individuals change (e.g. with individual age)?*

No, individuals' genotypes are fixed: with "show-genotype?" switched on, each bug keeps the same three numbers for its whole life. The change in the population occurs because of selection of individuals to reproduce. Better-adapted individuals are more likely to reproduce.

- *Increasing the "max-mutation-step" makes bug offspring less like their parents. How do you think this will influence the speed of adaptation of the bugs?*

Increasing the amount of variation in offspring will tend to speed up adaptation. However, there are limits: too much variation can cause maladaptation of offspring. For example, if a parent is well-adapted (perfectly camouflaged), too much variation in offspring could lead to some offspring being poorly-camouflaged and therefore vulnerable to predation.

- *Where do you see each of the four conditions in the model? Where does new variation come from?*

1. **Individuals vary in a trait:** the bugs differ in colour.
2. **Individuals vary in reproductive success:** bugs that you catch leave no more offspring; survivors may be chosen to reproduce.
3. **The trait affects reproductive success:** colours that contrast with the background are caught first, so they are less likely to survive and reproduce.
4. **The trait is inherited:** offspring get their colour genes from their parent.

New variation comes from mutation: each offspring's colour genes can differ slightly from its parent's (the "max-mutation-step" slider). Mutation is not one of the four conditions; it supplies the variation that condition 1 needs.

- *Newborn bugs start very small. Could this let some bugs escape for reasons that have nothing to do with their colour? How might that affect the results?*

Yes. A newborn bug is hard to see because it is small, whatever its colour, so for a short time it escapes for a reason that has nothing to do with camouflage. This adds chance to who survives and weakens selection on colour slightly. It does not stop camouflage evolving, because every bug is small only briefly and colour still matters once it has grown.

- *How does this model differ from a real predator–prey system? Give at least two differences.*

Possible answers: there is only one predator, and it does not evolve (although you do learn!); reproduction is asexual, with one parent and no recombination; the population size is fixed, so there is no population growth or decline; bugs do not move or choose where to rest; the only selection is predation on colour, whereas real prey face many other pressures (temperature, food, mates); generations overlap in an unusual way, because one offspring is born each time a bug is caught.

- *Optional extension: gene flow and local adaptation*

With a small "offspring-distance", offspring stay close to their parents, so bugs in each patch of background are mostly the descendants of bugs that survived in that patch: each patch can evolve its own matching colour (local adaptation). With a large "offspring-distance", offspring are scattered across the whole background. This is like frequent gene flow between populations: well-camouflaged colours from one patch keep arriving in patches where they stand out, so local adaptation is weaker or absent. Whether gene flow swamps local adaptation depends on how much movement there is compared with the strength of selection.

Solutions: Geometric growth

You can find the completed Excel file here.

Take home messages:

- The rate of exponential/geometric growth depends on the population growth rate.
- Lambda (λ) values < 1 lead to a exponential *decrease*, while values > 1 lead to exponential growth. A λ of 1 indicates no change.
- Exponential growth is unlimited. This is unrealistic because very soon competition (what Darwin called "The struggle for existence") will soon take effect as the population runs out of space or resources. Nevertheless, the model is useful and realistic for some limited situations (e.g. early population growth for new invaders)
- Growth trajectories can be plotted on the *natural scale* or on a *log scale*. Exponential growth (or decline) plotted on a log scale is a straight line. This could be useful to infer biological processes (or the lack of them). For example, exponential growth in population size indicates that there are no forces currently limiting the population (e.g. predators/competition/management actions).
- The slope of the population growth on the natural log scale can tell you what λ is. The "opposite" of *ln* is *exp*, so you can ask Excel to calculate `=EXP(slope)`.

Answers to the task:

- **Steps 1-2:** with $N_0 = 10$ and $\lambda = 1.1$, the population reaches 16.1 at $t = 5$ and 67.3 at $t = 20$. Plotted against time, N curves upwards ever more steeply.
- **Steps 3-4:** plotting ln(N) against time turns the curve into a straight line. A constant *proportional* increase each time step becomes a constant *absolute* increase on the log scale.
- **Step 5:** with $\lambda = 1.2$ the curve rises faster (steeper straight line on the log plot); with $\lambda = 0.8$ the population declines towards zero (straight line sloping downwards on the log plot); with $\lambda = 1$ the population stays at 10 (flat line). $\lambda = 1$ is the boundary between growth and decline: the population exactly replaces itself each time step ($R_m = 0$).
- **Step 6:** the trendline slope is $0.0953 = \ln(1.1)$, and `=EXP(0.0953)` returns 1.1, the λ you used. The slopes for $\lambda = 0.8$, 1 and 1.2 are -0.223, 0 and 0.182. If you get a different slope, check that you fitted the trendline to the `logN` (natural log) column, not to N on a logarithmic axis.
- **Step 7 (doubling time):** with $\lambda = 1.1$ the population passes 20 between $t = 7$ ($N = 19.5$) and $t = 8$ ($N = 21.4$), and passes 40 about 7 time steps later. The doubling time is $t_d = \ln 2 / \ln 1.1 = 0.693/0.0953 = 7.27$ time steps. It does not depend on the starting population size: a geometrically growing population doubles at a constant interval, whatever its size.
- **Geometric series:** $N_5 = 10 \times 1.1^5 = 16.1$, the same as the spreadsheet. After 900 generations, $N_{900} = 10 \times 1.1^{900} \approx 1.8 \times 10^{38}$. That is absurdly large (far more individuals than there are atoms in a human body, around 10^{27}), which shows why unlimited geometric growth cannot continue for long in reality.

Answers to the questions:

- **Constant growth rate:** the population is multiplied by the same factor each time step, so it grows by the same *proportion* each step and by ever-larger *numbers*. For example, with $\lambda = 1.1$ the population grows by 10% each step: from 10 to 11 in the first step, but from 61.2 to 67.3 between steps 19 and 20. It also doubles at a constant interval: $\ln 2 / \ln 1.1 \approx 7.3$ time steps.
- **Why unrealistic:** it assumes density independence (unlimited resources and space; competition, predation or disease do not increase as the population grows), constant birth and death rates and so a constant λ every year (no environmental variation), no time lags, no age or stage structure (all individuals are identical), and a closed population (no immigration or emigration). Real populations eventually run out of resources and growth slows.
- **Where it might be suitable:** the early stages of growth when a population is small relative to its resources, e.g. an invasive species newly arrived in a suitable habitat, a population recovering after a crash or after protection (e.g. some seal or raptor populations), or bacteria in fresh culture medium.

- **Decline of 25% per year:** this means $\lambda = 0.75$, so $N_4 = 800 \times 0.75^4 = 253.1$, i.e. about 253 individuals.
- λ **per half-year:** $r_m = \ln 1.1 = 0.0953$ per year, so $r_m = 0.0477$ per half-year, and $\lambda = e^{0.0477} = 1.049$ per half-year. Check: $1.049^2 = 1.1$. (Simply halving the 10% increase to 5% would be slightly too much, because growth compounds.)

Solutions: Estimating Population Growth Rate

You can find the completed Excel file here.

1. What does the log-transformed plot of population size over time tell you about the population's growth trend? Does the population grow exponentially?

- The log-transformed plot should show a straight line if the population is growing exponentially, because the natural logarithm of an exponentially growing population will linearise the exponential curve. The slope of this line represents the natural log of the population growth rate $log(\lambda)$. If the line has a positive slope, it indicates the population is growing; if it's flat, the population is stable; and if it's negative, the population is shrinking.
- In this exercise, assuming $\lambda > 1$, the population is indeed growing exponentially, which will be reflected by a positively sloped line in the log-transformed plot.

2. What is the estimated population growth rate (λ) based on your regression analysis?

- From the linear regression performed on the log-transformed data, the slope of the line corresponds to $log(\lambda)$. To find λ, exponentiate the slope: $\lambda = e^{\text{slope}}$.
- With the data provided, the slope of the regression of ln(N) on year is about 0.070 (0.0696), so $\lambda = e^{0.0696} \approx 1.07$. The population grows by about 7% per year. (Small differences in the third decimal place are fine; they come from rounding in the trendline label.)
- In fact, the data were simulated with an average λ of 1.09, with random year-to-year variation in λ and measurement error in the counts. Our estimate is lower than 1.09 for two reasons: year-to-year variation in λ lowers long-run growth (growth follows the geometric mean of λ, which is smaller than the arithmetic mean), and with only 25 years of noisy counts the estimate has considerable sampling error. This is a useful reminder that an estimate from real data is never exactly the "true" value.

3. What assumptions does this model make? Discuss any potential real-world factors that might affect the accuracy of your estimate for λ.

- The model assumes a constant growth rate over time. It does not account for environmental changes, resource limitations, or carrying capacity (which would lead to logistic rather than exponential growth).
- The model also assumes no immigration, emigration, or catastrophic events that might disrupt the population trend.
- **Environmental variability**: In reality, populations experience fluctuations due to weather, food availability, disease, and other stochastic factors, which can cause growth rates to change over time.
- **Density dependence**: As populations grow larger, they may face resource limitations or increased competition, leading to a slowdown in growth, which would not be captured by a simple geometric model.
- **Measurement errors**: In practice, the measurement of population sizes may not be precise, as discussed earlier with the introduction of noise. These errors can lead to inaccuracies in estimating λ.

Solutions: Stochastic population growth

You can find the completed Excel file here.

Take home messages:

- Population growth rates are unlikely to be constant through time.
- Environmental stochasticity will cause changes through time in vital rates (birth/death) which will influence λ (and r_m).
- We can model environmental stochasticity using a distribution of population growth rates instead of a single deterministic value.
- We then can randomly draw growth rate values from this distribution for each time-step in a population projection.
- The "spread" or uncertainty in the population size at a future time point depends on (i) the spread (variance) of the distribution and (ii) the amount of time elapsed.
- We can repeatedly simulate a population projection (e.g. in R, or more laboriously in Excel) to get a feel for the likely fate of a population. E.g. "what percent of the projections show extinction within 50 years?".
- We can use this relatively simple and "unrealistic" model to nevertheless get an intuition about general trends in population dynamics and extinction risk.
- Extinction probability increases as (i) initial population size decreases; (ii) variance increases; (iii) growth rates decrease; (iv) the time frame increases.
- Long-run growth follows the mean of r_m (the geometric mean of λ), not the arithmetic mean of λ. With mean $r_m = 0.05$ and variance 0.1, the typical (median) population grows by $e^{0.05} = 1.051$ per year, but the average λ is $e^{0.05+0.1/2} = 1.105$. In the default R run, after 50 generations the median population size is about 116 (theory: $10 \times e^{0.05 \times 49} = 116$), while the mean is about 1,900, pulled up by a few runs that boom.

Things to try

Your numbers will differ from these, because each run draws new random numbers. The values below are from the example run shown in the chapter, followed in brackets by the range seen over 100 further runs of 1,000 trials each. Differences of a few percentage points between runs are just chance.

1. **Default settings** (mean 0.05, variance 0.1, `startPop` = 10, 50 generations): extinction probability **0.058** (range 0.04 to 0.08). Most trajectories grow, but they spread out enormously: after 50 generations some populations are near 1 and others are in the tens or hundreds of thousands.
2. **`var.r` = 0.5**: extinction probability rises to **0.418** (range 0.39 to 0.47), about seven times higher, even though the mean growth rate is unchanged. The trajectories also spread out much more.
3. **`startPop` = 3**: extinction probability rises to **0.235** (range 0.21 to 0.28). Starting size matters a lot: a small population needs only a short run of bad years to fall below 1 individual, whereas a larger one has a buffer.
4. **`mean.r` = 0**: extinction probability is **0.249** over 50 generations (range 0.24 to 0.30). So extinction is not certain within 50 generations. But it is certain in the long run: with a mean of zero the log population size is a random walk with no upward drift, and such a walk eventually hits any lower boundary. You can see this by increasing `nGen`: with `nGen = 200` the extinction probability is **0.567** (range 0.54 to 0.61), and it keeps rising towards 1 as time increases.
5. **Challenge:** keep `mean.r = 0.05` and increase `var.r` to **1**: extinction probability is **0.576** (range 0.55 to 0.62), so more than half of the populations go extinct even though they grow on average (and the median population still grows in the long run). Other combinations work too (e.g. `mean.r = 0.02` with `var.r = 0.5` gives about 0.5). This tells us that a positive average growth rate does not guarantee persistence: with enough year-to-year variation, chance runs of bad years drive many populations below the extinction threshold before the positive trend can carry them to safety. The variance matters as much as the mean.

If you compare these results with the rule from the lecture (extinction is almost certain if $\sigma^2 > 2\bar{r}$), remember that the rule uses $\bar{r}$ on the λ scale and describes the very long run. In this simulation r_m is

drawn on the log scale, so the long-run growth rate is $\bar{r}_m$ itself, and certain long-run extinction needs $\bar{r}_m \leq 0$ (see the box in the chapter).

Questions

- **Deterministic vs. stochastic:** a deterministic model always gives the same trajectory for the same parameters; a stochastic model draws the growth rate at random each time step, so each run gives a different trajectory and the model predicts a *distribution* of possible outcomes (including a probability of extinction) rather than a single number.
- **Why more realistic:** real growth rates vary from year to year with weather, food supply, disease and so on, and small populations are also affected by chance births and deaths. A stochastic model captures this variation, and so shows the uncertainty in forecasts and the risk of extinction, which a deterministic model cannot.
- **Two populations with identical growth rates:** run the R code (or the Excel sheet) twice with the same settings, or look at any two of the trajectories in the plot. With mean 0.05 and variance 0.1, one population may grow into the thousands while another, with exactly the same parameters, falls below 1 and goes extinct. The implication is that the fate of a single real population is partly a matter of luck, so we should describe it with probabilities, and that two populations that differ in outcome need not differ in their underlying biology.
- **Extinction risk and population size:** small populations are at much higher risk (in the default runs, about 0.06 for 10 individuals against about 0.24 for 3), because a short run of bad years can take them below the threshold.
- **Extinction risk and environmental variation:** more variable environments (higher variance in r_m) increase extinction risk, even if the average growth rate does not change (about 0.06 at variance 0.1 against about 0.42 at variance 0.5). This is why increasing environmental variability, for example more frequent extreme weather, is a conservation concern.

Solutions: Basic logistic population growth

You were asked to use the Excel sheet to explore the logistic growth model by changing its parameters and examining the two graphs.

Graph 1: Population size through time

Step 1 (default settings: `Initial N` = 10, $r_m = 0.8$, $K = 200$). The maximum population size is the carrying capacity, $K = 200$. The population grows quickly at first and then slows as it approaches K, giving an S-shaped (sigmoidal) curve. It reaches about 99% of K after around 9 time steps.

Step 2 ($K = 300$). The maximum population size increases to 300, but the time taken to approach it stays roughly the same (around 10 time steps). K sets *where* the population ends up. It does not change *how fast* it gets there.

Step 3 ($r_m = 0.4$). The population grows more slowly and takes roughly twice as long to approach K (around 18 time steps rather than 9). The time taken to approach K is related to r_m: as you increase r_m this time decreases, and halving r_m roughly doubles it.

Step 4 ($r_m = 1.8$). The population *overshoots* the carrying capacity (to about 218), then oscillates around K with damped oscillations that gradually converge on K. It is a common misconception that K is the maximum population size. When r_m is markedly greater than 1, the population grows so fast that it can exceed K.

This means that if a population has even a moderate population growth rate and density dependence, it can fluctuate in numbers without any extrinsic factor (like weather) acting.

Step 5 (higher r_m). As you increase r_m, the nature of the oscillations changes. In fact, this is predictable:

- $r_m < 1$: the population approaches K smoothly.

- $1 < r_m < 2$: damped oscillations of decreasing amplitude around K.
- $2 < r_m < 2.449$: a stable 2-point limit cycle, in which the population bounces between two values around K (e.g. $r_m = 2.1$ or 2.4).
- $2.449 < r_m < 2.57$: the cycle doubles to 4 points (e.g. $r_m = 2.5$), then 8 points, and so on (period doubling).
- $r_m > 2.57$: the population fluctuates chaotically, with no repeating pattern (e.g. $r_m = 2.8$, 2.9 or 3.0). Within the chaotic region there are narrow windows of regular cycles (for example a 3-point cycle near $r_m = 2.83$), but you are unlikely to hit one by chance.
- r_m above 3 (e.g. 3.1): the population overshoots so far that a later time step gives $N \leq 0$ (with `Initial N` $= 10$ and $K = 200$, at $t = 4$), i.e. extinction. In the spreadsheet this shows up as negative population sizes, which are a mathematical artefact of the model. At exactly $r_m = 3.0$ the population stays positive and chaotic.

So yes, there is a limit to how high r_m can be.

Step 6 (sensitivity). When r_m is high (> 2.57) the dynamics are chaotic, and the population becomes extremely hard (in practice impossible) to predict. Changing r_m from 2.8 to 2.81, or changing the initial population size by 1, produces trajectories that soon become completely different. Because a change this small could simply be measurement error, this would cause big problems for a population manager.

Many pest species have very high population growth rates, for example outbreak insects such as locusts and some forest moths. High growth rates with density dependence are one reason why it can be hard to predict when the next pest outbreak will happen (real outbreaks are also driven by weather and other external factors).

Chaos is not the same as randomness. The model contains no random element: run it again with exactly the same parameters and you get exactly the same trajectory. The unpredictability comes from extreme sensitivity to the starting conditions and parameter values.

Graph 2: Per capita growth rate vs. population size

Step 1. The per capita growth rate declines linearly as population size increases. It crosses the x-axis (population size) at K (here, 200) and crosses the y-axis at r_m (here, 0.8).

Step 2 ($K = 300$). The x-axis intercept moves to 300. The y-axis intercept stays at r_m, so the line becomes shallower. The line always crosses the x-axis at K, whatever the growth rate.

Step 3 ($r_m = 1.8$). The y-axis intercept rises to 1.8 and the line becomes steeper, but it still crosses the x-axis at $K = 200$.

Step 4. The two parameters control the two intercepts: r_m sets the y-axis intercept (the per capita growth rate of a very small population) and K sets the x-axis intercept (the population size at which per capita growth is zero). The slope of the line is $-r_m/K$.

Reflection

Step 5. The two graphs are different views of the same model. As the population in Graph 1 approaches K, the per capita growth rate in Graph 2 approaches zero, so population growth slows and levels off. When N is above K, the per capita growth rate is negative and the population shrinks.

For exponential growth, the per capita growth rate does not depend on population size, so the equivalent of Graph 2 would be a horizontal line at r_m. Graph 1 would show unlimited, ever-accelerating (J-shaped) growth with no levelling off.

Solutions: Deeper into logistic growth

Relationship between Logistic and Exponential growth equations

The logistic growth equation simplifies to become the exponential growth equation if you set K to be infinite (∞). The simplification goes like this:

1. $\frac{dN}{dt} = r_m N\left(1 - \frac{N}{K}\right)$
2. $\frac{dN}{dt} = r_m N\left(1 - \frac{N}{\infty}\right)$
3. $\frac{dN}{dt} = r_m N\left(1 - 0\right)$
4. $\frac{dN}{dt} = r_m N \times 1$
5. $\frac{dN}{dt} = r_m N$

This simplification also works with the discrete version of the equation used in the Excel workbook, $N_{t+1} = N_t + r_m N_t\left(1 - \frac{N_t}{K}\right)$, which becomes $N_{t+1} = N_t + r_m N_t$.

The discrete version uses the same r_m but applies it once per whole time step. It is therefore an approximation of the continuous model: a small population is multiplied by $1 + r_m$ each time step rather than by $\lambda = e^{r_m}$. The approximation is good for small r_m and poor for large r_m. This is why the discrete logistic model can overshoot K, oscillate and become chaotic at high r_m, while the continuous logistic model always approaches K smoothly.

Task 1: Type of dynamics depends on r_m

As you increase r_m, the dynamics of the population change. At low r_m (< 1, e.g. 0.8) population size (N) simply converges smoothly on the carrying capacity, K. As r_m increases ($1 < r_m < 2$, e.g. 1.2 or 1.8) you will see damped oscillations, whereby N overshoots K and then oscillates and converges to K. Then ($2 < r_m < 2.57$) you will see stable cycles where the population size oscillates *around* K indefinitely. Between 2 and 2.449 (e.g. 2.4) these are 2-point cycles, where N bounces between two values. Between 2.449 and 2.57 they are 4-point, then 8-point (and so on) cycles. Above $r_m \approx 2.57$ (e.g. 2.7) the dynamics are chaotic: they never repeat and are unpredictable in practice. The critical values of r_m that define the type of dynamics are fixed: if we know r_m we can predict the type of dynamics.

The type of dynamics depends only on r_m, not on K or starting population size (N_0) (as long as $0 < N_0 < K$).

Task 2: Per capita growth rate vs. population size (Figure 2)

The line crosses the x-axis at K and the y-axis at r_m. The y-intercept is the per capita growth rate of a very small population, which is the maximum per capita growth rate r_m. The x-intercept is the population size at which per capita growth is zero, which is the carrying capacity K. The slope of the line is $-r_m/K$. Increasing r_m raises the y-intercept and makes the line steeper. Increasing K moves the x-intercept to the right and makes the line shallower.

For $r_m = 1.5$ and $K = 250$, your sketch should be a straight line falling from 1.5 on the y-axis to 250 on the x-axis.

Having some information about the parameters therefore allows you to sketch the dynamics. Conversely, if you have a graph showing aspects of the population dynamics, you can work out the parameters of the logistic equation.

Task 3: Population growth rate vs. population size (Figure 3)

The population growth rate is zero ($dN/dt = 0$) at $N = 0$ and at $N = K$. At $N = 0$ there are no individuals to reproduce. At $N = K$ the per capita growth rate is zero (Figure 2), so births balance deaths. Between these two points dN/dt is positive, and above K it is negative (the population shrinks).

The curve is a parabola because dN/dt is the population size multiplied by the per capita growth rate: $dN/dt = N \times r_m\left(1 - \frac{N}{K}\right) = r_m N - \frac{r_m}{K}N^2$. When N is small there are few individuals, so growth is slow even though each individual reproduces quickly. When N is close to K there are many individuals, but each reproduces slowly. Growth is fastest in between, at exactly $N = K/2$, where $dN/dt = r_m K/4$. Changing K moves the peak (always to $K/2$). Changing r_m changes the height of the peak, but not its position.

The peak of this curve is the *maximum sustainable yield* (MSY): the largest number of individuals that could be harvested each time step without the population declining. For $r_m = 0.8$ and $K = 200$, MSY =

$0.8 \times 200/4 = 40$ individuals per time step, achieved by holding the population at $N = 100$. Harvesting at exactly MSY is risky because r_m and K are only estimates and vary between years. If the true MSY is lower than estimated, or the population falls below $K/2$, the harvest exceeds what the population can replace and the population declines further.

Task 4: Comparison with exponential growth

For exponential growth, population size is unlimited and will grow exponentially if $r_m > 0$ (or $\lambda > 1$). Logistic growth always shows some kind of limit (smooth convergence, damped oscillation, cycles or chaos), as the population size is attracted to K in some way.

The equivalent figures for exponential growth are:

- **Fig 1** (N vs. t): a J-shaped curve that accelerates without levelling off.
- **Fig 2** (per capita growth rate vs. N): a horizontal line at r_m. The per capita growth rate is constant and unaffected by N, unlike the negative slope of the logistic model.
- **Fig 3** (dN/dt vs. N): a straight line through the origin with slope r_m. The more individuals there are, the faster the population grows, unlike the logistic parabola.

In the `Exponential` worksheet the parameter labelled "r_m intrinsic growth rate" (0.1) is applied once per time step, $N_{t+1} = N_t(1 + 0.1)$, so it acts as the per-step rate R_m (geometric growth with $\lambda = 1.1$). dN/dt increases linearly with N (Figure 3) but exponentially with time.

Task 5: Time lag

Time-lagged density dependence results in oscillations. With the default $r_m = 0.6$, the ordinary logistic model converges smoothly on K. With a 1-step lag, the population overshoots K and then shows damped oscillations. At r_m above about 1 (e.g. 1.2), the oscillations become sustained cycles. This happens because the population responds to its density one time step ago. It keeps growing after it has passed K, and keeps declining after it has fallen below K.

The lecture's thresholds for lags ($r_m\tau < e^{-1}$ smooth, up to $\pi/2$ damped, above that cycles with a period of about 4τ) apply to the continuous-time lag model. In this discrete version with a 1-step lag the approach is smooth only for r_m below about 0.25, oscillations are damped between about 0.25 and 1, and above 1 there are sustained cycles with a period of about 6 to 7 time steps (7 at $r_m = 1.2$, where N cycles between about 67 and 647 with $K = 400$). The message is the same: lags cause overshoot and cycles.

Therefore, if we see oscillations in population dynamics, they could be due to a high r_m value or to lagged density dependence. Time lags make populations harder to predict and manage, because the population's response to a change (e.g. a harvest or a cull) is delayed.

N_{t+1} vs. N_t

Because $N_{t+1} = N_t + dN/dt$, a plot of N_{t+1} against N_t is also a parabola: it is the Figure 3 parabola with the line $N_{t+1} = N_t$ added to it. It passes through the origin and crosses the line $N_{t+1} = N_t$ at $N_t = 0$ and $N_t = K$. These are the equilibria, where $dN/dt = 0$ and the population does not change from one time step to the next. How steeply the parabola crosses the line at K depends on r_m. This slope determines whether the population approaches K smoothly, oscillates, or behaves chaotically. The slope at K is $1 - r_m$: for $r_m < 1$ it is between 0 and 1 and the population approaches K smoothly; for $1 < r_m < 2$ it is between -1 and 0 and the population overshoots and oscillates back towards K (damped oscillations); for $r_m > 2$ it is steeper than -1, so K is unstable and the population cycles or behaves chaotically. This one rule explains all the dynamics in Task 1.

Chaotic dynamics

Chaotic dynamics are deterministic in the sense that if you have the same initial conditions you will get the same results (dynamics). However, small changes in any parameters (N_0, K, or r_m) will lead to very different dynamics. In that sense, because of measurement error, the population dynamics would be very hard to predict in practice.

Solutions: Life tables and survivorship types

The take home messages for this exercise are that life tables are a rich source of information on the life history of a population or species. From a simple accounting of survivors and fertility at each age we can obtain information about the "shape" of the survivorship trajectory: do most deaths happen early or late? Is mortality risk constant with age? We can also calculate measures including life expectancy, **generation time**, R_0 and population growth rate. You should know what these measures mean.

Life table. For the cohort in the *Life table* sheet:

- $R_0 = \sum l_x m_x = 4.06$ daughters per female, so the population is growing.
- $T = \sum x l_x m_x / R_0 = 15.35/4.06 = 3.78$ years.
- $r_c = \ln(R_0)/T = \ln(4.06)/3.78 = 0.37$ per year ($\lambda \approx e^{r_c} = 1.45$). The exact intrinsic rate of increase from the Euler-Lotka equation is $r_m = 0.41$ ($\lambda = 1.50$): the capacity for increase, r_c, slightly underestimates r_m when generations overlap.
- Life expectancy at birth is $e_0 = 3.76$ years. It rises to $e_1 = 4.14$ years: once the risky first year is survived, an individual can expect to live longer than a newborn.
- p_x and q_x for each age, e.g. $p_2 = 254/298 = 0.852$ and $q_2 = 0.148$; $q_9 = 0.88$ (only 5 of 43 nine-year-olds survive to age 10).

Declining force of selection (optional task). Adding one daughter at age x (ages 1 to 10) raises R_0 by exactly l_x: by 0.70 (17%) at age 1, 0.60 (15%) at age 2, 0.13 (3%) at age 8 and 0.01 (0.2%) at age 10. The effect falls with age because fewer and fewer females survive to use the extra reproduction. Age 0 is left out because newborns cannot breed at birth. (The capacity for increase $r_c = \ln(R_0)/T$ behaves oddly here: it actually *falls* slightly when a daughter is added at ages 7 to 10, because T increases more than $\ln(R_0)$. The exact r_m always increases, by 40% at age 1 but by less than 1% from age 7 onwards. This is why we use R_0 for the experiment.)

Survivorship curves. Population 1 is Type I (q_x rises with age, from 0.02 to over 0.6); Population 2 is close to Type II (q_x stays roughly between 0.25 and 0.4, so $\ln(l_x)$ falls in a nearly straight line); Population 3 is Type III (q_x falls from 0.73 in the first year to about 0.1 in old age).

You should understand the relationship between mortality risk and survivorship curve. In other words, you should be able to describe, or do a rough sketch of mortality risk with age, if you have a survivorship curve (or vice versa).

The completed Excel spreadsheet is here.

Answers to the questions

1. What are the three main types of survivorship curves, and which organisms typically exhibit each type?

Type I curves show high survival in early life and high mortality at older ages (e.g. humans in developed countries, zoo and domestic animals). Type II curves show roughly constant mortality throughout life (e.g. many small birds). Type III curves show high juvenile mortality, followed by lower mortality for the few that survive the vulnerable early period (e.g. many fish and trees).

2. What does a net reproductive rate (R_0) of less than 1 signify for a population?

On average, each female is replaced by fewer than one daughter over her lifetime, so the population is declining.

3. How does the shape of a survivorship curve reflect trade-offs in life history strategies, typically?

Type I curves suggest heavy investment in each offspring, with low juvenile mortality: typical of species with few offspring and much parental care. Type III curves reflect species that produce many offspring but give each one little care, leading to high early mortality.

4. How is survivorship (l_x) different from survival probability (p_x)?

- **Survivorship (l_x)** is the proportion of newborns that survive from birth to age x. It is cumulative across all the age classes up to x, starts at $l_0 = 1$ and can only fall with age.
- **Survival probability (p_x)** is the probability that an individual alive at age x survives to age $x+1$. It measures survival over a single time step, and can rise or fall with age.

The two are linked: $l_{x+1} = l_x \times p_x$. In the worked example, $l_2 = 0.4$ and $p_2 = 0.25$, so $l_3 = 0.4 \times 0.25 = 0.1$.

5. How does altering reproduction or survival/mortality at different ages affect R_0 and population growth rate?

Changes early in life have a larger effect on R_0 and population growth rate than changes later in life, because fewer individuals survive to experience late-life changes (adding one daughter at age x raises R_0 by l_x). This is the declining force of selection.

6. Two populations have the same R_0, but one has a shorter generation time. Which grows faster per year, and why?

The one with the shorter generation time. R_0 is growth per generation, and $r_c = \ln(R_0)/T$ is (approximately) growth per year, so a shorter T means the population multiplies by R_0 more often. For example, with $R_0 = 2$: $T = 1$ year gives $r_c = 0.69$ per year; $T = 2$ years gives $r_c = 0.35$ per year.

7. The probability of dying in a year, q_x, can never be greater than 1, but the force of mortality, μ_x, can. Why?

q_x is a probability over an interval, so it lies between 0 and 1. μ_x is an instantaneous rate per individual still alive (like a speed), so it has no upper limit. For a rate that is constant over a year, $q_x = 1 - e^{-\mu_x}$: at low mortality q_x is close to μ_x, but as μ_x increases, q_x approaches 1 without reaching it (e.g. $\mu_x = 2$ gives $q_x = 0.86$).

Solutions: Matrix population modelling

The main take home messages for this class are that you should: (1) understand, and be able to draw or interpret, a life cycle diagram; (2) be able to (roughly) parameterise a model shown in the life cycle diagram, remembering that a reproduction entry must include the survival needed for newborns to be counted at the next census; (3) be able to write out a matrix model (the square of numbers) for a given life cycle diagram (and *vice versa*); (4) understand how population projection works, and therefore be able to interpret a figure or description showing a population projection; (5) understand what an elasticity matrix tells us (the relative influence of a proportional change in each transition on population growth), and how it differs from sensitivity (the effect of an absolute change); (6) understand the logic of *in silico* experiments (changing the matrix and comparing λ) to investigate management options.

Model answers for the matrix modelling practical

Does the starting population matter? The early (transient) dynamics depend strongly on the starting population structure, and the lines may zigzag or even decline temporarily. Eventually the lines become parallel and straight on the log scale, with the same slope as before. The slope is the same because the long-term growth rate (λ) depends only on the matrix, not on the starting population. The gaps between the lines are also the same, because the stage proportions converge to the same stable stage distribution. Starting structure affects the short term, but not the long-term growth rate or structure (given constant vital rates).

An evolutionary experiment (homework). For the example matrix, $\lambda = 1.346$. A mutant that has one extra baby as an adult (fecundity 8 to 9) but lower juvenile-to-adult survival (0.10 to 0.08) has

$\lambda = 1.295$, so the mutant has lower fitness and would not spread: the small loss of juvenile survival outweighs the extra baby. If juvenile-to-adult survival only falls to 0.09, the mutant's λ is 1.3466, almost exactly the same as the original (1.3458), so selection on this trade-off would be very weak. Juvenile-to-adult survival has the highest elasticity in this matrix, which is why a small proportional cost there matters so much. Your own organism will give different numbers, but the logic is the same: compare λ for the original and mutant matrices.

Question 1. On a log scale, straight lines mean the population is growing (or shrinking) exponentially at a constant rate, i.e. the growth rate has stabilised at λ once the transient phase has passed.

Question 2. Elasticity measures the proportional change in λ caused by a small proportional change in each matrix element. It identifies which transitions have the greatest influence on population growth. If adult survival has the highest elasticity, conservation might focus on protecting adults (e.g. habitat protection, reducing human-induced mortality). If juvenile survival has a high elasticity, efforts might target juveniles (e.g. habitat restoration or predator control). Managers can then allocate limited resources to where they are likely to matter most, bearing in mind that high elasticity does not mean a rate is easy to change in practice.

Question 3. Key assumptions include:

- constant vital rates (no environmental variation from year to year);
- no density dependence (vital rates do not depend on population size);
- a closed population, with no immigration or emigration;
- all individuals within a stage are identical;
- only females are modelled, and there is no shortage of mates;
- a fixed time step (here one year) with the census at a fixed point in the annual cycle.

Several of these are shared with the exponential/geometric growth models.

Solutions: Pre- and Post-reproduction census

Why `A1` and `A2` give the same λ. Both give $\lambda = 1.0703$ (to 4 decimal places), and they are exactly identical, not just equal to rounding error. Both matrices have the same characteristic equation, $\lambda^3 = p_0p_1m_2\lambda + p_0p_1p_2m_3$: each term is one route from birth to reproduction (survive to age 2 and breed, or survive to age 3 and breed), and each route contains the same survival and reproduction terms in both census types. The net reproductive rate is also the same in both: $R_0 = p_0p_1m_2 + p_0p_1p_2m_3 = 0.54 + 0.648 = 1.188$. Census timing only changes *when* we count the population (and so which age classes are in the matrix), not what the animals do, so it cannot change how fast the population grows. What *does* change is anything that depends on the classes counted: the stable age distribution (post-breeding: 74% newborns of age 0; pre-breeding: 43% age 1, with no newborns counted), the reproductive values and the sensitivities.

`A2` vs `A3`. The matrices differ only in the top row (the fertilities): `A2` has $m_2p_0 = 0.6$ and $m_3p_0 = 1.2$, whereas `A3` has $m_2p_1 = 2.7$ and $m_3p_2 = 3.6$. In `A3`, first-year survival p_0 appears nowhere in the matrix: it was removed from the sub-diagonal (correctly) but never added to the fertilities. So every newborn is effectively assumed to survive its first year, and $p_0 = 0.2$ is the lowest survival rate in the whole life cycle. Instead, the parent's survival (p_1 or p_2) is counted twice: once on the sub-diagonal and once in the fertility. The result is $\lambda = 1.864$ instead of 1.070: a rapidly growing population instead of a slowly growing one. (The mistaken matrix also gives $R_0 = 4.374$ instead of 1.188.)

Changing a survival probability. The correct conversion always agrees, whatever values you use, because the characteristic equations are identical. The error from the mistake depends on which parameter you change:

Change	λ for `A1` and `A2`	λ for `A3` (mistake)	Error
None ($p_0 = 0.2$, $p_1 = 0.9$)	1.070	1.864	0.79
$p_1 = 0.7$	0.969	1.562	0.59

Change	λ for `A1` and `A2`	λ for `A3` (mistake)	Error
$p_1 = 0.5$	0.850	1.266	0.42
$p_0 = 0.5$	1.548	1.864	0.32

Lowering p_1 shrinks the error a little, because p_1 is counted twice in `A3`, but the error stays large (at $p_1 = 0.5$ the correct matrices say the population is declining, while the mistake says it is growing by 27% per year). Changing p_0 shows what really drives the error: `A3` does not respond to p_0 at all (its λ stays at 1.864), and the error shrinks as p_0 gets closer to 1. The mistake is equivalent to assuming that all newborns survive their first year, so it does the most damage in species with low first-year survival, which is most birds, mammals, fish and plants.

Solutions: Life Table Response Experiments

Predicting the contributions. Each contribution is (treatment element − reference element) × (sensitivity at the midpoint matrix). Sensitivities are always positive here, so the sign of each contribution is the sign of the difference in the element:

Element	Difference (treatment − reference)	Sensitivity at midpoint	Contribution
Juvenile stasis (top left)	0	0.336	0
Fertility (top right)	$1.3 - 2.3 = -1.0$	0.208	−0.208
Juvenile → adult (bottom left)	$0.2 - 0.5 = -0.3$	1.071	−0.321
Adult survival (bottom right)	$0.7 - 0.6 = +0.1$	0.664	+0.066

The largest difference in an element is in fertility, but the largest contribution is from maturation (juvenile → adult), because λ is about five times more sensitive to maturation than to fertility.

Checking the sum. The contributions sum to −0.463, while the observed difference is $\lambda_{\text{treat}} - \lambda_{\text{ref}} = 0.992 - 1.451 = -0.460$. They are close but not identical, because the LTRE is a linear approximation: it assumes λ changes in a straight line with each element, with the slope (sensitivity) taken from the midpoint matrix, whereas in reality the sensitivities change as the elements change.

Changing one element of `A_treat`. For example:

- Treatment fertility 1.8 instead of 1.3: the fertility difference halves (−0.5 instead of −1.0), so the fertility contribution roughly halves, to −0.098. The maturation and adult survival contributions change slightly (−0.345 and +0.065), because the midpoint matrix, and therefore the sensitivities, have changed. The sum is −0.378 (observed $\Delta\lambda = -0.380$).
- Treatment maturation 0.4 instead of 0.2: the maturation difference falls from −0.3 to −0.1, so its contribution drops to −0.096, and fertility (−0.239) becomes the largest contribution. The sum is −0.270 (observed $\Delta\lambda = -0.270$).

In both cases the element you changed shows the biggest change in contribution, and the change can be explained as (new difference) × (sensitivity), with small knock-on changes in the other contributions because the sensitivities are recalculated at the new midpoint matrix.

Interpretation. The Location B population has a lower λ (0.99, slowly declining) than Location A (1.45) mainly because fewer juveniles survive and mature into adults; lower fertility is the second most important reason; and slightly higher adult survival at Location B partly offsets these. Negative contributions lower λ at Location B relative to Location A, positive ones raise it.

Why LTRE alone does not tell a manager what to target. LTRE is retrospective: it explains an observed difference in λ that has already happened. A contribution is large if the element *did* differ

a lot between the populations and/or if λ is sensitive to it. A vital rate can make a large contribution because it varied a lot for reasons managers cannot influence (e.g. spring weather), and a rate with a high sensitivity (here juvenile stasis, 0.336) can make no contribution at all because it did not differ. To ask what *would* happen if a vital rate were changed, use prospective tools (sensitivity, elasticity, *in silico* experiments), together with knowledge of which rates can realistically be changed, by how much, and at what cost.

Solutions: How many eggs should a bird lay?

You can find a completed Excel spreadsheet here.

The expected numbers of surviving chicks for clutches of 1 to 10 eggs are 0.9, 1.7, 2.46, **2.8**, 2.5, 1.8, 1.54, 1.2, 0.9 and 0.5, so the optimum is **4 eggs**. Laying 10 eggs gives only 0.5 surviving chicks on average.

The take-home message is that the trade-off between number of eggs and juvenile survival leads to an optimum number of eggs required to maximise reproductive success (i.e. number of surviving chicks).

You can move this optimum if the trade-off is changed. For example, if you increase survival of the chicks from larger clutches, then the optimum number of eggs will increase.

This fact reminds us that the environment (e.g. predation risk and resource abundance) will be an important factor in determining the evolution of life history parameters (like average number of babies).

1. What is Lack's hypothesis? Does the big bird's optimum support it, given that most big birds lay 3 eggs?

Lack's hypothesis is that clutch size evolves to the number from which, on average, the most young survive. For the big bird that is 4 eggs ($4 \times 0.70 = 2.8$ surviving chicks). But most big birds lay only 3 eggs (2.46 surviving chicks), one fewer than the single-season optimum, so these data do not fully support Lack's hypothesis. Many real birds also lay slightly fewer eggs than the clutch that produces the most young in one season. Question 3 shows one reason why.

2. A good food year.

Survival in clutches of 5 or more rises by 0.2, so the expected numbers of surviving chicks for 4 to 8 eggs become 2.8, **3.5**, 3.0, 2.94 and 2.8. The optimum rises to **5 eggs**. When food is plentiful, chicks in larger broods are fed better and survive better, so larger clutches pay. Many birds do lay more eggs in good years or rich habitats.

3. A cost to the parents.

Parent survival is 0.72, 0.64, 0.56, 0.48 and 0.40 for 1 to 5 eggs, so the expected number of breeding seasons (counting this season as the first) is 3.57, 2.78, 2.27, 1.92 and 1.67. Multiplying by the surviving chicks per season gives lifetime totals of 3.21, 4.72, **5.59**, 5.38 and 4.17. The optimum falls from 4 eggs to **3 eggs**, which is exactly the clutch that most big birds lay.

Lack counted the young raised in a single season. Once the cost to the parents' future breeding is included, the best clutch is smaller than the one that produces the most surviving young in one season. This is the argument behind the kestrel brood manipulations in the lecture: parents given extra chicks paid for them later. Neal's chapter *Evolution of Life Histories* discusses the same idea.

4. Egg number versus egg size.

You would need a second relationship, describing how the size of an egg (or offspring) affects its chance of survival, and a fixed amount of resources to divide among the eggs, so that more eggs means smaller eggs. The best combination is the one that maximises the number of surviving offspring (Smith & Fretwell 1974).

Solutions: Trade-offs and the force of selection

Simulate this by adding 1 to fertility (mx) at age 1 (the benefit), but setting Survival to 0 in the age-25 row, so that all die before age 25 (the cost). What is the fitness of this strategy?

Fitness (R0) would be 50.76 (compared to 49.81).

By setting survival to 0 at other ages, determine how many years of life could be lost before this cost is no longer worth bearing. Is this surprising?

You could "kill" the species at 18 (so that nobody reaches age 18), losing 8 years of life (ages 18 to 25), before the new strategy of a small increase in fertility at the expense of years of life at the end of life becomes unfavourable. In other words, a cost of losing 8 yrs of life is worth bearing if the benefit is an extra baby in year 1. With death at 18, $R_0 = 49.87$, still just above 49.81; with death at 17 it falls to 49.59.

Now reset everything (mx =10; survival = 0.8). Recall what fitness was when you added 1 to mx at age 1 (R0=50.811). If, instead of adding to mx at age 1 you were to increase mx at age 25, how much would you need to increase it to reach this figure?

At age 25 you would need to increase Fertility from 10 to 222 to reach the same level of improvement in fitness as increasing Fertility from 10 to 11 at age 1.

What about at age 20? 10 to 80 *Age 15?* 10 to 33 *Age 10?* 10 to 18 *Age 5?* 10 to 13

What do you notice?

You should notice that changes made at younger ages have a bigger effect than those at older ages. The plot below shows the increase in fertility that would be required to raise fitness (R_0) from 49.81 to 50.81 at ages 1 to 25 (the figures you estimated above). Adding Δm offspring at age x adds $l_x \Delta m$ to R_0, so the extra fertility needed is exactly $1/l_x$: 2.4, 7.5, 22.7, 69.4 and 211.8 at ages 5, 10, 15, 20 and 25 (the figures above are rounded up). You only need to change fertility a little at young ages, but you would need to make BIG changes at older ages.

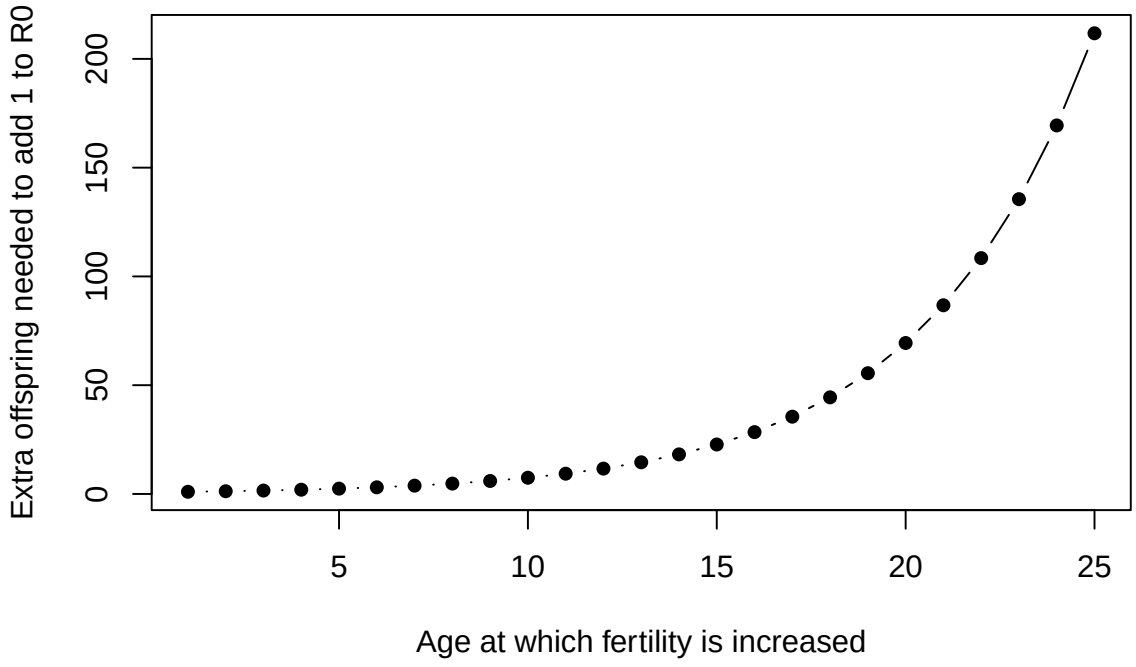


Figure 14: Extra offspring needed at each age to raise R0 by 1, when annual survival is 0.8.

Reset everything again (mx =10; survival = 0.8). Set mx from age 15 onwards to be 0. Now alter survival rate after this point (at ages 15-25). What happens to fitness?

You should notice that any changes you make to survival AFTER reproduction has been completed will

have no effect on fitness. In other words evolution "doesn't care about you" after you have finished with reproduction.

This is only true for an individual's *own* offspring. If older individuals help their relatives to survive and reproduce, survival after reproduction can still raise their inclusive fitness: see the grandmother hypothesis and killer whales in the lecture.

Questions

- *The declining force of selection.* A change in survival or reproduction at age x only affects the individuals still alive at that age, so its effect on fitness shrinks as survivorship (l_x) falls and as less of the lifetime's reproduction remains. Selection therefore acts strongly on early life and weakly on late life. Early reproduction, and genes with early benefits and late costs, are favoured, and senescence can evolve.
- *Age-related disease in humans.* Many diseases of old age (e.g. many cancers, heart disease, dementia) have their main effects after most reproduction has happened, so selection against them is weak. Harmful mutations with late effects can build up (mutation accumulation), and some genes that cause late disease may have been favoured for their early benefits (antagonistic pleiotropy).
- *Predation risk.* Where predation is high, an individual is unlikely to survive to old age whatever it invests in its body, so resources spent on maintenance and repair are less valuable. Selection should favour putting more into early reproduction and less into maintenance.
- *Survival after reproduction.* Older individuals can raise inclusive fitness by helping relatives. Post-reproductive killer whale females lead their groups to salmon when food is scarce, and in humans, grandmothers can improve their grandchildren's survival (the grandmother hypothesis).

Solutions: Matrix Population Models (MPMs): Projection and Simulation

The simulation results below come from 2,000 replicates per scenario. They are stochastic: in another set of runs, probabilities of extinction typically change by about ±0.02 and medians by a few individuals. Your results with 200 replicates will vary more. The deterministic λ values are exact.

Exercise 1: Projection interval.

1. $\Delta t = 1$ year means that one multiplication by $\mathbf{A}$ moves the population forward by one year, so every entry in the matrix is a rate per year (e.g. $P_A = 0.8$ is the probability that an adult survives one whole year).
2. With $\Delta t = 1$ month, P_A would be the probability of surviving one *month*. If mortality is spread evenly through the year, the monthly survival that gives the same annual survival is $0.8^{1/12} = 0.982$. F_A would be the number of new juveniles (counted at the next monthly census) per adult per *month*. For most animals and plants this is not constant: reproduction happens in a breeding season, so F_A would be zero in most months, and the matrix would have to change from month to month (a seasonal model). Growth probabilities (G_J, G_S) would also have to be per month.
3. For example: the species is short-lived or breeds several times a year (insects, rodents), so a year is too coarse; you want to model seasonal events such as a breeding season, a hunting season or a harsh winter; or management (e.g. culling) happens at a particular time of year.

Exercise 2: Lower adult survival. With $P_A = 0.70$, $\lambda = 0.964$, so the population now declines by about 4% per year. You can tell because $\lambda < 1$, and because the deterministic projection of total population size eventually goes down (after the transient phase). `simulate_once()` does not need to be changed, because it reads the vital rates from the matrix `A` that you give it: rebuild `A` and the simulation uses the new P_A automatically. (This is why we never use `A` as the name of an adult count.)

Exercise 3: Culling as management.

1. With a 20% adult harvest, the deterministic prediction (adult column × 0.8) is $\lambda = 0.903$, compared with 0.960 for 10% and 1.019 for no harvest. In the replicates, the median final population size (year

30) falls from about 143 (no harvest) to about 42 (10%) and about 11 (20%), and the probability of extinction within 30 years rises from 0 to about 0.03. The 10-90% band at year 30 moves down from about 88-209 (no harvest) to about 20-69 (10%) and 3-24 (20%), so with a 20% harvest a sizeable minority of runs end with only a handful of individuals.

2. A fixed quota of 5 adults per year gives a probability of extinction of about 0.85 (median final size 0), and 10 adults per year about 1.00. A 10% harvest gives about 0. A fixed quota is much riskier because the *proportion* removed rises as the population shrinks: when there are 50 adults, 5 is 10%, but when there are 10 adults, 5 is half of them. This sets up a downward spiral. A proportional harvest automatically takes fewer individuals when the population is small. (With about 20 adults at the start of culling, removing 5 per year is more than the population can replace.)
3. Culling the same number or the same proportion from each stage:

Stage culled	Fixed quota, 5/year: median final N	Fixed quota: P(extinct)	10% harvest: λ	10% harvest: median final N
None	143	0	1.019	143
Juveniles	101	0	1.006	109
Subadults	23	0.05	0.992	83
Adults	0	0.85	0.960	42

Culling adults has by far the largest effect, then subadults, and juveniles the least. The reason is reproductive value: how many future juveniles an individual in each stage is expected to produce. Relative to a juvenile, the reproductive values are juvenile 1 : subadult 2.9 : adult 5.5. A juvenile is worth little, because it has a low chance (0.35) of even reaching the subadult stage, whereas an adult is already breeding and has a high survival rate. Removing an adult therefore removes much more future reproduction than removing a juvenile.

Exercise 4: Contraception. Add an argument `fert_mult = 1` to the function, and replace the births line with:

The deterministic prediction for 90% effective contraception is $\lambda = 0.843$ (multiply `A["Juvenile", "Adult"]` by 0.1 and recompute the dominant eigenvalue), a much stronger reduction than the 10% adult harvest ($\lambda = 0.960$). In the replicates, the median final population size is about 2 (10-90% range 0 to 5), with a probability of extinction within 30 years of about 0.16, compared with a median of about 42 for the 10% adult harvest and 143 for no treatment. Contraception gives the largest reduction in population growth here, because it cuts the fertility of *all* adults by 90%, whereas the harvest removes only 10% of them. Note the time lag: contraception acts only through future recruitment, so the adults and subadults already present decline only slowly at first. Whether it is better in practice also depends on cost, feasibility and animal welfare, which the model does not include.

Exercise 5 (optional): Environmental stochasticity. Add one line at the start of each year in a copy of the function (just after `n_now <- out[t, ]`):

With this change, the median final population size is about 136 (10-90% range about 71 to 235), compared with about 145 (about 90 to 214) for demographic stochasticity alone. The average adult survival is unchanged, but the spread of outcomes is wider, because in a run of bad years all adults have low survival at the same time. Variation between years also tends to reduce the long-run growth rate slightly (the stochastic growth rate λ_s is lower than the λ of the average matrix); see the chapter "Stochastic population growth".

Solutions: Hardy-Weinberg equilibrium

Answers to the Hardy-Weinberg questions.

Remember the basic formulas: $p^2 + 2pq + q^2 = 1$ and $p + q = 1$

p = frequency of allele A (where there is dominance, A is the dominant allele)

q = frequency of allele a

p^2 = frequency of homozygous AA individuals

q^2 = frequency of homozygous aa individuals

$2pq$ = frequency of heterozygous Aa individuals

A general rule for all these problems: keep full precision in your calculator and **round only at the end**. Rounding p or q early can change the final answer in the second decimal place, which matters on a multiple-choice exam.

Problem 1

The frequency of allele a: Since $p + q = 1$ and $p = 0.8$, $q = 1 - 0.8 = 0.2$.

The expected genotype frequencies:

- $AA = p^2 = 0.8 \times 0.8 = 0.64$ (64%)
- $Aa = 2pq = 2 \times 0.8 \times 0.2 = 0.32$ (32%)
- $aa = q^2 = 0.2 \times 0.2 = 0.04$ (4%)

Check: $0.64 + 0.32 + 0.04 \approx 1$ (displayed frequencies are rounded).

The phenotype frequencies if A is completely dominant over a: Both AA and Aa individuals show the dominant phenotype, so its frequency is $0.64 + 0.32 \approx 0.96$ (96%). Only aa individuals show the recessive phenotype, so its frequency is 0.04 (4%). Heterozygotes cannot be distinguished from AA homozygotes by phenotype alone. There are approximately 8 times as many heterozygotes as recessive homozygotes.

Problem 2

Assuming HWE at birth, the frequency of SS babies is $q^2 = 0.09$. The HbS allele frequency is therefore $q = \sqrt{0.09} \approx 0.3$, and the HbA allele frequency is $p = 1 - q \approx 0.7$. The expected frequency of carriers (AS) is $2pq \approx 2 \times 0.7 \times 0.3 \approx 0.42$, so **about 42% of babies are expected to be carriers**.

Problem 3

The allele frequencies: 200 diploid individuals carry 400 allele copies. Each AA individual carries two copies of A and each Aa individual carries one, so the number of A copies is $2 \times 90 + 70 = 250$.

$p = 250/400 \approx 0.625$, and $q = 1 - p \approx 0.375$. (Check: the number of a copies is $2 \times 40 + 70 = 150$, and $150/400 \approx 0.375$.)

The expected counts under HWE are the expected frequencies multiplied by the sample size, 200:

Genotype	Observed	Expected under HWE
AA	90	$p^2N \approx 0.625^2 \times 200 \approx 78.125$
Aa	70	$2pqN \approx 2 \times 0.625 \times 0.375 \times 200 \approx 93.75$
aa	40	$q^2N \approx 0.375^2 \times 200 \approx 28.125$

Comparison: There are fewer heterozygotes than expected, and more of both homozygotes: a heterozygote deficit. There are 70 observed heterozygotes, compared with 93.75 expected.

Possible explanations include inbreeding or other non-random mating, combining partly separated subpopulations (the Wahlund effect), selection against heterozygotes, or genotyping errors. The comparison alone does not identify a cause. In the Inbreeding lecture you will quantify departures in heterozygosity with $F = 1 - H_o/H_e$, here $1 - 0.35/0.46875 \approx 0.253$.

Optional extension: the χ^2 test.

Genotype	O	E	$(O-E)^2/E$
AA	90	78.125	1.8
Aa	70	93.75	6.02
aa	40	28.125	5.01
Total	200	200	$\chi^2 \approx 12.84$

There are three classes, so you might expect $3-1=2$ degrees of freedom (one is lost because the expected counts are forced to add up to the sample size, 200). But we also estimated one parameter, p, from the data to calculate the expected counts, which costs another degree of freedom. So df $= 3-1-1=1$. Using the unrounded statistic, $\chi^2 \approx 12.84$ gives $P \approx 0.0003$. The test provides evidence of a departure from HWE at the 5% significance level. It does not identify the cause.

Problem 4

Since white is recessive (genotype bb), $q^2 = 0.4$. Take the square root: $q = \sqrt{0.4} \approx 0.632456$. Then $p = 1-q \approx 0.367544$. Keep full precision for subsequent calculations; the displayed values are rounded.

The heterozygote frequency is $2pq \approx 2{\times}0.367544{\times}0.632456 \approx 0.464911$, or about 46.5%. The homozygous dominant frequency is $p^2 \approx 0.367544^2 \approx 0.135089$.

Extension: Brown butterflies have genotype BB or Bb, so their frequency is $1-0.4=0.6$. The proportion of brown butterflies that are heterozygous is $2pq/(1-q^2) \approx 0.464911/0.6 \approx 0.774852$, or about 77.5%. Compare this with 46.5% in the population as a whole: the denominator here includes only brown butterflies.

Solutions: Gene pool model

Part 2: Drift through time

1. **How does population size affect the variation through time in allele frequencies? Why do you see these patterns?**

   - Each generation is a random sample of 2N alleles from the previous gene pool. The smaller the sample, the larger the sampling error, so allele frequencies fluctuate much more in small populations. With N = 20 the frequency of A wanders widely and usually hits 0 or 1 within 100 generations; with N = 1000 it stays close to 0.5 over the same period.

2. **With 20 individuals, does *A* always end up fixed, or is it sometimes lost?**

   - Both happen, roughly equally often. Drift has no direction: starting from 0.5, A is fixed in about half of the populations and lost in the other half.

3. **When the population size is large (say 1000), can you see an allele become fixed within 100 generations? Could it happen if you ran the simulation for much longer?**

   - Not within 100 generations: in large populations drift is slow. But yes, given enough time an allele is eventually fixed even in a large population. Starting from p = 0.5, the expected time until one allele is fixed is about $2.8N$ generations (more precisely $-4N[p\ln p + q\ln q]$), i.e. around 55 generations for N = 20 and around 2,800 generations for N = 1000. With `n_gen <- 3000` you will see fixation in some runs but not all. (This is a different quantity from the time a *new mutation that does fix* takes to get there, which is about $4N$ generations, as in the mutation lecture.)

Part 3: Many populations

1. **What proportion of the populations have lost one of their alleles by generation 100 for N = 20, 100 and 1000?**
   - Roughly 90% for N = 20, around 10 to 15% for N = 100, and essentially none for N = 1000 (exact numbers will vary between runs).
2. **Of the populations where one allele was fixed, in what proportion was it *A*? Does this depend on population size?**
   - About half, at every population size. Starting from a frequency of 0.5, A and a have equal chances of being fixed.
3. **So what does population size change: which allele is fixed, or how quickly variation is lost?**
   - How quickly. For a neutral allele, the probability of fixation equals its starting frequency whatever the population size. Population size determines the *speed* of drift: small populations lose variation quickly, large populations slowly.
4. **Are rare alleles more or less likely to be lost than common ones?**
   - More likely. A neutral allele at frequency p is eventually fixed with probability p and lost with probability $1 - p$. So an allele at 0.1 is lost 90% of the time, and an allele at 0.75 is lost only 25% of the time.
5. **Compare the proportion of populations where the new allele was fixed with $1/(2N)$. What does this tell you about the fate of most new mutations?**
   - With N = 20, a single new copy starts at $1/40 = 0.025$, and roughly 2 to 3% of populations end up fixed for it (with 200 replicates you might see anywhere from about 0.5% to 5%). This matches $1/(2N)$. Most new neutral mutations, about 39 out of 40 here and 1,999 out of 2,000 in a population of 1000, are lost by chance, usually within a few generations. This is the "Model 1" result from the lecture.
6. **Can you find a simple rule linking starting frequency to probability of fixation?**
   - The probability of fixation of a neutral allele equals its starting frequency. For example, one run of the template with 200 replicates gave 0.095 for $p_0 = 0.1$, 0.26 for 0.25 and 0.675 for 0.75. Your numbers will differ by a few percentage points from run to run (the standard error with 200 replicates is about 0.02 to 0.03), but they cluster around the starting frequency.
7. **Going further: does the simulated decline in H_e agree with $H_t = H_0(1 - 1/(2N))^t$?**
   - Yes. Averaged over 500 populations, H_e declines smoothly and follows the theoretical curve closely. With N = 20 it falls by 2.5% per generation, from 0.5 to about 0.04 after 99 generations of drift (theory: $0.5 \times (1 - 1/40)^{99} = 0.041$; repeated runs of the code give between about 0.03 and 0.05). Individual populations are noisy, which is why the average is needed. Note that H_e declines in every case, even though the average allele frequency across populations stays at about 0.5: drift removes variation *within* populations while creating differences *between* them.

Part 4: Bottlenecks and founder effects

1. **How does a genetic bottleneck influence the genetic composition of the population?**
   - During the bottleneck the population is small, so drift is strong and allele frequencies can change a lot in a few generations. One allele may be lost altogether. The genetic composition after the bottleneck is therefore often quite different from that before it, and genetic diversity is usually reduced.
2. **How might a genetic bottleneck affect the probability that one allele is fixed and the other lost?**
   - It greatly increases it. With 10 individuals for 21 generations, roughly half of all runs lose one allele during the bottleneck, whereas in a constant population of 1000 this essentially never happens within 100 generations. Which allele is fixed is still a matter of chance.

3. **What happens to allele frequency after the bottleneck? Does the population "recover" its original allele frequency?**

   - No. Once the population is large again, drift is weak, so the frequency stays close to wherever the bottleneck left it. Nothing pulls it back to 0.5. If an allele was lost, it cannot return without mutation or migration. The effects of a bottleneck persist long after the population has recovered in size.

4. **Does the severity of the bottleneck (i.e. length and population size) matter?**

   - Yes. Smaller and longer bottlenecks cause larger changes and a higher chance of losing an allele. Each generation of the bottleneck multiplies heterozygosity by $1-1/(2N)$, so 21 generations at N = 10 leaves about $(1-1/20)^{21} \approx 0.34$ of the original heterozygosity, whereas 21 generations at N = 100 leaves about 0.90.

5. **Optional: effective population size. What are the arithmetic and harmonic means of N over the bottleneck run?**

   - The arithmetic mean is 792.1 (79 generations at 1000 and 21 at 10), but the harmonic mean is only about 46 (`1 / mean(1 / N)` = 45.9). So the population loses diversity as if it had about 46 individuals throughout, not 790. A few generations at small size dominate the long-term effect of drift: N_e is much closer to the smallest population sizes than to the average. (Compare the lecture's example, 1,000, then 10, then 1,000, which gives $N_e \approx 29$ against an arithmetic mean of 670.) This is one reason why N_e is usually much smaller than the census size in real populations.

6. **Founder effects: how different are the new populations from the source population, and from each other? How would this change with 50 founders?**

   - Although the source population has a frequency of 0.5, the founded populations typically range from about 0.3 to 0.7, and some are further off. They also differ from each other. The spread reflects sampling only 20 alleles (standard deviation about $\sqrt{0.5 \times 0.5/20} \approx 0.11$). With 50 founders (100 alleles), the spread is smaller (standard deviation about 0.05), so the new populations resemble the source population more closely.

Conclusions

1. **What are the limitations of this simulation?**

   - The model makes the assumptions listed at the start, and each is a limitation:
     - **No selection:** in reality, some alleles affect survival or reproduction.
     - **No mutation:** new alleles never arise, so variation can only be lost. In real populations, mutation replenishes variation (mutation-drift balance).
     - **No migration:** real populations exchange genes, which can restore lost alleles and counteract drift.
     - **Random mating:** real populations may show inbreeding or assortative mating.
     - **Constant population size and non-overlapping generations:** real populations fluctuate in size and have overlapping generations, and not all individuals breed. The relevant population size for drift (the *effective* population size) is usually much smaller than the census size.
     - **One locus with two alleles:** real genomes have many loci, many with more than two alleles, and linked loci are not independent.

2. **Why is genetic diversity important to a population, for example in a conservation context?**

   - Genetic diversity is crucial for the adaptability and survival of a population. In a conservation context:
     - **Resilience to diseases:** a genetically diverse population is less likely to be wiped out by a single disease, as some individuals might have resistance.
     - **Adaptation to changing environments:** diversity provides the variation that selection needs if conditions change.

  - **Reduced inbreeding:** high genetic diversity reduces the chance that individuals inherit two copies of harmful recessive alleles, which lowers fitness (inbreeding depression).
  - **Long-term survival:** small populations lose diversity through drift, which is why conservation programmes try to keep populations large and connected.

3. **Can you think of real-life scenarios where understanding gene pool dynamics would be valuable?**

   - **Conservation biology:** managing captive breeding and reintroductions to retain genetic diversity, or moving individuals between isolated populations ("genetic rescue").
   - **Medical genetics:** founder effects explain why some genetic disorders are unusually common in particular populations descended from small founding groups.
   - **Agriculture:** maintaining diversity in crop and livestock breeding programmes.

4. **Challenge: how could you simulate a population that is steadily shrinking through time?**

   - Inside the loop, set the population size as a function of the generation number before sampling, for example `pop_size <- round(1000 * 0.95^(i - 1))` for a 5% decline per generation, or `pop_size <- max(10, 1000 - 10 * i)` for a linear decline to a floor of 10. As the population shrinks, drift becomes stronger and the frequency of A fluctuates more and more until one allele is fixed.

Solutions: Neutral or adaptive evolution in humans

There is no single correct ordering: groups can defend different placements. What matters is the reasoning. A good answer for each trait goes through three questions:

1. **Is the variation heritable?** If not, the trait cannot evolve by drift *or* by selection.
2. **How strongly does the trait affect survival or reproduction, in this environment?** Strong effects put the trait at the selection end of the continuum; weak or no effects put it at the drift end.
3. **Is the population large enough for that selection to beat drift?** Selection is effective only when s is clearly larger than about $1/(2N_e)$.

The take-home message is that whether a trait evolves adaptively or neutrally depends on the environment (and on population size), not just on the trait. A useful comparison is antibiotic resistance in bacteria. With antibiotics, resistance genes are strongly selected *for*. Without antibiotics they are not neutral: resistance usually has a fitness cost, so it is usually selected *against* (which is why resistance can decline when an antibiotic is used less, although compensatory mutations can reduce the cost). The environment can change both the strength and the direction of selection.

Humans are still evolving. In industrialised societies such as Denmark, almost everyone survives to adulthood, so selection now acts mainly through differences in the number of children people have (and the age at which they have them), rather than through survival. Studies of modern populations have measured selection on traits such as height, age at first birth and cholesterol in exactly this way (e.g. Byars et al. 2010, *PNAS*; Sanjak et al. 2018, *PNAS*).

Suggested placements

Trait	Heritable variation?	As foragers	In Denmark today	Where on the drift-selection continuum, and why
Metabolic efficiency	Yes, partly	Probably important: food was often scarce, so efficient use of energy aided survival and reproduction	Weak, possibly reversed: food is abundant, and very efficient storage may contribute to obesity and type 2 diabetes	Towards selection as foragers; weaker today. *Debatable*: the "thrifty gene" idea has weak direct genetic support
Eyesight acuity	Partly	Probably important for hunting and gathering	Weak: glasses correct poor vision	Towards selection as foragers; near the drift end today
Height	Yes, highly heritable (many genes)	Probably selected, but the optimum depends on environment	Modern studies detect weak selection, mainly through number of children, with differences between the sexes (e.g. Stulp et al. 2015, *Proc. R. Soc. B*)	In the middle: weak but measurable selection
Birth canal (pelvic) shape	Yes	Stabilising selection: the canal must allow birth of a big-brained baby while still allowing efficient walking	Weaker: caesarean section reduces the cost of a mismatch	Selection keeps the shape within limits, but much of the variation between populations fits drift during serial founder effects as humans spread out of Africa (Betti & Manica 2018, *Proc. R. Soc. B*; see the drift lecture). A good example of both acting together
Resistance to bubonic plague	Yes, partly (immune genes)	Weak: plague is a disease of dense agricultural and urban societies, not of small foraging groups	Weak: plague is rare and treatable	Strongly selected in medieval Europe during the Black Death (Klunk et al. 2022, *Nature*, immune genes such as *ERAP2*), not in foragers and not today. A clear example of environment-dependent selection

Trait	Heritable variation?	As foragers	In Denmark today	Where on the drift-selection continuum, and why
Eye colour	Yes, strongly (mainly *OCA2*/*HERC2*)	Probably weak, except perhaps through mate choice	Weak	Blue eyes rose quickly in Europe, which suggests some selection (on pigmentation in general, or sexual selection), but drift and "hitch-hiking" with selected pigmentation alleles are possible. *Debatable*
Earwax type (*ABCC11*)	Yes, simple (one DNA change)	Unknown; a selection signal exists, perhaps linked to sweat or body odour	Weak	Some evidence of past selection (Ohashi et al. 2011, *Molecular Biology and Evolution*), but the trait it acted on is uncertain. *Debatable*
Face symmetry	Weakly; much is developmental noise	Possibly via mate choice, as a cue of health	Weak	Mostly not heritable, so it can evolve little by either process

Examples of traits that **cannot** evolve much because their variation is mostly not heritable: the fine texture of the iris, the detailed ridge minutiae of fingerprints (these differ between identical twins), and much of the variation in face symmetry.

Gene flow between populations

- Differences that gene flow would erase: neutral variation (most DNA differences, and traits close to the drift end of the continuum). Very few migrants per generation are enough to prevent large neutral differences between populations, which is one reason why most human genetic variation is found *within* populations rather than between them.
- Differences that local selection can maintain against gene flow: traits where the selection coefficient is large compared with the migration rate. Examples are skin pigmentation (UV intensity differs with latitude), lactase persistence (dairying history), and HbS and Duffy-null (malaria). These traits vary between populations much more than most neutral markers do.

Discussion questions

- **Traits categorised differently between the two societies.** Eyesight acuity and metabolic efficiency were probably more strongly selected in foraging societies than in Denmark today. Others (e.g. plague resistance) were strongly selected only in particular historical environments. In both settings, though, selection still acts, especially through the number of children.
- **Traits with little heritable variation.** Iris texture, fingerprint detail and much of face symmetry: they can vary a lot but cannot evolve much by either process.

- **Gene flow vs local selection.** See above.
- **Preconceived notions.** Common ones are that "evolution has stopped in humans" (it has not), that "every trait is an adaptation" (many traits drift, and some barely vary heritably), and that "drift only matters in tiny populations" (it acts in every finite population; it dominates whenever s is small compared with $1/(2N_e)$).
- **Other traits to add.** Sense of smell (many olfactory receptor genes have become non-functional, consistent with relaxed selection and drift), lactase persistence, skin pigmentation, blood groups, alcohol flush reaction, ability to taste bitter compounds (*TAS2R38*), and wisdom teeth are all good candidates with some genetic evidence.

Solutions: Heritability

Predictions. If offspring were on average exactly like their parents, the slope would be 1 ($h^2 = 1$). If offspring were no more like their own parents than like any other bird, the line would be flat (slope 0, $h^2 = 0$). A slope in between means that offspring resemble their parents partly: on average, they lie part of the way between their parents' value and the population mean ("regression towards the mean").

After downloading the file, you can read in the data like this (or use the "Import Dataset" wizard in RStudio):

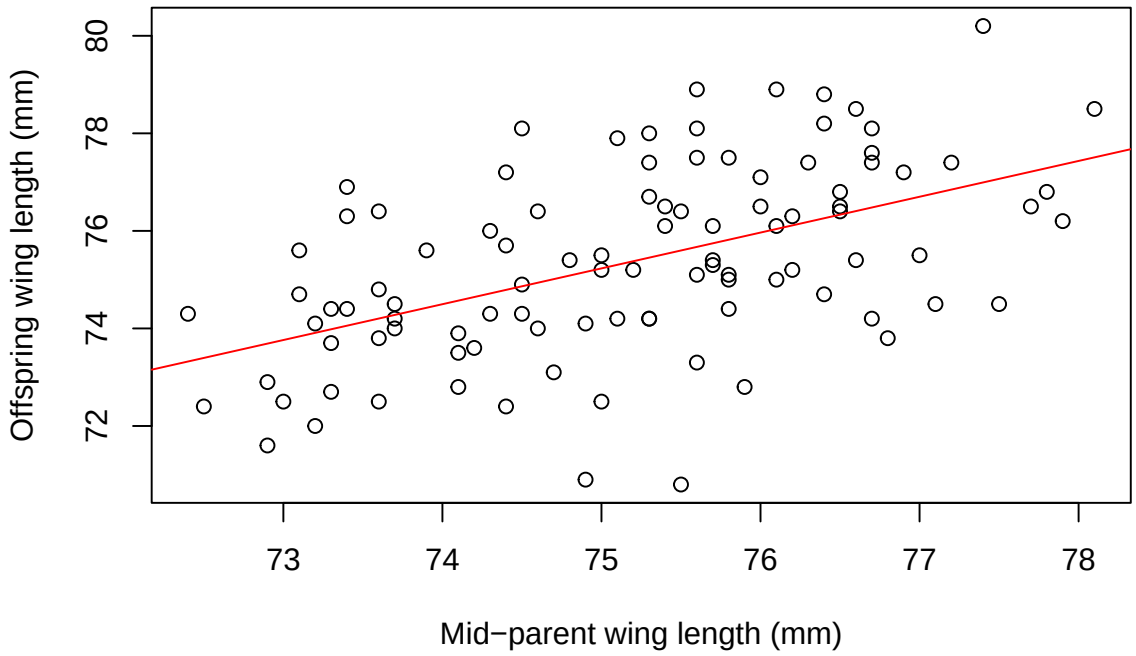
```
pop_1 <- read.csv("data/heritability_pop1.csv")
```

Plot the data and fit the model as follows.

```
# Plot the relationship
plot(pop_1$Midparent_value, pop_1$Offspring_value,
     xlab = "Mid-parent wing length (mm)",
     ylab = "Offspring wing length (mm)")

# Fit the linear regression model
mod1 <- lm(Offspring_value ~ Midparent_value, data = pop_1)

# Add the regression line to the plot
abline(mod1, col = "red")
```




```
# Summarise the model, and get the 95% confidence interval of the slope
summary(mod1)
```

```
##
## Call:
## lm(formula = Offspring_value ~ Midparent_value, data = pop_1)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.7982 -1.0608  0.0486  1.1242  3.2360
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)      20.1633     9.1736   2.198   0.0303 *
## Midparent_value   0.7342     0.1220   6.018 3.06e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.663 on 98 degrees of freedom
## Multiple R-squared:  0.2699, Adjusted R-squared:  0.2624
## F-statistic: 36.22 on 1 and 98 DF,  p-value: 3.059e-08
```

```
confint(mod1)[2, ]
```

```
##     2.5 %    97.5 %
## 0.4921315 0.9763443
```

What is your estimate of h^2, and how precise is it?

The heritability is the slope of the relationship: 0.73, with a 95% confidence interval of 0.49 to 0.98. This is a high heritability, but the interval is wide: anything from about 0.5 to nearly 1 is compatible with the data. (The data were simulated with a true value of 0.70.)

What does this heritability tell us about the variation in wing length in this population? What does it not tell us about any one bird?

About 73% of the variation in wing length among birds in this population is due to additive genetic differences between them. It does not mean that 73% of any one bird's wing length "comes from its genes": heritability describes variation in a population, not individuals. It also applies only to this population in this environment.

What else might explain the remaining variation?

The remaining 27% or so is due to differences in the environment that the birds grew up in (for example food supply, brood size, weather during the nestling period, parasites), to chance events during development, to measurement error, and to any non-additive genetic effects (dominance and interactions between genes).

Calculate V_P and V_A.

```
VP_1 <- var(pop_1$Offspring_value)
VA_1 <- coef(mod1)[2] * VP_1
VP_1
```

```
## [1] 3.7496
```

```
VA_1
```

```
## Midparent_value
##        2.753098
```

$V_P = 3.75$ mm^2 and $V_A = h^2 \times V_P = 0.73 \times 3.75 = 2.75$ mm^2. We use the offspring values for V_P because each offspring value is a single bird's phenotype. Each mid-parent value is the mean of two birds, and averaging two independent values halves the variance, so `var(pop_1$Midparent_value)` (here 1.88 mm^2) is only about half of V_P.

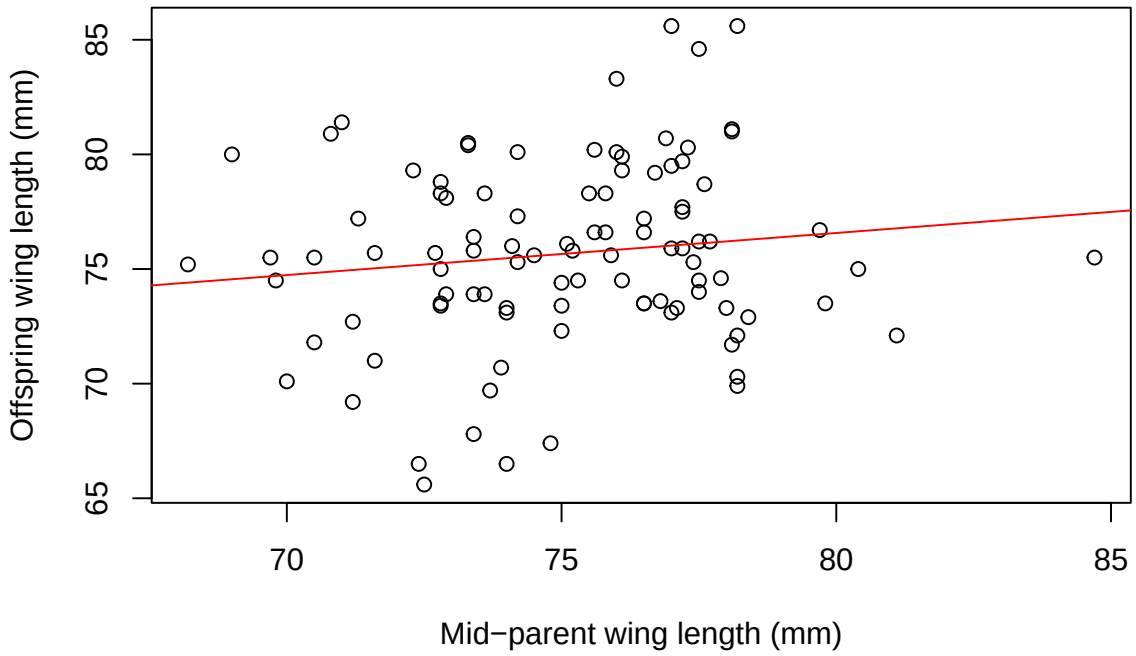
Predict the response to selection with $S = 2$ mm.

$R = h^2 S = 0.73 \times 2 = 1.47$ mm: the mean wing length of the next generation should increase by about 1.5 mm.

The graph for the second population looks like this:

```
pop_2 <- read.csv("data/heritability_pop2.csv")
plot(pop_2$Midparent_value, pop_2$Offspring_value,
     xlab = "Mid-parent wing length (mm)",
     ylab = "Offspring wing length (mm)")

mod2 <- lm(Offspring_value ~ Midparent_value, data = pop_2)
abline(mod2, col = "red")
```




```
summary(mod2)
```

```
##
## Call:
## lm(formula = Offspring_value ~ Midparent_value, data = pop_2)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.5957 -2.3746  0.0433  2.8487  9.5781
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)      61.8846    10.4816   5.904 5.11e-08 ***
## Midparent_value   0.1836     0.1394   1.317    0.191
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.969 on 98 degrees of freedom
## Multiple R-squared:  0.01738,    Adjusted R-squared:  0.007357
## F-statistic: 1.734 on 1 and 98 DF,  p-value: 0.191
```

```
confint(mod2)[2, ]
```

```
##      2.5 %     97.5 %
## -0.0931097  0.4603131
```

```
VP_2 <- var(pop_2$Offspring_value)
VA_2 <- coef(mod2)[2] * VP_2
VP_2
```

```
## [1] 15.8679
```

```
VA_2
```

```
## Midparent_value
##        2.913374
```

How does the heritability compare with population 1? Do the two confidence intervals overlap?

For this population the heritability is 0.18 (95% confidence interval -0.09 to 0.46), much lower than in population 1 (0.73). The two confidence intervals only just fail to overlap, and the interval for population 2 includes zero. That does not mean that h^2 is zero: the data are compatible with anything from no additive variance up to about 0.46. (The true values used in the simulation were 0.18 and 0.70.) Note how wide both intervals are, even with 100 families: heritability is hard to estimate precisely, and estimates from a few dozen families are very uncertain. Report the uncertainty, not just the slope. (Non-overlapping 95% confidence intervals are a conservative, informal check, not a formal test of the difference.)

Compare V_A and $V_P - V_A$ between the populations.

	Population 1	Population 2
h^2	0.73	0.18
95% confidence interval	0.49 to 0.98	-0.09 to 0.46
V_P (mm²)	3.75	15.87
V_A (mm²)	2.75	2.91
$V_P - V_A$ (mm²), an estimate of V_E	1.00	12.95

The estimates of the additive genetic variance are similar (both data sets were simulated with $V_A = 2.8$ mm²), but they are uncertain. Multiplying the confidence limits for h^2 by V_P gives roughly 1.85 to 3.66 mm² in population 1 and -1.48 to 7.30 mm² in population 2 (a variance cannot really be negative; this just shows how wide the interval is). So these data alone cannot show that V_A is equal. The clear difference is in the phenotypic variance: population 2 is about four times as variable. Because we simulated the data, we know that all of the extra variance is environmental, so heritability is lower in population 2 not because it has less genetic variation, but because its environment is more variable. With real data, this would be the most likely explanation rather than a certainty. Either way, heritability is a property of a population in an environment, not of a trait.

Can you suggest environmental differences that could explain this?

Anything that makes the conditions in which chicks grow up more variable in population 2 would do it. For example:

- **Food supply:** in a patchy habitat, some territories are rich and others poor, so chicks in different nests get very different amounts of food.
- **Weather:** cold, wet spells during the nestling period reduce growth in some broods but not others, especially if breeding is spread over a long season.
- **Brood size and competition:** chicks in large broods get less food each.
- **Parasites and disease:** infection during growth reduces final size in some individuals.

In each case V_E increases, so V_P increases and $h^2 = V_A/V_P$ falls, even though V_A is unchanged.

For the same selection differential, which population would respond faster to selection?

Population 1. With $S = 2$ mm, $R = 0.73 \times 2 = 1.47$ mm in population 1 but only $0.18 \times 2 = 0.37$ mm in population 2. In population 2, most of the difference between the selected parents and the rest of the population is environmental, and is not passed on.

Can you think of any real-world applications where understanding heritability would be important?

- Agriculture: plant and animal breeding programmes use heritability to predict how fast yield, disease resistance or growth rate will respond to selection.
- Conservation biology: the heritability of traits linked to survival and reproduction (for example breeding time, or tolerance of heat) indicates how fast a population might adapt to environmental change.
- Fisheries and hunting: harvesting the largest individuals is a form of selection; heritability predicts how fast body size or age at maturity will evolve in response (compare the bighorn sheep example from the lecture).
- Medicine: heritability estimates indicate how much of the variation in a disease risk is genetic, which guides the search for the genes involved (but a heritable condition can still be prevented or treated by changing the environment).

Violation of assumptions

Shared environment, correct parentage and measurement accuracy are the three set in class; the others are the optional extension.

- **Additive genetic effects:** dominance and interactions between genes are not passed on predictably, so the slope estimates only the additive part; genotype-by-environment interactions mean that the estimate applies only to the environment studied. These limit what the estimate means, rather than making it too high or too low.
- **No shared environment:** if parents and offspring share a good or a poor territory, they resemble each other for non-genetic reasons, and h^2 is overestimated. Cross-fostering removes this bias.
- **Correct parentage:** extra-pair paternity means that some chicks are not the offspring of their social father, which weakens the resemblance and underestimates h^2.
- **Linearity:** if the relationship is curved (for example because of dominance, or because extreme parents produce offspring that do poorly), a straight line is a poor summary, with no fixed direction of error.
- **Measurement accuracy:** errors in measuring the parents make the slope shallower (heritability is underestimated); errors in the offspring add noise and widen the confidence interval.
- **Random mating:** if birds choose mates of similar size (assortative mating), the mid-parent values vary more than under random mating, but their covariance with the offspring rises in step, so the slope on the mid-parent value still estimates h^2. This is an advantage of using both parents: the slope on a single parent would be steeper than $h^2/2$, so doubling it would overestimate h^2.
- **Representative sample:** it matters which birds are filtered. If selection acts on the offspring before they are measured (for example, only those that survived a harsh winter, with survival depending on wing length), the slope is too shallow and h^2 is underestimated. Choosing parents on the basis of the trait does not bias the slope, although a narrower range of parents gives a less precise estimate.
- **Independence:** if the same parent appears in several families, or several chicks from one nest are entered as separate points, the data points are not independent and the confidence interval will be too narrow.

Solutions: Lotka-Volterra competition

You can find a completed Excel spreadsheet here.

The full answers from the worksheet are here. Brief answers are given below. They use Neal's notation (*Interspecific Competition*): α is the PDF's α_{12} (effect of species 2 on species 1) and β is the PDF's α_{21} (effect of species 1 on species 2).

The general rule. Ask whether each species can invade when it is rare and the other species is at its carrying capacity:

- When species 2 is at K_2, a rare species 1 experiences crowding equivalent to αK_2 of its own individuals. It can grow if this is less than its own carrying capacity: $\alpha K_2 < K_1$, or $\alpha < K_1/K_2$. On the isocline graph, this means $K_2 < K_1/\alpha$.
- In the same way, a rare species 2 can grow when species 1 is at K_1 if $\beta K_1 < K_2$, or $\beta < K_2/K_1$. On the isocline graph, $K_1 < K_2/\beta$.

Species 1 can invade?	Species 2 can invade?	Outcome (question)
Yes	No	Species 1 excludes species 2 (Q1)
No	Yes	Species 2 excludes species 1 (Q2)
Yes	Yes	Stable coexistence (Q3)
No	No	Either species can win, depending on the starting densities (Q4)

Q1. For example $R_1 = 0.25$, $K_1 = 3000$, $\alpha = 1.25$; $R_2 = 0.5$, $K_2 = 1000$, $\beta = 0.5$; start $N_1 = 50$, $N_2 = 100$. Species 1 goes to 3000 and species 2 to zero (after 50 time steps the sheet shows about 2979 and 2.4). Species 1 can invade ($\alpha = 1.25 < K_1/K_2 = 3$) but species 2 cannot ($\beta = 0.5 > K_2/K_1 = 0.33$). This happens even though species 2 grows faster, starts larger and has the larger per capita effect ($\alpha > \beta$). Changing R or the starting numbers does not change the outcome. In ecological terms, species 1 can keep growing even when species 2 is at its own carrying capacity, but not the other way round.

Q2. For example $R_1 = 0.25$, $K_1 = 3000$, $\alpha = 2.25$; $R_2 = 0.25$, $K_2 = 2000$, $\beta = 0.25$; start 50 and 50. Species 2 goes to 2000 and species 1 to zero, even though $K_1 > K_2$: species 1 cannot invade ($\alpha = 2.25 > K_1/K_2 = 1.5$) but species 2 can ($\beta = 0.25 < K_2/K_1 = 0.67$).

Q3. For example $R_1 = 0.75$, $K_1 = 2500$, $\alpha = 0.75$; $R_2 = 0.5$, $K_2 = 2000$, $\beta = 0.5$. Both species can invade ($0.75 < 1.25$ and $0.5 < 0.8$), and the populations settle at $N_1 = 1600$, $N_2 = 1200$. You can check this with the equilibrium formulae $N_1^* = (K_1 - \alpha K_2)/(1 - \alpha\beta)$ and $N_2^* = (K_2 - \beta K_1)/(1 - \alpha\beta)$. The equilibrium is stable: starting from (1000, 1), (1, 1000), (500, 2500), (1000, 3000) or (250, 3000), the populations always return to (1600, 1200). The worksheet's default values also give stable coexistence, at (720, 640). In ecological terms, each species limits its own growth more than it limits its competitor's, once you allow for differences in carrying capacity. If the carrying capacities are similar, this means $\alpha < 1$ and $\beta < 1$: each individual competes more strongly with its own species than with the other species, for example because the two species partly use different resources (niche differences).

Q4. For example $R_1 = R_2 = 0.5$, $K_1 = K_2 = 1000$, $\alpha = \beta = 1.25$. Neither species can invade ($1.25 > 1$), so the isoclines cross at an *unstable* equilibrium, $N_1 = N_2 = 444.4$. Starting at (50, 50) the populations go to this point and stay there, but whichever species starts larger wins: (100, 50) and even (445, 444) end at (1000, 0); (50, 100) and (444, 445) end at (0, 1000). The growth rates matter too: starting at (50, 50) with $R_1 = 0.8$, $R_2 = 0.4$, species 1 wins, and with the R values swapped species 2 wins. In ecological terms, each species affects the other more than it affects itself: interspecific competition is stronger than intraspecific competition, for example through interference or mutual predation.

Fifth case. If the isoclines lie exactly on top of each other ($K_1 = K_2/\beta$ and $K_2 = K_1/\alpha$), every point on the line is an equilibrium. For example, with identical species ($\alpha = \beta = 1$, $K_1 = K_2 = 1000$, $R_1 = R_2 = 0.5$), starting at (100, 300) the populations end at (250, 750) and starting at (300, 100) they end at (750, 250): they keep their starting ratio. So the basic model does not predict that identical species exclude each other; that conclusion comes from Gause's experiments (see Neal, *Interspecific Competition*).

Q5 (optional). (a) *Rhizopertha* and *Oryzaephilus*: both can invade ($\alpha = 0.235 < K_1/K_2 = 0.75$; $\beta = 0.12 < K_2/K_1 = 1.33$), so they coexist, at $N_1 = 233$, $N_2 = 412$. Neal reports that this is what Crombie observed. (b) *Tribolium confusum* and *T. castaneum*: neither can invade ($\alpha = 1 > 0.65$; $\beta = 2.2 > 1.55$), so the winner depends on the starting densities. With $R_1 = R_2 = 0.5$, starting at (100, 20) *T. confusum* wins (220, 0), and starting at (20, 100) *T. castaneum* wins (0, 340). Park found the same in his experiments, with an "indeterminate zone" of starting densities where either species could win by chance.

Note that comparing a coefficient with 1 is not enough when the carrying capacities differ. In Q1, species 1 wins even though $\alpha = 1.25 > 1$, because its carrying capacity is three times larger.

Solutions: Lotka-Volterra predator-prey dynamics

You can find a completed Excel spreadsheet here.

The worksheet's own answers are here. **One of them is wrong:** the answer to question 3 says that smaller values of f delay extinction and larger values hasten it. That is not what the spreadsheet does (see below). Use the answers here instead.

All numbers come from the spreadsheet model ($V_{t+1} = V_t + RV_t - aC_tV_t$, $C_{t+1} = C_t + afV_tC_t - qC_t$, with populations not allowed to go below zero), with the default values $R = 0.25$, $a = 0.01$, $f = 0.008$, $q = 0.1$, starting at $V = 1000$, $C = 20$, copied down far enough to see the prey hit zero. In Neal's notation, $V = H$, $C = P$, $R = r$, $af = c$ and $q = d$.

Change from default	Time step at which the prey hit zero
None	220
$R = 0.15$ / 0.4	416 / 55
$q = 0.05$ / 0.2	398 / 58
$f = 0.004$ / 0.006 / 0.010 / 0.012 / 0.016	129 / 173 / 200 / 203 / 154
$a = 0.005$ / 0.0075 / 0.015 / 0.02	78 / 122 / 257 / 175

1. **R.** A larger prey growth rate makes the system *less* stable: the cycles grow faster and extinction comes sooner. Faster prey growth allows faster predator growth, and each population responds one whole time step late, so the overshoot is bigger.
2. **q.** Predators that starve faster also make the system less stable (extinction sooner); slower starvation delays it. With high q the predators crash when prey are scarce, so the prey recover to very high numbers; the predators then explode and wipe them out.
3. **f.** The effect of conversion efficiency is **not simple**. From the default start, both the smallest ($f = 0.004$) and the largest ($f = 0.016$) values give the earliest extinction, and intermediate values the latest. The reason is that f only moves the predator isocline ($V = q/(af)$: 2500 prey at $f = 0.004$, 1250 at the default, 625 at $f = 0.016$). It does not change how fast the discrete model spirals outwards. What changes is how far the fixed starting point (1000, 20) is from the new equilibrium: the further away it is, the bigger the first cycle and the sooner the crash. If you start at the same relative position (80 % of the equilibrium numbers), $f = 0.004$, 0.008 and 0.016 all give extinction at time step 220.
4. **a.** The attack rate moves both isoclines ($C = R/a$ and $V = q/(af)$), so its effect also depends on where you start: from the default start, both a much lower ($a = 0.005$) and a somewhat higher ($a = 0.02$) attack rate bring extinction forward, but $a = 0.015$ delays it.
5. **Starting numbers.** Yes. Starting exactly where the isoclines cross ($V = 1250$, $C = 25$), nothing changes. The further from that point you start, the larger the cycles and the sooner the extinction.
6. **Ultimate outcome.** In the spreadsheet, both species always go extinct in the end: the cycles grow until the prey hit zero, and then the predators starve. With the defaults this takes 220 time steps, longer than the 100 steps in the original sheet, but the growing cycles show it coming. This is caused by the discrete time steps (each population responds to the other one step late), not by the biology: in the continuous-time model the cycles keep the same size for ever (see the solutions to the continuous- and discrete-time R chapters). Near the equilibrium, the distance from the equilibrium grows by a factor of about $\sqrt{1 + Rq}$ per time step. This contains R and q but not a or f, which is why R and q have simple effects in the table and a and f do not. Real predator and prey populations rarely show either growing or perfectly constant cycles. Factors missing from the model include prey carrying capacity, refuges for the prey, predators that become satiated (a Type II functional response), alternative prey, predator interference or territoriality, spatial structure and environmental variation. Parts 2 to 4 of the worksheet add some of these: refuges stop the prey going extinct, and a prey carrying capacity turns the expanding oscillations into damped ones that settle at an equilibrium (Neal discusses this in *Predator-Prey Interactions*).

Solutions: Continuous-time Lotka-Volterra predator-prey model in R

All numbers below come from the chapter code with the default parameters ($R = 0.25$, $a = 0.01$, $f = 0.008$, $q = 0.1$), start $V = 1000$, $C = 20$, run to time 200, unless stated otherwise.

1. Oscillations

- Both populations cycle regularly and for ever. The prey peak first (at $t \approx 13.7, 53.8, 93.9, ...$) and the predators peak about a quarter of a cycle later ($t \approx 22.4, 62.5, 102.6, ...$). Prey increase while predators are scarce; predators then increase because food is plentiful; the larger predator population pushes the prey down; the predators then starve and decline, and the cycle repeats.
- With prey on the x-axis the orbit runs **anticlockwise**. Below the prey isocline ($C < R/a = 25$) prey increase (move right); above it they decrease (move left). To the right of the predator isocline ($V > q/(af) = 1250$) predators increase (move up); to the left they decrease (move down). Following these arrows round the four regions gives right, up, left, down: anticlockwise.

2. Neutral stability and starting numbers

- Starting exactly at $V = 1250$, $C = 25$ nothing changes: this is the equilibrium. Starting just next to it ($V = 1300$, $C = 25$) gives a very small cycle (prey 1201 to 1300, predators 24.4 to 25.6), which does not grow or shrink.
- Further away, the cycles are bigger: from $V = 2000$, $C = 20$ the prey range is 632 to 2180; from $V = 1000$, $C = 40$ it is 475 to 2601 (predators 14 to 41). In every case the size of the cycle stays the same for the whole run: the oscillations neither grow nor die away. This is **neutral stability**: the model has no tendency to return to the equilibrium or to move away from it, so the size of the cycles is set entirely by where you start.
- The period changes a little with the size of the cycle: about 39.7 time units for the tiny cycle, about 40 at the default start, about 40.5 to 41.5 for the larger ones. For small cycles the period is $2\pi/\sqrt{Rq} = 2\pi/\sqrt{0.025} \approx 39.7$.

3. Equilibrium and time-averages

- $V^* = q/(af) = 0.1/(0.01 \times 0.008) = 1250$ and $C^* = R/a = 0.25/0.01 = 25$.
- The populations never settle there, but `mean(output_df$V)` = 1250.6 and `mean(output_df$C)` = 25.01: the **averages over the cycles equal the equilibrium** (the small difference is because the run does not end at the end of a complete cycle).

4. Moving the isoclines

- Doubling R (to 0.5) raises the prey isocline to $C = 50$ (more predators are needed to stop the prey growing). Doubling a (to 0.02) lowers the prey isocline to $C = 12.5$ *and* moves the predator isocline left to $V = 625$. Doubling f (to 0.016) moves only the predator isocline, to $V = 625$. Doubling q (to 0.2) moves the predator isocline right, to $V = 2500$. Because the starting point (1000, 20) is now a different distance from the new equilibrium, the cycles also change size.
- With $R = 0.2$ and $q = 0.12$ the equilibrium moves to $V^* = 1500$, $C^* = 20$. The run gives average prey 1511 and average predators 19.95. So killing both species **increases the average number of prey** and decreases the average number of predators. This is Volterra's principle: an indiscriminate mortality factor (pesticide, fishing) can benefit the prey because it hits their predators.

5. Extinction and reality

- From $V = 1000$, $C = 100$ the predators fall to about 2.0 and the prey to about 8. In the model neither population can ever reach exactly zero (the growth rates are proportional to population size, so the populations approach zero but never reach it). In a real population, a handful of individuals, or a fraction of one, means extinction, and chance events (demographic stochasticity) would make it very likely.

- Unrealistic assumptions include: prey grow exponentially without predators (no carrying capacity); each predator eats in proportion to prey density without limit (no handling time or satiation, i.e. a Type I functional response); predator death rate is constant; no time lags, no refuges, no other prey or predators; individuals meet at random; populations are large enough to treat as continuous. Prey carrying capacity, predator satiation, refuges, alternative prey, spatial structure and environmental noise will all change the cycles: some damp them, others destabilise them, so real cycles are rarely neutrally stable.

Solutions: Discrete-time Lotka-Volterra predator-prey model in R

All numbers below come from the chapter code with the default parameters ($R = 0.25$, $a = 0.01$, $f = 0.008$, $q = 0.1$), start $V = 1000$, $C = 20$ and `total_time = 300`. The model is deterministic, so you should get exactly these numbers.

- **Default run ($\Delta t = 1$).** Instead of a closed loop, the phase plot spirals **outwards**: each cycle is a little larger than the one before. By the last cycle the prey reach about 16,000 and the predators about 230. The prey hit zero at $t = 220$; after that the predators have nothing to eat and dwindle towards zero (about 0.05 by $t = 300$). The populations "react one step too late": the change over each step is calculated from the numbers at the start of the step, so the prey keep growing for a whole step after predators have become numerous enough to stop them, and so on round the cycle.
- **Smaller time steps.** With $\Delta t = 0.1$ the spiral still grows, but much more slowly: over 300 time units the prey stay between about 670 and 2120 and neither population goes extinct. With $\Delta t = 0.01$ the orbit is almost closed (prey up to about 1850), very close to the continuous-time model (prey 812 to 1823). Small time steps are just a more accurate way of calculating the continuous model.
- **Larger time step.** With $\Delta t = 2$ the spiral grows faster and the prey hit zero earlier, at $t = 142$.
- **Which parameters control the spiral?** R and q. Doubling R (to 0.5) brings the prey crash forward from $t = 220$ to $t = 64$; doubling q (to 0.2) brings it forward to $t = 58$. Doubling a or f also changes the time of the crash from the default start ($t = 175$ and $t = 154$), but only because the equilibrium moves and the start is then a different distance from it. If you start the same relative distance from the equilibrium (e.g. at 80 % of V^* and C^*), doubling a or f makes no difference (crash at $t = 220$, as with the defaults), while doubling R ($t = 79$) or q ($t = 132$) still speeds it up.
- **Why.** Near the equilibrium, each step multiplies the distance from the equilibrium by about $\sqrt{1 + \Delta t^2 Rq}$. With the defaults this is $\sqrt{1 + 0.025} \approx 1.012$, about 1 % growth per step. The factor contains R, q and Δt but not a or f, and it is always bigger than 1, so the discrete model always spirals out eventually; it gets closer to 1 (no growth, as in the continuous model) as Δt gets smaller.

Solutions: Game theory: Hawks and doves

Your own class results will differ from year to year. A worked analysis of real class data (2022) is in the Results of the hawk-dove games chapter.

- *Predictions before playing*

Game Two (new partner every round): with B = 4 and C = 3, Hawk earns more than Dove whatever the opponent plays (4 > 2 against a Dove, 0.5 > 0 against a Hawk), so theory predicts mostly Hawk play, and that more hawkish players earn more.

Game One (same partner for 15 rounds): a good prediction is that Hawk play will be less common than in Game Two, and that pairs who settle into mutual Dove (2 each per round) will earn far more than pairs stuck in mutual Hawk (0.5 each per round). Taking turns (Hawk/Dove, then Dove/Hawk) also averages 2 per round each, so it is just as good as mutual Dove with these payoffs. In round 15 there is no future round in which to be punished, so theory predicts more Hawk play in the last round. (In practice people often keep cooperating; in the 2022 class, Hawk play in round 15 was no higher than average.)

- *Did the % Hawk trend in Game One and Game Two match what you predicted?*

Compare your class's charts with your predictions. In the 2022 class, neither game showed a trend in Hawk play across rounds (p = 0.9 for Game 1 and 0.72 for Game 2), and Hawk play was only slightly lower in Game 1 (0.62 vs 0.69 per player; not significant). A flat line does not mean nothing happened: many pairs settle on a pattern within the first few rounds and then keep to it.

- *Did the total-benefit-vs-hawkishness pattern support the one-shot theory prediction (Hawk pays off best), or the repeated-game reciprocity argument (Dove pays off best when you keep meeting the same partner)?*

Both, depending on the game and on how you look at it.

- Within a pair in Game 1, the more hawkish player *always* earns at least as much as their partner, because Hawk earns more than Dove in every round against the same opponent. So comparing partners always favours Hawk.
- Across pairs in Game 1, more hawkish *pairs* earn less: escalation provokes escalation, and both players end up with 0.5 per round instead of 2. This supports the reciprocity argument (2022 class: pair mean payoff fell by about 2 per round from an all-Dove to an all-Hawk pair).
- In Game 2, more hawkish individuals earned more (2022 class: slope 1.21 per round, p = 0.006), as the one-shot theory predicts.
- *Which game, if either, better matches what you'd expect from a species whose individuals encounter each other only once versus a species that lives in small, stable social groups?*

Game 2 is like a species in which individuals rarely meet the same opponent twice: the one-shot logic applies, and selection should favour Hawk when B > C. Game 1 is like a small, stable social group in which the same individuals meet repeatedly: restraint can pay because escalation is answered in later encounters (reciprocity, the "shadow of the future"). Being able to recognise individuals and remember past encounters is required for this.

- *Did you or your partner take turns? What happened in round 15?*

Taking turns gives each player (4 + 0)/2 = 2 per round, exactly the same as mutual Dove, so it is a perfectly good way to cooperate in this game. (With Neal's payoffs, which include a display cost, taking turns is actually better than mutual Dove.) In round 15, a player who has cooperated so far gains by switching to Hawk (4 instead of 2) because the partner cannot retaliate. If your class showed a jump in Hawk play in the last round, that is the end-game effect; if not, people cooperated beyond what strict theory predicts.

- *What would happen if you increased or decreased the value of the benefit (B) relative to the cost (C)? Would that change which strategy dominates in a one-off encounter?*

Yes. Let p be the frequency of Hawks in the population (with no display cost, D = 0). The average payoffs are:

- Hawk: $p\frac{B-C}{2} + (1-p)B$
- Dove: $(1-p)\frac{B}{2}$

If B > C (as in the class game), Hawk always earns more than Dove, so Hawk is the ESS: a population of Hawks cannot be invaded by Doves. Making B larger relative to C only strengthens this.

If C > B, Hawks do well when rare (they beat the Doves) but badly when common (they keep meeting and injuring each other), and Doves do the opposite. Neither pure strategy is stable. Setting the two payoffs equal and solving gives the mixed ESS:

$$p^* = \frac{B}{C}$$

Worked example: B = 4, C = 8. Then $p^* = 4/8 = 0.5$. Check: Hawk earns $0.5 \times (4-8)/2 + 0.5 \times 4 = -1 + 2 = 1$; Dove earns $0.5 \times 4/2 = 1$. The payoffs are equal, so neither strategy can do better by becoming more common. If Hawks were more common than 0.5, Doves would earn more and increase, and vice versa.

The more costly fighting is relative to the prize, the fewer Hawks there are at equilibrium (with B = 4: C = 5 gives $p^* = 0.8$, C = 8 gives 0.5, C = 16 gives 0.25). Neal's version, which includes a cost of display D for Doves, gives $p^* = (B/2 + D)/(C/2 + D)$; with B = 50, C = 100 and D = 10 this is 7/12, the example used in the lecture.

Note that the ESS is not the best outcome for the group: at the mixed ESS with B = 4, C = 8, everyone earns 1 per encounter, whereas a population of Doves would earn 2 each. A population of Doves cannot resist invasion by Hawks.

Take-home messages:

- In a one-off contest with B > C, Hawk is the ESS, even though everyone would do better as Doves.
- When C > B, the ESS is a mix of Hawks and Doves with Hawk frequency p* = B/C.
- Repeated interaction with the same partner changes the incentives: escalation is answered in kind, so restraint can pay.

Solutions: The legend of Ambalappuzha

You can find the completed Excel file here.

How much rice? The 64 squares hold $2^{64} - 1 = 1.84 \times 10^{19}$ grains of rice. At 25 mg (0.000025 kg) per grain, that is about 4.6×10^{14} kg (461 billion tonnes): hundreds of times the world's annual rice production.

Great tits (optional exercise). Each female lays about 6 eggs, so she produces about 3 daughters per year. We start with 100 females (one per breeding pair).

- If every female survives every year, and daughters breed the year after they hatch, each female this year becomes herself plus 3 daughters next year: $F_{t+1} = F_t + 3F_t = 4F_t$, so $\lambda = 4$. After 5 years there are $100 \times 4^5 = 102{,}400$ females.
- If instead adults die after breeding once, only the 3 daughters per female are left next year, so $\lambda = 3$ and after 5 years there are $100 \times 3^5 = 24{,}300$ females.

Either way, numbers explode within a few years. The answer depends strongly on the assumptions, which is the point of the discussion prompts: real great tits have far from 100% survival (most fledglings die in their first year), and density dependence (territories, food, nest sites) stops growth long before these numbers are reached.

Take home messages:

- Exponential/geometric growth is extremely powerful and can very quickly lead to very large numbers.
- The multiplication rate, λ, depends on both reproduction and survival, and on assumptions about the life cycle (e.g. whether adults survive to breed again).

Solutions: From population biology to fitness

The matrix model with a trade-off (increased juvenile survival, but old-adult survival reduced to 0.05) looks like this.

```
A3 <- matrix(c(0.00, 4.00, 2.00,
               0.11, 0.80, 0.00,
               0.00, 0.10, 0.05),
             byrow = TRUE, nrow = 3)

eigs(A3)$lambda
```

We can compare the λ value from this model (1.187) with the original one (1.166) to ask whether the large cost is worth paying for the small benefit. It turns out that it is. You can then ask how MUCH cost would be worth bearing, by repeating the exercise and slowly reducing old-adult survival until you reach the original λ value.

You should find that the cost can be huge, and still worth bearing. In fact, you could reduce old-adult survival to 0 and it is still worth doing ($\lambda = 1.186$)! You can even reduce old-age adult reproduction without much effect on fitness: with old-adult survival at 0.05 and old-adult reproduction set to 0, λ is 1.175, still above the original.

HOWEVER, changes to prime-age adults are much more important. With juvenile survival at 0.11, you can only reduce prime-age adult survival to about 0.76 (from 0.8) before the cost is not worth bearing (population growth rate falls below that of the baseline model).

Thus, old-age costs are "worth it" while prime-age costs are much more important. This finding is central to the evolution of senescence and life span, which you met in the practical *Trade-offs and the declining force of selection* (the declining force of selection).

Solutions: From plain English to a matrix model

Matrix. With stages hatchling (H), juvenile (J) and adult (A), and the census once a year:

- Fertilities (top row): juveniles 1, adults 3 hatchlings per year (the text says these are hatchlings, so we treat them as already counted at the next census).
- Hatchling to juvenile: 0.20.
- Juveniles survive with probability 0.70 (30% mortality). Of the survivors, 5% stay juvenile and 95% become adults: stasis $0.70 \times 0.05 = 0.035$, growth $0.70 \times 0.95 = 0.665$.
- Adults survive with probability 0.82 (18% mortality) and stay adults.

  A_hawk <- matrix(c(0.00, 1.000, 3.00, 0.20, 0.035, 0.00, 0.00, 0.665, 0.82), byrow = TRUE, nrow = 3) popdemo::eigs(A_hawk, what = "lambda") # 1.172 pr <- popdemo::project(A_hawk, vector = c(1000, 150, 5), time = 20)

Assumption about juvenile stasis. "5% of juveniles fail to transition" can also be read as 5% of *all* juveniles: stasis 0.05 and growth 0.65 (so survival is still 0.70). This gives $\lambda = 1.171$, almost the same as 1.172, so the assumption barely matters here. What matters is that you state it, and that stasis plus growth for juveniles does not exceed their survival of 0.70.

Interpretation. $\lambda \approx 1.17$, so the population grows by about 17% per year. After the transient period (the starting population is dominated by hatchlings), the projection settles into steady exponential growth, reaching about 14,300 individuals after 20 years.

Solutions: Continuous traits from discrete genes

Shape of the histograms. Both the classroom histogram and the R histogram should be roughly bell-shaped and centred near 47.5 (35 + 12.5). With only 30 individuals the classroom histogram will be lumpy; with 100 it is smoother. With genes alone, every value is a whole number, so the distribution has many classes but is still discrete.

Does the environment make the trait continuous? Yes. After adding the environmental effect, the values are no longer whole numbers, and the gaps between the genetic classes are filled in: individuals with the same number of ON genes now differ slightly, and individuals with different numbers of ON genes overlap. This is exactly Neal's point about wheat kernel colour: with little environmental effect there are only a few discrete classes, but environmental (and other small genetic) effects blur them into a continuous distribution.

Variances. In the example run shown in the chapter, `var(trait_value)` is 6.29 (expected 6.25) and `var(trait_value_env)` is 8.24 (expected $6.25 + 2.25 = 8.5$). These values vary a lot between runs: with 100 individuals, the genetic variance typically lies between about 4.7 and 8.0, and the phenotypic variance between about 6.3 and 11.1.

Heritability. All the genetic variance here is additive, so $h^2 = V_A/V_P = 6.25/8.5 = 0.74$. With an environmental standard deviation of 4, $V_E = 16$, $V_P = 22.25$ and $h^2 = 6.25/22.25 = 0.28$. The genes are exactly the same, but the more variable environment lowers the heritability, just as for the two bird populations in the heritability practical.

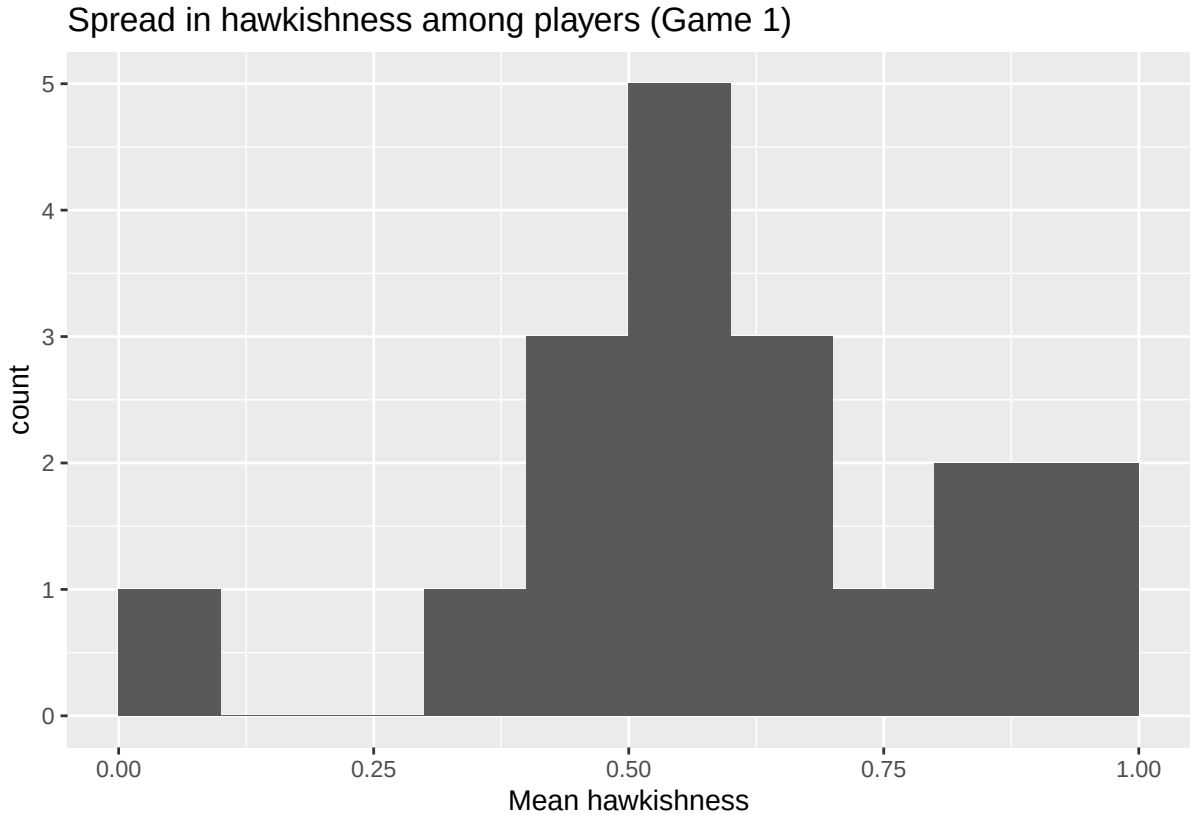
Results of the hawk-dove games

This section summarises class outcomes for Game 1 (same partner) and Game 2 (changing partners). Use it to compare with your own observations and results.

These results are from the 2022 class, who played both games.

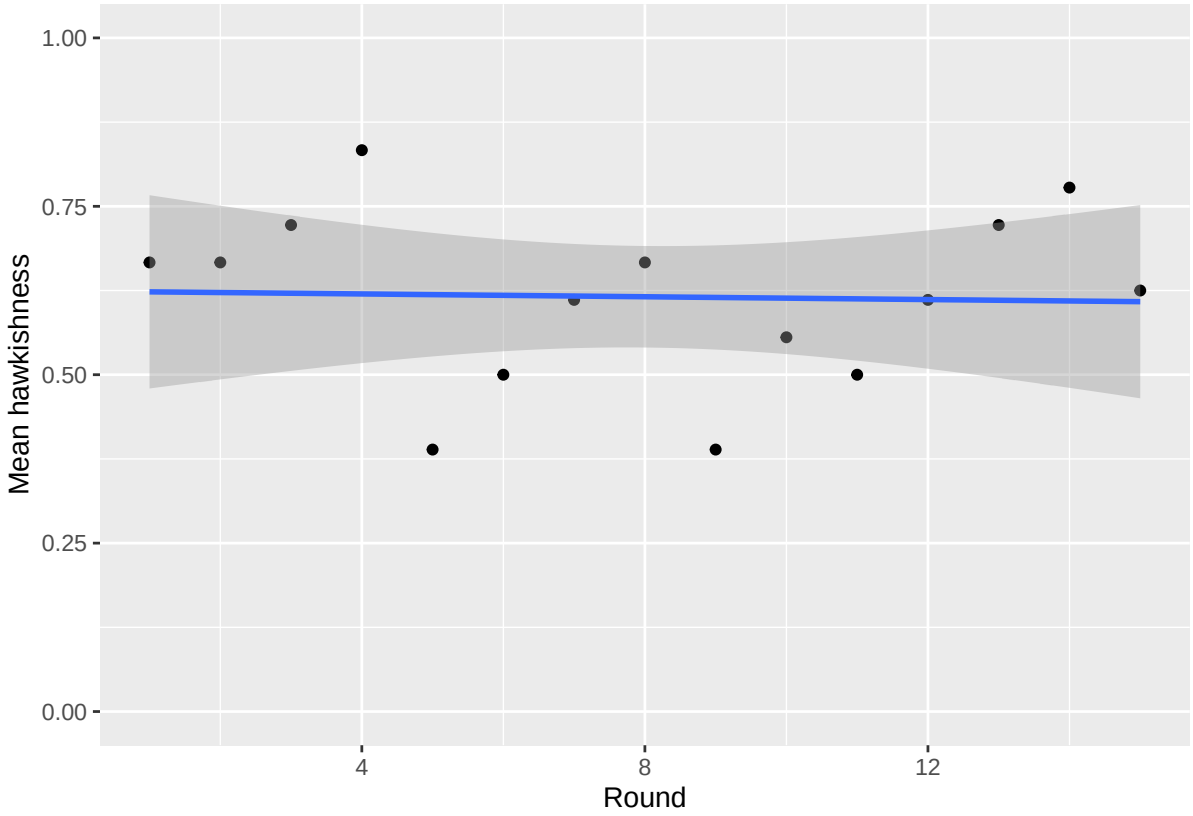
Game 1: the same partner for 15 rounds

The average hawkishness in Game 1 (the proportion of rounds in which Hawk was played) was 0.62 (averaged over players), but players differed.



Is this more spread than we would expect by chance? The lecture showed that a mixed ESS can be reached in two ways: a mixture of pure Hawks and pure Doves, or every individual playing Hawk with the same probability. If every player had played Hawk with the same probability (0.62) in each of their 15 rounds, chance alone would give a standard deviation in hawkishness among players of about 0.13. The observed standard deviation is 0.22, so these players really did differ in strategy: some were consistently hawkish, others consistently dovish.

Did the level of hawkishness change as the game went on?



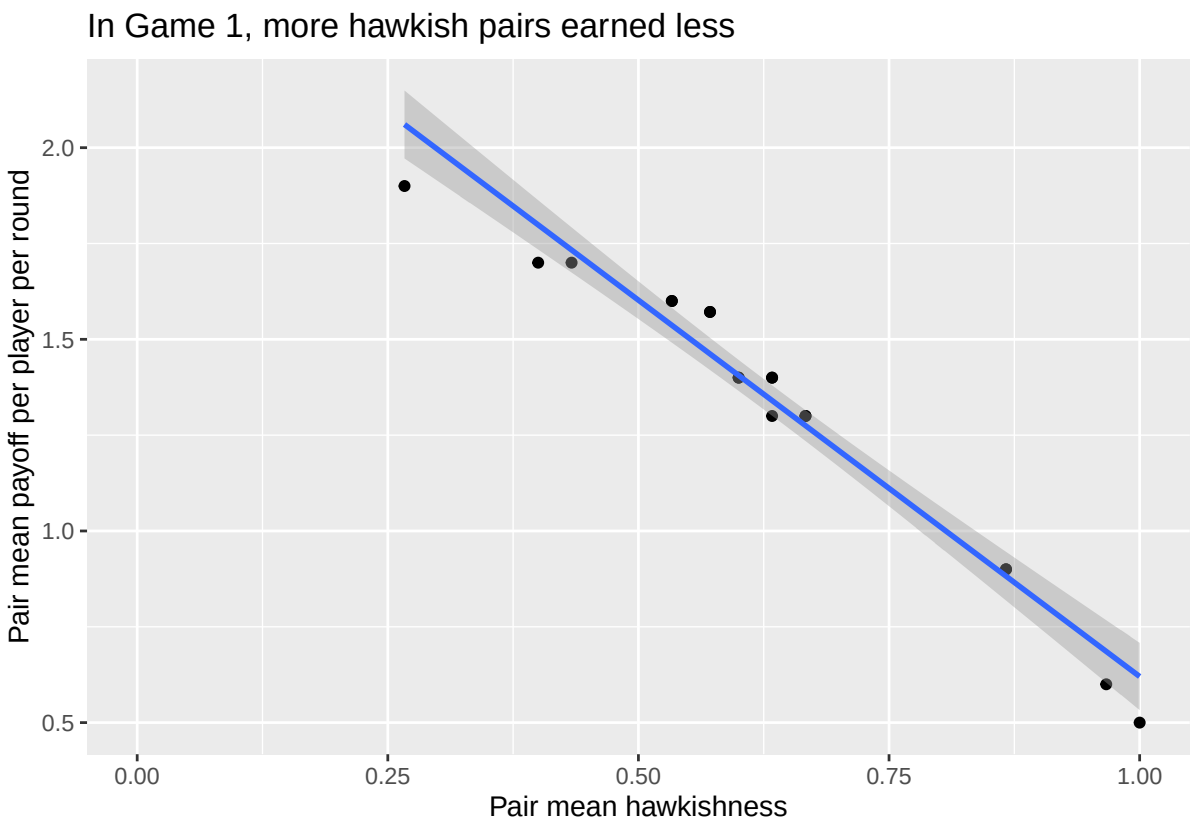
No: there is no detectable trend across the 15 rounds ($p = 0.9$). Nor was there a jump in Hawk play in the final round (round 15: 0.62), even though there was no next round in which to be punished.

Does being hawkish pay with a fixed partner?

With the payoffs used in this game ($B = 4$, $C = 3$), a *single, anonymous* encounter has a simple answer. Because $B > C$, Hawk beats Dove regardless of what your opponent plays (Hawk-vs-Dove gives $4 > 2$ for Dove-vs-Dove, and Hawk-vs-Hawk gives $0.5 > 0$ for Dove-vs-Hawk), so the one-shot ESS here is pure Hawk, not an intermediate mix. (The stable mix of Hawks and Doves, with Hawk frequency $p^* = B/C$, only arises when the cost of fighting, C, exceeds the value of the resource, B: the opposite of the situation here. With $B > C$, the payoff ranking Hawk-vs-Dove > Dove-vs-Dove > Hawk-vs-Hawk > Dove-vs-Hawk has the same structure as the classic Prisoner's Dilemma, with Hawk playing the role of "Defect".)

This has an important consequence: **within a pair, the more hawkish player always earns at least as much as their partner**, because in every round Hawk earns more than Dove against the same opponent. So asking "who won in each pair?" always favours Hawk. The interesting question for Game 1 is a different one: do pairs that play Hawk a lot end up worse off, *both of them*, than pairs that hold back?

To answer it, we look at each pair as a unit: the pair's mean hawkishness (both players' cards) against the pair's mean payoff per player per round.



The relationship is strongly negative (slope -1.96 per round): pairs that escalated a lot got stuck in Hawk-Hawk rounds worth only 0.5 each, while pairs that held back earned close to the 2 per round of mutual Dove. (The data file records players rather than pair IDs, so most pairs appear twice in this plot, once from each partner's records, and the two points usually lie on top of each other. Because of this duplication we do not quote a p-value.)

That is the pattern expected if hawkishness provokes hawkishness: with a fixed partner, playing Hawk does not make you lose to your partner, but it lowers what *both* of you earn.

Reciprocity in repeated games

This idea is not covered in the lecture, but it explains why repeated play can change the outcome.

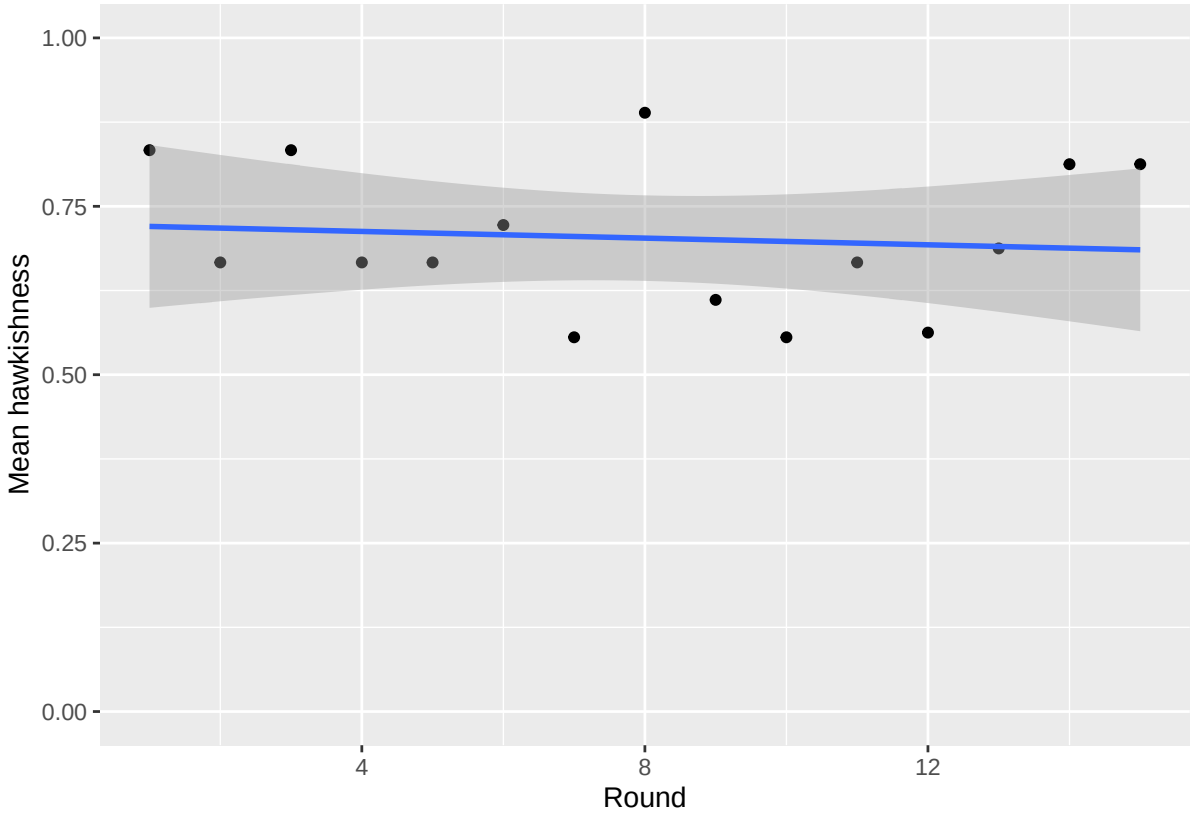
When you meet the same partner again and again, what you do now affects how they treat you later. This is sometimes called the *shadow of the future*. A simple strategy that does well in repeated Prisoner's Dilemma games is **tit-for-tat**: cooperate (Dove) in the first round, then copy whatever your partner did in the previous round. Tit-for-tat is never the first to escalate, punishes escalation immediately, and forgives as soon as the partner returns to Dove (Axelrod & Hamilton 1981, *Science* 211: 1390-1396).

Two caveats. First, in a game of known, fixed length, defecting (Hawk) in the last round always pays, because there is no future to protect, so cooperation is not guaranteed in theory (in practice people often cooperate anyway). Second, with the class payoffs, taking turns (Hawk/Dove, then Dove/Hawk) pays exactly as well as mutual Dove, 2 per round each, so there is more than one way to cooperate.

Game 2: a new partner every round

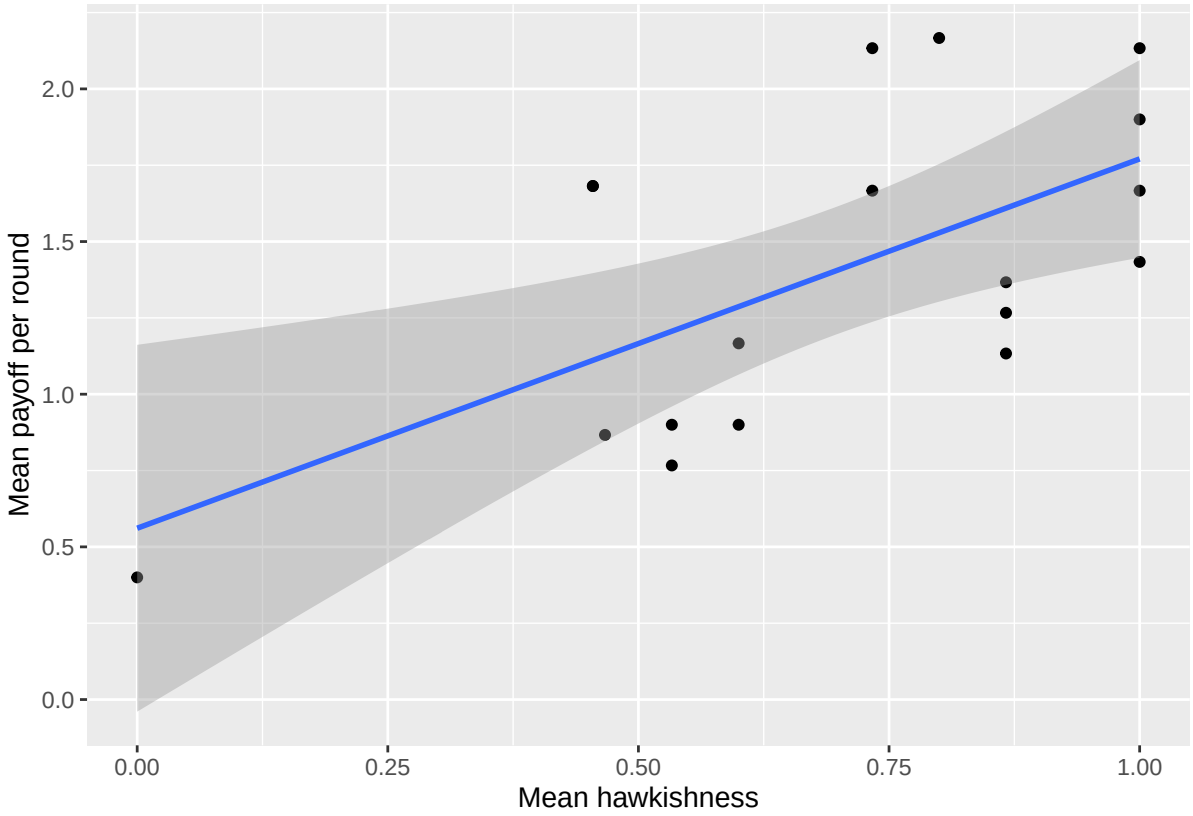
The students were asked to compete against a new opponent every round. The idea was that this would make it much harder to learn your opponent's strategy, and harder to come to an agreement that reduces aggression.

The average hawkishness in Game 2 was 0.69, a little higher than in Game 1 (0.62).



Again, hawkishness did not change systematically over the game ($p = 0.72$).

Finally, we can look at what the best strategy is in Game 2. Here each point is one player, and we use the mean payoff per round, because a few players missed some rounds.



The best strategy has changed: now the more hawkish players earned more (slope 1.21 per round, p =

0.006).

This matches the one-shot theoretical prediction much better than Game 1 did. In Game 2 you faced a new partner every round, so there was no opportunity to build reciprocity, and no way to be "punished" for playing Hawk against the same individual again. With no repeated interaction, no "shadow of the future", the game reduces to the simple one-shot logic, and since B > C, Hawk pays off best.

In this setting we might expect natural selection to drive towards the evolution of more aggressive individuals.

Comparing the two games

Our hypothesis was that, compared with Game 2, players in Game 1 would play Hawk less often and earn more per round. In this class, hawkishness was slightly lower in Game 1 (0.62 vs 0.69 per player; p = 0.34), but mean payoff per round was *not* higher (1.25 vs 1.4; p = 0.36). Neither difference is statistically significant, so with one class of 18 players we cannot reject the null hypothesis. Remember too that Game 2 was always played after Game 1, so any difference between the games could also reflect learning or fatigue.

What did differ clearly between the games was *which* strategy paid: within Game 1, pairs that held back did best, while within Game 2, individual Hawks did best.

Put together, the two games illustrate a general result from game theory: repeated interaction with the same partner can favour restraint even in a game where Hawk wins any single encounter, while anonymous, one-off interactions favour the individually dominant strategy. This is one reason why reputation, memory, and repeated encounters matter so much for the evolution of cooperation, both in nature and in human societies.

Think about what this means in terms of human cooperation and conflict avoidance.

Can you see the value of understanding your opponent/competitor?

Part 7: Appendix - extras

Exponential growth in detail

One of the foundational concepts in population dynamics is the exponential growth model. This model is a simplified representation of how populations grow when resources are not limiting. While it's rare for populations to experience uninhibited growth for extended periods, understanding this basic model is crucial for understanding more complex dynamics that include factors like resource limitations, predation, and disease.

In this chapter, we will explore the mathematical framework that describes exponential growth in both discrete and continuous time settings. We will introduce key parameters such as the per-capita birth rate, death rate, and intrinsic rate of increase. We will also demonstrate how to apply these models using R programming, offering a practical dimension to these theoretical concepts.

Learning outcomes:

- Understand the fundamental equations that underpin exponential growth in populations.
- Differentiate between discrete and continuous time models.
- Interpret the role and significance of various parameters in exponential growth models.
- Use R programming to analyse exponential growth scenarios with real data.

Nomenclature

Symbol	Meaning	Alternatives
R_m	Per capita rate of increase per time step, per capita population growth rate (discrete time)	R, r_d (Gotelli)
r_m	Intrinsic rate of increase (continuous time)	r. From a life table, $r_m \approx \ln(R_0)/T$, which Neal calls r_c, the capacity for increase
λ	Population growth rate, population multiplication rate	
R_0	Net reproductive rate: the expected number of female offspring per female, per generation ($\sum l_x m_x$)	Not the same as the "basic reproduction number" of epidemiology, which uses the same symbol

Discrete time model

We'll first look at *discrete* population growth, which is the population growth that is considered to grow over distinct time steps (e.g., 1 year intervals). This is also called **Geometric Population Growth**.

A population at time t has size N: N_t.

After one time interval, there will be some births (B), and some deaths (D). Births will have the effect of increasing the population size at time $t+1$ while deaths will decrease population size. The population at the next time step (N_{t+1}) is thus:

eqn. 1. $N_{t+1} = N_t + B - D$

We can think about the birth/death processes on a per-individual basis, and use per-capita birth rate (b) and per-capita death rate (d). We can think of b as the average number of offspring produced by an individual during the time interval starting at N_t and ending at N_{t+1}. Similarly, d is the probability that an individual alive at N_t will die at some point during the interval.

The **total** number of births within a time interval depends on the number of individuals there are at the start of the interval (N_t), and the total number of births in the population during the time interval is bN_t. Similarly, the total number of deaths is dN_t. Thus:

eqn. 2. $N_{t+1} = N_t + bN_t - dN_t$.

This equation can be simplified to:

eqn. 3. $N_{t+1} = N_t + (b - d)N_t$.

Furthermore, because the expression $(b - d)$ is important, we can give it its own symbol, R_m (as in Neal). R_m is called the *per capita rate of increase*. (Note that the nomenclature varies depending on the book/paper! In other places this is called R or r_d.)

So we now have:

eqn. 4. $N_{t+1} = N_t + R_m N_t$.

Or, if we are only interested in the **change** in population size:

eqn. 5. $\frac{\Delta N}{\Delta t} = R_m N_t$.

Equation 4 can be simplified again, by factoring out the N_t on the right hand side:

eqn. 6. $N_{t+1} = (R_m + 1)N_t$.

The quantity $(R_m + 1)$ is given its own symbol: λ, the **population multiplication rate** (also known as the "finite rate of increase").

eqn. 7. $N_{t+1} = \lambda N_t$.

By rearranging this equation (eqn. 7) we can see that λ is simply the ratio of population size at time $t + 1$ and t:

eqn. 8. $\lambda = N_{t+1}/N_t$.

It follows, therefore, that when the population is neither growing nor declining (when $N_{t+1} = N_t$), $\lambda = 1$ (and $R_m = 0$).

Calculating N for any future time point

Assuming the population growth rate remains constant, we can calculate the population at any time in the future.

As a starting point, consider equation 7: $N_{t+1} = \lambda N_t$.

If we want to calculate N_{t+2}, we would need to plug in N_{t+1} instead of N_t: $N_{t+2} = \lambda N_{t+1}$,

and, since we know that $N_{t+1} = \lambda N_t$,: $N_{t+2} = \lambda\lambda N_t$.

Similarly, if we wanted to calculate N_{t+3}, we'd end up with: $N_{t+3} = \lambda\lambda\lambda N_t$.

This can be simplified by raising λ to a suitable power, and using the starting population at time = 0, N_0:

eqn. 9. $N_t = \lambda^t N_0$.

This should be familiar to those of you that did (or remember!) the concept of geometric series which was covered in MM554 Mathematics for Biology.

Applying the model

If we plot exponential growth on a log scale we can see that it is straight line. For example, in the plot below I show the sequence for a population with a starting population of 1 and a λ (population multiplication rate) of 1.2 (i.e., the population increases by 20% each year). In (A) the time series is plotted on the natural scale while in (B) it is plotted on the log scale.

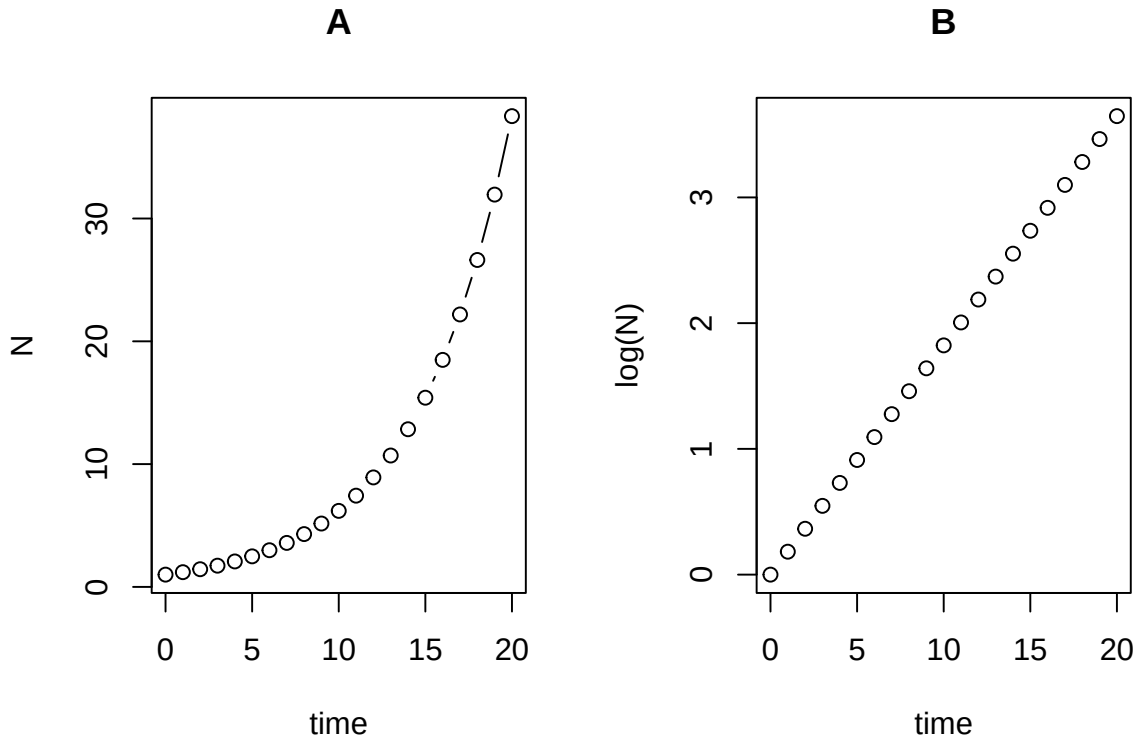
```
# Set initial population and lambda
startPop <- 1
lambda <- 1.2

# Create a data frame with time and corresponding population
df1 <- data.frame(time = 0:20)
df1$N <- lambda^df1$time * startPop
```

```
# Plotting the data
par(mfrow=c(1,2))

# Plot on natural scale
plot(df1$time, df1$N, type = "b", xlab = "time", ylab = "N")
title("A")

# Plot on log scale
plot(df1$time, log(df1$N), type = "b", xlab = "time", ylab = "log(N)")
title("B")
```



In fact, we can linearise the relationship by log transforming both sides of equation 9:

$\ln N_t = \ln(\lambda^t N_0)$,

which can be re-written as:

$\ln N_t = \ln(\lambda)t + \ln(N_0)$.

This looks familiar. Indeed, the equation of a straight line ($y = ax + b$) maps onto this. In this equation, the slope a is equivalent to $\ln(\lambda)$ and the intercept (b) is equivalent to $\ln(N_0)$.

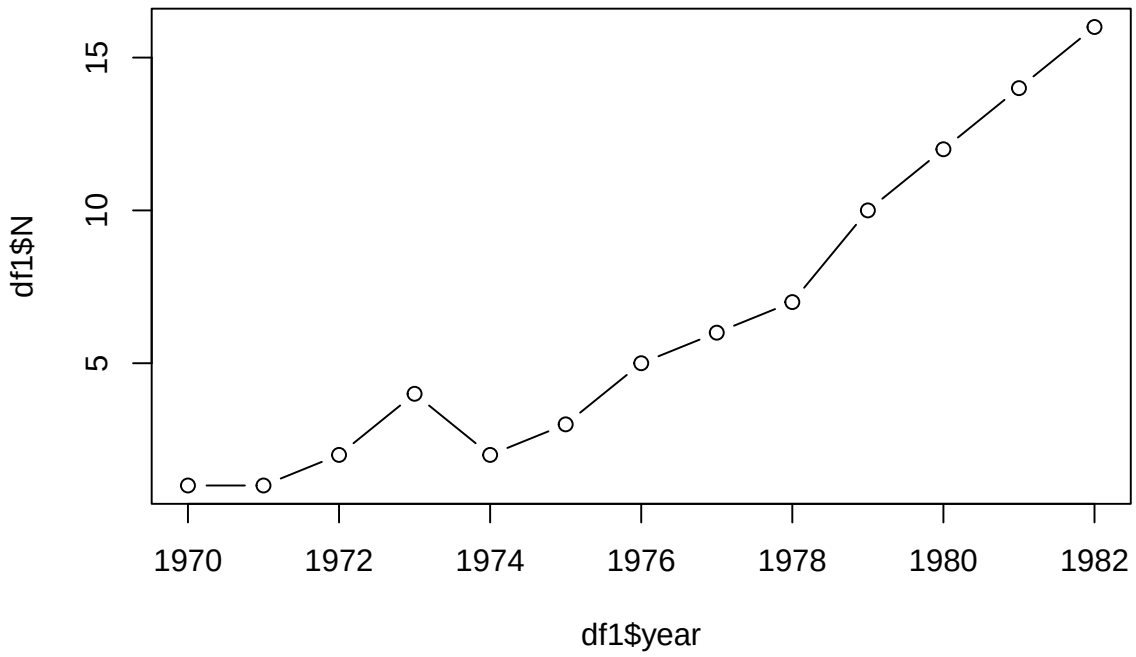
This is very convenient because now we can use simple regression methods to estimate $\ln(\lambda)$ (the slope) of the relationship, and therefore the value of λ (or R_m, which is $\lambda - 1$).

Real-World Application: Breeding Pairs of Merlin (Falco columbarius)

To bring these mathematical concepts to life, let's consider an example from the Neal textbook that focuses on the population of breeding pairs of Merlin, a species of small falcon. The data spans from 1970 to 1982 and shows the following population sizes (breeding pairs): `1,1,2,4,2,3,5,6,7,10,12,14,16`

This is not much data, so we can simply put it into R manually, and plot it, like this:

```
df1 <- data.frame(year = 1970:1982,
                  N = c(1,1,2,4,2,3,5,6,7,10,12,14,16))
plot(df1$year,df1$N, type = "b")
```



Observations and Context

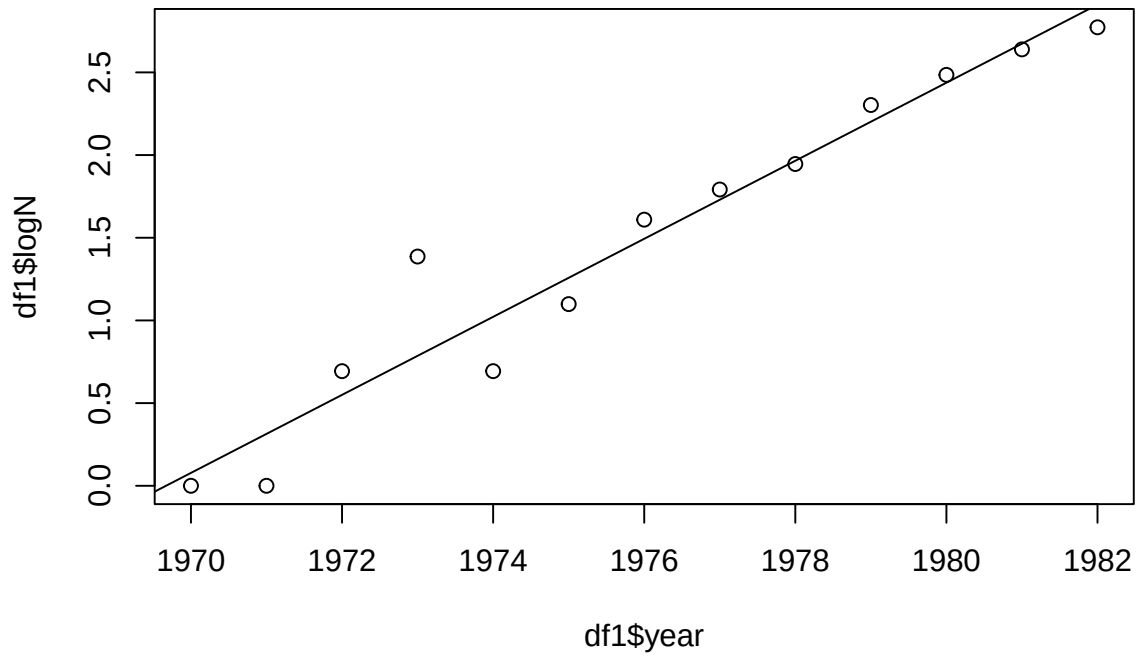
It is clear from the graph that the number of breeding pairs is increasing over the years, albeit with some fluctuations. These fluctuations could be due to various factors such as changes in food availability, predation, or human activity. However, the general trend suggests growth.

Applying the Exponential Growth Model

To analyse this data, we'll use the exponential growth model. We'll plot the data on a log scale and fit a regression model:

```
df1$logN <- log(df1$N)
plot(df1$year,df1$logN)

mod1 <- lm(logN~year,data = df1)
abline(mod1)
```



Model summary output:

```
##
## Call:
## lm(formula = logN ~ year, data = df1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.32858 -0.13684 -0.01967  0.10104  0.60054
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -464.77072   36.12270  -12.87 5.66e-08 ***
## year           0.23596    0.01828   12.91 5.48e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2466 on 11 degrees of freedom
## Multiple R-squared:  0.9381, Adjusted R-squared:  0.9324
## F-statistic: 166.6 on 1 and 11 DF,  p-value: 5.478e-08
```

Interpretation

The summary of the model indicates that the slope of the relationship between year and logN is 0.236. This slope is $\ln(\lambda)$, so the population multiplication rate (λ) is $e^{\text{slope}} = 1.266$, the intrinsic rate of increase (r_m) is the slope itself = 0.236, and the per capita rate of increase ($R_m = \lambda - 1$) is 0.266.

Strictly, this slope is the *realised* average growth rate over 1970-1982. It equals the intrinsic rate r_m only if growth was not slowed by density dependence during that period (Neal returns to the merlins in the chapter *Logistic Growth*).

This suggests that the number of breeding pairs is indeed growing exponentially over the time period studied, with a specific rate of increase. Understanding this rate is crucial for conservation efforts, because it provides insights into the population's resilience, its demands on the ecosystem, and how it might respond to future changes in environmental conditions or conservation policies.

Continuous time model

In contrast to the discrete time model, the continuous time model allows us to understand population growth without the constraint of clear time intervals. This is particularly useful for studying populations that don't have breeding seasons or for those that experience continuous births and deaths (e.g. bacteria, humans).

In continuous time, the population changes smoothly according to

eqn. 10. $\frac{dN}{dt} = r_m N$,

where r_m is the intrinsic rate of increase: the instantaneous per capita birth rate minus death rate ($r_m = b - d$, with b and d now instantaneous rates rather than per time step). Solving this equation gives the population size at any time t:

eqn. 11. $N_t = N_0 e^{r_m t}$.

Compare this with the discrete-time result, eqn. 9: $N_t = \lambda^t N_0$. The two describe the same growth if $\lambda = e^{r_m}$, or equivalently

eqn. 12. $r_m = \ln(\lambda)$: "r_m is the natural log of lambda", which is the same as saying $r_m = \ln(N_{t+1}/N_t)$ for a population growing at a constant rate.

To back-transform from a natural log, we use the exponential. Therefore, $\lambda = e^{r_m}$: "lambda is the exponential of r_m".

This allows us to relate λ and r_m directly. From the discrete model (above), we know that $\lambda = 1 + R_m$, so:

$1 + R_m = e^{r_m}$

$R_m = e^{r_m} - 1$

Doubling time

A useful way to express a growth rate is the time it takes the population to double. Setting $N_t = 2N_0$ in eqn. 11 gives $2 = e^{r_m t_d}$, so the doubling time is

eqn. 13. $t_d = \ln(2)/r_m = \ln(2)/\ln(\lambda)$.

For the merlins, $t_d = \ln(2)/0.236 = 2.9$ years.

```
r_m <- as.vector(coef(mod1)[2])  # slope of the merlin regression
log(2) / r_m                     # doubling time in years
```

```
## [1] 2.937516
```

Changing the time scale

Because r_m is a rate per unit time, it is easy to convert between time scales: just divide or multiply by the number of time units. For example, a population with $\lambda = 2$ per week has $r_m = \ln(2) = 0.693$ per week, which is $0.693/7 = 0.099$ per day, so λ per day is $e^{0.099} = 1.104$. (You cannot simply divide λ by 7.)

Zero population growth

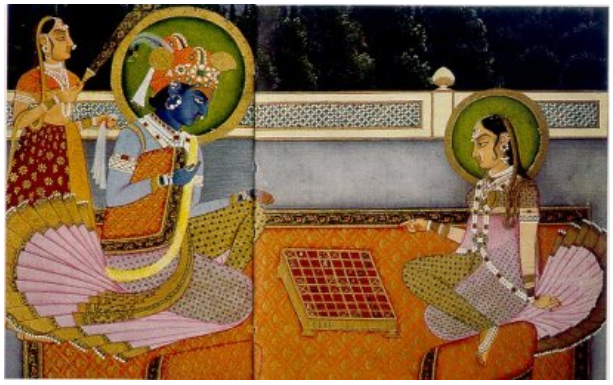
When the population is steady with zero growth, $r_m = 0$ and $\lambda = 1$. It's important to note that the relationship $\lambda = r_m + 1$ only holds true when the population is not growing or shrinking, and approximately when r_m is small (e.g. $r_m = 0.05$ gives $\lambda = 1.051$). It should not be generalised. Do not make the common mistake to think that λ is simply $r_m + 1$!

The legend of Ambalappuzha

Exponential growth is a powerful concept. To help us grasp it better let's use an ancient Indian chess legend as an example.

Learning outcomes

- Use of Excel formulae for mathematical modelling.
- Understanding the multiplicative process of exponential/geometric growth.



According to legend, Lord Krishna once appeared in the form of a wise man in the court of the king and challenged him to a game of chess. The king was a chess enthusiast and naturally accepted the invitation.

The king asked the wise man to choose a prize in case he won. The old man told the king that he had few material needs and that all he wished was a few grains of rice.

He added that the amount of rice itself should be determined using the chessboard in the following manner: one grain of rice would be placed in the first square, two grains in the second square, four in the third square, and so on. Every square would have double the number of grains of its predecessor.

Upon hearing this, the king was unhappy, since the man requested only a few grains of rice instead of any of the other riches of the kingdom, which the king would have been happy to donate (he was a generous guy). He requested the old man to add other items to his prize, but the man declined.

So the game of chess started and, needless to say, the king lost the game so it was soon time to pay the old man his prize. As he started adding grains of rice to the chessboard, the king soon realised the true nature of the wise man's demand. The royal granary soon ran out of rice and the king realised that even if he provided all the rice in his kingdom and even the whole of India, he would never be able to fulfil the promised reward. He was distraught that he could not fulfil his promise!

Seeing the king upset, the wise man appeared to the king in his true form, that of Lord Krishna. He told the king that he did not have to pay the debt immediately but could pay him over time. The king would serve rice in the temple freely to the pilgrims every day until the debt was paid off. And that is why the Ambalappuzha Temple in India still serves rice pudding (palpayasam) to pilgrims: the debt is still being paid off.

Use the Excel sheet RiceOnAChessboard.xlsx to calculate the quantity of rice that the king owed.

A grain of rice weighs 25mg, what weight of rice did the king owe in total, in kg?

Animals/plants, not grains of rice

Imagine that instead of rice, we were talking about population growth (e.g. bacteria or rabbits) reproducing every time step.

Quick exercise (optional)

We have ~100 breeding pairs of great tits (*Parus major*, musvit) in woodland around SDU. Each female lays ~6 eggs per year with an assumed 50:50 sex ratio, so each female produces 3 daughters per year.

- **If survival were 100%**, how many females would there be after 5 years? Assume that every female (adults and all their daughters) survives every year, that daughters breed in the year after they hatch, and that every female breeds every year. What is λ for the females?
- How does your answer change if, instead, adults die after breeding once (so only the daughters are left to breed next year)?
- **Now relax that assumption:** allow realistic survival and recruitment. How do your answers change?

Discussion prompts

- **Is this simple multiplicative model realistic?**
  - Why / why not?
- **Key processes to consider:**
  - Survival < 100%; age/stage-specific survival and fecundity
  - Density dependence (resources, territories, disease, predation)
  - Carrying capacity and feedbacks on reproduction/survival
- **Variation over time and space:**
  - Stochasticity (good/bad years), seasonality, environmental drivers
  - Dispersal, immigration/emigration
- **Model structure:**
  - Age/stage structure (e.g. juvenile vs adult), sex ratio, mating system
  - Time-step choice (daily/weekly/annual) and what processes occur within a step

Optional: Try these calculations in R

You can do this kind of calculation easily in *R*. Try this.

```
myData <- data.frame(Squares = 1:64,nRice = NA)
myData$nRice[1] <- 1

for (i in 2:64){
  myData$nRice[i] <- myData$nRice[i-1]*2
}
```

Now we can look at the top and bottom of the 64 row data frame like this:

```
head(myData)
```

```
##   Squares nRice
## 1       1     1
## 2       2     2
## 3       3     4
## 4       4     8
## 5       5    16
## 6       6    32
```

```
tail(myData)
```

```
##    Squares        nRice
## 59      59 2.882304e+17
## 60      60 5.764608e+17
## 61      61 1.152922e+18
## 62      62 2.305843e+18
## 63      63 4.611686e+18
## 64      64 9.223372e+18
```

Output preview:

```
##   Squares nRice
## 1       1     1
## 2       2     2
## 3       3     4
## 4       4     8
## 5       5    16
## 6       6    32
```

```
##    Squares        nRice
## 59      59 2.882304e+17
## 60      60 5.764608e+17
## 61      61 1.152922e+18
## 62      62 2.305843e+18
## 63      63 4.611686e+18
## 64      64 9.223372e+18
```

And we can sum up the total number of grains of rice on the 64 squares of the board like this:

```
sum(myData$nRice)
```

```
## [1] 1.844674e+19
```

Output:

```
## [1] 1.844674e+19
```

To put that *HUGE* number in context, if a grain of rice weighs 25mg (0.000025kg), then we'd have 461,168,601,842,739 kg. That's 461 gigatonnes, or roughly 600 to 900 times annual world rice production (depending on whether you use paddy or milled weight)!

From population biology to fitness

The purpose of this practical is to draw clear links between the first part of the course (population biology) and the second part of the course (evolution).

Learning outcomes:

- Understanding the relationship between population growth and the concept of fitness.
- Understanding the concept of an evolutionary trade-off.

We will focus on the concept of **fitness**.

Fitness is a slippery concept, but it is widely accepted that it is closely related to population growth rate. In this class you will explore this concept using some mathematical modelling.

This practical uses RStudio (R). It is similar to the previous exercise on matrix population models, but ask for help if you get stuck!

An *in silico* experiment

As you learned in the classes on age- and stage-structured population dynamics, differences in survival and reproduction can be modelled using matrix population models (MPMs). These models can be simple or complex, and can be thought of as mathematical descriptions of the life history of the species (or population) in a particular environment.

Earlier in the course you will have played with construction and analyses of these models by creating MPMs for species with different life histories such as high juvenile survival, or low juvenile survival etc.

We will first need to load the `popdemo` package like this. Note that if you have not installed this package you should first install it with the command `install.packages("popdemo")`.

Let's set up our baseline model. This model describes a population of some mammal species which we have divided into 3 stages: juvenile, adult and senescent (old).

You can "read" the matrix by looking at the columns and rows: a value in column **j** and row **i** tells you the "transition" **from** stage j **to** stage i. For example, column **2**, row **1** says that an individual in the adult age class produces an average of 4 babies, and column **3**, row **1** says that one from the senescent age class produces 2 babies. Juveniles have a probability of 0.1 (10% chance) to survive to adulthood (and they reach maturity in 1 year, so there is no "stasis" where they can remain being juveniles). Adults can survive in 2 ways, they can survive and remain as adults (probability = 0.8) or they can survive and transition to being in the senescent age class (probability = 0.1). Therefore, the total survival probability is 0.9. Senescent adults survive less well (probability = 0.4).

We can project a population like this:

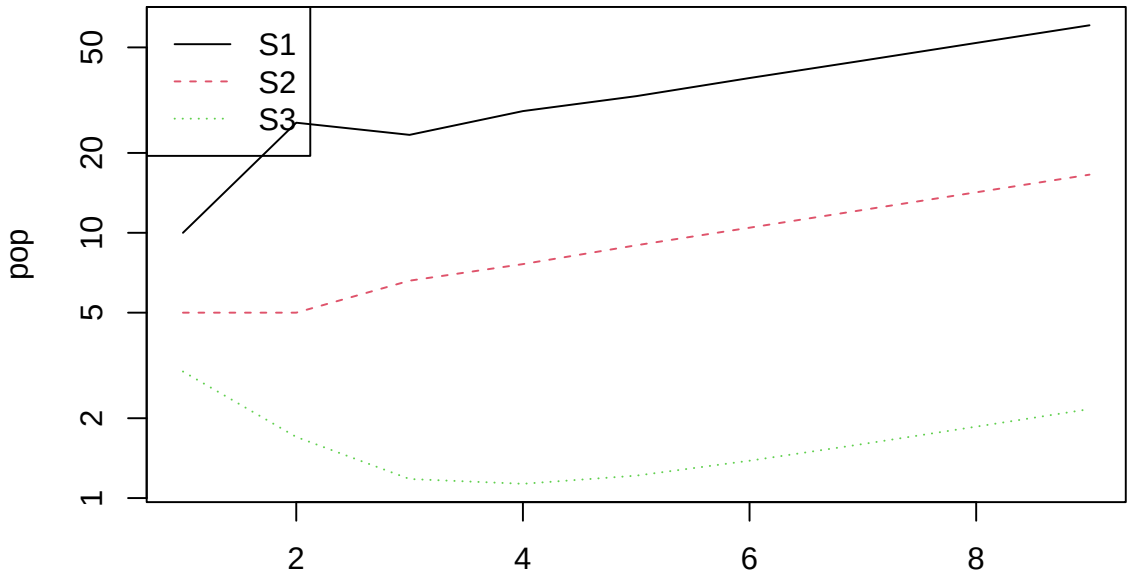
Take a look at `pr`, the projected population. This gives you the total population size, and below that the population sizes in each stage.

```
## 1 deterministic population projection over 8 time intervals.
##
## [1] 18.00000 32.70000 31.18000 37.51200 42.93080 50.15272 58.36605 68.04116
## [9] 79.29730
```

You can access the population sizes of the different stages using `vec(pr)`.

```
##             S1       S2       S3
##  [1,] 10.00000  5.00000 3.000000
##  [2,] 26.00000  5.00000 1.700000
##  [3,] 23.40000  6.60000 1.180000
##  [4,] 28.76000  7.62000 1.132000
##  [5,] 32.74400  8.97200 1.214800
##  [6,] 38.31760 10.45200 1.383120
##  [7,] 44.57424 12.19336 1.598448
##  [8,] 51.97034 14.21211 1.858715
##  [9,] 60.56588 16.56672 2.164697
```

Let's plot this… Check out how, after a "transient" period, there is exponential growth in all stages of the population. The population is growing steadily with a fixed population growth rate (λ).



You can see that the population is increasing and we can calculate the precise population growth rate (λ) like this:

```
## [1] 1.165587
```

Thus, the population is growing at 16.56% per year.

So where does evolution come in?

The link to fitness

In this population consider that suddenly a mutation arises in an individual parent that causes it to give more care to their offspring. For example, maybe they provide milk with a higher fat content, or build a safer nest. Whatever the mechanism, let's assume that it results in a small increase in juvenile survival.

We can simulate this by increasing the juvenile survival in the matrix model from 0.10 to 0.11.

What effect does that have on population growth rate?

```
## [1] 1.192317
```

The small increase in juvenile survival has resulted in a small increase in population growth rate, from 16.56% to 19.23% per year.

If you consider that the original population now consists of two genotypes – "ordinary" and "caring" – what do you think will happen to the percentage of the two genotypes through time?

You can be sure that the proportion of the caring genotype will grow faster than the ordinary genotype. It is the FITTER genotype. (Treating each genotype's λ as its fitness assumes that offspring inherit their parent's life history exactly, and that growth is density-independent, as in this model.)

Introducing a trade-off

It is common that apparently beneficial behaviours or innovations come at a cost. In evolutionary biology these are called **trade-offs**.

Let's explore such a trade-off now and see how it might affect fitness.

We'll stick with the same example above, but we'll introduce a new genotype that has a trade-off between juvenile survival and old-adult survival.

The benefit is clear: a change in adult behaviour or physiology increases juvenile survival a little bit. But such changes often come with a cost: The new genotype allocates extra resources to babies but this exhausts the adults causing older adults to have very small survival probability.

Is this new genotype viable? In other words, is the fitness of the genotype greater than that of the original genotype? If so, the new genotype will come to dominate the population.

Modify the matrix to reduce old adult survival to, say 0.05 (5% survival) and re-calculate the population growth rate.

Is this "trade-off genotype" fitter than the original one? i.e. is the small benefit worth the large cost?

Try doing the same thing for the prime-age adults. How much can you reduce survival before the cost is not worth bearing?

From plain English to a matrix model

Background

This activity is designed for quick in-class practice translating a verbal life-history description into a matrix population model. It connects directly to the workflow in the chapter Matrix population modelling.

Learning outcomes:

- Convert a verbal life-history description into a matrix population model.
- Build and interpret a simple projection from initial stage abundances.

Worked example

Inputs

Suppose juveniles do not reproduce, each adult contributes 2 offspring per year that survive to be counted as juveniles at the next census, juvenile survival is 0.4, and adult survival is 0.7. A simple stage matrix would be:

Steps

1. Put fecundity terms in the top row (when all newborns enter stage 1). A fecundity term is the number of offspring counted at the *next* census per individual now, so it must include any survival needed to get there (see the fecundity box in the chapter "Matrix population modelling"). If a description gives you the number of babies born, multiply by the survival needed to reach the next count.
2. Put survival/transition terms in the subdiagonal/diagonal entries.
3. Assemble the matrix in row = stage at $t + 1$, column = stage at t form.

Output and interpretation

$$A = \begin{bmatrix} 0 & 2 \\ 0.4 & 0.7 \end{bmatrix}$$

The key idea is to map each biological statement to a matrix entry: fecundity goes in the top row, survival and stage transitions go on the sub-diagonal or main diagonal.

Your task

Read this description of the life history of the red-tailed hawk (*Buteo jamaicensis*):

> We assumed that the Red-tailed hawk life history could be described in terms of three major life stages: hatchling, juvenile, and adult (generally the third year of life and beyond). Adults are the primary reproductive stage, and produce an average of 3 new hatchlings each year. Juveniles tend to produce only 1 new hatchling per year on average. We assumed that adults experienced an average of 18% mortality each year. Juvenile mortality was set at 30% per year. Approximately 5% of juveniles fail to transition to adults, remaining in the juvenile phase. Finally, hatchlings had a 20% chance of surviving and transitioning to become juveniles. We initialized the population with 1000 hatchlings, 150 juveniles, and 5 adults.

Assume that the population is counted once a year, and that the numbers of "new hatchlings" per juvenile and per adult are already the numbers counted as hatchlings at the next census (so they can go straight into the matrix as fertilities).

1. Draw a 3-stage life cycle diagram (hatchling, juvenile, adult).
2. Translate the verbal description into a 3x3 projection matrix. State any assumptions you make about the juvenile stasis and transition rates.
3. Use the initial population sizes given (1000 hatchlings, 150 juveniles, 5 adults) and project the population for 20 time steps.
4. Calculate the dominant eigenvalue and interpret whether the population grows or declines.
5. Compare your approach with the workflow in the chapter Matrix population modelling.

Takeaways

- Verbal life-history descriptions can be translated directly into matrix entries with clear assumptions.
- Simple projections reveal whether the implied population is growing or declining.

Continuous traits from discrete genes

Background

This activity uses a fast classroom simulation to show how many small genetic effects, together with some environmental variation, can generate a continuous, approximately normal trait distribution. It connects to the heritability practical in Chapter @ref(heritability-from-a-linear-regression).

Reading: Neal, chapter *Polygenic Inheritance and Quantitative Genetics* (Nilsson-Ehle's wheat kernel colour example makes the same point).

Learning outcomes:

- Explain how many genes with small additive effects create many trait classes, and how environmental variation blurs these classes into a continuous distribution.
- Compare a classroom simulation with a simple R simulation.
- Relate the simulation to the partition $V_P = V_G + V_E$ and to heritability.

Worked example

Inputs

- Baseline trait value: `35`
- Number of genes: `25`
- Each gene contributes `1` unit when ON, and each gene is ON with probability 0.5

Steps

1. Compute expected number of ON genes under 50:50 ON/OFF: `25 * 0.5 = 12.5`.
2. Add expected genetic contribution to baseline.

Output and interpretation

Expected trait value is `35 + 12.5 = 47.5`. Individual values vary around this because each gene is ON or OFF at random. The genetic variance is $25 \times 0.5 \times 0.5 = 6.25$ (the variance of a binomial distribution). Notice that, with genes alone, every trait value is a whole number: there are many classes (35 to 60), but they are still discrete.

Your task

You will need the instructions and the Excel spreadsheet (download them):

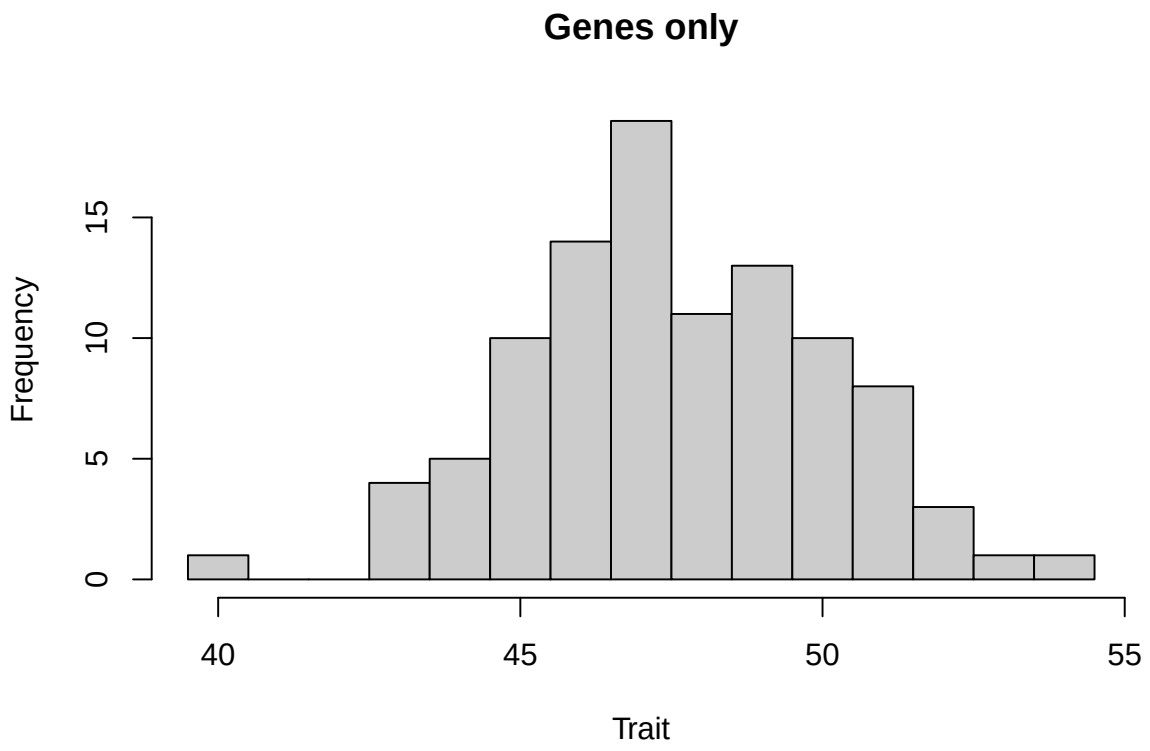
- Continuous_Traits_From_Discrete_Genes.docx (instructions)
- Continuous_Traits_From_Discrete_Genes.xlsx (spreadsheet: each time you press Enter, it re-randomises 25 ON/OFF genes and gives the trait value of a new individual)

1. **Classroom simulation.** Each student is a gene. In each round, every student flips a coin: heads = ON (1), tails = OFF (0). Count the number of ONs in the class and add it to a baseline of 35. That is the trait value of one individual. Repeat for at least 30 rounds (30 individuals), and record each value. (Alternatively, use the spreadsheet to generate at least 30 individuals.)
2. Plot a quick histogram of the observed trait values. Describe the shape.
3. Now simulate the same process in R for 100 individuals and 25 genes (first chunk below). Compare your histogram with the classroom version.
4. Real traits are also affected by the environment (food, temperature, chance events during development). Add a small random environmental effect to each individual (second chunk below) and compare the two histograms.
5. Relate the outcome to the concepts of genetic and environmental variance in Chapter @ref(heritability-from-a-linear-regression).

```
baseline <- 35
n_genes <- 25
n_individuals <- 100

gene_on <- rbinom(n_individuals, size = n_genes, prob = 0.5)
trait_value <- baseline + gene_on

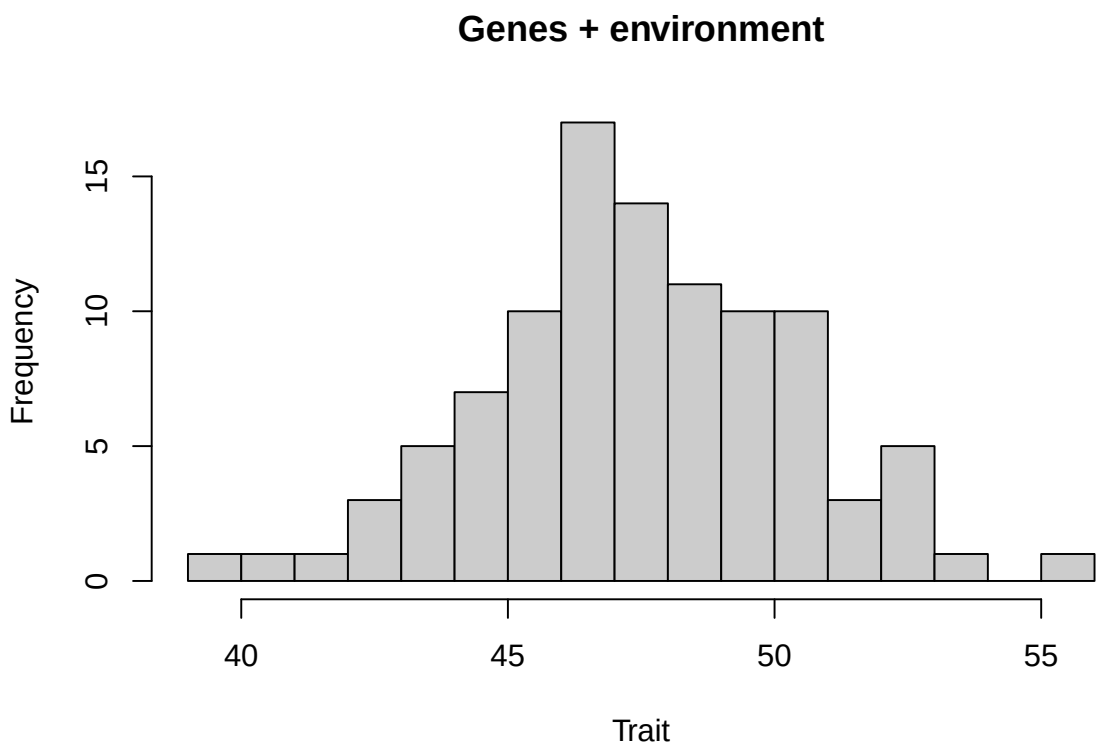
hist(trait_value, breaks = seq(min(trait_value) - 0.5, max(trait_value) + 0.5, 1),
     col = "gray80", main = "Genes only", xlab = "Trait")
```



Now add an environmental effect, drawn from a normal distribution with mean 0 and standard deviation 1.5 (so the environmental variance, V_E, is $1.5^2 = 2.25$):

```
environment_effect <- rnorm(n_individuals, mean = 0, sd = 1.5)
trait_value_env <- trait_value + environment_effect

hist(trait_value_env, breaks = 15, col = "gray80",
     main = "Genes + environment", xlab = "Trait")
```




```
var(trait_value)       # genetic variance (expected 6.25)
```

```
## [1] 6.290404
```

```
var(trait_value_env)   # phenotypic variance, V_P = V_G + V_E (expected 8.5)
```

```
## [1] 8.240729
```

- Are the trait values in the second histogram still whole numbers? Why does the environment make the distribution continuous?
- In this model all the genetic variance is additive. What is the expected heritability, $h^2 = V_A/V_P$? (Use the expected variances: 6.25 and 8.5.)
- What happens to the heritability if you increase the standard deviation of the environmental effect to 4?

Takeaways

- Many genes with small additive effects produce many trait classes and a roughly bell-shaped distribution; environmental variation blurs the classes into a continuous distribution. (Normality is an approximation, not a law.)

- Phenotypic variance is the sum of genetic and environmental variance ($V_P = V_G + V_E$), so the same genes give a lower heritability in a more variable environment.
- Random genetic variation produces a spread of trait values even with identical environments.

Building a phylogenetic tree

Background

This activity builds intuition for phylogenetic trees by using discrete morphological characters. It follows the classroom activity by Wojciech Grajkowski, *Building a phylogenetic tree* (Bioscience Explained 6(2)), which you can download here as a PDF. The PDF includes the organism cards and a worksheet.

Reading: Neal, chapter *Darwin's Evolutionary Theories* (common descent and Darwin's "tree of life").

Learning outcomes:

- Build a simple character table for a set of organisms.
- Compute pairwise differences and translate them into a tree.
- Relate morphology-based trees to the idea of evolutionary relatedness, and recognise when similarity may not reflect relatedness.

Key idea

With six organisms and seven characters, you can compute a pairwise difference table, then group the most similar organisms first. The internal nodes of the tree represent inferred common ancestors: the pairs joined first represent the *most recent* common ancestors, and the deeper splits represent older ones.

A caveat: similarity is not the same as relatedness. Grouping organisms by overall similarity (the method used here, which is called UPGMA when done by computer) recovers the true tree only if characters change at a roughly constant rate in all lineages and the same character state never evolves more than once. In reality, unrelated organisms can evolve the same feature independently (**convergence**, e.g. wings in birds and bats, or loss of thorns in several plant lineages), and features can be lost again (**reversal**). Some shared features are also simply inherited from a distant ancestor of the whole group (ancestral states) and say nothing about who is most closely related. Modern phylogenetics therefore groups organisms by shared *derived* characters (new features that arose in their common ancestor), uses an outgroup to decide which states are ancestral, and increasingly uses DNA sequences. Keep this in mind when you interpret your tree.

Worked example

The PDF's example has six beetle-like organisms (A to F), described by seven characters:

```
beetles <- data.frame(
  row.names    = c("A", "B", "C", "D", "E", "F"),
  body_shape   = c("Oval", "Oval", "Round", "Oval", "Round", "Round"),
  colour       = c("Yellow", "Yellow", "Orange", "Pink", "Pink", "Orange"),
  head_colour  = c("White", "Black", "Black", "White", "Black", "Black"),
  antennae     = c("No", "No", "No", "Yes", "Yes", "No"),
  jaws_visible = c("Yes", "Yes", "Yes", "No", "No", "Yes"),
```

```
  spots        = c("Yes", "Yes", "Yes", "No", "No", "Yes"),
  spot_colour  = c("Black", "Black", "Black", "None", "None", "White")
)
beetles
```

```
##   body_shape colour head_colour antennae jaws_visible spots spot_colour
## A       Oval Yellow       White       No          Yes   Yes       Black
## B       Oval Yellow       Black       No          Yes   Yes       Black
## C      Round Orange       Black       No          Yes   Yes       Black
## D       Oval   Pink       White      Yes           No    No        None
## E      Round   Pink       Black      Yes           No    No        None
## F      Round Orange       Black       No          Yes   Yes       White
```

Next, count the number of characters that differ between each pair of organisms. This is the "data matrix" in the PDF:

```
n <- nrow(beetles)
differences <- matrix(0, nrow = n, ncol = n,
                      dimnames = list(rownames(beetles), rownames(beetles)))
for (i in 1:n) {
  for (j in 1:n) {
    differences[i, j] <- sum(beetles[i, ] != beetles[j, ])
  }
}
differences
```

```
##   A B C D E F
## A 0 1 3 5 7 4
## B 1 0 2 6 6 3
## C 3 2 0 7 5 1
## D 5 6 7 0 2 7
## E 7 6 5 2 0 5
## F 4 3 1 7 5 0
```

Finally, join the most similar organisms first (Figure @ref(fig:beetle-tree)). The function `hclust` with `method = "average"` does this automatically: at each step it joins the two closest groups, measuring the distance between groups as the average number of differences between their members.

```
tree <- hclust(as.dist(differences), method = "average")
plot(tree, main = "", xlab = "", sub = "",
     ylab = "Average number of differences")
```

A and B differ in only one character, as do C and F, so these pairs are joined first (at a height of 1): they share the most recent common ancestors. D and E (2 differences) are joined next, then the A+B and C+F groups (an average of 3 differences), and finally the two main groups. This is the same tree as the one drawn by hand in the PDF.

Now look back at the character table. On this tree, no single change can explain body shape or head colour. E is round like C and F, but groups with D, and A has a white head like D, but groups with B. At least one of these characters must have changed more than once (convergence or reversal). Which one, and in which direction, cannot be decided without knowing the ancestral states, for example from an outgroup.

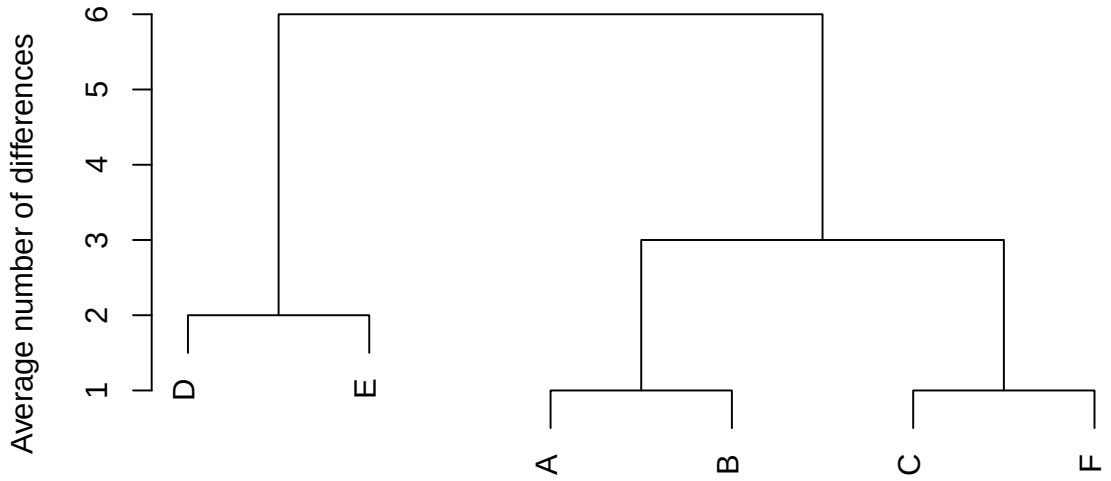


Figure 15: Tree for the six beetle-like organisms in the PDF, built by joining the most similar organisms first (UPGMA). The height of each join is the average number of character differences between the two groups.

Your task

Use the PDF as a guide for the classroom activity.

1. Pick six organisms (use the cards in the PDF: plants, birds or butterflies, or invent six organisms with clear traits).
2. Identify seven discrete characters that differ among the organisms (presence/absence, colour class, tail length category, etc.).
3. Fill in a character table and then compute a pairwise difference table.
4. Draw a phylogenetic tree by grouping the most similar organisms first, then adding deeper splits.
5. Identify any characters that look convergent on your tree (the PDF's "Additional questions").
6. Optional: compare your hand-drawn tree with an R result, by replacing the `beetles` table in the worked example with your own character table.

Takeaways

- Morphological characters can be converted into a simple distance matrix.
- The most similar pairs in the distance matrix are joined first, and represent the most recent common ancestors in the tree.
- Overall similarity is not the same as relatedness: convergence and reversals can mislead, which is why phylogenetics relies on shared derived characters and on many characters, including DNA.